cft-anyons Lab Book

From category data to provably rigorous CFTs

cft-anyons contributors

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Contents

1 Frontmatter, North Star, and Claim Status **15**
1.1 The Two Success Criteria . . . 15
1.2 Claim Status . . . 15
1.3 Evidence Chain . . . 15

2 Programme Map **16**
2.1 The Pipeline . . . 16
2.2 Stage 1 — Category Input . . . 16
2.3 Stage 2 — Microscopic Models . . . 16
2.4 Stage 3 — Lattice Symmetry Generators . . . 16
2.5 Stage 4 — Provable Continuum Limit . . . 16
2.6 Stage 5 — Rigorous (C)FT Target . . . 17
2.7 First Decisions . . . 17

3 Many-Particle Hilbert Spaces: the AND/OR/Vacuum Grammar **18**
3.1 Sources and Dependencies . . . 18
3.2 The Three Primitives . . . 18
3.3 Two Monoidal Structures and Their Units . . . 18
3.4 The Grammar and its Carrier . . . 19
3.5 Particle-Number Grading . . . 19
3.6 Lamport Dependency Skeleton . . . 19
3.7 Scope and Deferral . . . 20

4 Indefinite Particle Number: the Full Fock Completion **21**
4.1 Sources and Dependencies . . . 21
4.2 Two Direct Sums — the Load-Bearing Distinction . . . 21
4.3 Completed Tensor Powers . . . 21
4.4 The Full Fock Space . . . 21
4.5 The Grammar Generates the Indefinite-Particle Space . . . 22
4.6 Exchange Sectors Come Later . . . 22
4.7 Lamport Dependency Skeleton . . . 23
4.8 Scope and Pitfalls . . . 23

5 A Hilbert-Space Compiler Contract **24**
5.1 Status and Local Anchors . . . 24
5.2 Mini North Star . . . 24
5.3 Input Record . . . 24
5.4 Output Record . . . 25
5.5 Observable-Algebra Policies . . . 25
5.6 Lamport Dependency Skeleton . . . 25
5.7 Boundary . . . 25

6 Symmetry-Decorated Many-Particle Grammar **26**
6.1 Sources and Dependencies . . . 26
6.2 Decorated Atoms . . . 26
6.3 Propagation Through OR and AND . . . 26
6.4 The Derived Full-Fock Representation . . . 26
6.5 Intertwiners and Symmetry-Preserving Compilation . . . 27
6.6 Lamport Dependency Skeleton . . . 27
6.7 Boundary . . . 27

7 Sectors, Projections, Subspaces, and Quotients **28**
7.1 Sources and Dependencies . . . 28
7.2 The Triangle . . . 28
7.3 Finite-Group Invariants . . . 28
7.4 Coinvariant Quotient . . . 29
7.5 Observable-Algebra Distinctions . . . 29
7.6 Checked Example . . . 29
7.7 Lamport Dependency Skeleton . . . 29
7.8 Boundary . . . 30

8 Exchange Symmetry and Statistics as Sector Selection **31**
8.1 Sources and Dependencies . . . 31
8.2 Two Symmetry Layers . . . 31
8.3 Irrep-Labeled Exchange Projectors . . . 31
8.4 Quotient Presentations . . . 32
8.5 Para and Anyonic Boundaries . . . 32
8.6 Lamport Dependency Skeleton . . . 32
8.7 Compiler Consequence . . . 32

9 Scoping the Hilbert-Space Compiler **33**
9.1 Status, Sources, and Dependencies . . . 33
9.2 Compile Judgement . . . 33
9.3 Surface Language for the First Compiler . . . 33
9.4 Constructor Rules . . . 33
9.5 Example 1: A Maybe Qubit . . . 34
9.6 Example 2: Two Distinguishable Qubits . . . 35
9.7 Example 3: Derived Tensor-Power Closure . . . 35
9.8 Example 4: Two-Qubit Exchange-Irrep Sectors . . . 35
9.9 Example 5: A Finite Invariant Quotient . . . 36
9.10 Example 6: A Required Compile Error . . . 36

9.11 Lamport Dependency Skeleton . . . 36
9.12 Acceptance Criteria and Boundary . . . 36

10 Fibonacci Kinematics as a Fusion-Rule Sector **37**
10.1 Status, Sources, and Dependencies . . . 37
10.2 One Physical Atom, Two Charge Labels . . . 37
10.3 Ambient Path Space and Admissibility Projector . . . 37
10.4 Dimension Recurrence . . . 38
10.5 Small Examples . . . 38
10.6 Compiler Consequence . . . 38
10.7 Lamport Dependency Skeleton . . . 39
10.8 Boundary . . . 39

11 Why Lattice Symmetry Generators Are a Problem **40**
11.1 Status, Sources, and Dependencies . . . 40
11.2 The Minimal Quantum-Mechanical Input . . . 40
11.3 What the Lattice Adds . . . 40
11.4 The Continuum Target . . . 40
11.5 Intuition to Be Checked . . . 41
11.6 Source Boundary . . . 41

12 The Continuum Bulk Symmetry Target **42**
12.1 Status, Sources, and Dependencies . . . 42
12.2 Group-Level Target . . . 42
12.3 Vector-Field Derivation . . . 42
12.4 Boost-Time Relation . . . 43
12.5 Boundary . . . 43

13 A One-Dimensional Boost Produces a Momentum Current **44**
13.1 Status, Sources, and Dependencies . . . 44
13.2 Setup . . . 44
13.3 Commutator Calculation . . . 44
13.4 Uniform Spacing . . . 45
13.5 Interpretation . . . 45
13.6 Boundary . . . 45

14 Position-Dependent Bulk Generators **46**
14.1 Status, Sources, and Dependencies . . . 46
14.2 Continuum Picture . . . 46
14.3 Scalar Ramps . . . 46
14.4 Boost Candidates . . . 46
14.5 Rotation Candidates . . . 47
14.6 Compiler Moral . . . 47
14.7 Boundary . . . 47

15 Koo–Saleur as the Prototype **48**
15.1 Status, Sources, and Dependencies . . . 48
15.2 Continuum Side . . . 48
15.3 Lattice Side . . . 48
15.4 Scaling Representation Matters . . . 48
15.5 What This Foreshadows . . . 49
15.6 Boundary . . . 49

16 Acceptance Tests for Lattice Symmetry Candidates **50**
16.1 Status, Sources, and Dependencies . . . 50
16.2 Local Finite-Scale Checks . . . 50
16.3 Algebraic Checks . . . 50
16.4 Scaling Checks . . . 50
16.5 Observable Covariance Checks . . . 50
16.6 Acceptance Payload . . . 51
16.7 Boundary . . . 51

17 A Compiler Interface for Lattice Symmetry Candidates **52**
17.1 Status, Sources, and Dependencies . . . 52
17.2 Input Contract . . . 52
17.3 Candidate Generation . . . 52
17.4 Output Contract . . . 52
17.5 Failure Modes . . . 53
17.6 Compiler Principle . . . 53

18 Small Examples and the Lattice-Symmetry Research Queue **54**
18.1 Status, Sources, and Dependencies . . . 54
18.2 Example 1: Black-Box Quantum System . . . 54
18.3 Example 2: Open Uniform Chain . . . 54
18.4 Example 3: Nonuniform Chain . . . 54
18.5 Example 4: Periodic Chain . . . 54
18.6 Example 5: Square Lattice . . . 55
18.7 Example 6: Koo–Saleur Modes . . . 55
18.8 Research Queue . . . 55
18.9 Landing Point . . . 55

19 Why Galilean Symmetry Is a Different Lattice Target **56**
19.1 Status, Sources, and Dependencies . . . 56
19.2 The New Target . . . 56
19.3 Immediate Consequence for Lattices . . . 56
19.4 Compiler Implication . . . 56
19.5 Why Projective Data Matter Earlier . . . 57
19.6 Boundary for CA-18–CA-22 . . . 57

20 The Galilei Algebra as Newtonian Vector Fields **58**
20.1 Status, Sources, and Dependencies . . . 58
20.2 Generators . . . 58
20.3 Translation and Boost Brackets . . . 58
20.4 Rotation on Vectors . . . 58

20.5 Rotation-Rotation Bracket . . . 59
20.6 Checked Representatives . . . 59
20.7 Boundary . . . 59

21 Projective Galilean Symmetry and the Mass Central Term 60
21.1 Status, Sources, and Dependencies . . . 60
21.2 Why This Belongs in the Grammar . . . 60
21.3 Canonical One-Particle Calculation . . . 60
21.4 Compatibility with Time Evolution . . . 60
21.5 Central Extension Versus Vector Fields . . . 61
21.6 Source Gap . . . 61
21.7 Compiler Payload . . . 61

22 Lattice Galilean Boosts from Mass-Density First Moments 62
22.1 Status, Sources, and Dependencies . . . 62
22.2 Input Data . . . 62
22.3 Discrete Continuity Assumption . . . 62
22.4 First-Moment Derivation . . . 62
22.5 Relation to Center of Mass . . . 63
22.6 What Is and Is Not Proven . . . 63

23 Compiler Rules and Examples for Galilean Targets 64
23.1 Status, Sources, and Dependencies . . . 64
23.2 Target Declaration . . . 64
23.3 Required Input Slots . . . 64
23.4 Generated Candidates . . . 64
23.5 Example 1: One Free Particle . . . 65
23.6 Example 2: Number-Conserving Open Chain . . . 65
23.7 Example 3: Hamiltonian Density Only . . . 65
23.8 Example 4: Spin Chain with No Particle Number . . . 65
23.9 Example 5: Direct Sum of Mass Sectors . . . 65
23.10Next Obligations . . . 66

24 Gaussian Boson Lorentz Generator Roadmap 67
24.1 Status, Sources, and Dependencies . . . 67
24.2 Problem Statement . . . 67
24.3 North Star . . . 67
24.4 Shard Roadmap . . . 68
24.5 Side Quests . . . 69
24.6 Numerical Verification Plan . . . 69
24.7 Acceptance Boundary . . . 69

25 Gaussian Boson Symbol Calculus and the Boost-Time Residual 70
25.1 Status, Sources, and Dependencies . . . 70
25.2 Coefficient Tier . . . 70
25.3 Nearest-Neighbour Klein-Gordon Example . . . 70
25.4 Momentum-Space Position . . . 71
25.5 Boost-Time Residual . . . 71
25.6 Lattice Residual . . . 71

25.7 Rotation Residual . . . 72
25.8 Boundary . . . 72

26 Gaussian Boson Diagonalization and the One-Particle Symbol 73
26.1 Status, Sources, and Dependencies . . . 73
26.2 Scalar Coefficient Tier . . . 73
26.3 Mode Decoupling . . . 73
26.4 Complex Structure and Fock Space . . . 74
26.5 What Is Not Yet Included . . . 74
26.6 Compiler Consequence . . . 74

27 One-Particle Generator Algebra for Gaussian Bosons 75
27.1 Status, Sources, and Dependencies . . . 75
27.2 Momentum Core . . . 75
27.3 Exact Translation and Rotation Brackets . . . 75
27.4 Boost-Momentum Bracket . . . 76
27.5 Boost-Time Residual . . . 76
27.6 Rotation-Hamiltonian Residual . . . 76
27.7 Boost-Boost Residual . . . 77
27.8 Second Quantization . . . 77

28 Lorentz Residual Conditions on Gaussian Coefficients 78
28.1 Status, Sources, and Dependencies . . . 78
28.2 Gradient Residual . . . 78
28.3 Low-Energy Hessian Condition . . . 78
28.4 Nearest-Neighbour Klein-Gordon . . . 79
28.5 Anisotropic Counterexample . . . 79
28.6 Hessian-Passing Instability Witness . . . 79
28.7 Doubler Counterexample . . . 80
28.8 Compiler Rule . . . 80

29 Numerical Verification Suite for Gaussian Boson Examples 81
29.1 Status, Sources, and Dependencies . . . 81
29.2 Why These Tests . . . 81
29.3 Numerical Tolerance Policy . . . 81
29.4 Structural Coefficient Validation . . . 82
29.5 Periodic Stiffness Matrix . . . 82
29.6 Implemented Example Systems . . . 82
29.7 Current Julia Surface . . . 84
29.8 Next Numerical Milestones . . . 85

30 Gaussian Boson Real-Space Energy Density 86
30.1 Status, Sources, and Dependencies . . . 86
30.2 Cell Volume and Notation . . . 86
30.3 Equal-Endpoint Bond Split . . . 86
30.4 Boundary Policy . . . 87
30.5 Vacuum-Energy Policy . . . 87
30.6 Improvement Policy . . . 87
30.7 Compiler Consequence . . . 87

31 One-Dimensional Gaussian Boson Stress-Energy Candidates **88**
31.1 Status, Sources, and Dependencies . . . 88
31.2 Continuum Motivation and Boundary . . . 88
31.3 Finite Open Chain Setup . . . 88
31.4 Candidate Dictionary . . . 88
31.5 Boundary, Subtraction, and Improvement . . . 89
31.6 Compiler Payload and Acceptance . . . 90

32 Gaussian Current-Symbol Equivalence in One Dimension **91**
32.1 Status, Sources, and Dependencies . . . 91
32.2 Real-Space Current Orientation . . . 91
32.3 Klein-Gordon Equality . . . 91
32.4 Julia Witness . . . 92
32.5 Open Obligations . . . 92

33 Higher-Dimensional Gaussian Cell-Current Proposal **93**
33.1 Status, Sources, and Dependencies . . . 93
33.2 Compiler Input: Oriented Support Data . . . 93
33.3 Energy Density and Energy Flux . . . 93
33.4 Momentum Densities and Stress Currents . . . 94
33.5 Boost and Symbol Diagnostics . . . 94
33.6 Rotation and Antisymmetric-Stress Residual . . . 94
33.7 Trace, Improvement, and Boundary Alternatives . . . 95

34 First Gaussian Stress-Energy Numerical Suite **96**
34.1 Status, Sources, and Dependencies . . . 96
34.2 Finite Open-Chain Continuity . . . 96
34.3 Current-Symbol Slope Residual . . . 96
34.4 What This Falsifies . . . 97
34.5 Open Obligations . . . 97

35 Qubit Nearest-Neighbour Symmetry Quest **98**
35.1 Status, Sources, and Dependencies . . . 98
35.2 Input Tier . . . 98
35.3 Question . . . 98
35.4 Investigation Plan . . . 99
35.5 Immediate Rule-Outs . . . 99

36 One-Dimensional Qubit Boost-Current Obstructions **100**
36.1 Status, Sources, and Dependencies . . . 100
36.2 Adjacent Current in Pauli Coefficients . . . 100
36.3 Fully On-Site Terms . . . 100
36.4 Other Immediate Rejections . . . 101
36.5 The Next Residual . . . 101
36.6 Boundary . . . 102

37 Two-Dimensional Qubit Plaquette Symmetry Diagnostics **103**
37.1 Status, Sources, and Dependencies . . . 103
37.2 Edge-Density Input . . . 103
37.3 Patch Diagnostics . . . 103
37.4 Proposed Rejection Filters . . . 103
37.5 Rotation Is Not Free . . . 104
37.6 Rule-Out Classes . . . 104

38 Qubit Nearest-Neighbour Witt/Virasoro Diagnostics **105**
38.1 Status, Sources, and Dependencies . . . 105
38.2 Fourier Density and Current Modes . . . 105
38.3 First Failure Witness . . . 105
38.4 Chiral-Rank Diagnostic . . . 105
38.5 Compressed Witt Residuals . . . 106
38.6 Boundary . . . 106

39 Qubit Local-Algebra Equation Framework **107**
39.1 Status, Sources, and Dependencies . . . 107
39.2 Pauli Word Algebra . . . 107
39.3 Bulk Equality Modulo Divergence . . . 107
39.4 What "Necessary" Means Here . . . 108

40 One-Dimensional Qubit Poincare Necessary Equations **109**
40.1 Status, Sources, and Dependencies . . . 109
40.2 Input and Current . . . 109
40.3 Translation Conservation . . . 109
40.4 Boost-Translation Relation . . . 110
40.5 Boundary . . . 110

41 Computer Algebra for One-Dimensional Qubit Residuals **111**
41.1 Status, Sources, and Dependencies . . . 111
41.2 Operator and Coefficient Roundtrip . . . 111
41.3 Checked Residual Helpers . . . 111
41.4 Sentinels . . . 111
41.5 Current Gap . . . 112

42 Two-Dimensional Qubit Poincare Equation Schema **113**
42.1 Status, Sources, and Dependencies . . . 113
42.2 Edge Densities and Vertex Currents . . . 113
42.3 Divergence-Relaxed Residuals . . . 113
42.4 Necessary Bulk Equations . . . 114
42.5 Implementation Boundary . . . 114

43 Qubit Witt/Virasoro Necessary Equations **115**
43.1 Status, Sources, and Dependencies . . . 115
43.2 Exact Density-Mode Algebra . . . 115
43.3 Chiral Residuals . . . 115
43.4 Necessary Filter . . . 116

44 Vacuum Moment Constraints for Qubit Residuals **117**
44.1 Status, Sources, and Dependencies . . . 117
44.2 Finite Moment Data . . . 117
44.3 Zero Expectations . . . 117
44.4 GNS Relation Constraints . . . 117
44.5 Highest-Weight Caveat . . . 118

45 SDP Exclusion Hierarchy for Qubit Hamiltonians **119**
45.1 Status, Sources, and Dependencies . . . 119
45.2 Level Data . . . 119
45.3 SDP Constraints . . . 119
45.4 Exclusion Statement . . . 119
45.5 Non-Implications . . . 120

46 Roadmap for Qubit Symmetry Exclusion **121**
46.1 Status, Sources, and Dependencies . . . 121
46.2 What We Have Now . . . 121
46.3 Implementation Steps . . . 121
46.4 Acceptance for an Exclusion Claim . . . 121
46.5 Open Gaps . . . 122

47 Qubit SDP Implementation Contract **123**
47.1 Status, Sources, and Dependencies . . . 123
47.2 Package Split . . . 123
47.3 Verdicts . . . 123

48 Pauli Word Moment Basis for Infinite Qubit Chains **124**
48.1 Status, Sources, and Dependencies . . . 124
48.2 Positioned Words . . . 124
48.3 Translation Quotient . . . 124

49 Moment Matrices and Positivity Constraints **125**
49.1 Status, Sources, and Dependencies . . . 125
49.2 Affine Moment Matrix . . . 125
49.3 Tiny Feasibility Tests . . . 125

50 Compiling Poincare Residuals into Moment Equations **126**
50.1 Status, Sources, and Dependencies . . . 126
50.2 Algebraic Witnesses . . . 126
50.3 Moment Relations . . . 126
50.4 Important Boundary . . . 126

51 SDP Levels and the Mosek Backend **127**
51.1 Status, Sources, and Dependencies . . . 127
51.2 Level Data . . . 127
51.3 Solver Status . . . 127
51.4 Smoke Run . . . 127

52 Sentinel Hamiltonians for the SDP Hierarchy **128**
52.1 Status, Sources, and Dependencies . . . 128
52.2 Matrices . . . 128
52.3 Gate Outcomes . . . 128

53 Candidate Hamiltonian Selection Roadmap **129**
53.1 Status, Sources, and Dependencies . . . 129
53.2 Screening Order . . . 129
53.3 Next Implementation Steps . . . 129

54 Qubit Candidate Scan Contract **130**
54.1 Status, Sources, and Dependencies . . . 130
54.2 Contract . . . 130
54.3 Verdict Boundary . . . 130

55 Pauli Coefficients for Scan Families **131**
55.1 Status, Sources, and Dependencies . . . 131
55.2 Implemented Maps . . . 131
55.3 Synthetic Families . . . 131

56 Locally Sourced Spin-Chain Inputs **132**
56.1 Status, Sources, and Dependencies . . . 132
56.2 Allowed Physical Labels . . . 132
56.3 Excluded From Physical Labeling . . . 132

57 Synthetic and Stress-Test Hamiltonian Grids **133**
57.1 Status, Sources, and Dependencies . . . 133
57.2 Grids . . . 133
57.3 Purpose . . . 133

58 Current, Conservation, and Boost Witness Gates **134**
58.1 Status, Sources, and Dependencies . . . 134
58.2 Gates . . . 134
58.3 Interpretation . . . 134

59 Optional SDP Levels for Scan Survivors **135**
59.1 Status, Sources, and Dependencies . . . 135
59.2 Level Data . . . 135
59.3 First-Run Boundary . . . 135

60 Reproducible Scan Bundle Format **136**
60.1 Status, Sources, and Dependencies . . . 136
60.2 Files . . . 136
60.3 Acceptance . . . 136

61 First Candidate Scan Results **137**
61.1 Status, Sources, and Dependencies . . . 137
61.2 Headline Table . . . 137
61.3 Notable Points . . . 137

62 From Scan Failures to Scaling Work **138**
62.1 Status, Sources, and Dependencies . . . 138
62.2 Queue . . . 138
62.3 SDP Upgrade . . . 138

63 The Categorical Borchers–Uhlmann Pipeline: a Programme Refinement **139**
63.1 Status, Sources, and Dependencies . . . 139
63.2 Refined North Star . . . 139
63.3 Five Stages . . . 139
63.4 Two Load-Bearing Constraints . . . 140
63.5 Functoriality and Fidelity Assessment . . . 141
63.6 Boundary . . . 141

64 All-Level Moment Feasibility Gives a Qubit Reference State **142**
64.1 Status, Sources, and Dependencies . . . 142
64.2 Canonical Full-Window Levels . . . 142
64.3 Finite Moments Are Finite States . . . 142
64.4 Compactness . . . 143
64.5 Residual Relations in the GNS Representation . . . 143
64.6 Exact Alternative . . . 143
64.7 Non-Implications . . . 144
64.8 Boundary . . . 144

65 Quantified Relaxations of the Qubit Symmetry Gates **145**
65.1 Status, Sources, and Dependencies . . . 145
65.2 Coefficient Norm and Gauge . . . 145
65.3 Algebraic Profiles . . . 145
65.4 State-Level Residuals . . . 146
65.5 Scaling Reading . . . 146
65.6 Sentinel Expectations . . . 146
65.7 Boundary . . . 146

66 Anyonic Word Algebra from a Site Object **147**
66.1 Status, Sources, and Dependencies . . . 147
66.2 Site Object First . . . 147
66.3 Placed Matrix Blocks . . . 147
66.4 Inclusions, Shift, and AF Limit . . . 148
66.5 The BU-Style Free Pre-Layer . . . 148
66.6 Defect Algebra Is a Different Layer . . . 149
66.7 Functoriality Target . . . 149
66.8 Lamport Dependency Skeleton . . . 149
66.9 Boundary . . . 149

67 Anyonic States and Variable-N Fock Space **150**
67.1 Status, Sources, and Dependencies . . . 150
67.2 The Finite Hard-Core Carrier . . . 150
67.3 Action of the Word Algebra . . . 150
67.4 Birth and Death Operators . . . 151
67.5 Hollands' Dense Chain as a Corner . . . 151

67.6 Fibonacci Test Targets . . . 152
67.7 Lamport Dependency Skeleton . . . 152
67.8 Boundary . . . 153

68 Refinement-Map Requirements and Obstructions 154
68.1 Status, Sources, and Dependencies . . . 154
68.2 Strict OAR Requirements . . . 154
68.3 Fallback and Soft Relaxations . . . 154
68.4 The No-Go Mechanism as a Checklist . . . 154
68.5 Where Fixed Anyon Number Enters the Braiding RG . . . 155
68.6 Boundary . . . 155

69 Refinement Maps for the Variable-N Anyonic Word Algebra 156
69.1 Status, Sources, and Dependencies . . . 156
69.2 Vacuum Insertion . . . 156
69.3 Algebra Map . . . 156
69.4 State Consistency . . . 156
69.5 Density Reading . . . 157
69.6 Virasoro Compatibility Target . . . 157
69.7 Dead Prior Art: the Deficit/Pair-Creation Precedent . . . 157
69.8 Fibonacci $H_2 \to H_4$ Test . . . 157
69.9 Comparison and Boundary . . . 158

70 The Dilute Temperley–Lieb Algebra and the Anyonic Word Algebra 159
70.1 Status, Sources, and Dependencies . . . 159
70.2 The Dilute Algebra . . . 159
70.3 The Evaluation Map . . . 159
70.4 The Obstruction: No Surjection for Fibonacci . . . 160
70.5 Consequences for the Programme . . . 160
70.6 Boundary . . . 160

71 The Dilute Koo–Saleur Ansatz 161
71.1 Status, Sources, and Dependencies . . . 161
71.2 What the Dense Template Requires . . . 161
71.3 The Dilute Ansatz . . . 161
71.4 Two Sourced Critical Anchors . . . 161
71.5 Proof Obligations . . . 162
71.6 Validation Plan (CA-71 Tier) . . . 162
71.7 Boundary . . . 162

72 The Refinement Placement Category and Exact Composition 163
72.1 Status, Sources, and Dependencies . . . 163
72.2 Placement Injections and Bare Refinements . . . 163
72.3 Exact Composition . . . 163
72.4 Cellwise Refinement and Cell-Shift Compatibility . . . 164
72.5 The Forest-Functor Reading . . . 164
72.6 The Ore Dichotomy and Fallback Homes . . . 164
72.7 Julia Invariants . . . 165
72.8 Boundary . . . 165

73 GNS Descent and the Corner Calculus of Placement Refinements **166**
73.1 Status, Sources, and Dependencies . . . 166
73.2 Setting . . . 166
73.3 The Corner Calculus . . . 166
73.4 The GNS Descent Lemma . . . 166
73.5 Residual Descent: the Contract for the Categorical Residual Set . . . 167
73.6 The Unital CP Completion and Its Exact Defect . . . 167
73.7 The BU-Level Square . . . 168
73.8 Julia Invariants . . . 168
73.9 Boundary . . . 168

74 The Categorical Residual Set on the Anyonic Word Tower **169**
74.1 Status, Sources, and Dependencies . . . 169
74.2 Setting . . . 169
74.3 The Ported Local Residual Calculus . . . 169
74.4 Open-Window Mode Calculus . . . 169
74.5 Witt/Virasoro Residuals with Two Central Constants . . . 170
74.6 The Categorical Residual Set . . . 170
74.7 Exact Transport and the Stretched Dichotomy . . . 170
74.8 Julia Invariants . . . 171
74.9 Boundary . . . 171

75 The Jones–Wenzl Kernel Decision for the Dilute Evaluation Map **172**
75.1 Status, Sources, and Dependencies . . . 172
75.2 Setting and the Decision Data . . . 172
75.3 Parity-Refined Multiplicities . . . 172
75.4 The Kernel Theorem . . . 173
75.5 The Upgraded Conjecture . . . 173
75.6 Julia Invariants . . . 174
75.7 Gram-Rank Deferral and Boundary . . . 174

76 The CFT Machine and Its Quantitative OAR Contract **175**
76.1 Sources, Dependencies, and Status . . . 175
76.2 The Actual Construction Target . . . 175
76.3 A Quantitative Wilson-State Theorem . . . 175
76.4 The Generator Contract and Its Boundary . . . 176

77 A Supported Ising Modular-Data Compiler **177**
77.1 Sources, Dependencies, and Scope . . . 177
77.2 Exact Input and Concrete Output . . . 177
77.3 The Local Observable Theory . . . 177
77.4 Symmetry and Modular Fidelity . . . 178

78 Coset Lattice Stress Generators with Exact Core Stabilization **179**
78.1 Sources, Dependencies, and Scope . . . 179
78.2 An Explicit Microscopic Polynomial . . . 179
78.3 Exact Stabilization on Free-Energy Cores . . . 179
78.4 Exact Witnesses and the Remaining Interface . . . 180

79 Coset Positivity and a Scalar-Augmented OAR Tower **181**
79.1 Sources, Dependencies, and Scope . . . 181
79.2 The Raw Finite Zero Mode Is Not Positive . . . 181
79.3 Computable Vacuum-Descendant Spaces . . . 181
79.4 An Independent Scalar Gives Genuine Unital OAR . . . 182
79.5 What the Construction Does Not Identify . . . 182

80 Smooth Coset Symmetry Limits and Local Reconstruction **183**
80.1 Sources and Scope . . . 183
80.2 Three Cutoffs and an Ambient Operator Theorem . . . 183
80.3 Local Fields in the Descendant OAR Representation . . . 184
80.4 The Local Fusion-Space Question . . . 184

81 String-Net Histories as Trotter Expansions: the Spacetime Dictionary **185**
81.1 Status, Sources, and Dependencies . . . 185
81.2 The Levin–Wen Rules and What They Fix . . . 185
81.3 The Trotter Dictionary . . . 185
81.4 Dense Calibration . . . 186
81.5 Dilute Calibration . . . 186
81.6 What the Calibration Says About the Principle . . . 187
81.7 Beyond Loops: the Fibonacci Net Gas (Proposal) . . . 187

82 The Net Case: the Dilute Fibonacci Chain Is the Golden Chain Blocked **189**
82.1 Status, Sources, and Dependencies . . . 189
82.2 Blocking: When Dilute Is Dense in Disguise . . . 189
82.3 The Temperley–Lieb Generators in Tile Form . . . 189
82.4 The Isotropic Net Gas Is the $Q = \varphi + 1$ Potts Model . . . 190
82.5 What This Does to the Principle . . . 191

1 Frontmatter, North Star, and Claim Status

This is a reproducible research lab book. Its guiding goal — the *north star* — is a single, concrete construction problem:

> Given a (unitary) fusion or modular tensor category $\mathcal{C}$, together with whatever additional data is needed (OPE coefficients / conformal data), construct a family of microscopic (lattice) models and their symmetry generators whose continuum limit *provably* yields a mathematically rigorous QFT/CFT that (i) realises the input category data and (ii) carries a full projective unitary representation of the symmetries.

The working belief is that this is achievable at least for *rational* CFTs, and perhaps for all QFTs. This is a programme, not a settled theory: every step toward the goal is a question, proposal, or hypothesis until a local source, a local derivation, or a reproducible run supports it.

1.1 The Two Success Criteria

A construction is only a candidate solution if it meets both, non-negotiably:

1. **Provability.** The continuum limit is established rigorously (an operator-algebraic / wavelet renormalisation limit with theorems, not a heuristic scaling argument).
2. **Fidelity.** The realised continuum content faithfully reproduces the input category data $\mathcal{C}$, and the symmetry representation is *full* and projective-unitary.

1.2 Claim Status

No claim is treated as established until it is marked *Source-local* or *Checked* and linked to a local source, derivation, test, or run artifact. The status labels are fixed in `CONVENTIONS.md`: *Question*, *Proposal*, *Source-local*, *Checked*, and *Rejected*. These labels are part of the scientific record; they separate programme-level conjecture from results.

1.3 Evidence Chain

Every substantive assertion should be traceable through:

1. a local source manifest under `references/` (the kept papers are the ground truth),
2. a convention entry in `CONVENTIONS.md`,
3. a derivation, script, test, or run bundle when computation is involved,
4. an index row in `INDEX.md`, and
5. a report shard in this lab book.

Missing links are allowed during exploration only when they are explicit.

2 Programme Map

This shard is a map of questions, not a list of results. The north star (§1 / CA-00) factors into a pipeline of stages; each stage is an open problem with its own conventions, sources, and checkable invariants.

2.1 The Pipeline

$$\underbrace{\mathcal{C}\ (+\,\text{OPE/conformal data})}_{\text{input}} \longrightarrow \underbrace{\text{microscopic models}}_{\text{lattice Hilbert space + Hamiltonian}} \longrightarrow \underbrace{\text{lattice symmetry generators}}_{\text{e.g. Koo–Saleur } L_n}$$

$$\longrightarrow \underbrace{\text{provable continuum limit}}_{\text{OAR / wavelets}} \longrightarrow \underbrace{\text{rigorous (C)FT}}_{\text{conformal net / VOA / Wightman}}$$

2.2 Stage 1 — Category Input

- Which categories are in initial scope: multiplicity-free unitary fusion categories (Fibonacci, Ising/TL), then modular tensor categories?
- What auxiliary data beyond $\mathcal{C}$ is genuinely required (OPE coefficients, central charge, conformal weights), and for which target CFTs?
- Which gauge and indexing conventions are fixed before any F/R-symbol is written (see `CONVENTIONS.md`)?

2.3 Stage 2 — Microscopic Models

- Which family realises $\mathcal{C}$ on the lattice: anyon chains, string-nets, or another construction — and how is the family indexed by scale?
- What is the lattice Hilbert space (fusion-tree basis) and the Hamiltonian, and which invariants pin them (fusion-rule consistency, pentagon, projector idempotence)?

2.4 Stage 3 — Lattice Symmetry Generators

- Which lattice operators approximate the continuum symmetry generators (Koo–Saleur lattice Virasoro L_n, $\bar{L}_n$, and kin)?
- What convergence statement is the target, and what is checkable at finite size (commutator algebra, central charge extraction)?

2.5 Stage 4 — Provable Continuum Limit

- Which rigorous renormalisation framework carries the limit (operator- algebraic renormalisation, wavelet / multi-scale entanglement)?
- What must be proved: existence of the limit, the limiting Hilbert space, and convergence of the symmetry generators to a projective unitary representation?

2.6 Stage 5 — Rigorous (C)FT Target

- What is the target object: a conformal net, a vertex operator algebra, a Wightman theory, or an Osterwalder–Schrader theory?
- What is the fidelity statement: that the target's superselection / representation category recovers $\mathcal{C}$?

2.7 First Decisions

The first technical session should not start with formulas. It should register the sources for one worked input category (Fibonacci or Ising/TL), fix the minimum conventions (F/R-symbol gauge, vacuum index, fusion-tree ordering), and identify the smallest invariant that can be checked end-to-end on one stage of the pipeline.

3 Many-Particle Hilbert Spaces: the AND/OR/Vacuum Grammar

3.1 Sources and Dependencies

This is a pre-topic foundation for the programme (see CA-01, the programme map): how to build the Hilbert space of an indefinite number of particles of arbitrary types. The guiding idea is that two logical connectives map cleanly onto two Hilbert-space constructions, and the no-particle system is the complex line.

This shard treats the *finite* grammar and its semantics; the rigorous *indefinite*-particle (Hilbert-completed) case is CA-03. It harmonises with, and re-proves nothing from, the finite "AND/OR many-particle grammar" developed by the author in the sibling project (notation $\mathcal{V}$, $\mathcal{V}_n$, AND $= \otimes$, OR $= \oplus$, $\mathbf{1} = \mathbb{C}$); see that project's shard `AQM-70`. Statistics (Bose / Fermi / para) are deferred throughout.

3.2 The Three Primitives

Let particle *types* be (separable, complex) Hilbert spaces; a type's space is the state space of one particle of that type. Three primitives generate composite systems:

- **OR = direct sum $\oplus$.** "Either a particle of type 1 or of type 2" is $\mathcal{H}_1 \oplus \mathcal{H}_2$.
- **AND = tensor product $\otimes$.** "A particle of type 1 *and* a particle of type 2" is $\mathcal{H}_1 \otimes \mathcal{H}_2$.
- **vacuum $= \mathbb{C}$.** The no-particle system is the complex line.

Crucially, this grammar fixes *indefinite particle number*; exchange symmetry (statistics) is a *separate, later* step, realised as a quotient or subspace (CA-03, §"Symmetries come later"). Keeping the two apart is the whole point.

3.3 Two Monoidal Structures and Their Units

On the category Hilb of complex Hilbert spaces (with the obvious unitary coherence isomorphisms) there are two symmetric monoidal structures:

$$(\text{Hilb},\ \oplus,\ \{0\}) \qquad \text{and} \qquad (\text{Hilb},\ \otimes,\ \mathbb{C}),$$

with the two units *distinct*:

$$A \oplus \{0\} \ \cong\ A \qquad (\oplus\text{-unit is the } \textit{zero space}\ \{0\}),$$

$$A \otimes \mathbb{C} \ \cong\ A \qquad (\otimes\text{-unit is the } \textit{vacuum}\ \mathbb{C}).$$

The two are linked by a *distributivity* natural isomorphism, which on Hilbert spaces is *unitary*:

$$\delta :\ A \otimes (B \oplus C) \ \xrightarrow{\cong}\ (A \otimes B) \oplus (A \otimes C), \qquad A \otimes \{0\} \ \cong\ \{0\}.$$

Together $(\text{Hilb}, \oplus, \{0\}, \otimes, \mathbb{C})$ is a symmetric *rig* (bimonoidal) category. Its coherence — that all diagrams built from associators, unitors, symmetries, and δ commute — is standard (Laplaza; Kelly) and is *not* re-proved here; downstream we use only the elementary, manifestly unitary associativity/unitor isomorphisms and the degree-0 identification $\mathbb{C} \otimes A \cong A$.

Vacuum $= \mathbb{C}$, not $\{0\}$. The vacuum is the $\otimes$-unit $\mathbb{C}$ (the empty tensor product, one particle's worth of "nothing"), a *one-dimensional* space carrying the unit vector Ω. It is *not* the $\oplus$-unit $\{0\}$ (the empty direct sum, the impossible system). Empty $\otimes$ is $\mathbb{C}$; empty $\oplus$ is $\{0\}$. Conflating them is the first pitfall this grammar guards against (cf. CONVENTIONS.md (a), which fixes the vacuum *object* as $\mathbb{C}$).

3.4 The Grammar and its Carrier

Fix a set $\{K_x\}_{x\in X}$ of single-particle type spaces (the *atoms*). Grammar expressions are

$$E ::= 0 \;\mid\; \mathbf{1} \;\mid\; K \;\mid\; \mathrm{AND}(E,F) \;\mid\; \mathrm{OR}(E,F),$$

where 0 is the zero atom, $\mathbf{1}$ is the vacuum atom ($\mathbf{1} = \mathbb{C}$), and K ranges over the atoms. The *carrier* $\mathcal{V}(E)$ is defined recursively:

$$\mathcal{V}(0) = \{0\}, \qquad \mathcal{V}(\mathbf{1}) = \mathbb{C}, \qquad \mathcal{V}(K) = K,$$
$$\mathcal{V}(\mathrm{AND}(E,F)) = \mathcal{V}(E) \otimes \mathcal{V}(F), \quad \mathcal{V}(\mathrm{OR}(E,F)) = \mathcal{V}(E) \oplus \mathcal{V}(F).$$

Thus AND is independent coexistence and OR is alternatives. Parenthesisation and order are part of the written expression; the rig coherence above guarantees that re-bracketing or re-ordering a *fixed* expression changes $\mathcal{V}(E)$ only by a canonical unitary isomorphism, so $\mathcal{V}$ is well defined up to canonical unitary isomorphism. (Morally $\mathcal{V}$ is the evaluation of the free symmetric rig category on $\{K_x\}$ into Hilb; we use this only as motivation, not as a load-bearing theorem.)

3.5 Particle-Number Grading

Declare each atom K to be *one* particle and $\mathbf{1}$ to be *zero* particles. The carrier then carries a particle-number grading $\mathcal{V}_n$:

$$\mathcal{V}_0(\mathbf{1}) = \mathbb{C}, \quad \mathcal{V}_n(\mathbf{1}) = \{0\}\ (n \neq 0), \qquad \mathcal{V}_1(K) = K, \quad \mathcal{V}_n(K) = \{0\}\ (n \neq 1), \quad \mathcal{V}_n(0) = \{0\},$$
$$\mathcal{V}_n(\mathrm{OR}(E,F)) = \mathcal{V}_n(E) \oplus \mathcal{V}_n(F), \qquad \mathcal{V}_n(\mathrm{AND}(E,F)) = \bigoplus_{i+j=n} \mathcal{V}_i(E) \otimes \mathcal{V}_j(F).$$

For a *finite* expression the parse tree is finite, so only finitely many sectors are nonzero and $\mathcal{V}(E) = \bigoplus_{n\geq 0} \mathcal{V}_n(E)$ is a *finite* direct sum. The integer n is a grammar count of one-particle atoms; it is not a dynamics, a statistics rule, or (yet) an indefinite-particle space.

Worked atoms. A one-particle "menu" over a finite type set X is $P = \mathrm{OR}_{x\in X} K_x$, with $\mathcal{V}(P) = \bigoplus_{x\in X} K_x$ (the infinite, ℓ^2-completed menu is CA-03). The expression $\mathbf{1}\,\mathrm{OR}\,K$ has carrier $\mathbb{C} \oplus K$ — a "maybe" particle (either zero or one particle of type K), as in the sibling shard's $\mathbb{C} \oplus \mathbb{C}^d$. Tensoring N "maybe" atoms, $\mathrm{AND}_{j=1}^{N}(\mathbf{1}\,\mathrm{OR}\,K)$, has carrier $(\mathbb{C} \oplus K)^{\otimes N}$, whose grade-$n$ sector (by distributivity) is $\bigoplus_{|S|=n} \bigotimes_{x\in S} K$ over n-subsets of the N slots — the finite "occupation" picture.

3.6 Lamport Dependency Skeleton

$D1$: atoms $\{K_x\}_{x\in X}$ (single-particle Hilbert spaces); $\mathbf{1} = \mathbb{C}$ (vacuum), $0 = \{0\}$.
$D2$: $E ::= 0 \mid \mathbf{1} \mid K \mid \mathrm{AND}(E,F) \mid \mathrm{OR}(E,F)$.
$D3$: $\mathcal{V}$: $\mathcal{V}(0) = \{0\}$, $\mathcal{V}(\mathbf{1}) = \mathbb{C}$, $\mathcal{V}(\mathrm{AND}) = \otimes$, $\mathcal{V}(\mathrm{OR}) = \oplus$.
$D4$: $\mathcal{V}_0(\mathbf{1}) = \mathbb{C}$, $\mathcal{V}_1(K) = K$; $\mathcal{V}_n(\mathrm{OR}) = \oplus$, $\mathcal{V}_n(\mathrm{AND}) = \bigoplus_{i+j=n} \mathcal{V}_i \otimes \mathcal{V}_j$.
$D5$: $(\mathrm{Hilb}, \oplus, \{0\}, \otimes, \mathbb{C})$ a symmetric rig category, with
$\delta : A \otimes (B \oplus C) \xrightarrow{\cong} (A \otimes B) \oplus (A \otimes C)$ unitary and natural.
$T1$: (D5) coherence (Laplaza; Kelly) $\Rightarrow \mathcal{V}$ well defined up to canonical unitary iso.
$T2$: (D4) finite E : $\mathcal{V}(E) = \bigoplus_{n\geq 0} \mathcal{V}_n(E)$ a finite sum (finite particle number).

3.7 Scope and Deferral

Two extensions are deliberately out of this shard:

- **Indefinite particle number.** A genuine superposition over *unboundedly* many particle numbers has infinite support and does not live in the finite direct sum above; it requires the Hilbert (ℓ^2) completion. That is the content of CA-03, and is the new rigorous part of this pre-topic.
- **Statistics.** AND and OR choose no exchange symmetry. Bosonic / fermionic / para sectors arise later, by acting with the permutation group on tensor slots and passing to an invariant subspace or quotient (CA-03). This grammar only sets up the carrier those symmetries act on.

Pitfalls. (i) The $\oplus$-unit is $\{0\}$; the $\otimes$-unit (the vacuum) is $\mathbb{C}$ — distinct objects. (ii) Distributivity is an *isomorphism*, not an equality; all sector identities below hold up to canonical unitary iso. (iii) Everything here is *finite*; do not silently read the grading as an indefinite-particle space.

4 Indefinite Particle Number: the Full Fock Completion

4.1 Sources and Dependencies

This shard completes CA-02's finite AND/OR/vacuum grammar to a genuine *indefinite*-particle Hilbert space. It depends on CA-02 (the two monoidal structures, the units $\{0\}$ and $\mathbb{C}$, the carrier $\mathcal{V}$ and grading $\mathcal{V}_n$). The named Fock construction is anchored locally:

The sibling project's finite grammar (`AQM-70`) is recovered as the finite-truncation of the object built here.

4.2 Two Direct Sums — the Load-Bearing Distinction

Let $\{W_n\}_{n\geq 0}$ be a countable family of Hilbert spaces.

- The **algebraic direct sum** $\bigoplus_n^{\mathrm{alg}} W_n$ consists of finite-support sequences (w_n) ($w_n \in W_n$, $w_n = 0$ for all but finitely many n). It is a pre-Hilbert space, generally *incomplete*.
- The **Hilbert (ℓ^2) direct sum** $\widehat{\bigoplus}_n W_n$ consists of sequences with $\sum_n \|w_n\|_{W_n}^2 < \infty$, with inner product $\langle (w_n), (v_n)\rangle = \sum_n \langle w_n, v_n\rangle_{W_n}$ (absolutely convergent by Cauchy–Schwarz). It is *complete*, hence a Hilbert space.

$\bigoplus_n^{\mathrm{alg}} W_n$ is a *dense* subspace of $\widehat{\bigoplus}_n W_n$, and the latter is its completion (standard functional analysis).

Why indefinite particle number forces $\widehat{\bigoplus}$. A state with nonzero amplitude in *unboundedly* many particle-number sectors (a genuine superposition over indefinitely many particles) has infinite support: it lies in $\widehat{\bigoplus}$ but not in $\bigoplus^{\mathrm{alg}}$. "Indefinite particle number" $\Leftrightarrow$ "not confined to finitely many sectors" $\Rightarrow$ the ℓ^2 completion is *required*. This is exactly what CA-02's finite grammar lacks.

4.3 Completed Tensor Powers

For Hilbert spaces H, K, the *completed* (Hilbert) tensor product $H \mathbin{\widehat{\otimes}} K$ is the completion of the algebraic tensor product under the inner product determined by $\langle a\otimes b,\, c\otimes d\rangle = \langle a, c\rangle_H \langle b, d\rangle_K$ (well defined and positive definite). For finite-dimensional factors $\mathbin{\widehat{\otimes}} = \otimes$; the completion bites only for infinite-dimensional one-particle spaces. Define the *completed tensor powers*

$$H^{\widehat{\otimes} 0} := \mathbb{C} \ \text{ (the vacuum sector)}, \qquad H^{\widehat{\otimes} n} := \underbrace{H \mathbin{\widehat{\otimes}} \cdots \mathbin{\widehat{\otimes}} H}_{n},$$

the bracketing immaterial up to the canonical unitary associativity isomorphism.

4.4 The Full Fock Space

Definition. For a one-particle Hilbert space H, the *full Fock space* is

$$\boxed{\mathfrak{F}(H) \;:=\; \widehat{\bigoplus_{n\geq 0}} H^{\widehat{\otimes} n}}, \qquad H^{\widehat{\otimes} 0} = \mathbb{C},$$

i.e. $\psi = (\psi_n)_{n\geq 0}$ with $\psi_n \in H^{\widehat{\otimes} n}$ and $\|\psi\|^2 = \sum_n \|\psi_n\|^2 < \infty$. The unit vector $\Omega = (1, 0, 0, \dots)$ in the degree-0 sector $\mathbb{C}$ is the *vacuum*. This is the *distinguishable* Fock space (the author's term in the local source `1910.10619`); in operator algebra it is also called the *free* or *Boltzmann* Fock

space (standard terminology, not anchored in a local source), namely the Hilbert completion of the tensor algebra $T(H) = \bigoplus_n H^{\otimes n}$. It is *not* the symmetric (bosonic) or antisymmetric (fermionic) Fock space; see "Exchange sectors come later" below.

Compiler status. In the compiler grammar this is not an additional primitive constructor. It is the Hilbert completion of the ordinary tensor-power expressions $\mathbf{1}, H, H^{\widehat{\otimes}2}, \ldots$: a countable OR over all finite AND powers. If a one-particle symmetry representation is present, CA-05 makes this same object a graded direct-sum representation.

Theorem (Fock is a Hilbert space). $\mathfrak{F}(H)$ is a Hilbert space; (i) the algebraic direct sum $\bigoplus_n^{\mathrm{alg}} H^{\widehat{\otimes}n}$ (the finite-particle vectors) is *dense*; (ii) distinct-n sectors are *mutually orthogonal*, $\langle \psi, \varphi \rangle = \sum_n \langle \psi_n, \varphi_n \rangle$, so $H^{\widehat{\otimes}m} \perp H^{\widehat{\otimes}n}$ for $m \neq n$; (iii) on the degree-0 sector the inner product is the standard one on $\mathbb{C}$ and Ω is a unit vector. *Proof.* An instance of the ℓ^2-direct-sum facts above with $W_n = H^{\widehat{\otimes}n}$: completeness, density of the algebraic sum, and summand orthogonality are exactly those statements. □

The $1/N!$ caveat (a convention trap). Schottenloher's inner product on $D = \bigoplus_N H_N$ weights sector N by $1/N!$; that factor is a *symmetric*-Fock normalisation correcting the $N!$ overcounting of symmetric tensors. The full Fock space above uses the *plain* ℓ^2 sum with *no* combinatorial weight. Do not copy the $1/N!$ into the distinguishable case.

4.5 The Grammar Generates the Indefinite-Particle Space

Let the one-particle menu over a countable type set X be $P = \mathrm{OR}_{x \in X} K_x$, with carrier $\mathcal{V}(P) = \widehat{\bigoplus}_{x \in X} K_x$ (the ℓ^2 direct sum; an ordinary finite direct sum when X is finite). The indefinite-particle space over this menu is $\mathfrak{F}(\mathcal{V}(P))$. Expanding each completed tensor power by distributivity (CA-02's δ, extended unitarily to the completions) gives the $(n, \text{type-word})$-graded decomposition

$$\mathcal{V}(P)^{\widehat{\otimes}n} \;\cong\; \widehat{\bigoplus}_{w \in X^n} K_w, \quad K_w := K_{x_1} \mathbin{\widehat{\otimes}} \cdots \mathbin{\widehat{\otimes}} K_{x_n}, \qquad \Longrightarrow \qquad \mathfrak{F}(\mathcal{V}(P)) \;\cong\; \widehat{\bigoplus}_{n \geq 0} \widehat{\bigoplus}_{w \in X^n} K_w.$$

Each of the n tensor slots independently chooses a type $x_i \in X$ (two slots may choose the same type; nothing is symmetrised). The grading is $\mathcal{V}_n = \widehat{\bigoplus}_{w \in X^n} K_w$ (n = particle number, $w \in X^n$ = ordered type-word) — the ℓ^2-completed, all-n analogue of CA-02's finite grade-n sector. The sibling project's finite expressions ($P_X^{\mathrm{AND}\,m}$, the truncation $T_{\leq M}$) are precisely the finite-support / finite-M restrictions of this object; that is the exact harmonisation: `AQM-70` is the $\bigoplus^{\mathrm{alg}}$/finite-M truncation of $\mathfrak{F}$.

4.6 Exchange Sectors Come Later

The full Fock space already carries symmetry representations when H does: internal symmetries act sectorwise by tensor powers (CA-05), and for each n the permutation group S_n acts unitarily on $H^{\widehat{\otimes}n}$ by permuting tensor slots. What is deferred is the *sector selection*. Bosonic and fermionic Fock spaces are the images of the per-sector orthogonal projectors

$$\Gamma_{\mathrm{s}}(H) = P_{\mathrm{sym}}\, \mathfrak{F}(H) \quad \text{(symmetric / Bose)}, \qquad \Gamma_{\mathrm{a}}(H) = P_{\mathrm{anti}}\, \mathfrak{F}(H) \quad \text{(antisymmetric / Fermi)},$$

with para-statistics the other Young-symmetry isotypic components. This is the "impose an equivalence relation under particle exchange" step. We define no projector and prove nothing here:

the construction of these symmetry quotients/subspaces, and the genuinely anyonic case (where standard Fock space is not the right object), are deferred to later shards.

4.7 Lamport Dependency Skeleton

$D6$: $\bigoplus^{\mathrm{alg}}$(finite support) $\subset \widehat{\bigoplus}(\ell^2)$: $\langle (w_n),(v_n)\rangle = \sum_n \langle w_n, v_n\rangle$.
$D7$: completed tensor power $H^{\widehat{\otimes} n}$, $H^{\widehat{\otimes} 0} = \mathbb{C}$ (vacuum).
$D8$: $\mathfrak{F}(H) := \widehat{\bigoplus}_{n\geq 0} H^{\widehat{\otimes} n}$ (full / distinguishable Fock); $\Omega = (1, 0, \dots)$.
$D9$: $P = \mathrm{OR}_{x\in X} K_x$; $\quad \mathfrak{F}(\mathcal{V}(P)) = \widehat{\bigoplus}_{n\geq 0} \mathcal{V}(P)^{\widehat{\otimes} n}$.
$T3$: (D6,D7) $\mathfrak{F}(H)$ Hilbert; $\bigoplus^{\mathrm{alg}}$ dense; $H^{\widehat{\otimes} m} \perp H^{\widehat{\otimes} n}$ $(m \neq n)$.
$T4$: indefinite particle no. $\Leftrightarrow$ infinite support $\Rightarrow \widehat{\bigoplus}$ (not $\bigoplus^{\mathrm{alg}}$).
$T5$: (D8,δ) $\mathfrak{F}(\mathcal{V}(P)) \cong \widehat{\bigoplus}_{n\geq 0} \widehat{\bigoplus}_{w\in X^n} K_w$ ($(n$, type-word) grading).
$T6$: $\Gamma_{\mathrm{s}} = P_{\mathrm{sym}}\mathfrak{F}$, $\Gamma_{\mathrm{a}} = P_{\mathrm{anti}}\mathfrak{F}$ deferred (statistics later).

4.8 Scope and Pitfalls

In scope: the indefinite-particle direction, i.e. countable OR (the Fock completion). **Out of scope:** countable AND (an *infinite tensor product*), which is analytically a different object (von Neumann's infinite tensor product, needing a stabilising reference sequence) — not developed here.

Pitfalls: (i) *algebraic* $\bigoplus^{\mathrm{alg}}$ (finite particle number, dense, incomplete) vs. *completed* $\widehat{\bigoplus}$ (indefinite, complete) — the distinction this shard exists to make. (ii) For infinite-dimensional one-particle H, the AND of CA-02 must mean the *completed* $\widehat{\otimes}$, not the algebraic $\otimes$. (iii) The $1/N!$ symmetric-Fock weight must not be imported into the distinguishable case. (iv) All sector identities hold on dense subspaces and extend by continuity; states are ℓ^2-summable and cross-sector inner products vanish.

5 A Hilbert-Space Compiler Contract

5.1 Status and Local Anchors

Status: Proposal / contract shard. This shard introduces no new mathematical classification theorem. It records a design target for the pre-work sequence around CA-02 and CA-03: a typed grammar of many-particle expressions should be compilable to Hilbert-space data and observable-algebra data with explicit evidence status.

Local anchors are CA-02 for the finite AND/OR/vacuum grammar, CA-03 for the Hilbert direct-sum Fock completion, and `CONVENTIONS.md` entries (f) and (g) for symmetry and sector semantics. The first executable check for this sequence is the finite averaging projector in `test/runtests.jl`, backed by `src/CftAnyons.jl`.

5.2 Mini North Star

Proposal: build a *Hilbert-space compiler*. The user supplies an expression in a typed grammar, together with the atom table and required policies; the compiler returns the Hilbert carrier, its observable algebra policy, its symmetry representation data, and any sector projection or quotient presentation that was requested.

The motivating analogy is "regular expressions for quantum mechanics": a small syntax has compositional semantics. The analogy is only a prompt. Here the syntax is typed and semantics-bearing; every constructor must say what it does to Hilbert carriers, particle gradings, observable algebras, symmetries, and evidence metadata.

5.3 Input Record

A compiled expression is not just a string. It has an input record:

$$\mathsf{Input} = (\mathsf{Atoms},\ E,\ \mathsf{Completion},\ \mathsf{ObsPolicy},\ \mathsf{Symmetry},\ \mathsf{SectorPolicy}).$$

Atoms. A table of atoms. At minimum, each atom names a Hilbert space K_x. Later entries may include a group representation, a particle degree, a source locator, or category data.

E. A grammar expression built from $0, \mathbf{1}, K_x, \mathrm{AND}, \mathrm{OR}$, and later sector constructors. Fock-style all-n notation is treated as a derived completion of tensor powers, not as a primitive grammar constructor.

Completion. Whether $\oplus, \otimes$ are finite Hilbert operations, Hilbert completions $\widehat{\bigoplus}, \widehat{\otimes}$, or a finite cutoff. CA-03 fixes the first infinite completion of the derived all-tensor-power construction used here.

ObsPolicy. The observable-algebra choice. There is no silent default: full $\mathcal{B}(H)$, block-preserving, tensor-local, symmetry-invariant, and compressed sector algebras are different outputs.

Symmetry. Optional unitary or projective-unitary representation data, handled according to CONVENTIONS.md (f).

SectorPolicy. Optional projection, subspace, or quotient data; see CONVENTIONS.md (g).

5.4 Output Record

The intended output is a typed record, not only a Hilbert-space dimension:

$$\mathsf{Compile}(E) = (H_E,\ \{H_{E,n}\}_{n\geq 0},\ \mathcal{A}_E,\ U_E,\ P_E,\ Q_E,\ \mathsf{Evidence}_E).$$

H_E. The Hilbert carrier. CA-02 and CA-03 define the first carrier semantics.

$\{H_{E,n}\}$. The particle-number grading when the atom table declares particle degrees.

$\mathcal{A}_E$. The selected observable algebra. This slot is mandatory once any observable claim is made.

U_E. The propagated symmetry representation, if symmetry data were supplied.

P_E. The selected sector projection, if a sector constructor was used.

Q_E. The quotient presentation, if one was requested and its closed relation subspace was named.

$\mathsf{Evidence}_E$. Source-local, checked, proposal, question, or rejected evidence tags for the pieces above.

5.5 Observable-Algebra Policies

The carrier alone does not determine the observable algebra relevant to a model. For example, $H \oplus K$ supports at least:

$$\mathcal{B}(H \oplus K), \qquad \mathcal{B}(H) \oplus \mathcal{B}(K), \qquad \text{off-diagonal creation/annihilation blocks between } H \text{ and } K.$$

All three are legitimate in different contexts; none is forced by the word "OR" alone. Similarly, $H \otimes K$ supports the full algebra $\mathcal{B}(H \otimes K)$ and the tensor-local subalgebra generated by $\mathcal{B}(H) \otimes I$ and $I \otimes \mathcal{B}(K)$.

Compiler rule: no report statement may infer an observable algebra merely from a carrier constructor. The statement must name the observable policy, or remain a carrier-only statement.

5.6 Lamport Dependency Skeleton

$D10$: $\mathsf{Input} = (\mathsf{Atoms}, E, \mathsf{Completion}, \mathsf{ObsPolicy}, \mathsf{Symmetry}, \mathsf{SectorPolicy})$.
$D11$: $\mathsf{Compile}(E) = (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \mathsf{Evidence}_E)$.
$D12$: $\mathsf{ObsPolicy} \in \{\text{carrier-only}, \mathcal{B}(H), \text{block}, \text{tensor-local}, \text{invariant}, \text{compressed}\}$.
$T7$: A carrier claim needs H_E; an observable claim also needs $\mathcal{A}_E$.
$T8$: The compiler analogy is a proposal, not a theorem of canonical quantization.

5.7 Boundary

This contract does not yet compile fusion-category labels, anyonic braid statistics, Hamiltonians, continuum limits, or CFT data. It only fixes the pre-work target: future shards should be able to say exactly which input record they compile and which output slots they have actually justified.

6 Symmetry-Decorated Many-Particle Grammar

6.1 Sources and Dependencies

This shard extends CA-02 and CA-03 by decorating atoms with symmetry data. It uses CONVENTIONS.md (f).

6.2 Decorated Atoms

Fix a topological group G. A symmetry-decorated atom is a pair

$$(K_x, U_x), \qquad U_x : G \to U(K_x),$$

where K_x is the atomic Hilbert space and U_x is a unitary representation. If the intended symmetry is projective, the pre-work convention is to choose a central extension

$$1 \to U(1) \to E \to G \to 1$$

and use an honest unitary representation $U_x : E \to U(K_x)$. The expression is then compiled over E, with a note that the induced physical symmetry is projective over G.

The vacuum atom is $(\mathbb{C}, \mathbf{1})$, where $\mathbf{1}(g)z = z$. The zero atom carries the unique representation on $\{0\}$.

6.3 Propagation Through OR and AND

If E, F have compiled carriers and representations (H_E, U_E) and (H_F, U_F) for the same group or chosen central extension, define

$$H_{\mathrm{OR}(E,F)} = H_E \oplus H_F, \qquad U_{\mathrm{OR}(E,F)}(g) = U_E(g) \oplus U_F(g),$$

and

$$H_{\mathrm{AND}(E,F)} = H_E \mathbin{\widehat{\otimes}} H_F, \qquad U_{\mathrm{AND}(E,F)}(g) = U_E(g) \mathbin{\widehat{\otimes}} U_F(g).$$

For finite-dimensional factors, $\widehat{\otimes} = \otimes$. In infinite-dimensional Hilbert spaces, the completed tensor product is the one already required by CA-03.

These formulas are the Hilbert-space version of the monoidal structure on $\mathrm{Rep}^{\dagger}(G)$: direct sums are block actions and tensor products are diagonal actions. If two atoms use different groups, different central extensions, or incompatible projective multipliers, the compiler must stop and ask for a common symmetry object before forming a single U_E.

6.4 The Derived Full-Fock Representation

For an ordinary unitary representation $U : G \to U(H)$, the full-Fock object is itself a graded representation:

$$\mathfrak{F}(U)(g) = \bigoplus_{n \geq 0} U(g)^{\widehat{\otimes} n} \quad \text{on} \quad \mathfrak{F}(H) = \widehat{\bigoplus}_{n \geq 0} H^{\widehat{\otimes} n},$$

with $U(g)^{\widehat{\otimes} 0} = \mathrm{id}_{\mathbb{C}}$. This preserves the particle-number grading. This is not a new grammar constructor: it is the direct sum of the tensor-power representations already produced by OR and AND, completed as in CA-03. In this pre-work shard, no Shale–Stinespring-type implementability theorem is claimed for general canonical field algebras. Implementing a one-particle unitary on a chosen CAR/CCR observable algebra is a separate field-algebra question.

6.5 Intertwiners and Symmetry-Preserving Compilation

A morphism between decorated atoms or compiled expressions is symmetry-preserving only when it is an intertwiner:

$$TU_E(g) = U_F(g)T \quad \text{for all } g.$$

This matters for compiler optimisations. Rebracketing and distributivity isomorphisms used by CA-02 are acceptable in the decorated grammar only when they also intertwine the propagated actions. The standard Hilbert-space associators, unitors, symmetries, and distributivity maps do so for diagonal direct-sum and tensor-product actions.

6.6 Lamport Dependency Skeleton

$D13$: decorated atom $(K_x, U_x),\ U_x : G \to U(K_x)$.
$D14$: projective symmetry over G is compiled via $U_x : E \to U(K_x)$ for a named central extension E.
$D15$: $U_{E\,\mathrm{OR}\,F} = U_E \oplus U_F, \quad U_{E\,\mathrm{AND}\,F} = U_E \mathbin{\widehat{\otimes}} U_F$.
$D16$: $\mathfrak{F}(U) = \widehat{\bigoplus}_{n\geq 0} U^{\widehat{\otimes} n}, \quad U^{\widehat{\otimes} 0} = 1_{\mathbb{C}}$.
$T9$: The propagated symmetry preserves the CA-02/CA-03 particle grading.
$T10$: A single compiled U_E requires a common group or central extension.

6.7 Boundary

This shard does not yet impose invariance, select charge sectors, form quotient spaces, or define symmetry-invariant observable algebras. Those are sector operations, not mere symmetry propagation, and are treated in CA-06.

7 Sectors, Projections, Subspaces, and Quotients

7.1 Sources and Dependencies

This shard depends on CA-04 and CA-05 and fixes the first sector semantics for the Hilbert-space compiler. It uses CONVENTIONS.md (g).

7.2 The Triangle

Let H be a Hilbert space. The compiler treats a sector first as an orthogonal projection

$$P = P^* = P^2 \in \mathcal{B}(H).$$

It may then expose either of the equivalent Hilbert-space presentations

$$H_P := \operatorname{Ran}(P) \subset H, \qquad H/\ker(P) = H/\operatorname{Ran}(1-P).$$

The map

$$[v] \in H/\ker(P) \longmapsto Pv \in \operatorname{Ran}(P)$$

is a well-defined unitary isomorphism when the quotient has the Hilbert quotient norm. Thus the safe slogan is not "symmetry equals quotient" but:

$$\text{projection} \iff \text{closed subspace} \iff \text{quotient by the named orthogonal complement.}$$

7.3 Finite-Group Invariants

Let G be finite and $U : G \to U(H)$ a unitary representation. Define

$$P_G = \frac{1}{|G|} \sum_{g \in G} U(g).$$

Then P_G is the orthogonal projection onto the invariant subspace

$$H^G = \{v \in H : U(g)v = v \ \forall g \in G\}.$$

Proof. Idempotence follows by reindexing:

$$P_G^2 = \frac{1}{|G|^2} \sum_{g,h} U(gh) = \frac{1}{|G|} \sum_{k} U(k) = P_G.$$

Self-adjointness follows from unitarity and $g \mapsto g^{-1}$:

$$P_G^* = \frac{1}{|G|} \sum_{g} U(g)^* = \frac{1}{|G|} \sum_{g} U(g^{-1}) = P_G.$$

For every k, $U(k)P_G = P_G$, so $\operatorname{Ran}(P_G) \subset H^G$. If $v \in H^G$, then $P_G v = v$, so $H^G \subset \operatorname{Ran}(P_G)$. □

7.4 Coinvariant Quotient

For the same finite action, let

$$R_G = \operatorname{span}\{U(g)v - v : g \in G,\ v \in H\}.$$

Then $R_G \subseteq \ker(P_G)$ because $P_G U(g) = P_G$. Conversely, if $w \in \ker(P_G)$, then

$$w = \frac{1}{|G|} \sum_{g \in G} \big(w - U(g)w\big) \in R_G.$$

Therefore $R_G = \ker(P_G) = (H^G)^\perp$, and the coinvariant quotient

$$H_G := H/R_G$$

is canonically unitarily isomorphic to the invariant subspace H^G. For compact non-finite groups the analogous statement requires a Haar-integral and analytic hypotheses; this shard records only the finite checked case.

7.5 Observable-Algebra Distinctions

There are three different symmetry-related algebras that the compiler must not identify silently:

$$\mathcal{B}(H), \qquad \mathcal{B}(H)^G := \{A : U(g)AU(g)^* = A\ \forall g\}, \qquad P_G \mathcal{B}(H) P_G \cong \mathcal{B}(H^G).$$

The first is the full algebra before imposing symmetry. The second is the algebra of invariant observables on the original carrier. The third is the compressed algebra acting on the invariant sector. A model may choose one of these, but the choice is an observable policy in the sense of CA-04, not a consequence of the word "symmetry" alone.

7.6 Checked Example

The current Julia check uses $G = C_2$ acting on $\mathbb{C}^2$ by the swap matrix S. The averaging projection is

$$P = \frac{1}{2}(I + S) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

The test checks $P^* = P = P^2$, checks $I - P$ is the complementary orthogonal projection, verifies $P(I - P) = 0$, and verifies that $(1, 1)$ is fixed while $(1, -1)$ is killed. This is the first executable invariant for the sector compiler.

7.7 Lamport Dependency Skeleton

$D17$: sector projection $P = P^* = P^2$; $\quad H_P = \operatorname{Ran}(P)$.
$D18$: $H/\ker(P) \xrightarrow{[v] \mapsto Pv} \operatorname{Ran}(P)$.
$D19$: $P_G = |G|^{-1} \sum_g U(g) \quad (G \text{ finite})$.
$D20$: $R_G = \operatorname{span}\{U(g)v - v\}$.
$L1$: $P_G = P_G^* = P_G^2$, $\quad \operatorname{Ran}(P_G) = H^G$.
$L2$: $R_G = \ker(P_G) = (H^G)^\perp$, $\quad H/R_G \cong H^G$.
$T11$: Projection, subspace, and quotient are equivalent only after the relation subspace is named.
$T12$: $\mathcal{B}(H)^G$ and $P_G \mathcal{B}(H) P_G$ are different observable policies.

7.8 Boundary

The finite averaging theorem is not a statement about spontaneous symmetry breaking, gauge constraints, BRST cohomology, braid statistics, or superselection sectors in a continuum CFT. It is the first Hilbert-space sector rule needed by the compiler.

8 Exchange Symmetry and Statistics as Sector Selection

8.1 Sources and Dependencies

This shard discharges part of the deferral in CA-03: ordinary Bose/Fermi statistics are treated as choices of irreducible exchange-symmetry sectors on the full distinguishable Fock carrier. It depends on CA-05 for internal symmetry propagation and CA-06 for projection/subspace/quotient semantics.

8.2 Two Symmetry Layers

There are two distinct actions on many-particle carriers:

$$\text{internal symmetry } G, \qquad \text{exchange symmetry } S_n.$$

If $U : G \to U(H)$ is an internal one-particle representation, then on the n-particle distinguishable sector $H^{\widehat{\otimes} n}$ it acts diagonally:

$$U_n(g) = U(g)^{\widehat{\otimes} n}.$$

The permutation group S_n acts by permuting tensor slots:

$$\Pi_n(\sigma)(v_1 \otimes \cdots \otimes v_n) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)}.$$

The two actions commute:

$$\Pi_n(\sigma) U_n(g) = U_n(g) \Pi_n(\sigma).$$

Thus internal charge sectors and exchange-statistics sectors can be imposed in either order in the ordinary tensor-slot setting.

8.3 Irrep-Labeled Exchange Projectors

The representation $\Pi_n : S_n \to U(H^{\widehat{\otimes} n})$ is the primary datum. Sector projections are supplied by irreducible representation data of S_n, not by specialised grammar constructors. For a one-dimensional unitary character $\chi : S_n \to U(1)$, CA-06 applied to the twisted representation $\overline{\chi}\,\Pi_n$ gives

$$P_{\chi,n} = \frac{1}{n!} \sum_{\sigma \in S_n} \overline{\chi(\sigma)}\, \Pi_n(\sigma),$$

whose range is the χ-sector

$$\{ w \in H^{\widehat{\otimes} n} : \Pi_n(\sigma) w = \chi(\sigma) w \ \forall \sigma \in S_n \}.$$

For the trivial character 1, this is the symmetric projector:

$$P_{\mathrm{s},n} = \frac{1}{n!} \sum_{\sigma \in S_n} \Pi_n(\sigma),$$

and for the sign character sgn, this is the antisymmetric projector:

$$P_{\mathrm{a},n} = \frac{1}{n!} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \Pi_n(\sigma).$$

Their ranges are the symmetric and antisymmetric tensor powers:

$$\mathrm{Ran}(P_{\mathrm{s},n}) = H^{\widehat{\otimes}_s n}, \qquad \mathrm{Ran}(P_{\mathrm{a},n}) = \bigwedge^n H.$$

The corresponding Fock sectors are

$$\Gamma_{\mathrm{s}}(H) = \widehat{\bigoplus}_{n\geq 0} \mathrm{Ran}(P_{\mathrm{s},n}), \qquad \Gamma_{\mathrm{a}}(H) = \widehat{\bigoplus}_{n\geq 0} \mathrm{Ran}(P_{\mathrm{a},n}).$$

Thus "Bose" and "Fermi" name the trivial- and sign-irrep choices for ordinary exchange symmetry. The compiler should record the selected irrep/character and the resulting projection, not a special Bose/Fermi syntax node.

8.4 Quotient Presentations

For a one-dimensional character χ, the sector can also be presented as a quotient of the full tensor algebra by the closed linear span of relations

$$\Pi_n(\sigma)v - \chi(\sigma)v.$$

The trivial character recovers symmetric relations; the sign character recovers antisymmetric signed relations. In the compiler, these quotient presentations are secondary: the primary sector object is the orthogonal projector, because it fixes the inherited Hilbert structure and the compressed observable algebra.

8.5 Para and Anyonic Boundaries

Higher-dimensional irreducible S_n-sectors require central idempotents or matrix-block projectors for the group algebra of S_n. This is exactly the same representation/irrep principle, but this shard records it as a proposal rather than a completed construction; a later shard must source the convention and projector normalization before using it.

Genuinely anyonic exchange is not an S_n-projection problem on ordinary tensor slots. It involves braid-group or categorical data and, in this project, must be handled together with the fusion-category conventions for associators, braiding, and F/R-symbols. No braid-statistics projector is defined here.

8.6 Lamport Dependency Skeleton

$D21$: $U_n(g) = U(g)^{\widehat{\otimes}n}$ (internal symmetry on $H^{\widehat{\otimes}n}$).
$D22$: $\Pi_n : S_n \to U(H^{\widehat{\otimes}n})$ permutes tensor slots.
$D23$: $P_{\chi,n} = n!^{-1} \sum_\sigma \overline{\chi(\sigma)} \Pi_n(\sigma)$ (χ a one-dimensional S_n character).
$L3$: $\Pi_n(\sigma) U_n(g) = U_n(g) \Pi_n(\sigma)$.
$L4$: $P_{\chi,n}$ is an orthogonal projection by CA-06 finite averaging.
$T13$: $\Gamma_{\mathrm{s}}, \Gamma_{\mathrm{a}}$ are the trivial/sign irrep sectors inside full Fock.
$T14$: Anyonic exchange remains deferred to categorical braid data.

8.7 Compiler Consequence

The compiler now has its first nontrivial sector family: given the exchange representation $\Pi_n : S_n \to U(H^{\widehat{\otimes}n})$, a selected irrep character χ supplies $P_{\chi,n}$, its range, its quotient relations, and the chosen compressed observable algebra. There is no primitive Bose, Fermi, or Fock grammar node in this account; Fock is the derived graded representation of CA-05, and Bose/Fermi are names for particular irrep choices of exchange symmetry.

9 Scoping the Hilbert-Space Compiler

9.1 Status, Sources, and Dependencies

Status: specification shard. This shard turns CA-04's contract into a first partial compiler specification. Its scope is the pre-work grammar of CA-02–CA-07: finite Hilbert carriers, derived all-tensor-power completions, unitary/projective symmetries via central extensions, and irrep/projection sectors.

9.2 Compile Judgement

The compiler is a *partial* map. A successful judgement has the form

$$\Gamma; \Pi \vdash E \Downarrow (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \mathsf{Ev}_E),$$

where Γ is the atom table and Π is the policy table. A failed judgement has the form

$$\Gamma; \Pi \vdash E \Uparrow \mathsf{reason}.$$

The failure result is part of the specification: incompatible symmetry data, missing observable policy, or an ill-typed sector constructor must stop compilation.

The atom table entry for x has at least a Hilbert space K_x, a degree $d_x \in \mathbb{N}$ (usually 1), evidence metadata, and optionally symmetry data $U_x : G \to U(K_x)$ or a named projective central extension as in CA-05.

9.3 Surface Language for the First Compiler

The first surface language is deliberately small:

$$E ::= 0 \mid \mathbf{1} \mid x \mid \mathrm{OR}(E, F) \mid \mathrm{AND}(E, F) \mid \mathrm{Sector}_{G,\chi}(E).$$

Here G acts on the compiled carrier and χ is a one-dimensional unitary character; invariants are the case $\chi = 1$. Fock, finite truncation, symmetric sectors, and antisymmetric sectors are derived from OR/AND, representation propagation, and the generic sector operation.

For a one-particle atom x, define tensor-power expressions recursively by

$$x^{[0]} := \mathbf{1}, \qquad x^{[m+1]} := \mathrm{AND}(x^{[m]}, x).$$

Then

$$\mathsf{T}_{\le M}(x) := \mathrm{OR}_{m=0}^{M} x^{[m]}, \qquad \mathsf{T}_{\infty}(x) := \text{Hilbert completion of } \mathrm{OR}_{m\ge 0}\, x^{[m]}.$$

$\mathsf{T}_{\infty}(x)$ is the usual full/distinguishable Fock space $\widehat{\bigoplus}_{m\ge 0} K_x^{\widehat{\otimes} m}$, but as a derived graded representation.

9.4 Constructor Rules

The carrier, grading, and symmetry rules are recursive:

$$H_0 = \{0\}, \quad H_{\mathbf{1}} = \mathbb{C}, \quad H_x = K_x,$$

$$H_{\mathrm{OR}(E,F)} = H_E \oplus H_F, \qquad H_{\mathrm{AND}(E,F)} = H_E \mathbin{\widehat{\otimes}} H_F.$$

The grading rules are CA-02's:

$$H_{\mathrm{OR}(E,F),n} = H_{E,n} \oplus H_{F,n}, \qquad H_{\mathrm{AND}(E,F),n} = \bigoplus_{i+j=n} H_{E,i} \mathbin{\widehat{\otimes}} H_{F,j}.$$

If U_E, U_F are representations of the same group or named central extension, the symmetry rules are

$$U_{\mathrm{OR}}(g) = U_E(g) \oplus U_F(g), \qquad U_{\mathrm{AND}}(g) = U_E(g) \mathbin{\widehat{\otimes}} U_F(g).$$

If the symmetry objects are not the same, the carrier may still compile, but the symmetry output is rejected unless the requested output was explicitly carrier-only.

For the derived tensor-power closures,

$$H_{\mathsf{T}_\infty(x)} = \widehat{\bigoplus}_{m\geq 0} K_x^{\widehat{\otimes} m}, \qquad H_{\mathsf{T}_{\leq M}(x)} = \bigoplus_{m=0}^{M} K_x^{\otimes m}.$$

For finite examples the compiler may report dimensions:

$$\begin{aligned} \dim \mathrm{OR} &= \dim H_E + \dim H_F, \\ \dim \mathrm{AND} &= (\dim H_E)(\dim H_F), \\ \dim \mathsf{T}_{\leq M}(x) &= \sum_{m=0}^{M} (\dim K_x)^m. \end{aligned}$$

These follow from choosing orthonormal bases: disjoint union gives direct-sum bases, ordered pairs give tensor-product bases, and truncation is a finite sum.

Sector constructors return projection witnesses supplied by representation data. For finite G, a unitary character $\chi : G \to U(1)$, and representation $U_E : G \to U(H_E)$, set

$$P_\chi = |G|^{-1} \sum_{g\in G} \overline{\chi(g)}\, U_E(g), \qquad H_{\mathrm{Sector}_{G,\chi}(E)} = \mathrm{Ran}(P_\chi).$$

The quotient relation is generated by $U_E(g)v - \chi(g)v$. The invariant sector is $\chi = 1$. For exchange sectors on $K^{\widehat{\otimes} n}$, $G = S_n$ and $U_E = \Pi_n$; the trivial and sign characters give

$$P_{\mathrm{s},n} = n!^{-1} \sum_{\sigma\in S_n} \Pi_n(\sigma), \qquad P_{\mathrm{a},n} = n!^{-1} \sum_{\sigma\in S_n} \mathrm{sgn}(\sigma)\Pi_n(\sigma).$$

The observable algebra is never inferred from these formulas. It is the policy slot Π_{obs}: carrier-only, full, block, tensor-local, invariant, or compressed.

9.5 Example 1: A Maybe Qubit

Let $Q = \mathbb{C}^2$ with basis e_0, e_1. Compile

$$E_1 = \mathrm{OR}(\mathbf{1}, Q).$$

The rule gives

$$H_{E_1} = \mathbb{C} \oplus \mathbb{C}^2, \qquad H_{E_1,0} = \mathbb{C}, \quad H_{E_1,1} = \mathbb{C}^2, \quad \dim H_{E_1} = 3.$$

With full observable policy, $\mathcal{A}_{E_1} = \mathcal{B}(\mathbb{C} \oplus \mathbb{C}^2)$. With block policy,

$$\mathcal{A}_{E_1}^{\mathrm{block}} = \mathcal{B}(\mathbb{C}) \oplus \mathcal{B}(\mathbb{C}^2).$$

Thus the same carrier has two different compiled observable outputs.

9.6 Example 2: Two Distinguishable Qubits

Compile $E_2 = \mathrm{AND}(Q, Q)$. The carrier is

$$H_{E_2} = Q \otimes Q, \qquad \{e_0 \otimes e_0, e_0 \otimes e_1, e_1 \otimes e_0, e_1 \otimes e_1\} \text{ is a basis,}$$

so $\dim H_{E_2} = 4$, all in degree 2. The tensor-local policy is the subalgebra generated by

$$\mathcal{B}(Q) \otimes I, \qquad I \otimes \mathcal{B}(Q),$$

whereas the full policy is $\mathcal{B}(Q \otimes Q)$. The compiler records which one was requested; it does not identify them by syntax alone.

9.7 Example 3: Derived Tensor-Power Closure

Compile the macro-expanded expression

$$E_3 = \mathsf{T}_{\leq 2}(Q) = \mathrm{OR}(\mathbf{1}, \mathrm{OR}(Q, \mathrm{AND}(Q, Q))).$$

The output carrier is

$$H_{E_3} = \mathbb{C} \oplus Q \oplus (Q \otimes Q),$$

with sector dimensions

$$\dim H_0 = 1, \qquad \dim H_1 = 2, \qquad \dim H_2 = 4, \qquad \dim H_{E_3} = 1 + 2 + 4 = 7.$$

If $U : G \to U(Q)$ is supplied, symmetry propagation gives

$$U_{E_3}(g) = 1_{\mathbb{C}} \oplus U(g) \oplus (U(g) \otimes U(g)).$$

No Bose/Fermi relation has been imposed yet; this is still the distinguishable graded representation.

9.8 Example 4: Two-Qubit Exchange-Irrep Sectors

Let S be the swap on $Q \otimes Q$. Since

$$S(e_0e_0) = e_0e_0, \quad S(e_1e_1) = e_1e_1, \quad S(e_0e_1) = e_1e_0, \quad S(e_1e_0) = e_0e_1,$$

where e_ie_j abbreviates $e_i \otimes e_j$, the exchange representation of S_2 has two one-dimensional irrep characters, trivial and sign, supplying projectors with ranges:

$$P_{\mathrm{s}} = \frac{1}{2}(I + S), \qquad P_{\mathrm{a}} = \frac{1}{2}(I - S)$$

$$\mathrm{Ran}(P_{\mathrm{s}}) = \mathrm{span}\{e_0e_0,\ (e_0e_1 + e_1e_0)/\sqrt{2},\ e_1e_1\},$$
$$\mathrm{Ran}(P_{\mathrm{a}}) = \mathrm{span}\{(e_0e_1 - e_1e_0)/\sqrt{2}\}.$$

Thus the compiler reports dimensions 3 and 1 for the trivial- and sign-irrep sectors. The Julia test suite checks the projection, orthogonality, trace/dimension, and basis-vector claims for this example.

9.9 Example 5: A Finite Invariant Quotient

Let $C_2 = \{1, s\}$ act on $H = \mathbb{C}^2$ by the swap matrix. Then

$$P_{C_2} = \frac{1}{2}(I + S) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

Its range is $\mathrm{span}\{(1,1)\}$, and its kernel is $\mathrm{span}\{(1,-1)\}$. The quotient presentation is therefore

$$\mathbb{C}^2/\,\mathrm{span}\{(1,-1)\} \cong \mathrm{span}\{(1,1)\}.$$

This is $\mathrm{Sector}_{C_2,1}(H)$, the trivial-irrep sector of the C_2 representation. The invariant observable algebra on the original carrier consists of the operators commuting with S, while the compressed algebra on the invariant line is one-dimensional. These are different observable policies even in this small example.

9.10 Example 6: A Required Compile Error

Let A, B be atoms with projective symmetries over the same physical group G, but represented in the input by unrelated central extensions $E_A \to G$ and $E_B \to G$. The carrier of $\mathrm{OR}(A, B)$ is still $K_A \oplus K_B$. However, a request for one compiled symmetry representation must fail:

$$\Gamma; \Pi \vdash \mathrm{OR}(A, B) \Uparrow \text{“no common symmetry group or central extension specified”}.$$

This is not a missing feature; it is the strict semantics required by CA-05 and CONVENTIONS.md (f).

9.11 Lamport Dependency Skeleton

$D24$: $\Gamma; \Pi \vdash E \Downarrow (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \mathsf{Ev}_E)$ or $\Gamma; \Pi \vdash E \Uparrow \mathsf{reason}$.
$D25$: $E ::= 0 \mid \mathbf{1} \mid x \mid \mathrm{OR} \mid \mathrm{AND} \mid \mathrm{Sector}_{G,\chi}$.
$D26$: $\mathsf{T}_{\le M}(x) = \mathrm{OR}_{m=0}^{M} x^{[m]}, \quad H_{\mathsf{T}_{\le M}} = \bigoplus_{m=0}^{M} K_x^{\otimes m}$.
$D27$: $P_\chi = |G|^{-1} \sum_g \overline{\chi(g)} U_E(g), \quad H_{\mathrm{Sector}_{G,\chi}} = \mathrm{Ran}(P_\chi)$.
$T15$: Finite dimensions obey sum, product, and $\sum_{m=0}^{M} d^m$ by explicit basis constructions.
$T16$: Irrep/character choices supply projection/subspace/quotient witnesses.
$T17$: Compiler failure is a specified outcome when requested structure is ill typed or unsupported.

9.12 Acceptance Criteria and Boundary

Future implementation work must preserve these acceptance criteria: every constructor fills typed output slots; every nontrivial mathematical output has an evidence tag; every quotient names its relation subspace; every observable claim names its observable policy; and every executable example asserts an invariant, not merely successful execution.

This compiler does not yet parse fusion-category simples, anyonic braid data, Hamiltonians, lattice limits, conformal nets, or VOA data. Those enter only after their source conventions and invariants are fixed. CA-08 is the finite Hilbert-space kernel that later shards must extend without weakening the output record, the evidence discipline, or the failure semantics.

10 Fibonacci Kinematics as a Fusion-Rule Sector

10.1 Status, Sources, and Dependencies

Status: checked kinematic shard. This shard confirms the side conjecture: before braiding, Hamiltonians, or continuum limits, the Fibonacci fusion-path Hilbert spaces are a sector construction from one physical atom τ. The vacuum $\mathbf{1}$ is still present as the tensor unit and as a charge label; it is not a second physical particle atom.

Dependencies: CA-02 for AND/OR/vacuum, CA-06 for projection/subspace semantics, CA-08 for compiler output slots, and CONVENTIONS.md (c) for the left-associated kinematic path basis.

10.2 One Physical Atom, Two Charge Labels

The physical input word is n copies of one anyon type:

$$\tau, \tau, \ldots, \tau.$$

The charge labels used to describe intermediate fusion outcomes are

$$L = \{\mathbf{1}, \tau\}.$$

Thus the proposed slogan is correct only with this distinction:

$$\text{one physical atom } \tau \quad + \quad \text{charge labels } \{\mathbf{1}, \tau\} \quad + \quad \text{fusion-rule sector selection.}$$

The vacuum $\mathbf{1}$ is required because it is both the tensor unit and a possible total charge. It is not an additional non-vacuum particle species.

10.3 Ambient Path Space and Admissibility Projector

Use the left-associated path basis fixed in CONVENTIONS.md (c). For n physical τ-atoms and total charge $c \in \{\mathbf{1}, \tau\}$, define the ambient word space

$$W_{n,c} = \ell^2\{(x_0, \ldots, x_n) : x_i \in \{\mathbf{1}, \tau\},\ x_0 = \mathbf{1},\ x_n = c\}.$$

The basis vector $|x_0, \ldots, x_n\rangle$ records the cumulative charge after successively fusing the first i copies of τ.

The Fibonacci fusion rules needed here are

$$\mathbf{1} \otimes \tau = \tau, \qquad \tau \otimes \mathbf{1} = \tau, \qquad \tau \otimes \tau = \mathbf{1} \oplus \tau.$$

Let $N_{ab}^{c} \in \{0, 1\}$ be these multiplicity-free fusion coefficients. Define the diagonal operator

$$P_{n,c}|x_0, \ldots, x_n\rangle = \left(\prod_{i=1}^{n} N_{x_{i-1},\tau}^{x_i}\right)|x_0, \ldots, x_n\rangle.$$

Since every diagonal entry is 0 or 1, $P_{n,c} = P_{n,c}^{*} = P_{n,c}^{2}$. The kinematic Fibonacci Hilbert space is therefore the sector

$$H_{n,c}^{\mathrm{Fib}} := \mathrm{Ran}(P_{n,c}) \subseteq W_{n,c}.$$

This is exactly the CA-06 projection/subspace semantics, but the projector is not supplied by a group character. It is supplied by categorical fusion-rule admissibility.

10.4 Dimension Recurrence

Let

$$a_n = \dim H^{\mathrm{Fib}}_{n,\mathbf{1}}, \qquad b_n = \dim H^{\mathrm{Fib}}_{n,\tau}.$$

At $n = 0$, no physical τ-atom has been fused, so

$$a_0 = 1, \qquad b_0 = 0.$$

Adding one more τ gives the transition equations

$$a_{n+1} = b_n, \qquad b_{n+1} = a_n + b_n.$$

The first equation says total charge $\mathbf{1}$ can only come from $\tau \otimes \tau \to \mathbf{1}$. The second says total charge τ can come from $\mathbf{1} \otimes \tau \to \tau$ or from $\tau \otimes \tau \to \tau$. Hence

$$b_{n+1} = b_n + b_{n-1}, \qquad b_0 = 0, \quad b_1 = 1.$$

Thus the total-τ sector dimensions are Fibonacci numbers:

$$0, 1, 1, 2, 3, 5, 8, \ldots.$$

The Julia test suite checks the pair sequence

$$(a_n, b_n) = (1,0), (0,1), (1,1), (1,2), (2,3), (3,5), (5,8), (8,13)$$

for $0 \le n \le 7$, including the recurrence.

10.5 Small Examples

$n = 1$. For total charge τ, the ambient basis is $|\mathbf{1}, \tau\rangle$, and it is admissible. Thus $H^{\mathrm{Fib}}_{1,\tau} \cong \mathbb{C}$.

$n = 2$. The admissible paths are

$$|\mathbf{1}, \tau, \mathbf{1}\rangle \in H^{\mathrm{Fib}}_{2,\mathbf{1}}, \qquad |\mathbf{1}, \tau, \tau\rangle \in H^{\mathrm{Fib}}_{2,\tau}.$$

This is the kinematic content of $\tau \otimes \tau = \mathbf{1} \oplus \tau$.

$n = 3$. The total-τ ambient basis has two admissible paths:

$$|\mathbf{1}, \tau, \mathbf{1}, \tau\rangle, \qquad |\mathbf{1}, \tau, \tau, \tau\rangle.$$

The forbidden word $|\mathbf{1}, \mathbf{1}, \tau, \tau\rangle$, for example, is killed because $N^{\mathbf{1}}_{\mathbf{1},\tau} = 0$. Therefore $\dim H^{\mathrm{Fib}}_{3,\tau} = 2$.

10.6 Compiler Consequence

This is the first example where the Hilbert-space compiler wants category data. At the carrier level, it can compile the ambient finite word space $W_{n,c}$ and the diagonal sector projection $P_{n,c}$. But producing $P_{n,c}$ from the expression "n copies of τ" requires the fusion coefficients N^c_{ab}. That is already tensor-category input, not ordinary Hilbert-space grammar.

Moreover, the source makes clear that different fusion orders give different fusion-tree bases and are related by F-moves. Therefore a full Fibonacci compiler must eventually output not only

$$H^{\mathrm{Fib}}_{n,c}$$

but also the coherent recoupling data that identifies different bracketings: the F-symbols satisfying the pentagon equations, and later the R-symbols for braiding. The current shard is only kinematic: it builds the admissible sector and counts it.

10.7 Lamport Dependency Skeleton

$D28 : L = \{\mathbf{1}, \tau\}, \quad \mathbf{1} \otimes \tau = \tau, \quad \tau \otimes \tau = \mathbf{1} \oplus \tau.$
$D29 : W_{n,c} = \ell^2\{(x_0, \ldots, x_n) : x_0 = \mathbf{1},\ x_n = c,\ x_i \in L\}.$
$D30 : P_{n,c}|x\rangle = \prod_i N_{x_{i-1},\tau}^{x_i}|x\rangle.$
$D31 : H_{n,c}^{\mathrm{Fib}} = \mathrm{Ran}(P_{n,c}).$
$L5 : P_{n,c} = P_{n,c}^* = P_{n,c}^2.$
$T18 : \dim H_{n,\tau}^{\mathrm{Fib}}$ obeys $b_{n+1} = b_n + b_{n-1},\ b_0 = 0,\ b_1 = 1.$
$T19 :$ Full Fibonacci compilation needs F-coherent tensor-category data.

10.8 Boundary

This shard does not fix the Fibonacci F-matrix gauge, any R-symbols, braid representations, golden-chain Hamiltonians, or a continuum limit. It only realises the fusion-path Hilbert spaces kinematically as admissibility sectors.

11 Why Lattice Symmetry Generators Are a Problem

11.1 Status, Sources, and Dependencies

Status: proposal shard with sourced boundary. The claim that a lattice boost should look like a first moment of the Hamiltonian density is an intuition until a local source, derivation, or test is attached. This shard sets the boundary for CA-10–CA-17.

Dependencies: CA-05 for symmetry as unitary/projective-unitary representation, CA-08 for compiler output slots, and CONVENTIONS.md (h) for lattice candidate generator signs.

11.2 The Minimal Quantum-Mechanical Input

An unadorned quantum system gives a Hilbert space and, once dynamics is chosen, a self-adjoint Hamiltonian. By Stone's theorem this is the same kind of datum as a strongly continuous unitary representation of the additive group $\mathbb{R}$:

$$U(t) = \exp(-itH).$$

Thus the one symmetry forced by the bare dynamical presentation is time translation. Everything beyond this requires extra structure: spatial coordinates, locality, a metric, a lattice graph, or a continuum target.

11.3 What the Lattice Adds

A lattice Hamiltonian is not merely a self-adjoint operator. It usually comes as a decomposition

$$H = \sum_j h_j$$

where each h_j is local near a site, bond, cell, or other geometric support. The cited lattice-fermion source uses exactly this pattern: a Hamiltonian is a sum of self-adjoint finite-support terms, together with an observable algebra whose generators are intended as discretisations of continuum fields.

This decomposition is extra data. Two different decompositions of the same operator H can lead to different candidate densities and hence different candidate symmetry generators. Therefore the compiler target is not

$$\text{Hilbert space plus one operator } H,$$

but rather

$$(\text{Hilbert space},\ \text{lattice geometry},\ \{h_j\},\ \text{scaling map}).$$

11.4 The Continuum Target

If the intended continuum theory is Lorentz-invariant in $d+1$ dimensions, the target symmetry is not just $\mathbb{R}$-time translation. It is a projective-unitary representation of the Poincare group, lifted as necessary to a unitary representation of a covering group. Its translation subgroup has self-adjoint generators

$$P_0 = H, \qquad P_1, \ldots, P_d,$$

where the P_a are momenta.

Motivating question. If the microscopic input only supplies a lattice Hamiltonian density $H = \sum_j h_j$, how do we guess the lattice analogues of

$$P_a, \qquad M_{ab} \text{ rotations}, \qquad M_{0a} \text{ boosts?}$$

11.5 Intuition to Be Checked

Intuition 1: a rotation is a position-dependent spatial translation. In the plane, the vector field for an infinitesimal rotation about the origin is

$$v(x, y) = (-y, x).$$

Thus if we can identify local momentum densities, rotations should be spatial first moments of those densities.

Intuition 2: a boost is a position-dependent time translation. Since time translation is generated by energy, the one-dimensional boost candidate is a first moment of the energy density,

$$K_x \ \propto \ \sum_j x_j h_j.$$

Intuition 3: the commutator of a boost with time translation should produce spatial translation, up to sign and normalization. On the lattice this suggests that

$$i[H, K_x]$$

is a candidate momentum generator. CA-12 derives the nearest-neighbour formula from this ansatz and records the sign convention explicitly.

11.6 Source Boundary

Weinberg QFT Vol. 1 is now registered with targeted OCR anchors, and the original Koo–Saleur source is now registered from arXiv with source-TeX and PDF text anchors. The present shard still cites Schottenloher for continuum symmetry definitions, OAR/wavelet sources for scaling-limit translation recovery, and the local lattice-fermion source for rigorous free-fermion Koo–Saleur convergence. CA-14 cites the original Koo–Saleur paper for the source-local lattice Virasoro prototype.

12 The Continuum Bulk Symmetry Target

12.1 Status, Sources, and Dependencies

Status: sourced definitions plus checked local derivation. The Poincare group and the projective-unitary implementation are sourced. The commutator table below is a local vector-field derivation from the sourced metric-preserving description, and its signs are pinned by a Julia check.

Dependencies: CA-05, CA-10, and CONVENTIONS.md (h), (k).

12.2 Group-Level Target

For a $D = d + 1$ dimensional Lorentzian target, Schottenloher describes the Poincare group as the semidirect product of the Lorentz group and translations:

$$P = SO_0(1, d) \ltimes \mathbb{R}^{d+1},$$

with the usual replacement by the relevant covering group when the quantum symmetry is implemented projectively. The translation subgroup is abelian and has commuting self-adjoint generators P_μ. In the source convention,

$$P_0 = H, \qquad P_a \text{ are spatial momenta } (a = 1, \ldots, d).$$

For Euclidean two-dimensional CFT language, the Euclidean motion group is generated by rotations and translations. This gives the Wick-rotated analogue: rotations and translations are the geometric target before scale/conformal extensions are added.

12.3 Vector-Field Derivation

Let $\eta = \mathrm{diag}(1, -1, \ldots, -1)$, and lower indices by $x_\mu = \eta_{\mu\nu} x^\nu$. Use vector fields

$$P_\mu = \partial_\mu, \qquad M_{\mu\nu} = x_\mu \partial_\nu - x_\nu \partial_\mu, \qquad M_{\mu\nu} = -M_{\nu\mu}.$$

The source gives the metric-preserving Lie algebra condition $\omega^T \eta + \eta\omega = 0$. The $M_{\mu\nu}$ form the corresponding coordinate vector fields, while P_μ are translations.

Compute the Lie bracket of vector fields. Since the P_μ have constant coefficients,

$$[P_\mu, P_\nu] = 0.$$

For the mixed bracket,

$$\begin{aligned}[M_{\mu\nu}, P_\rho] &= -\partial_\rho(x_\mu)\partial_\nu + \partial_\rho(x_\nu)\partial_\mu \\ &= \eta_{\nu\rho} P_\mu - \eta_{\mu\rho} P_\nu.\end{aligned}$$

For Lorentz-Lorentz brackets,

$$\begin{aligned}[M_{\mu\nu}, M_{\rho\sigma}] = \eta_{\nu\rho} M_{\mu\sigma} - \eta_{\mu\rho} M_{\nu\sigma} \\ + \eta_{\mu\sigma} M_{\nu\rho} - \eta_{\nu\sigma} M_{\mu\rho}.\end{aligned}$$

The Julia check evaluates representative brackets using this convention. For example in $3 + 1$ dimensions,

$$[M_{01}, P_0] = -P_1, \qquad [M_{01}, P_1] = -P_0.$$

12.4 Boost-Time Relation

With $K_a := M_{0a}$, the vector-field convention gives

$$[K_a, P_0] = -P_a.$$

Equivalently, $[P_0, K_a] = P_a$. The passage from this vector-field statement to self-adjoint commutator notation is the separate sign convention in CONVENTIONS.md (k): Stone generators use $U_X(t) = \exp(-itA_X)$, the spacetime flow is implemented passively, and the resulting target is

$$i[H, K_a] = P_a, \qquad i[P_b, K_a] = \delta_{ab} H.$$

Spatial rotations carry the additional convention $J_{ab} = -A_{M_{ab}}$, which is why the Gaussian one-particle table has $i[K_a, K_b] = J_{ab}$ even though the active vector fields obey $[M_{0a}, M_{0b}] = -M_{ab}$. CA-12 keeps finite-lattice signs explicit under CONVENTIONS.md (h).

12.5 Boundary

This shard does not claim that a finite lattice has an exact Poincare representation. It only identifies the continuum algebraic target whose relations a sequence of lattice candidates should approximate after scaling.

13 A One-Dimensional Boost Produces a Momentum Current

13.1 Status, Sources, and Dependencies

Status: checked local derivation. The identity in this shard is not quoted from a source. It is derived from the local Hamiltonian-density ansatz and checked by a small coefficient invariant.

Dependencies: CA-10, CA-11, and CONVENTIONS.md (h).

13.2 Setup

Let a finite open one-dimensional lattice have sites or bonds labelled $j = 1, \ldots, N$, real positions x_j, and a self-adjoint local-density decomposition

$$H = \sum_{j=1}^{N} h_j.$$

Assume the density terms commute at distance greater than one:

$$[h_j, h_k] = 0 \qquad \text{if } |j-k| > 1.$$

This is the finite-range simplification appropriate for the first derivation; longer-range densities only add longer-range edge terms.

Define the position-weighted energy moment

$$K_x = \sum_{j=1}^{N} x_j h_j.$$

This is the lattice boost candidate in the sense of CA-10: a boost is treated as a position-dependent time translation, and time translation is generated by energy density.

13.3 Commutator Calculation

Compute

$$\begin{aligned} i[H, K_x] &= i\left[\sum_j h_j, \sum_k x_k h_k\right] \\ &= i\sum_{j,k} x_k [h_j, h_k]. \end{aligned}$$

The terms with $j = k$ vanish, and by the nearest-neighbour assumption only pairs $(j, j+1)$ survive. For each adjacent pair,

$$i x_{j+1}[h_j, h_{j+1}] + i x_j [h_{j+1}, h_j] = i(x_{j+1} - x_j)[h_j, h_{j+1}].$$

Hence

$$i[H, K_x] = \sum_{j=1}^{N-1} (x_{j+1} - x_j)\, i[h_j, h_{j+1}].$$

Each $i[h_j, h_{j+1}]$ is self-adjoint because h_j, h_{j+1} are self-adjoint.

13.4 Uniform Spacing

If $x_j = aj$, then $x_{j+1} - x_j = a$ and

$$i[H, K_x] = a \sum_{j=1}^{N-1} i[h_j, h_{j+1}].$$

For unit spacing this gives the candidate bulk momentum

$$P_{\text{bulk}}^{\text{lat}} := \sum_{j=1}^{N-1} i[h_j, h_{j+1}],$$

with endpoint terms deliberately visible. Periodic chains require an extra boundary convention because the position function j is not single-valued on a circle.

13.5 Interpretation

The continuum bracket in CA-11 says that boost plus time translation gives a spatial translation, up to sign convention. The lattice calculation says that the same operation applied to the first energy moment produces the commutator of neighbouring energy densities. Therefore $i[h_j, h_{j+1}]$ is not guessed from thin air: it is the local current forced by the first-moment ansatz.

13.6 Boundary

This is not yet a theorem that $P_{\text{bulk}}^{\text{lat}}$ converges to continuum momentum. The missing ingredients are a scaling map, a domain for the candidate operators, a state/sector sequence, and convergence of the commutator relations. CA-15 turns these into acceptance checks.

14 Position-Dependent Bulk Generators

14.1 Status, Sources, and Dependencies

Status: proposal shard with local derivation pattern. The sourced continuum content is geometric: vector fields, translations, and rotations. The higher-dimensional lattice formulas are compiler targets, not established convergence claims, and this shard does not define a stress-energy tensor or a lattice momentum density.

Dependencies: CA-10–CA-12 and CONVENTIONS.md (h).

14.2 Continuum Picture

Translations are flows of constant vector fields. In the Euclidean plane, the infinitesimal motion

$$X(z) = c + i\theta z$$

contains both translations (c) and rotations ($i\theta z$). Written in real coordinates, the rotation part is the position-dependent vector field

$$(x, y) \mapsto (-\theta y, \theta x).$$

This is the precise continuum content of the slogan:

$$\text{rotation} = \text{position-dependent translation}.$$

14.3 Scalar Ramps

Suppose lattice density terms are attached to positions $r \in \Lambda \subset \mathbb{R}^d$. For a scalar function $f : \Lambda \to \mathbb{R}$, define

$$K_f := \sum_{r \in \Lambda} f(r) h_r.$$

The same derivation as CA-12 gives, for commuting non-overlapping densities,

$$i[H, K_f] = \sum_{\{r,s\}} (f(s) - f(r))\, i[h_r, h_s],$$

where the sum is over unordered overlapping pairs after a choice of orientation. This is a discrete-gradient response. Linear ramps

$$f_a(r) = r^a$$

are therefore the first candidates for spatial momentum components.

14.4 Boost Candidates

Boosts can be attempted directly from energy density:

$$K_a^{\text{lat}} := \sum_{r \in \Lambda} r^a h_r.$$

Then $i[H, K_a^{\text{lat}}]$ gives the a-direction current candidate. This is the higher-dimensional version of the one-dimensional derivation, but it depends on the chosen density supports and orientations.

14.5 Rotation Candidates

Rotations require one more layer. In continuum notation,

$$J_{ab} = \int (x_a p_b(x) - x_b p_a(x))\, dx.$$

Here this formula is only a continuum mnemonic for why rotations should involve an angular-momentum density. The Schottenloher anchors above support the underlying source vector fields and rotations, not a QFT stress-energy tensor, not a momentum-density formula, and not a lattice dictionary for $p_a(x)$. Thus an actual lattice rotation candidate must be routed through a later momentum-density/current convention rather than built directly from h_r. Once such a convention is fixed, the provisional compiler shape is:

$$J^{\mathrm{lat}}_{ab} \approx \sum_r \big(r_a p^{\mathrm{lat}}_b(r) - r_b p^{\mathrm{lat}}_a(r)\big),$$

where p^{lat}_a must come with its own source, derivation, or checked continuity-equation evidence.

14.6 Compiler Moral

The lattice Hilbert-space compiler cannot infer these operators from the carrier alone. It needs a lattice geometry, a local Hamiltonian-density decomposition, and a target vector-field algebra. The output is not an exact symmetry by default; it is a candidate list with proof obligations.

14.7 Boundary

No stress-energy convention is fixed here. No cell-orientation convention is fixed here. No periodic wrapping convention is fixed here. No higher-dimensional convergence theorem is fixed here. Those choices must be made in a later shard before any numerical or categorical lattice model uses these candidates as checked symmetries.

15 Koo–Saleur as the Prototype

15.1 Status, Sources, and Dependencies

Status: source-local prototype plus later rigorous precedent. The original Koo–Saleur paper is registered here and supports the lattice Virasoro prototype as a conjectural construction with numerical and analytic checks for XXZ/RSOS/Temperley–Lieb settings. This shard keeps that separate from the local lattice-fermion/OAR treatment, which states and proves a free-fermion Koo–Saleur convergence result.

Dependencies: CA-10–CA-13.

15.2 Continuum Side

In a two-dimensional CFT on the circle, the Hamiltonian is expressed as an integral of a Hamiltonian density,

$$H = \int h(x)\, dx.$$

The source identifies $h(x)$ as a local quantum field built from the chiral and anti-chiral energy-momentum tensors $T, \bar{T}$. Its Fourier modes H_k are linear combinations of the Virasoro generators $L_k, \bar{L}_k$.

Thus in this special setting the desired spacetime symmetries are not abstract: they are generated by modes of the stress tensor.

15.3 Lattice Side

The Koo–Saleur move is to imitate the continuum construction using the lattice Hamiltonian density. From a lattice density

$$h^{(N)} : \Lambda_N \to \mathcal{A}_N$$

one forms Fourier modes $H_k^{(N)}$. In the original Temperley–Lieb presentation, the Hamiltonian-limit density is represented by ground-state-subtracted $e_j - e_\infty$ terms, while the second stress-tensor component is represented by nearest-neighbour commutators $[e_j, e_{j+1}]$. The lattice Virasoro candidates $l_n, \bar{l}_{-n}$ are therefore Fourier sums of

$$(e_j - e_\infty) \;\pm\; \text{constant} \cdot [e_j, e_{j+1}],$$

with the $c/24$ zero-mode shift. The later free-fermion/OAR Definition 4.1 uses the same structural pattern: density modes plus commutator corrections against the lattice Hamiltonian mode $H_0^{(N)}$ produce lattice analogues of L_k and $\bar{L}_k$.

The important structural point for our programme is:

$$\text{lattice symmetry generator} = \text{weighted Hamiltonian density} + \text{commutator correction}.$$

This is the same motif as CA-12, but now with Fourier weights and a Virasoro target rather than a first moment and a Poincare translation target.

15.4 Scaling Representation Matters

The original source already builds in ground-state subtraction, normal ordering, and a double-limit process; it also warns that finite-L lattice quantities do not close as the Virasoro algebra and that a scaling limit of a commutator is not generally the commutator of scaling limits. The

later rigorous free-fermion/OAR source similarly makes convergence depend on the correct scaling-limit representation and on subtracting ground-state expectation values, corresponding to normal ordering. Therefore the compiler must not output only formal sums. It must also output:

state sequence,
renormalisation/scaling maps,
vacuum subtraction policy,
operator topology target.

15.5 What This Foreshadows

Koo–Saleur is a prototype for the lattice-symmetry compiler:

$$\{h_j\} \longmapsto \{H_k^{(N)}, L_k^{(N)}, \bar{L}_k^{(N)}\} \longrightarrow \{L_k, \bar{L}_k\}.$$

The first arrow is algebraic and finite-scale. The second is analytic and requires a scaling theorem.

15.6 Boundary

The original Koo–Saleur source supports the prototype as a conjectural and numerically checked construction for XXZ/RSOS/Temperley–Lieb lattice models; it does not prove a general rigorous convergence theorem for arbitrary microscopic models. The convergence theorem cited here is still the later free-fermion/OAR result, and arbitrary anyon chains remain outside both sourced claims.

16 Acceptance Tests for Lattice Symmetry Candidates

16.1 Status, Sources, and Dependencies

Status: specification shard. This shard does not prove convergence for new models. It states the checks required before a candidate generator is promoted from proposal to checked continuum symmetry data.

Dependencies: CA-10–CA-14.

16.2 Local Finite-Scale Checks

At fixed lattice size N, a candidate generator G_N must first pass finite operator checks:

$$G_N = G_N^*, \qquad G_N \text{ is built from named local density terms,} \qquad \operatorname{supp}(G_N) \text{ policy is explicit.}$$

For unbounded thermodynamic-limit candidates, the finite check is only the first layer; domains must later be named.

16.3 Algebraic Checks

Given a target Lie algebra with generators X_a, the lattice candidates $G_a^{(N)}$ should satisfy commutator closure in a controlled sense:

$$i[G_a^{(N)}, G_b^{(N)}] \approx f_{ab}{}^c G_c^{(N)} + \text{central or boundary terms.}$$

The approximation relation must be specified: exact finite-size equality, matrix-element convergence, strong resolvent convergence, strong operator convergence on a core, or convergence after compression to a low-energy sector.

16.4 Scaling Checks

The OAR source shows that continuous spatial translations can be reconstructed from discrete dyadic translations in a suitable scaling limit, with the usual momentum operators as generators. It also states that analogous convergence for Lorentz or conformal transformations requires further work. This is the main warning for this programme:

$$\text{translation recovery} \not\Rightarrow \text{Poincare recovery.}$$

Therefore every candidate boost or rotation needs its own scaling statement. For a sequence G_N, the report must say:

$$G_N \xrightarrow{?} G$$

and name the topology, state sequence, renormalisation maps, and dense domain or core on which the statement is intended.

16.5 Observable Covariance Checks

The target is not only an operator on states. It is a symmetry action on the observable system. The evidence payload should include

$$\alpha_N^G(A) = e^{itG_N} A e^{-itG_N}$$

or the infinitesimal commutator $i[G_N, A]$, together with the expected transformation of local algebras. For translations this means moving supports. For rotations it means rotating supports. For boosts it means mixing time and space in the continuum representation.

16.6 Acceptance Payload

A lattice generator candidate is accepted into the compiler only with:

1. density provenance: the exact h_j and supports,
2. finite algebra checks: self-adjointness and commutators,
3. renormalisation maps and state sequence,
4. operator topology or matrix-element convergence target,
5. observable covariance statement,
6. boundary/central/vacuum-subtraction policy.

16.7 Boundary

Time-zero net axioms and Lieb-Robinson bounds give important locality control, but local causality is not the same as full Poincare covariance. The compiler must keep these evidence levels separate.

17 A Compiler Interface for Lattice Symmetry Candidates

17.1 Status, Sources, and Dependencies

Status: compiler specification shard. This shard extends CA-08 at the level of output slots. It is not yet an implementation.

Dependencies: CA-08 and CA-10–CA-15.

17.2 Input Contract

The lattice symmetry compiler receives a compiled Hilbert/QM system plus:

Λ_N	: finite lattice, graph, or cell complex,
x	: $\Lambda_N \to \mathbb{R}^d$ position map,
$\mathcal{A}_N$	: observable algebra with locality map,
$H_N = \sum_r h_r$	: self-adjoint local Hamiltonian density,
T	: target symmetry algebra or group,
S_N	: state/sector sequence and scaling maps, if known.

If the Hamiltonian is supplied only as a black-box matrix, the compiler must fail for boost/rotation generation: the density decomposition is part of the mathematical data.

17.3 Candidate Generation

For one-dimensional open chains, the first candidate layer is deterministic:

$$K_x = \sum_j x_j h_j, \qquad P_x = i[H, K_x].$$

For uniform nearest-neighbour density this reduces to the CA-12 current sum.

For higher-dimensional lattices, scalar ramps $f_a(r) = r^a$ generate candidate momentum components

$$P_a^{\mathrm{lat}} := i[H, K_{f_a}].$$

Rotation candidates are then built only after the momentum-density/current layer has its own convention and evidence:

$$J_{ab}^{\mathrm{lat}} \approx \sum_r (r_a p_b^{\mathrm{lat}}(r) - r_b p_a^{\mathrm{lat}}(r)).$$

Until that layer is fixed, this is a deferred proposal slot rather than a formula emitted by the compiler. Koo–Saleur-style modes use Fourier weights instead of polynomial weights and include the commutator corrections specified by the cited Definition 4.1.

17.4 Output Contract

The compiler output for each candidate generator must include:

name	: for example $P_a, K_a, J_{ab}, L_k, \bar{L}_k$,
formula	: finite expression in h_r and commutators,
domain policy	: finite matrix, core, or unbounded deferral,
locality support	: site/bond/cell support accounting,
status	: Proposal, Source-local, or Checked,
evidence	: source lines, derivation, test, or run artifact,
continuum target	: target generator and expected bracket.

17.5 Failure Modes

The compiler must fail loudly when:

1. H has no density decomposition,
2. Λ_N has no position map but a boost/rotation is requested,
3. periodic boundary positions are requested without a branch convention,
4. the target algebra requires a source/convention not yet fixed,
5. a convergence claim is requested without scaling maps or states.

17.6 Compiler Principle

The finite expression and the continuum theorem are different compilation products. The first can often be generated algebraically from $(\Lambda_N, x, \{h_r\})$. The second is a proof obligation involving scaling maps, states, and operator topology. The compiler should output both slots, but it must never fill the proof slot with intuition.

18 Small Examples and the Lattice-Symmetry Research Queue

18.1 Status, Sources, and Dependencies

Status: worked examples and queue. The examples are deliberately small. They illustrate compiler behaviour, not new convergence theorems.

Dependencies: CA-10–CA-16 and CONVENTIONS.md (h), (j).

18.2 Example 1: Black-Box Quantum System

Input: a Hilbert space and a self-adjoint Hamiltonian H.

Output: time translation $U(t) = e^{-itH}$.

Failed request: infer boosts, rotations, or spatial translations. There is no lattice geometry, no position map, and no Hamiltonian density. The compiler must stop.

18.3 Example 2: Open Uniform Chain

Input:

$$H = \sum_{j=1}^{N} h_j, \qquad x_j = j, \qquad [h_j, h_k] = 0 \text{ for } |j-k| > 1.$$

Candidate boost:

$$K = \sum_j j h_j.$$

Candidate momentum:

$$P = i[H, K] = \sum_{j=1}^{N-1} i[h_j, h_{j+1}].$$

Status: checked finite derivation; no continuum convergence yet.

18.4 Example 3: Nonuniform Chain

Input: positions $x_1 < \cdots < x_N$. The same derivation gives

$$P_x = i[H, K_x] = \sum_{j=1}^{N-1} (x_{j+1} - x_j) i[h_j, h_{j+1}].$$

The coefficients are discrete spacings. The Julia test pins this invariant for uniform and nonuniform examples.

18.5 Example 4: Periodic Chain

Input: a circle with periodic density terms.

Problem: $x_j = j$ is not a single-valued function on the circle. A periodic boost-like first moment requires a branch cut, a sawtooth, Fourier modes, or a different construction. The same warning applies to finite periodic momentum-coordinate tests: a finite grid supports periodic symbols and Fourier eigenvalue checks, not an exact differential operator $X = i\partial_k$ with a canonical commutator. Koo–Saleur Fourier modes are the better sourced periodic prototype.

18.6 Example 5: Square Lattice

Input: positions $r = (m, n)$. Scalar ramps $f_x(r) = m$, $f_y(r) = n$ give two candidate momentum components via

$$P_a^{\text{lat}} = i[H, \sum_r r^a h_r].$$

Rotation about the origin should then be built as a position-dependent translation from the two momentum-density components. Status: proposal until an orientation and density-localisation convention is fixed.

18.7 Example 6: Koo–Saleur Modes

Input: periodic one-dimensional critical Hamiltonian density. Fourier modes of the density and commutator corrections give candidates for $L_k, \bar{L}_k$. In the cited free-fermion/OAR setting, smeared approximants converge to smeared Virasoro generators. Status outside that setting: source-local precedent, not a general theorem.

18.8 Research Queue

1. Use registered Weinberg anchors before promoting exact Lorentz-generator formulas.
2. Use the registered original Koo–Saleur anchors before promoting model-specific KS claims.
3. Source or derive a general local energy-current theorem.
4. Fix periodic branch/sawtooth/Fourier first-moment conventions.
5. Define higher-dimensional oriented density supports.
6. Add numerical examples checking commutator closure in small chains.
7. Connect anyon-chain Hamiltonian densities to the same candidate layer.

18.9 Landing Point

The next compiler upgrade is therefore not only a Hilbert-space compiler and not yet only a tensor-category compiler. It is a lattice-QM compiler layer:

$$(\text{carrier}, \mathcal{A}_N, \Lambda_N, \{h_r\}, x, S_N) \longmapsto (\text{candidate bulk symmetries}, \text{evidence obligations}).$$

Only after those obligations are discharged can the continuum symmetry representation be claimed.

19 Why Galilean Symmetry Is a Different Lattice Target

19.1 Status, Sources, and Dependencies

Status: sourced boundary plus proposal shard. The Galilei group as a classical nonrelativistic symmetry group is sourced locally. The general projective-representation and central-extension mechanism is sourced locally. The specific mass central extension of the Galilei group is not yet sourced in this repository; CA-20 treats its coefficient as a checked canonical-commutator derivation, not as a literature claim about a named group.

Dependencies: CA-05 for projective symmetries via central extensions, CA-10–CA-17 for the Lorentzian lattice-symmetry programme, and CONVENTIONS.md (i).

19.2 The New Target

The previous block targeted Lorentz-invariant continuum systems. Its first lattice intuition was:

$$\text{Lorentz boost} \simeq \text{position-dependent time translation}, \qquad K_a^{\mathrm{Lor}} \sim \sum_x x_a h_x.$$

That is appropriate only when the continuum target mixes space and time in the Lorentzian way and the boost is controlled by the energy density.

The nonrelativistic target is different. Schottenloher lists the Galilei group as the symmetry group for free particles in classical nonrelativistic mechanics. For the lab-book compiler this means that a Galilean target is not obtained by renaming the Poincare block. It has absolute time, spatial translations, rotations, and boosts whose quantum implementation may be projective.

19.3 Immediate Consequence for Lattices

The Lorentzian boost ansatz used only the Hamiltonian density h_x. A Galilean boost needs more input. At time $t = 0$, the familiar nonrelativistic one-particle expression is controlled by the mass-weighted position, not by the energy-weighted position:

$$G_a(0) \sim m X_a.$$

For a many-body lattice this suggests a mass-density or particle-number-density first moment:

$$G_a^{\mathrm{lat}}(0) \sim \sum_{x \in \Lambda} x_a \, \mu_x, \qquad \mu_x = m \, n_x \quad \text{when the particles have common mass } m.$$

This is a proposal-level lattice formula until the model supplies the density μ_x, the conservation law, and the continuity/current relation.

19.4 Compiler Implication

The compiler rule is therefore stricter than in CA-16:

$$\begin{aligned} &\text{Lorentzian candidate layer:} && (\Lambda, x, \{h_x\}) \text{ may generate } K_a \sim \textstyle\sum_x x_a h_x, \\ &\text{Galilean candidate layer:} && (\Lambda, x, \{h_x\}) \text{ is not enough.} \end{aligned}$$

For Galilean boosts the input must also name:

$$\{\mu_x\} \text{ or } \{n_x\}, \qquad M = \sum_x \mu_x, \qquad [H, M] = 0,$$

plus, for a momentum candidate, a current witness for the discrete continuity equation. If those data are absent, the compiler must fail loudly rather than silently using the energy first moment.

19.5 Why Projective Data Matter Earlier

In the Hilbert-space grammar, projective symmetries were handled by passing to a central extension. Galilean symmetry is the first lattice target where that choice becomes operational: the mass parameter is expected to sit in the projective/central part of the quantum representation. Locally we can support only this precise statement:

projective quantum symmetries are represented by unitary representations of central extensions.

The stronger statement identifying the Galilei extension by its standard literature name remains a source-acquisition task.

19.6 Boundary for CA-18–CA-22

The follow-up block does four things:

CA-19	derives the unextended Galilei vector-field algebra,
CA-20	derives the mass central coefficient on a canonical core,
CA-21	derives the lattice first-moment continuity formula,
CA-22	turns the data into compiler rules and small examples.

No shard in this block claims that a finite lattice has exact Galilean symmetry unless the corresponding finite-dimensional representation and commutator checks are actually supplied.

20 The Galilei Algebra as Newtonian Vector Fields

20.1 Status, Sources, and Dependencies

Status: checked local derivation. The source anchor is that the Galilei group is the nonrelativistic free-particle symmetry group. The bracket table below is derived here from vector fields and checked by the Julia test suite.

Dependencies: CA-18 and CONVENTIONS.md (i).

20.2 Generators

Use Newtonian spacetime coordinates

$$(t, x_1, \ldots, x_d).$$

The convention fixed in CONVENTIONS.md (i) is:

$$\begin{aligned} H &= \partial_t, && \text{time translation,} \\ P_a &= \partial_a, && \text{spatial translation,} \\ G_a &= t\,\partial_a, && \text{Galilean boost,} \\ J_{ab} &= x_a\partial_b - x_b\partial_a, \quad a<b, && \text{spatial rotation.} \end{aligned}$$

Here $a, b, c \in \{1, \ldots, d\}$. The boost vector field says that a boost changes position by a velocity times time while leaving absolute time fixed.

20.3 Translation and Boost Brackets

The translation fields have constant coefficients, hence

$$[H, P_a] = 0, \qquad [P_a, P_b] = 0.$$

Since $G_a = t\partial_a$,

$$\begin{aligned} [H, G_a] &= [\partial_t, t\partial_a] \\ &= \partial_t(t)\partial_a \\ &= P_a. \end{aligned}$$

The reverse bracket is therefore $[G_a, H] = -P_a$. Spatial translations do not differentiate the coefficient t, so

$$[P_a, G_b] = 0, \qquad [G_a, G_b] = 0.$$

This is the unextended vector-field algebra. It is not yet the projective quantum algebra; CA-20 adds the central term that is invisible to vector fields on spacetime.

20.4 Rotation on Vectors

For $J_{ab} = x_a\partial_b - x_b\partial_a$, compute against a translation:

$$\begin{aligned} [J_{ab}, P_c] &= -\partial_c(x_a)\partial_b + \partial_c(x_b)\partial_a \\ &= \delta_{bc}P_a - \delta_{ac}P_b. \end{aligned}$$

The same calculation applies to boosts because J_{ab} does not act on t:

$$[J_{ab}, G_c] = \delta_{bc}G_a - \delta_{ac}G_b.$$

Thus both P_c and G_c transform as spatial vectors.

20.5 Rotation-Rotation Bracket

For two rotations, expanding the vector-field bracket gives

$$\begin{aligned}[J_{ab}, J_{cd}] = \delta_{bc} J_{ad} - \delta_{ac} J_{bd} \\ + \delta_{ad} J_{bc} - \delta_{bd} J_{ac},\end{aligned}$$

with the convention $J_{ba} = -J_{ab}$ and $J_{aa} = 0$. In $d = 3$, one representative checked bracket is

$$[J_{12}, J_{23}] = J_{13}.$$

20.6 Checked Representatives

The test suite pins the signs with the following representative outcomes:

$$\begin{aligned}[H, G_1] &= P_1, \\ [G_1, H] &= -P_1, \\ [J_{12}, P_1] &= -P_2, \\ [J_{12}, P_2] &= P_1, \\ [J_{12}, G_1] &= -G_2, \\ [J_{12}, J_{23}] &= J_{13}.\end{aligned}$$

It also checks that $[P_1, G_2] = 0$ and $[G_1, G_2] = 0$ in the unextended vector-field algebra.

20.7 Boundary

The signs here are signs for vector fields. A quantum model uses self-adjoint generators and unitary flows, so translating this table into operator commutators requires the Stone convention and the chosen $i[A, B]$ observable action. The report therefore cites this shard for the geometric algebra and CA-20 for the central quantum coefficient.

21 Projective Galilean Symmetry and the Mass Central Term

21.1 Status, Sources, and Dependencies

Status: sourced projective framework plus checked local derivation. Schottenloher supplies the representation-theoretic framework: projective symmetries need not lift to the original group, but lift through a central extension. The coefficient $m\delta_{ab}$ below is a local canonical commutator derivation checked in Julia. A local source specifically identifying the Galilei mass central extension by its standard name has not yet been registered.

Dependencies: CA-05, CA-18, CA-19, and CONVENTIONS.md (f), (i).

21.2 Why This Belongs in the Grammar

CA-05 made a conservative grammar choice: a projective symmetry is represented by an honest unitary representation of a named central extension

$$1 \longrightarrow U(1) \longrightarrow E \longrightarrow G \longrightarrow 1.$$

For Galilean symmetry this is not a decorative detail. The central parameter is physically interpreted as mass in the canonical nonrelativistic model, and it controls how boosts and momenta compose at the quantum level.

21.3 Canonical One-Particle Calculation

Work formally on a common invariant core where the position and momentum operators obey

$$[X_a, P_b] = i\delta_{ab} I.$$

For a particle of mass m, the time-zero boost generator is represented by

$$G_a(0) = mX_a.$$

Then

$$\begin{aligned} [G_a(0), P_b] &= [mX_a, P_b] \\ &= m[X_a, P_b] \\ &= i\, m\, \delta_{ab} I. \end{aligned}$$

The Julia check records exactly the real coefficient $m\delta_{ab}$.

21.4 Compatibility with Time Evolution

For the free Hamiltonian

$$H = \frac{1}{2m} \sum_b P_b^2,$$

the same canonical commutator gives

$$\begin{aligned} [H, mX_a] &= \frac{1}{2} \sum_b [P_b^2, X_a] \\ &= \frac{1}{2} \sum_b \big(P_b [P_b, X_a] + [P_b, X_a] P_b \big) \\ &= \frac{1}{2} \big(-iP_a - iP_a \big) = -iP_a. \end{aligned}$$

Thus

$$i[H, G_a(0)] = P_a.$$

This matches the CA-19 vector-field relation $[H, G_a] = P_a$ after applying the self-adjoint-generator convention used for observable evolution.

21.5 Central Extension Versus Vector Fields

The unextended vector-field algebra had $[G_a, P_b] = 0$. The canonical quantum calculation has instead

$$[G_a, P_b] = i\, m\, \delta_{ab} I.$$

The additional term is central: it is a scalar multiple of the identity on a fixed-mass sector. This is the algebraic reason the compiler must attach a central-extension witness to Galilean symmetry data rather than store only the spacetime vector fields.

21.6 Source Gap

The local source collection includes Bargmann's theorem about lifting projective representations under a Lie-algebra cohomology hypothesis, and it includes the general central-extension theorem for projective representations. It does not yet include a registered source for the standard mass central extension of the Galilei group. Until that source is acquired, this shard uses the cautious name *mass central term* and cites the checked canonical calculation as its local evidence.

21.7 Compiler Payload

A Galilean projective symmetry record must therefore include:

base group/algebra	Galilei target from CA-19,
central generator	M or I on a fixed-mass sector,
central charge	m or a mass operator,
raw commutator witness	$[G_a, P_b] = i\delta_{ab} M$,
sign convention	conversion to $i[A, B]$ stated explicitly.

An OR of sectors with different masses can still be a Hilbert direct sum, but the central element must then be represented as a block-diagonal operator rather than a single scalar charge.

22 Lattice Galilean Boosts from Mass-Density First Moments

22.1 Status, Sources, and Dependencies

Status: proposal plus checked local derivation. Local sources support Galilean invariance as a physical assumption in nonrelativistic fermion models and give examples where conservation laws produce continuity equations. The finite open-chain first-moment formula below is a local derivation checked by the Julia test suite.

Dependencies: CA-18–CA-20 and CONVENTIONS.md (h), (i).

22.2 Input Data

Let $\Lambda = \{1, \ldots, N\}$ be a finite open chain with positions x_j. A Galilean lattice candidate needs, in addition to the Hamiltonian, a conserved mass or particle-number density:

$$M = \sum_j \mu_j, \qquad [H, M] = 0.$$

For identical particles of mass m, a common special case is

$$\mu_j = m n_j,$$

where n_j is a number density.

The time-zero Galilean boost candidate is the first mass moment

$$G^{\mathrm{lat}}(0) = \sum_j x_j \mu_j.$$

This is the Galilean analogue of mX, not the Lorentzian energy moment $\sum_j x_j h_j$.

22.3 Discrete Continuity Assumption

Assume a nearest-neighbour open-chain continuity equation with bond currents $J_{j+1/2}$:

$$i[H, \mu_j] = J_{j-1/2} - J_{j+1/2},$$

where endpoint currents are either absent or treated as boundary terms. This assumption is model data. It does not follow merely from writing $H = \sum_j h_j$.

22.4 First-Moment Derivation

Multiply the continuity equation by x_j and sum:

$$\begin{aligned} i\left[H, \sum_j x_j \mu_j\right] &= \sum_j x_j (J_{j-1/2} - J_{j+1/2}) \\ &= \sum_{j=1}^{N-1} (x_{j+1} - x_j) J_{j+1/2} \quad + \quad \text{boundary terms.} \end{aligned}$$

For a unit-spaced open chain away from endpoints,

$$i[H, G^{\mathrm{lat}}(0)] \simeq \sum_{j=1}^{N-1} J_{j+1/2}.$$

This is the lattice momentum candidate associated with the Galilean boost. For nonuniform positions the coefficients are the spacings

$$\Delta x_j = x_{j+1} - x_j,$$

and the test suite records a coefficient check for this shard.

22.5 Relation to Center of Mass

The local paired-fermion source uses Galilean invariance to explain a response in terms of center-of-mass acceleration while relative motion is unaffected. That is exactly the intuition this shard turns into compiler data:

$$\text{Galilean boost} \quad \rightsquigarrow \quad \text{center-of-mass / mass-first-moment generator.}$$

The source does not prove the lattice formula above. It supports the physical role of center-of-mass motion under Galilean invariance; the lattice identity is the checked local derivation.

22.6 What Is and Is Not Proven

The finite identity proves only that, given a local continuity equation, the time derivative of the mass first moment is a current sum. It does not prove:

1. that the current sum converges to continuum momentum,
2. that the finite lattice has exact Galilei symmetry,
3. that the mass current equals momentum density in every model,
4. that boundary terms vanish without a stated boundary condition.

Those are acceptance-test obligations for later shards or model-specific examples.

23 Compiler Rules and Examples for Galilean Targets

23.1 Status, Sources, and Dependencies

Status: compiler specification shard. This shard does not implement the compiler. It specifies the extra fields a Galilean target requires, using the checked algebra and first-moment derivations from CA-19–CA-21.

Dependencies: CA-08, CA-16, and CA-18–CA-21.

23.2 Target Declaration

A Galilean compiler target is a structured request:

$$\text{Target}(\text{Gal}(d),\ \text{projective},\ \text{mass}).$$

It asks for:

H	time translation,
P_a	spatial translations / momenta,
J_{ab}	rotations,
G_a	Galilean boosts,
M	central mass generator.

The finite candidate layer is allowed to output formulas and witnesses. It is not allowed to assert continuum Galilean covariance without scaling-limit evidence.

23.3 Required Input Slots

Compared with CA-16, the Galilean target adds the following required slots:

μ_x or n_x	local mass or particle-number density,
$M = \sum_x \mu_x$	conserved total mass/number,
$[H, M] = 0$	finite commutator witness,
$J_{x,e}$	current witness for a discrete continuity equation,
m or M	central charge/operator for the projective representation,
boundary policy	open, periodic with branch, or thermodynamic-limit handling.

If any of the first four are absent, boosts and momenta for a Galilean target are underdetermined.

23.4 Generated Candidates

For an open chain, the deterministic candidate formulas are

$$G^{\text{lat}}(0) = \sum_j x_j \mu_j,$$

and, when a continuity witness is supplied,

$$P^{\text{lat}} := i[H, G^{\text{lat}}(0)] = \sum_{j=1}^{N-1} (x_{j+1} - x_j) J_{j+1/2} \quad + \quad \text{boundary terms.}$$

Rotations in higher dimension require momentum or current-density data:

$$J^{\text{lat}}_{ab} \approx \sum_x (x_a p_b(x) - x_b p_a(x)).$$

Thus rotations are second-stage candidates: first compile current/momentum density, then form the spatial first moment.

23.5 Example 1: One Free Particle

Input:

$$\mathcal{H} = L^2(\mathbb{R}^d), \qquad H = \frac{P^2}{2m}, \qquad [X_a, P_b] = i\delta_{ab} I.$$

Output at $t = 0$:

$$G_a = mX_a, \qquad [G_a, P_b] = im\delta_{ab} I, \qquad i[H, G_a] = P_a.$$

Status: checked formal derivation on a common core, with unbounded-operator domain work deferred.

23.6 Example 2: Number-Conserving Open Chain

Input:

$$H, \quad n_j, \quad [H, \sum_j n_j] = 0, \quad i[H, n_j] = J_{j-1/2} - J_{j+1/2}.$$

For identical mass m,

$$\mu_j = mn_j, \qquad G^{\text{lat}}(0) = m \sum_j x_j n_j.$$

The compiler may output the current-sum momentum candidate with coefficient $m(x_{j+1} - x_j)$. It must still mark continuum Galilean covariance as a proof obligation.

23.7 Example 3: Hamiltonian Density Only

Input:

$$H = \sum_j h_j$$

with no n_j, no μ_j, and no current witness. For a Lorentzian target, CA-12 permits the energy-moment proposal. For a Galilean target, the compiler must return:

```
CompileError:  missing mass/number density and continuity witness.
```

Using $\sum_j x_j h_j$ here would be a convention bug.

23.8 Example 4: Spin Chain with No Particle Number

A spin chain may have a perfectly good Hamiltonian density and exact lattice translations. Unless a conserved density is explicitly declared as mass or particle number, it does not compile as a Galilean particle system. The correct output is either a failed Galilean target or a different target declaration.

23.9 Example 5: Direct Sum of Mass Sectors

For

$$\mathcal{H} = \mathcal{H}_{m_1} \oplus \mathcal{H}_{m_2},$$

the central generator is not one scalar if $m_1 \neq m_2$. It is

$$M = m_1 I_{\mathcal{H}_{m_1}} \oplus m_2 I_{\mathcal{H}_{m_2}}.$$

The compiler can carry the OR carrier, but an irreducible fixed-mass sector selection is a separate projection/sector step in the sense of CA-06.

23.10 Next Obligations

The next Galilean work needs:

1. a registered source for the standard mass central extension,
2. model examples with explicit lattice number-current continuity,
3. finite commutator tests for those models,
4. scaling maps proving that current sums converge to continuum momenta,
5. a decision on periodic boundary branch conventions.

Only after these are present should the compiler promote Galilean lattice candidates from formulas to continuum symmetry evidence.

24 Gaussian Boson Lorentz Generator Roadmap

24.1 Status, Sources, and Dependencies

Status: roadmap shard with checked seed. This shard scopes a multi-shard programme. CA-24 supplies the first checked symbol calculation. The stronger continuum conclusion is not yet proved for a general coefficient class.

Dependencies: CA-10–CA-17 for lattice Lorentz motivation and acceptance tests, and CONVENTIONS.md (j). This shard starts the Gaussian-boson block.

24.2 Problem Statement

We now restrict the lattice-symmetry problem to a solvable class: translation-invariant Gaussian bosonic lattice systems in spatial dimensions

$$d = 1, 2, 3, \qquad D = d + 1.$$

The physical target is Lorentz/Poincare covariance in the scaling limit. The expected endpoint is the scalar free field:

$$\omega(k)^2 = m^2 + |k|^2$$

after the correct normalization of lattice spacing, velocity, field strength, and mass.

The inputs are not merely a matrix Hamiltonian. The compiler receives:

$\Lambda_{\epsilon,L}$	periodic hypercubic lattice or infinite lattice,
q_x, p_x	canonical bosonic field variables,
V_r	translation-invariant quasi-local quadratic coefficients,
ϵ	lattice spacing,
S_ϵ	state/scaling-map data, when a continuum claim is requested.

The first coefficient tier is the scalar canonical Hamiltonian

$$H = \frac{1}{2}\sum_x p_x^2 + \frac{1}{2}\sum_{x,r} q_x V_r q_{x+r}.$$

Full bosonic BdG/pairing systems are a later tier, not silently included in the first convention.

24.3 North Star

For each admissible coefficient family $V_r^{(\epsilon)}$, compute the candidate generators as functions of the coefficients:

$$H, \quad P_a, \quad J_{ab}, \quad K_a,$$

then compute the residuals in the Poincare algebra. The residuals become conditions on the symbol

$$\omega_\epsilon(k)^2 = \sum_r V_r^{(\epsilon)} e^{i\epsilon k\cdot r}.$$

The end goal is a theorem-shaped statement:

Gaussian lattice coefficients satisfying locality, positivity,
single low-energy minimum, and isotropic Klein-Gordon Taylor data
$\Longrightarrow$ scaling limit is the free scalar/Klein-Gordon Poincare system.

CA-15 proof boundary. This is a theorem-shaped target, not a theorem already obtained from the symbol checks. A continuum claim still needs the CA-15 payload: scaling maps, state sequence, topology, operator domains, local observable covariance, and boundary/renormalization policy. The value of the Gaussian block is that the algebraic residuals can be made explicit before attempting that proof.

24.4 Shard Roadmap

The current Gaussian block is:

CA-24. Symbol calculus for the scalar coefficient tier, including the boost-time residual and the checked nearest-neighbour Klein-Gordon seed.

CA-25. Diagonalization into one-particle symbols, with the sourced free-scalar Fock construction separated from the local coefficient-symbol derivation and BdG pairing deferred.

CA-26. Smooth-patch one-particle generator algebra for H, P, J, K, exact brackets, dispersion-dependent residuals, and only formal $\mathrm{d}\Gamma$ bookkeeping on the algebraic finite-particle core.

CA-27. Lorentz residual conditions on coefficients: Taylor constraints, anisotropic failures, and doubler counterexamples.

CA-28. Numerical invariant suite for scalar symbols and finite periodic stiffness matrices, covering Klein-Gordon, anisotropic, and doubler examples.

CA-29. Real-space energy-density convention for the scalar Gaussian Hamiltonian: cell volume, bond sharing, boundary policy, vacuum subtraction, and improvement terms before any position-weighted use.

CA-30. One-dimensional stress-energy candidate dictionary: T_{00} from CA-29, checked finite-open-chain T_{01} from A4, and proposal-level T_{10} and T_{11} witnesses left for later work.

CA-31. One-dimensional current-symbol bridge: the translation-invariant integrated T_{01} current symbol equals the nearest-neighbour Klein-Gordon boost-time symbol, while finite open-chain $\mathrm{d}\Gamma(k)$ equivalence remains open.

CA-32. Higher-dimensional cell-current proposal: oriented $d = 2, 3$ support data, energy-flux equations, momentum-density witnesses, stress-current witnesses, and rotation/boundary diagnostics as compiler inputs rather than convergence claims.

The remaining queued follow-up block is proposal work, not completed stress-energy material:

Open-chain momentum equivalence. Upgrade the CA-31 symbol bridge to finite open-chain wave-packet and $\mathrm{d}\Gamma(k)$ comparison data, including boundary residuals.

Stress-energy numerical suite. Add numerical checks only after candidate formulas and source-backed expected identities exist.

Source acquisition. Use the newly registered discrete Gaussian/free-field Virasoro source only within its square-grid dGFF change-of-measure scope. Separate source anchors are still required for harmonic-chain energy-current formulas and any continuum stress-energy/Poincare-generator claim.

24.5 Side Quests

Source acquisition. The registered local OAR sources cover the free scalar lattice scaling limit and spacetime translations. They explicitly warn that Lorentz/conformal convergence requires further work. The discrete Gaussian free-field Virasoro source is now local; it supports lattice-level change-of-measure Virasoro claims for the square-grid dGFF, not a continuum stress-energy or Poincare-generator formula.

Generator ambiguity. A local Hamiltonian density is not unique:

$$h_x \mapsto h_x + \nabla \cdot b_x$$

changes first moments by boundary or current terms. The Gaussian block must track which density convention produced $K_a = \sum_x x_a h_x$.

Periodic positions. Numerical tests use periodic lattices and finite Brillouin grids. First moments, position coordinates, and finite-grid momentum-coordinate generators need a branch, sawtooth, Fourier interpolation, or another explicit periodic coordinate convention. Until that is fixed, early finite tests are symbol/eigenvalue checks on the periodic grid, or open-chain first-moment checks.

BdG pairing. A general quasi-local Hamiltonian in creation and annihilation operators may include pairing terms. The scalar q, p tier captures Klein-Gordon directly; BdG requires symplectic positivity and a stable bosonic Bogoliubov convention before generator formulas are promoted.

24.6 Numerical Verification Plan

The Julia layer should stay at the invariant level:

1. symbol from coefficients equals the analytic dispersion,
2. $\frac{1}{2}\nabla\omega_\epsilon(k)^2$ equals the smooth-patch boost-time residual,
3. $\frac{\sin(\epsilon k_a)}{\epsilon} \to k_a$ for the KG lattice,
4. anisotropic Hessians fail the Lorentz residual checks,
5. finite Fourier matrices diagonalize periodic stiffness matrices.

The first three checks are implemented with CA-24.

24.7 Acceptance Boundary

The roadmap rows above remain scoped by the CA-15 proof boundary stated next to the theorem-shaped goal. Passing symbol or finite-matrix tests is not yet a proof of Lorentz invariance.

25 Gaussian Boson Symbol Calculus and the Boost-Time Residual

25.1 Status, Sources, and Dependencies

Status: sourced model class plus checked flat local derivation. The standard lattice scalar-field Hamiltonian and dispersion are sourced. The boost-time residual calculation is a local one-particle derivation on flat $L^2(dk)$, checked by Julia on positive-dispersion patches. It is not yet a sourced OAR or mass-shell Poincare representation theorem, and massless zero-mode examples are coefficient-level tests only.

Dependencies: CA-11, CA-12, CA-23, and CONVENTIONS.md (h), (j), (k).

25.2 Coefficient Tier

Fix $d \in \{1, 2, 3\}$. On a periodic hypercubic lattice with spacing ϵ, use canonical fields

$$[q_x, q_y] = 0, \qquad [p_x, p_y] = 0, \qquad [q_x, p_y] = i\delta_{xy}.$$

The first scalar Gaussian tier is

$$H = \frac{1}{2}\sum_x p_x^2 + \frac{1}{2}\sum_{x,r} q_x V_r q_{x+r}, \qquad V_r = V_{-r} \in \mathbb{R}.$$

Quasi-locality means that V_r has enough decay, or finite range in the first examples, so that the Fourier symbol and its derivatives used below exist.

Define

$$\omega_\epsilon(k)^2 := \sum_r V_r e^{i\epsilon k\cdot r}.$$

For a stable scalar model, this symbol is non-negative, with the zero-mode policy stated separately in the massless case. The helper parameter m is a nonnegative mass label; the generator statements below use only the stricter case $\omega_\epsilon(k) > 0$ on the chosen patch.

25.3 Nearest-Neighbour Klein-Gordon Example

The sourced lattice scalar-field example has, in the notation of this shard,

$$V_0 = m^2 + \frac{2d}{\epsilon^2}, \qquad V_{\pm e_a} = -\frac{1}{\epsilon^2}.$$

Therefore

$$\begin{aligned}\omega_{\epsilon,m}(k)^2 &= m^2 + \frac{2d}{\epsilon^2} - \frac{1}{\epsilon^2}\sum_{a=1}^{d}(e^{i\epsilon k_a} + e^{-i\epsilon k_a}) \\ &= m^2 + \frac{2}{\epsilon^2}\sum_{a=1}^{d}(1 - \cos(\epsilon k_a)).\end{aligned}$$

For fixed physical k and $\epsilon \to 0$,

$$\omega_{\epsilon,m}(k)^2 = m^2 + |k|^2 + O(\epsilon^2|k|^4).$$

25.4 Momentum-Space Position

After diagonalization, this shard uses the flat local momentum space $L^2(\mathcal{U}, dk)$. The one-particle Hamiltonian is multiplication by $\omega(k)$, with $\omega(k) > 0$ on $\mathcal{U}$, and

$$X_a = i\partial_{k_a}$$

on a common smooth core. The time-zero boost candidate corresponding to the energy first moment is the flat-measure symmetrized operator

$$K_a = \frac{1}{2}(X_a\omega + \omega X_a).$$

The symmetrization is the one-particle version of taking a self-adjoint first moment of the energy density. If the one-particle scalar product is changed, the adjoint and hence the boost formula must be rederived; no unitary equivalence with the OAR weighted space or Schottenloher mass-shell realization is asserted here.

25.5 Boost-Time Residual

For a test wavefunction ψ,

$$\begin{aligned}[K_a, \omega]\psi &= \frac{1}{2}(X_a\omega^2 - \omega X_a\omega)\psi \\ &= i\,\omega\,(\partial_a\omega)\psi.\end{aligned}$$

Thus

$$i[\omega, K_a] = \omega\,\partial_a\omega = \frac{1}{2}\partial_a\omega(k)^2.$$

This is the algebraic output of the boost-time commutator. With the vector-field-to-Stone sign map of CONVENTIONS.md (k), the continuum Poincare target is $i[H, K_a] = P_a$. Taking $P_a = k_a$, one needs

$$\frac{1}{2}\partial_a\omega(k)^2 = k_a.$$

Integrating over k gives

$$\omega(k)^2 = m^2 + |k|^2$$

on the connected low-energy patch. With speed c, the right side is $c^2 k_a$, giving $\omega(k)^2 = m^2 + c^2|k|^2$.

25.6 Lattice Residual

For the nearest-neighbour lattice Klein-Gordon symbol,

$$\frac{1}{2}\partial_a\omega_{\epsilon,m}(k)^2 = \frac{\sin(\epsilon k_a)}{\epsilon}.$$

Therefore the boost-time commutator has the correct continuum limit:

$$\frac{\sin(\epsilon k_a)}{\epsilon} = k_a + O(\epsilon^2 k_a^3).$$

The Julia tests check this in $d = 1, 2, 3$, both from the closed formula and by evaluating the coefficient symbol $\sum_r V_r e^{i\epsilon k\cdot r}$.

25.7 Rotation Residual

For the one-particle rotation candidate

$$J_{ab} = X_a P_b - X_b P_a, \qquad P_a = k_a,$$

one obtains

$$i[\omega, J_{ab}] = k_b \partial_a \omega - k_a \partial_b \omega.$$

Thus rotations commute with the Hamiltonian exactly when the dispersion is radial on the momentum region under consideration. For a hypercubic lattice this is not exact over the full Brillouin zone, but the Klein-Gordon expansion is isotropic to quadratic order at $k = 0$.

25.8 Boundary

This shard is a one-particle symbol calculation. The many-body generators are second quantizations plus vacuum-energy and domain policies. Real-space locality of $K_a = \sum_x x_a h_x$, boundary terms, and convergence of the second-quantized operators are deferred to later shards.

26 Gaussian Boson Diagonalization and the One-Particle Symbol

26.1 Status, Sources, and Dependencies

Status: sourced template plus flat local coefficient derivation. The standard lattice free-scalar Hamiltonian, Fock representation, continuum one-particle space, and second-quantized Hamiltonian are sourced. The extension from nearest-neighbour coefficients to an arbitrary real translation-invariant scalar symbol is a local algebraic calculation on the flat $L^2(dk)$ symbol space fixed in CONVENTIONS.md (j).

Dependencies: CA-23–CA-24 and CONVENTIONS.md (a), (f), (j).

26.2 Scalar Coefficient Tier

Fix $d \in \{1, 2, 3\}$. The first Gaussian tier is

$$H = \frac{1}{2}\sum_x p_x^2 + \frac{1}{2}\sum_{x,r} q_x V_r q_{x+r}, \qquad V_r = V_{-r} \in \mathbb{R}.$$

On an infinite translation-invariant lattice, or on a periodic lattice before taking volume limits, the Fourier transform sends the potential kernel to

$$\omega_\epsilon(k)^2 = \sum_r V_r e^{i\epsilon k\cdot r}.$$

The reality condition $V_r = V_{-r}$ makes this symbol real:

$$\omega_\epsilon(k)^2 = V_0 + \sum_{r>0} 2V_r \cos(\epsilon k \cdot r).$$

The stability condition for this tier is

$$\omega_\epsilon(k)^2 \geq 0$$

on the Brillouin zone. The `mass` parameter is a nonnegative label; in the nearest-neighbour Klein-Gordon family, $m > 0$ gives the positive-dispersion case used by the current generator statements. For a general coefficient symbol the required condition remains $\omega_\epsilon(k) > 0$ on the chosen patch. When $m = 0$, zero modes are coefficient data only until a zero-mode convention excludes or otherwise controls them.

26.3 Mode Decoupling

The Hamilton equations are a local derivation from the canonical commutators:

$$\dot{q}_x = p_x, \qquad \dot{p}_x = -\sum_r V_r q_{x+r}.$$

Fourier transforming gives

$$\dot{\widehat{q}}(k) = \widehat{p}(k), \qquad \dot{\widehat{p}}(k) = -\omega_\epsilon(k)^2 \widehat{q}(k),$$

and hence

$$\ddot{\widehat{q}}(k) + \omega_\epsilon(k)^2 \widehat{q}(k) = 0.$$

Thus each momentum mode is an oscillator with frequency $\omega_\epsilon(k)$. This is the algebraic reason the compiler can reduce the scalar Gaussian tier to multiplication by a dispersion symbol.

26.4 Complex Structure and Fock Space

For $\omega > 0$ on the chosen momentum patch, the oscillator complex structure is

$$(q, p) \longmapsto (-\omega p, \omega^{-1} q).$$

The sourced free-scalar construction uses this pattern with $\gamma_m(k) = \sqrt{|k|^2 + m^2}$ in the continuum and with $\gamma_m^{(N)}(k)$ on the lattice. It also realizes the Weyl algebra in a bosonic Fock representation by creation and annihilation operators. The source scalar product is the OAR one on real Cauchy data, with $\gamma_m^{-1/2}\widehat{q} + i\gamma_m^{1/2}\widehat{p}$; the text then realizes the representation on a mass-independent flat L^2 Fock space. This shard uses only the resulting flat symbol bookkeeping and does not prove that the boost formula of CA-24/CA-26 intertwines the OAR or mass-shell representation.

The compiler-level output for the positive-dispersion scalar tier is therefore:

$\mathfrak{h}_\epsilon$	flat one-particle momentum space on the chosen patch or grid,
h_ϵ	multiplication by $\omega_\epsilon(k)$,
$\mathcal{F}_+(\mathfrak{h}_\epsilon)$	symmetric Fock completion,
H_ϵ	$d\Gamma(h_\epsilon) + E_{\epsilon,0}$ if a vacuum-energy policy is named.

The additive constant $E_{\epsilon,0}$ is not part of the commutator algebra for the generators considered here, but it is part of a Hamiltonian-density normal-ordering policy and must be recorded before using ground-state energies as numerical claims.

26.5 What Is Not Yet Included

This shard does not define the full bosonic BdG tier. A general quadratic Hamiltonian in creation and annihilation operators may include pairing terms, and then the correct diagonalization is a symplectic/Bogoliubov problem with positivity, implementability, and domain conditions. Those conventions are not fixed here.

The current compiler rule is deliberately narrower:

$$\text{scalar } (q, p) \text{ tier} \quad \Rightarrow \quad \omega_\epsilon(k)^2 \text{ symbol} \quad \Rightarrow \quad h_\epsilon = \omega_\epsilon(k).$$

BdG inputs must fail closed or route to a later convention until that tier is written.

26.6 Compiler Consequence

For the positive-dispersion scalar Gaussian block, later candidate generators can be constructed first on the one-particle smooth momentum core and then second-quantized. The coefficient data enter only through $\omega_\epsilon(k)^2$ and its derivatives, while the continuum proof still requires the CA-15 payload: scaling maps, state convergence, domains, and observable covariance.

27 One-Particle Generator Algebra for Gaussian Bosons

27.1 Status, Sources, and Dependencies

Status: flat local derivation with limited Julia checks. The Poincare target, Stone convention, and free-boson Fock realization are sourced. The commutator identities in this shard are local differential-operator derivations on a flat $L^2(dk)$ smooth momentum-space core. The Julia suite currently checks coefficient/symbol consequences: Klein-Gordon symbols, boost-time symbols, sampled rotation-Hamiltonian residuals, sampled boost-boost residual coefficient data, low-energy Hessian examples, finite periodic massive spectra, the massless doubler coefficient-rejection witness, and the Gaussian tolerance policy. It does not check the full generator bracket table, finite coordinate commutators, or any second-quantized identity. The smooth-core derivation is not yet a sourced OAR or mass-shell Poincare representation theorem.

Dependencies: CA-11, CA-24–CA-25, and CONVENTIONS.md (h), (j), (k).

27.2 Momentum Core

There are two separate settings.

Smooth patch. Work on a connected smooth momentum patch $\mathcal{U}$ where $\omega(k) > 0$, with the flat measure dk. Let

$$\mathcal{D}_0 = C_c^\infty(\mathcal{U}) \subset L^2(\mathcal{U}, dk)$$

and define

$$X_a = i\partial_{k_a}, \qquad h = \omega(k), \qquad p_a = k_a.$$

The one-particle Hamiltonian and momenta are multiplication operators:

$$H^{(1)} = h, \qquad P_a^{(1)} = p_a.$$

The rotation and time-zero boost candidates are

$$J_{ab}^{(1)} = X_a P_b - X_b P_a, \qquad K_a^{(1)} = \frac{1}{2}(X_a h + h X_a).$$

These are written in the self-adjoint sign convention of CONVENTIONS.md (k); in particular J_{ab} is the Gaussian rotation sign, not the raw lowered-index CA-11 vector field M_{ab}. All commutators below are identities on $\mathcal{D}_0$; promoting them to self-adjoint generator statements is a separate domain theorem. Promoting this flat formula to the OAR one-particle Hilbert space or Schottenloher's mass-shell/free-boson Poincare representation additionally requires a sourced or locally derived unitary intertwiner.

Finite periodic grids. On a finite periodic Gaussian lattice, the dual grid is used for multiplication symbols and Fourier eigenvalue checks. The current finite suite does not place the differential operator $X_a = i\partial_{k_a}$ on that grid and does not claim an exact finite $[X_a, p_b] = i\delta_{ab}$ commutator. Periodic first-moment or momentum-coordinate generator tests require an explicit branch, sawtooth, Fourier-interpolation, or comparable periodic-coordinate convention before they can be stated. Thus the exact differential commutators below are smooth-patch algebra; the finite periodic checks remain symbol/eigenvalue checks.

27.3 Exact Translation and Rotation Brackets

On the smooth core, because h and p_a are multiplication operators,

$$i[h, p_a] = 0.$$

The canonical differential identity is the smooth-patch identity

$$[X_a, p_b] = i\delta_{ab}.$$

Therefore

$$i[P_c, J_{ab}] = \delta_{ac}P_b - \delta_{bc}P_a.$$

This is independent of the dispersion. It is kinematical on the smooth patch: it comes from the chosen momentum coordinate and the differential representation of rotations, in the rotation sign convention of CONVENTIONS.md (k).

27.4 Boost-Momentum Bracket

For the boost candidate,

$$\begin{aligned}[P_b, K_a] &= \frac{1}{2}\big([P_b, X_a]h + h[P_b, X_a]\big) \\ &= -i\delta_{ab}h.\end{aligned}$$

Hence

$$i[P_b, K_a] = \delta_{ab}H.$$

This is the boost-momentum line of the self-adjoint target table in CONVENTIONS.md (k). On the smooth patch, this part of the Poincare algebra is exact for every positive scalar dispersion because it only uses K_a as the symmetrized first moment of energy.

27.5 Boost-Time Residual

CA-24 derived

$$i[H, K_a] = \frac{1}{2}\partial_a\omega(k)^2.$$

The continuum target with speed $c = 1$ wants $P_a = k_a$. The residual is

$$R_a(k) := \frac{1}{2}\partial_a\omega(k)^2 - k_a.$$

For speed c, replace k_a by c^2k_a. This is the first algebraic filter on Hamiltonian coefficients. Julia now exposes it both as the boost-time symbol $B_a = \frac{1}{2}\partial_a\omega^2$ and as the residual vector $R_a = B_a - c^2k_a$, still only at coefficient-symbol level.

27.6 Rotation-Hamiltonian Residual

The Hamiltonian commutes with rotations iff the dispersion is radial on the patch:

$$i[h, J_{ab}] = k_b\partial_a\omega - k_a\partial_b\omega.$$

Equivalently,

$$k_b\partial_a\omega^2 - k_a\partial_b\omega^2 = 0$$

where $\omega > 0$. Hypercubic lattice symbols usually fail this globally on the Brillouin zone but may satisfy it to leading order at a low-energy minimum. The Julia helper records the sampled antisymmetric matrix

$$2(k_bB_a - k_aB_b), \qquad B_a = \frac{1}{2}\partial_a\omega^2,$$

so the factor of two follows the boost-time convention instead of being re-derived independently.

27.7 Boost-Boost Residual

Write

$$B_a(k) := \frac{1}{2}\partial_a \omega(k)^2, \qquad R_a(k) := B_a(k) - k_a.$$

Equivalently $B_a = h\partial_a h$. Since

$$K_a = i\left(h\partial_a + \frac{1}{2}\partial_a h\right),$$

the commutator of the first-order differential parts gives

$$i[K_a, K_b] = J_{ab} + R_b X_a - R_a X_b.$$

The leading $+J_{ab}$ is the sign dictated by CONVENTIONS.md (k): CA-11 has $[M_{0a}, M_{0b}] = -M_{ab}$, while the self-adjoint rotation here is $J_{ab} = -A_{M_{ab}}$. Here the possible divergence term vanishes because R is a gradient residual: $\partial_a R_b = \partial_b R_a$. Thus the boost-boost bracket closes to the rotation bracket whenever the boost-time residual R_a vanishes on the patch. The Julia helper records only the sampled coefficient tensor for the first-order residual $R_b X_a - R_a X_b$: its (a, b, c) entry is the coefficient of X_c for the ordered boost pair (a, b). It is not a finite-grid coordinate commutator and does not check the boost-boost operator bracket. The remaining caveat is analytic, not algebraic: all of these are unbounded operators and still require a common self-adjointness/core theorem before this becomes a continuum representation statement.

27.8 Second Quantization

Let

$$\mathcal{F}_{\mathrm{alg}}(\mathcal{D}_0) := \bigoplus_{n=0}^{\mathrm{fin}} \mathcal{D}_0^{\otimes_s n} \subset \mathcal{F}_+\big(L^2(\mathcal{U}, dk)\big)$$

be the algebraic finite-particle core. The formal many-body candidates are

$$H = d\Gamma(h), \quad P_a = d\Gamma(p_a), \quad J_{ab} = d\Gamma(J_{ab}^{(1)}), \quad K_a = d\Gamma(K_a^{(1)}),$$

with a separate vacuum-energy convention if absolute Hamiltonian values are used. The formal core identity is

$$[d\Gamma(A), d\Gamma(B)]\psi = d\Gamma([A, B])\psi, \qquad \psi \in \mathcal{F}_{\mathrm{alg}}(\mathcal{D}_0),$$

whenever the one-particle commutator identity is already established on $\mathcal{D}_0$ and the displayed products preserve that algebraic domain. This is a local $d\Gamma$ bookkeeping statement on the algebraic finite-particle core, not a Julia-checked invariant and not a self-adjoint many-body representation theorem. The current tests do not lift residuals to Gaussian Fock space; domain/self-adjointness for the unbounded $d\Gamma$ generators and the sourced OAR/mass-shell bridge remain open proof obligations.

28 Lorentz Residual Conditions on Gaussian Coefficients

28.1 Status, Sources, and Dependencies

Status: checked local coefficient calculus. The sourced massive free scalar model supplies the target dispersion. The general coefficient residuals are local Taylor calculations from $\omega_\epsilon(k)^2 = \sum_r V_r e^{i\epsilon k\cdot r}$. The examples in this shard are checked in Julia; the massless doubler is only a coefficient-level rejection witness, not a positive-dispersion generator example.

Dependencies: CA-24–CA-26 and CONVENTIONS.md (j).

28.2 Gradient Residual

For a scalar coefficient family,

$$\omega_\epsilon(k)^2 = \sum_r V_r e^{i\epsilon k\cdot r}.$$

The boost-time residual from CA-26 is

$$R_a(k) = \frac{1}{2}\partial_a\omega_\epsilon(k)^2 - c^2 k_a.$$

Explicitly,

$$\frac{1}{2}\partial_a\omega_\epsilon(k)^2 = \frac{i\epsilon}{2}\sum_r r_a V_r e^{i\epsilon k\cdot r}.$$

When $V_r = V_{-r}$, this is real and can be written as a sine sum:

$$\frac{1}{2}\partial_a\omega_\epsilon(k)^2 = -\epsilon\sum_{r>0} r_a V_r \sin(\epsilon k\cdot r),$$

where $r > 0$ denotes any fixed choice of one representative from each $\{r, -r\}$ pair.

28.3 Low-Energy Hessian Condition

At a symmetric minimum $k = 0$, the first derivative vanishes. The Hessian is

$$\partial_a\partial_b\omega_\epsilon(0)^2 = -\epsilon^2\sum_r r_a r_b V_r.$$

Therefore the quadratic Lorentz-speed condition is

$$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r = c^2\delta_{ab}.$$

This is the first finite-dimensional coefficient test. In compiler form:

$$\mathsf{HessResidual}_{ab} := -\frac{\epsilon^2}{2}\sum_r r_a r_b V_r - c^2\delta_{ab}.$$

It must vanish at the chosen low-energy point before the model can be routed to a Klein-Gordon continuum target.

It is not a positivity or patch certificate. The Hessian only sees the Taylor coefficient at the chosen point; a finite-grid symbol can still be negative elsewhere or have additional low-energy minima.

28.4 Nearest-Neighbour Klein-Gordon

For

$$V_0 = m^2 + \frac{2d}{\epsilon^2}, \qquad V_{\pm e_a} = -\frac{1}{\epsilon^2},$$

one obtains

$$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r = \delta_{ab}.$$

The gradient residual is the stronger finite-ϵ expression

$$R_a(k) = \frac{\sin(\epsilon k_a)}{\epsilon} - k_a,$$

which vanishes as $O(\epsilon^2 k_a^3)$ for fixed physical momentum.

28.5 Anisotropic Counterexample

Let the nearest-neighbour coefficients carry different speeds c_a:

$$\omega_\epsilon(k)^2 = m^2 + \frac{2}{\epsilon^2}\sum_a c_a^2(1 - \cos(\epsilon k_a)).$$

Then

$$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r = c_a^2 \delta_{ab}.$$

Unless all c_a agree with the target speed, the boost-time residual cannot match a single Lorentz cone. The Julia test suite checks this as a deliberate failure case, not as an approximate success.

28.6 Hessian-Passing Instability Witness

In $d = 1$, at $\epsilon = 1$, consider the scalar coefficients

$$V_0 = 0, \qquad V_{\pm 1} = 1, \qquad V_{\pm 2} = -\frac{1}{2}.$$

The symbol is

$$\omega(k)^2 = 2\cos k - \cos(2k).$$

Its low-energy Hessian data pass the speed-one condition:

$$-\frac{1}{2}\sum_r r^2 V_r = 1.$$

However, the $L = 4$ periodic sample has values

$$\omega(0)^2 = 1, \quad \omega(\pi/2)^2 = 1, \quad \omega(\pi)^2 = -3, \quad \omega(3\pi/2)^2 = 1.$$

The checked minimum-data helper records the negative minimum -3, with uncentered label 2 and momentum π, plus centered label -2 and momentum $-\pi$. Enabling its nonnegative finite-grid gate makes the helper raise an error. Thus the Hessian residual can vanish while the finite periodic symbol is unstable.

28.7 Doubler Counterexample

The Hessian condition is necessary but not sufficient. The symbol

$$\omega_\epsilon(k)^2 = m^2 + \frac{1}{\epsilon^2}\sum_a \sin^2(\epsilon k_a)$$

has the correct quadratic Hessian at $k = 0$ when $c = 1$. The checked example sets $m = 0$, so it is massless and has zero modes; by CONVENTIONS.md (j), this is outside the current positive-dispersion generator theorem. Its role is coefficient-level rejection: on even periodic grids it has multiple minima at Brillouin-zone points with $\sin(\epsilon k_a) = 0$. Therefore the compiler also needs a *single-patch* or *sector-selection* condition before a continuum relativistic scalar target is attached.

The current tests record this in $d = 1, 2, 3$ on the $L_a = 4$ periodic grids: the Hessian residual still vanishes at speed 1, while `symbol_minimum_data` finds 2^d zero minima with each centered label coordinate in $\{0, -2\}$. This is a finite-grid witness of extra patches, not a continuum species-count theorem.

28.8 Compiler Rule

For the scalar Gaussian tier, a Lorentz-target compile attempt must emit at least:

mass term	$\omega(0)^2$,
speed matrix	$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r$,
boost residual	$R_a(k)$,
rotation residual	$k_b\partial_a\omega^2 - k_a\partial_b\omega^2$,
boost-boost residual	$R_b X_a - R_a X_b$,
patch data	finite-grid minimum value, labels, momenta, and count.

Only the mass and speed entries are finite Taylor data. The rotation and boost-boost entries are now sampled smooth-symbol residuals in Julia, with the boost-boost entry represented as first-order X_c-coefficient data rather than as a finite-grid operator commutator. The residual and patch entries are what prevent an anisotropic or doubler model from being mislabeled as a single Klein-Gordon field.

29 Numerical Verification Suite for Gaussian Boson Examples

29.1 Status, Sources, and Dependencies

Status: checked Julia shard. This shard records the numerical invariants now implemented for the scalar Gaussian tier. The checks are not continuum proofs; they are finite algebraic witnesses that the coefficient compiler is computing the intended symbols and residuals. The massless doubler check is a coefficient-level rejection test, not a Fock-generator check. The finite periodic checks below do not assert an exact differential $[X_a, P_b] = i\delta_{ab}$ commutator.

The current numerical surface is deliberately narrow: it checks scalar coefficient structure, finite-periodic symbol values, finite stiffness spectra, assigned Fourier eigenvectors, periodic minimum data, and the implemented coefficient residuals/Hessians, including sampled rotation-Hamiltonian and boost-boost residual data on smooth-symbol momenta. It also checks centered periodic momentum labels for branch-aware low-energy windows, a loud finite-grid nonnegativity gate when requested, and test-local central finite-difference boost-time derivative oracles. It does not test finite boost generators, finite rotation generators, coordinate-generator commutators, or full generator closure.

Dependencies: CA-24–CA-27, INDEX.md, and CONVENTIONS.md (j).

29.2 Why These Tests

The numerical layer is not allowed to assert that a plot "looks relativistic". Each check must compare two independently constructed objects:

$$\text{real-space coefficient data} \quad \longleftrightarrow \quad \text{momentum-space symbol data.}$$

For Gaussian systems this is unusually powerful because the finite periodic stiffness matrix is exactly diagonalized by Fourier modes.

The boost-time derivative oracle follows the same rule. The production boost-time helper is compared to $\frac{1}{2}\nabla_k \omega(k)^2$ obtained by central finite differences of the separate scalar-symbol evaluator. Those tests use generic symmetric finite-range coefficient sets in $d = 1, 2, 3$, with off-axis terms in the higher-dimensional cases and sample momenta away from high-symmetry points.

The rotation and boost-boost residual tests are also smooth-symbol checks. They sample coefficient formulas at named momenta and compare residual sizes or tensor entries; they do not introduce $X_a = i\partial_{k_a}$ as a finite periodic matrix.

29.3 Numerical Tolerance Policy

CONVENTIONS.md (j) fixes the tolerance names used by this shard. With the `GAUSSIAN_` prefix and `_ATOL`/`_RTOL` suffix understood, the default pairs are

`SYMBOL_IMAG`	$(10^{-10}, 10^{-10})$,
`SYMBOL_VALUE`	$(10^{-10}, 10^{-10})$,
`EIGENVALUE`	$(10^{-10}, 10^{-10})$,
`MINIMUM_COUNT`	$(10^{-10}, 10^{-10})$,
`SMALL_SPACING_RESIDUAL`	$(5 \cdot 10^{-4}, 10^{-10})$.

The symbol-imaginary validator compares the complex symbol with its real part after the dimension and spacing preconditions are checked. The small-spacing pair is a named finite-sample tolerance, not a continuum error bound. The finite-difference derivative oracle is test-local and uses $(2 \cdot 10^{-8}, 2 \cdot 10^{-8})$; it is not part of the production tolerance surface.

29.4 Structural Coefficient Validation

The scalar Gaussian helpers now fail closed by default on coefficient data outside CONVENTIONS.md (j). The validator requires nonempty data, integer offsets, a common dimension $d \in \{1, 2, 3\}$, real coefficients, and aggregate paired equality $V_r = V_{-r}$ after duplicate offsets are summed. Negative tests cover a missing partner, unequal paired aggregates, complex coefficients, dimension mismatch, and non-integer offsets.

29.5 Periodic Stiffness Matrix

For finite periodic sizes $L_1, \ldots, L_d$, define the stiffness matrix

$$V_{xy} = \sum_r V_r \, \mathbf{1}_{y=x+r \bmod L}.$$

For the Fourier vector

$$\varphi_k(x) = |\Lambda|^{-1/2} e^{i\epsilon k \cdot x},$$

one has the exact identity

$$(V\varphi_k)(x) = \left(\sum_r V_r e^{i\epsilon k \cdot r} \right) \varphi_k(x) = \omega_\epsilon(k)^2 \varphi_k(x).$$

Therefore each dual label n gives a tested eigenvector with its assigned symbol value, and the sorted eigenvalues of the finite stiffness matrix must equal the sorted symbol values on the periodic Brillouin grid. These finite Fourier invariants are checked in Julia for $d = 1, 2, 3$, including mixed odd/even sizes with every $L_a > 2R$ and an off-axis symmetric pair $(1, 1), (-1, -1)$. They are not finite periodic first-moment or momentum-coordinate commutator tests; those require a branch, sawtooth, Fourier-interpolation, or other coordinate convention before they can become numerical claims.

The eigenvector suite also includes a one-sided shift sentinel to pin the implementation direction $y = x + r$. That sentinel is not a scalar Gaussian Hamiltonian claim. The strict default rejects it, and the test reaches this low-level matrix path only through an explicit non-validating keyword.

The eigenvalue oracle deliberately keeps the uncentered grid helper, because the tested symbol spectra are branch-insensitive. The separate centered grid helper returns the centered integer dual label and its physical representative

$$k_a = \frac{2\pi n_a}{\epsilon L_a}.$$

For even $L = 2r$, the branch is $n_a \in \{-r, \ldots, 0, \ldots, r - 1\}$, matching the centered dual lattice in the OAR source. Thus, at $L = 4$ and $\epsilon = 1$, the uncentered representative $3\pi/2$ is recorded on the centered branch as $n = -1$ and $k = -\pi/2$. This is only a momentum-label helper; it is not a periodic position-coordinate or generator construction.

29.6 Implemented Example Systems

Nearest-neighbour Klein-Gordon. The tests check the coefficient symbol against

$$m^2 + \frac{2}{\epsilon^2} \sum_a (1 - \cos(\epsilon k_a)),$$

the boost-time symbol against $\sin(\epsilon k_a)/\epsilon$, the residual vector against $\sin(\epsilon k_a)/\epsilon - k_a$, and the low-energy Hessian residual against zero. The sampled rotation and boost-boost residual helpers are

small on fixed low-energy momenta as ϵ is made small. The boost-time helper is also checked on non-Klein-Gordon finite-range symbols by the independent finite-difference oracle above.

Anisotropic cones. The tests build coefficients with speeds (c_1, c_2, c_3). They verify that the Hessian is

$$2\,\mathrm{diag}(c_1^2, c_2^2, c_3^2)$$

and that the Lorentz residual is nonzero when the target speed is 1. A separate sampled residual test checks that anisotropy also gives nonzero rotation-Hamiltonian and boost-boost residual data at a generic smooth momentum.

Hessian-passing unstable scalar. The tests build the one-dimensional coefficient witness

$$V_0 = 0, \qquad V_{\pm 1} = 1, \qquad V_{\pm 2} = -\frac{1}{2}.$$

Its speed-one Hessian residual vanishes, but finite periodic sampling at $L = 4$ has minimum -3. The minimum-data helper records one minimum at location 3, uncentered label 2, centered label -2, uncentered momentum π, and centered momentum $-\pi$. Requesting a nonnegative finite-grid symbol raises an error, so this example cannot pass as a stable scalar symbol.

Doubler symbol. The tests build

$$\omega_\epsilon(k)^2 = m^2 + \epsilon^{-2} \sum_a \sin^2(\epsilon k_a)$$

with $m = 0$. In $d = 1, 2, 3$, the speed-one Hessian residual vanishes on the chosen patch, but the $L_a = 4$ periodic grid has 2^d zero minima with centered label coordinates in $\{0, -2\}$. This catches the case where the Hessian at one minimum is correct but the compiler would need a multi-species, sector-selection, or zero-mode policy output. It is deliberately outside the positive-dispersion generator statements. The J6 symbol-residual test also samples a doubler momentum where the rotation residual is globally nonzero and an extra zero where R_a is nonzero relative to the selected $k = 0$ branch.

29.7 Current Julia Surface

The checked helper layer is:

`kg_nearest_neighbor_coefficients`	standard scalar KG coefficients,
`anisotropic_kg_nearest_neighbor_coefficients`	anisotropic coefficient witness,
`doubler_quadratic_coefficients`	coefficient-level doubler witness,
`validate_scalar_coefficients`	structural scalar-coefficient validator,
`scalar_quadratic_symbol`	$\sum_r V_r e^{i\epsilon k\cdot r}$,
`boost_time_symbol_from_coefficients`	$\frac{1}{2}\nabla\omega^2$,
`boost_time_residual_from_coefficients`	$\frac{1}{2}\nabla\omega^2 - c^2 k$,
`rotation_hamiltonian_residual_from_coefficients`	$k_b\partial_a\omega^2 - k_a\partial_b\omega^2$,
`boost_boost_residual_coefficients_from_coefficients`	coefficients of $R_bX_a - R_aX_b$,
`nearest_neighbor_integrated_energy_current_symbol`	1D integrated A4 current symbol,
`low_energy_hessian_from_coefficients`	$\partial_a\partial_b\omega^2(0)$,
`lorentz_hessian_residual`	$\frac{1}{2}\mathrm{Hess}(\omega^2)(0) - c^2 I$,
`periodic_stiffness_matrix`	finite periodic real-space matrix,
`periodic_momentum_grid`	uncentered spectrum grid,
`centered_periodic_momentum_grid`	centered labels and momenta,
`periodic_fourier_vector`	Fourier vector for a dual label,
`periodic_symbol_values`	finite Brillouin-grid eigenvalue oracle,
`symbol_minimum_data`	finite-grid minimum value, labels, momenta, and count,
`count_periodic_symbol_minima`	finite-grid minima counter.

This list is a coverage boundary, not a generator-algebra interface: no helper currently constructs a finite coordinate generator, a finite boost, a finite rotation, or their commutator table.

29.8 Next Numerical Milestones

Proposed next suite additions, none claimed as achieved here, are:

1. systematic sampled residual scans on shrinking low-energy windows,
2. comparison of open-boundary first moments with periodic symbol tests.

Any finite periodic coordinate-generator version of the first-moment comparison requires a branch or Fourier-interpolation convention before it can be promoted beyond a local experiment.

30 Gaussian Boson Real-Space Energy Density

30.1 Status, Sources, and Dependencies

Status: convention, not a theorem. This shard fixes how the scalar Gaussian Hamiltonian is split into site-attached real-space energy densities. It preserves the total Hamiltonian used in the sourced nearest-neighbour scalar model and in CA-24–CA-28, but it does not claim a continuum tensor object, a conservation law, or covariance.

Dependencies: CA-23–CA-28 and CONVENTIONS.md (j), (l).

30.2 Cell Volume and Notation

Let the hypercubic lattice spacing be ϵ, so one spatial cell has volume ϵ^d. In this shard h_x is the physical energy density attached to the cell at x:

$$H_\epsilon = \epsilon^d \sum_x h_x.$$

The integrated site energy is

$$e_x := \epsilon^d h_x, \qquad H_\epsilon = \sum_x e_x.$$

This distinction matters because the generic first-moment notation of CONVENTIONS.md (h) writes $H = \sum_x h_x$; for the Gaussian block, the object to insert there is e_x, not the density h_x.

For fields Φ_x, Π_x and a real symmetric translation-invariant kernel $V_r = V_{-r}$, use

$$H_\epsilon = \epsilon^d \left(\frac{1}{2} \sum_x \Pi_x^2 + \frac{1}{2} \sum_{x,r} \Phi_x V_r \Phi_{x+r} \right).$$

For the nearest-neighbour Klein-Gordon source, the renormalized coefficients are

$$V_0 = m^2 + \frac{2d}{\epsilon^2}, \qquad V_{\pm e_a} = -\frac{1}{\epsilon^2},$$

which give the dispersion recorded in CA-24.

30.3 Equal-Endpoint Bond Split

The chosen local density is

$$h_x = \frac{1}{2}\Pi_x^2 + \frac{1}{2} V_0 \Phi_x^2 + \frac{1}{4} \sum_{r \neq 0} V_r \big(\Phi_x \Phi_{x+r} + \Phi_{x-r} \Phi_x \big).$$

Each nonzero displacement bond is assigned half to each endpoint cell. Summing over x gives

$$\sum_x h_x = \frac{1}{2} \sum_x \Pi_x^2 + \frac{1}{2} \sum_{x,r} \Phi_x V_r \Phi_{x+r},$$

so this convention reproduces the total scalar Hamiltonian above after the cell-volume factor is restored.

The q, p notation in CA-24–CA-28 absorbs the cell volume into the coefficient bookkeeping. In that notation the same displayed formula defines e_x, and the physical density is $h_x = e_x/\epsilon^d$.

30.4 Boundary Policy

The default exact identities are infinite-lattice identities with summable tails, or finite periodic total-energy identities. On a finite open region, the density is obtained from the finite Hamiltonian actually named: include only bonds present in that Hamiltonian, and share each included bond equally between its included endpoints. No exterior ghost sites, omitted half-bonds, or boundary counterterms are implicit.

For periodic finite lattices, wrap-around bonds are included in the total energy split. This does not create a periodic position convention. Any periodic first moment still needs the branch, sawtooth, or interpolation policy required by CONVENTIONS.md (j).

30.5 Vacuum-Energy Policy

The density above is bare. No normal-ordering or vacuum subtraction is implicit. Once a state ω is named, the expectation-subtracted version may be recorded as

$$:h_x:_\omega := h_x - \omega(h_x)\mathbf{1}.$$

This changes H_ϵ by a scalar and therefore drops out of commutators, but it matters for absolute energies, expectation values, and Fourier modes of quadratic densities. The local subtraction source is the free-fermion Koo-Saleur setting, where the scaling-limit representation relies on subtracting vacuum or ground-state expectation values; this shard only records the analogous policy needed before such quantities are used in the scalar Gaussian block.

30.6 Improvement Policy

The default convention has no divergence improvement:

$$h'_x = h_x$$

unless a later shard explicitly names an edge field b, a discrete divergence $(\nabla_\epsilon \cdot b)_x$, and a boundary convention. Such a rewrite

$$h'_x = h_x + (\nabla_\epsilon \cdot b)_x$$

may leave the total energy unchanged on a periodic lattice or for decaying fields, but it is not harmless for position-weighted generators. A discrete summation-by-parts calculation shifts first moments by an edge-field term plus boundary contributions. For example, in one dimension with

$$(\nabla_\epsilon b)_n = \frac{b_{n+1/2} - b_{n-1/2}}{\epsilon}, \qquad x_n = \epsilon n,$$

one has

$$\epsilon \sum_n x_n (\nabla_\epsilon b)_n = -\epsilon \sum_n b_{n+1/2} + \text{boundary terms}.$$

Therefore improvements are separate data, not gauge choices to be silently modded out.

30.7 Compiler Consequence

A scalar Gaussian real-space generator proposal must now carry the tuple

$$\big(\epsilon^d,\ h_x,\ e_x = \epsilon^d h_x,\ \text{boundary policy},\ \text{subtraction policy},\ \text{improvement policy}\big).$$

The total symbol $\omega(k)^2$ remains sufficient for CA-24–CA-28, but it is not sufficient once a construction uses position weights or local quadratic modes. Those later constructions must cite this density convention or perform an explicit convention sweep.

31 One-Dimensional Gaussian Boson Stress-Energy Candidates

31.1 Status, Sources, and Dependencies

Status: proposal with checked T_{01} seed. This shard fixes a candidate dictionary for the finite open one-dimensional scalar Gaussian chain. Only the energy-current entry has a checked local witness. No convergence theorem, no continuum stress tensor, and no Virasoro representation is claimed.

Dependencies: CA-12, CA-23–CA-29, CONVENTIONS.md (j), (k), (l), and the A4 finite open-chain current tests.

31.2 Continuum Motivation and Boundary

The continuum notation $T_{\mu\nu}$ is used here as a target shape, not as an imported lattice theorem. Weinberg supplies the QFT motivation that spacetime translations lead to conserved currents grouped as an energy-momentum tensor and that translation generators are spatial integrals of time components. The same registered OCR also records two caveats that matter for the lattice dictionary: the canonical raised tensor need not be symmetric, while the Belinfante tensor is a modified conserved tensor with the same integrated translation generators and symmetry. The OCR around formulas is noisy, so this shard uses those anchors only for motivation and caveats, not for exact component signs. The self-adjoint commutator signs for any later Poincare comparison are the project convention in CONVENTIONS.md (k), not raw OCR import.

Schottenloher's Axiom 5 is narrower but sharper for the CFT endpoint: in a two-dimensional conformal field theory there are four fields $T_{\mu\nu}$, $\mu, \nu \in \{0, 1\}$, with symmetry and conservation. The later Luescher–Mack theorem gives tracelessness and two Virasoro mode families under the full Axioms 1–5. We do not use those CFT conclusions to define the lattice T_{10} or T_{11}. They are acceptance targets for a future continuum proof.

31.3 Finite Open Chain Setup

Work first with a finite open chain of N cells and spacing ϵ. The CA-29 density convention distinguishes the physical density h_j from the integrated cell energy

$$e_j = \epsilon h_j, \qquad H_\epsilon = \sum_{j=1}^{N} e_j.$$

Finite commutators use e_j. This avoids inserting an accidental factor of ϵ into a discrete continuity identity.

For the nearest-neighbour scalar Gaussian chain, the A4 helper implements the CA-29 equal-endpoint split of the finite Hamiltonian: kinetic and onsite terms belong to the site, and each present bond $q_j q_{j+1}$ is split equally between its two endpoints. Open-chain means no exterior ghost cells, no missing half-bonds, and no boundary counterterms are implicit.

31.4 Candidate Dictionary

T_{00}^{lat}. The lattice energy-density entry is the CA-29 density:

$$T_{00}^{\mathrm{lat}}(j) := h_j.$$

For exact finite open-chain algebra, the cell-integrated version is

$$E_j := e_j = \epsilon h_j.$$

Thus T_{00}^{lat} is the physical density label, while E_j is the object that appears in the tested quadratic commutators.

T_{01}^{lat}**.** The energy-current or energy-flux seed lives on open-chain edges:

$$J_{j+1/2} := i[e_j, e_{j+1}], \qquad 1 \le j < N.$$

Set endpoint currents to zero,

$$J_{1/2} = J_{N+1/2} = 0.$$

The checked finite continuity identity is

$$i[H_\epsilon, e_j] = J_{j-1/2} - J_{j+1/2}, \qquad 1 \le j \le N.$$

This is the only stress-energy component in this shard with a Julia witness. It is the lattice candidate corresponding to T_{01} in the cell-integrated continuity law

$$\frac{d}{dt} \int_{\text{cell } j} T_{00}\, dx = T_{01}(x_{j-1/2}) - T_{01}(x_{j+1/2}).$$

No continuum normalization or convergence of $J_{j+1/2}$ is claimed.

T_{10}^{lat}**.** Do not identify T_{10}^{lat} with T_{01}^{lat} at this stage. Define it instead as the future integrated momentum-density comparison target:

$$M_j \quad \text{with} \quad P^{\mathrm{lat}} = \sum_j M_j.$$

The required next witness is that the chosen M_j, or at least its total P^{lat}, matches the Gaussian one-particle flat-symbol momentum target after the appropriate $\mathrm{d}\Gamma(k)$ comparison is made. That is the S3 task, not an assumption in this shard.

T_{11}^{lat}**.** Define T_{11}^{lat} operationally as the future current of the chosen momentum density. Once M_j is fixed, the candidate stress component must come with edge terms $S_{j+1/2}$ and a checked equation

$$i[H_\epsilon, M_j] = S_{j-1/2} - S_{j+1/2}$$

with an explicit boundary convention. Until that witness exists, there is no lattice T_{11} formula. In particular, T_{11}^{lat} is not defined as $-T_{00}^{\mathrm{lat}}$, not derived from tracelessness, and not inferred from CFT holomorphic factorization.

31.5 Boundary, Subtraction, and Improvement

This candidate dictionary is open-boundary first. Endpoint currents are part of the statement, not small errors:

$$i[H_\epsilon, e_1] = -J_{3/2}, \qquad i[H_\epsilon, e_N] = J_{N-1/2}.$$

Periodic total-energy density splits are allowed by CA-29, but periodic first moments and periodic coordinate-dependent generator tests remain blocked until CONVENTIONS.md (j) fixes a coordinate branch, sawtooth, Fourier interpolation, or equivalent policy.

The default density is bare. A state-dependent subtraction $:h_j:_\omega = h_j - \omega(h_j)\mathbf{1}$ may later be named; it changes the Hamiltonian by a scalar and does not change commutators, but it is part of any claim about absolute energy or density modes. The default improvement is zero. Any later replacement $h'_j = h_j + (\nabla_\epsilon b)_j$ must name the edge field, discrete divergence, and boundary convention before first moments or currents are recomputed.

31.6 Compiler Payload and Acceptance

A compiler output that cites this shard must carry:

density convention	$h_j,\ e_j = \epsilon h_j,$ CA-29 split,
boundary policy	finite open chain, $J_{1/2} = J_{N+1/2} = 0$,
subtraction policy	bare unless a state ω is named,
improvement policy	$b = 0$ unless a discrete divergence is named,
checked T_{01} witness	$i[H_\epsilon, e_j] = J_{j-1/2} - J_{j+1/2}$.

The acceptance boundary is equally explicit. The checked witnesses are the finite open-chain continuity residual in `src/GaussianBosonCurrents.jl` and `test/runtests.jl`, plus the CA-31 translation-invariant current-symbol equality for the nearest-neighbour Klein-Gordon orientation. Still pending are the finite open-chain wave-packet comparison with the flat $\mathrm{d}\Gamma(k)$ momentum target; the T_{11} witness consisting of a chosen momentum density M_j and checked current $S_{j+1/2}$; and the continuum witness: scaling maps, domains, state sequence, and covariance convergence. Without those witnesses, the dictionary remains a proposal with checked T_{01} evidence, not a stress-tensor theorem.

32 Gaussian Current-Symbol Equivalence in One Dimension

32.1 Status, Sources, and Dependencies

Status: checked translation-invariant symbol bridge. This shard adds the first bridge between the finite real-space current convention and the momentum-symbol layer. It is deliberately one-dimensional and nearest-neighbour. The checked statement is the equality of the translation-invariant integrated-current symbol with the CA-26 boost-time symbol for the scalar Klein-Gordon coefficients. No finite open-chain momentum operator, no finite periodic first moment, no $d\Gamma(k)$ theorem, and no continuum convergence statement is claimed.

Dependencies: CA-24, CA-26, CA-29–CA-30, and CONVENTIONS.md (j), (k), (l).

32.2 Real-Space Current Orientation

The finite open-chain A4 witness fixes the local current on an open bond by

$$J_{j+1/2} = i[e_j, e_{j+1}]$$

with endpoint currents zero. For a nearest-neighbour scalar chain with bond coefficient b, the checked closed form is

$$J_{j+1/2} = \frac{b}{2}\big(q_{j+1}p_j - q_j p_{j+1}\big).$$

This sign is not adjustable in this shard: it is the CONVENTIONS.md (l) orientation already used in the open-chain continuity test.

To compare with the boost-time symbol, pass from the local edge current to the first-moment integrated current density. In uniform spacing ϵ, the nearest-neighbour first-moment coefficient contributes the spacing factor ϵ. The translation-invariant integrated current therefore has symbol

$$\mathcal{J}_\epsilon(k) = -\epsilon b \sin(\epsilon k).$$

This is a Fourier-symbol identity for the infinite or translation-invariant nearest-neighbour expression. It does not introduce a periodic coordinate branch and does not construct an open finite-chain momentum operator.

32.3 Klein-Gordon Equality

For the one-dimensional nearest-neighbour lattice Klein-Gordon coefficients,

$$b = -\epsilon^{-2}, \qquad \omega_{\epsilon,m}(k)^2 = m^2 + 2\epsilon^{-2}(1 - \cos(\epsilon k)).$$

The integrated-current symbol becomes

$$\mathcal{J}_\epsilon(k) = \frac{\sin(\epsilon k)}{\epsilon}.$$

The CA-26 boost-time symbol is

$$B_\epsilon(k) = \frac{1}{2}\partial_k \omega_{\epsilon,m}(k)^2 = \frac{\sin(\epsilon k)}{\epsilon}.$$

Thus, at this checked symbol level,

$$\mathcal{J}_\epsilon(k) = B_\epsilon(k).$$

The mass drops out because the current and the derivative of the onsite mass term are both insensitive to m^2.

32.4 Julia Witness

The helper

`nearest_neighbor_integrated_energy_current_symbol`

implements

$$-\epsilon b \sin(\epsilon k)$$

for one-dimensional data only. The Klein-Gordon wrapper inserts $b = -\epsilon^{-2}$. The tests compare that wrapper against

`kg_lattice_boost_time_symbol`

for several momenta and spacings. They also include a sign sentinel: for positive k and the Klein-Gordon negative bond, the current symbol is positive and agrees with the boost-time target; reversing the bond sign gives a negative symbol and fails the target comparison. A small-spacing check compares the current symbol with the centered continuum momentum representative k near zero using the existing Gaussian small-spacing tolerance.

32.5 Open Obligations

This shard closes only the first symbol-level bridge. The following remain pending:

1. a finite open-chain wave-packet comparison with boundary residuals,
2. an identification with the one-particle $\mathrm{d}\Gamma(k)$ momentum target,
3. a periodic first-moment construction, which still needs a branch policy,
4. a scaling or continuum convergence theorem.

Any compiler output citing this shard must therefore say "current-symbol equality" rather than "finite-chain momentum" or "continuum stress tensor".

33 Higher-Dimensional Gaussian Cell-Current Proposal

33.1 Status, Sources, and Dependencies

Status: proposal and compiler-input specification. This shard states the support data and local equations a higher-dimensional Gaussian stress-energy candidate must carry in $d = 2, 3$. It is not a theorem, does not define a global orientation convention, and makes no convergence claim.

Dependencies: CA-13, CA-23, CA-29–CA-31, and CONVENTIONS.md (j), (k), (l).

33.2 Compiler Input: Oriented Support Data

For each finite or infinite hypercubic support in $d = 2$ or $d = 3$, the compiler input must include an oriented incidence structure:

$$\mathcal{C}, \qquad \mathcal{F}, \qquad \sigma(c, f) \in \{-1, 0, +1\}.$$

Here $\mathcal{C}$ is the set of energy cells, attached to sites $x_c \in \epsilon\mathbb{Z}^d$. The set $\mathcal{F}$ contains oriented codimension-one faces. In $d = 2$, these faces are oriented edges between neighbouring square cells. In $d = 3$, they are oriented plaquettes between neighbouring cube cells. If an interior face f is oriented from its minus cell $c_-(f)$ to its plus cell $c_+(f)$, set

$$\sigma(c_-(f), f) = +1, \qquad \sigma(c_+(f), f) = -1,$$

and set $\sigma(c, f) = 0$ for all other cells. This sign convention is local input for this proposal, not a repository-wide orientation convention.

Boundary faces are separate data. For an open box, $\mathcal{F}_\partial$ contains oriented exterior faces with only one adjacent cell; their incidence sign is recorded as $\sigma_\partial(c, f) = +1$ for outward flux. Setting a boundary flux to zero is then a named boundary condition, not an omitted term. For a periodic support, exterior faces are absent, but any first moment or coordinate-weighted test still needs the branch or Fourier policy required by CONVENTIONS.md (j).

33.3 Energy Density and Energy Flux

The energy component starts from CA-29:

$$T_{00}^{\mathrm{lat}}(c) := h_c, \qquad e_c := \epsilon^d h_c, \qquad H_\epsilon = \sum_{c \in \mathcal{C}} e_c.$$

All finite commutator identities use the integrated cell energy e_c.

An energy-current candidate is face data

$$J_0(f) \quad (f \in \mathcal{F}),$$

with optional boundary data $J_0^\partial(f)$ on $f \in \mathcal{F}_\partial$. If f has normal direction a, then $J_0(f)$ is the proposed lattice T_{0a} flux through that oriented face. The required energy-continuity witness is the cell equation

$$i[H_\epsilon, e_c] + \sum_{f \in \mathcal{F}} \sigma(c, f) J_0(f) + \sum_{f \in \mathcal{F}_\partial} \sigma_\partial(c, f) J_0^\partial(f) = 0.$$

For an open no-leak boundary this specializes to $J_0^\partial(f) = 0$. For a driven or truncated-region diagnostic, the same displayed equation records the boundary residual explicitly. This shard does not propose a universal formula for $J_0(f)$; a later numerical suite must derive or solve for it from the chosen Gaussian density and then test the equation above.

33.4 Momentum Densities and Stress Currents

The momentum-density components are additional witnesses:

$$M_a(c) \quad (a = 1, \dots, d), \qquad P_a^{\mathrm{lat}} = \sum_c M_a(c).$$

They are not inferred from the energy flux $J_0(f)$. The reason is already visible in CA-30: T_{10} was kept as a momentum-density target, not as a consequence of T_{01}. In higher dimensions this separation is mandatory because rotations require M_a before angular momentum even has a candidate density.

For each chosen momentum density $M_a(c)$, the stress components are its currents. If f has normal direction b, write

$$S_a(f) \equiv T_{ab}^{\mathrm{lat}}(f).$$

The required momentum-continuity witness is

$$i[H_\epsilon, M_a(c)] + \sum_{f \in \mathcal{F}} \sigma(c, f) S_a(f) + \sum_{f \in \mathcal{F}_\partial} \sigma_\partial(c, f) S_a^\partial(f) = 0$$

for every a and every cell c. Until M_a and S_a are supplied with this witness, there is no lattice T_{a0} or T_{ab} claim.

33.5 Boost and Symbol Diagnostics

The boost first moment can be formed from the integrated energy cells:

$$K_a^{\mathrm{lat}} := \sum_c x_c^a e_c.$$

Using the energy-continuity equation, $i[H_\epsilon, K_a^{\mathrm{lat}}]$ reduces by discrete summation by parts to the integrated a-normal energy flux plus boundary terms. The higher-dimensional analogue of CA-31 is then a test, not an assumption:

$$\text{integrated } T_{0a} \text{ symbol} \quad \stackrel{?}{=} \quad \frac{1}{2} \partial_a \omega(k)^2.$$

Only after this equality is checked in the relevant local or symbol regime may it be compared with the flat one-particle momentum target k_a and the residuals of CA-26–CA-27. Passing the symbol equality would still not prove a continuum stress tensor.

33.6 Rotation and Antisymmetric-Stress Residual

For $a < b$, a rotation candidate requires the momentum-density witnesses:

$$J_{ab}^{\mathrm{lat}} := \sum_c \big(x_c^a M_b(c) - x_c^b M_a(c) \big).$$

This formula is unavailable if only T_{00} and T_{0a} have been supplied. Applying the momentum-continuity equation gives a discrete summation-by-parts diagnostic of the form

$$i[H_\epsilon, J_{ab}^{\mathrm{lat}}] = \epsilon \sum_{f \perp a} S_b(f) - \epsilon \sum_{f \perp b} S_a(f) + \text{boundary terms},$$

with the displayed sign tied to the incidence convention above. In continuum language this is the antisymmetric-stress obstruction: rotation conservation targets $T_{ab} = T_{ba}$, after boundary terms and any spin/improvement data have been named. Weinberg's registered anchors motivate that a symmetric Belinfante tensor can carry the same integrated translations, but the canonical tensor need not be symmetric. Therefore stress symmetry is a target or witness, not an assumption baked into the lattice proposal.

33.7 Trace, Improvement, and Boundary Alternatives

Trace diagnostics are also target-dependent. A compiler may report the diagonal stress combination built from h_c and the $S_a(f)$, but it must name the continuum target and metric/sign convention before calling it a trace residual. No massless-CFT tracelessness test, massive Klein-Gordon trace test, or improvement formula is imported here.

The CA-29 improvement policy remains active:

$$h'_c = h_c + (\nabla_\epsilon \cdot b)_c.$$

Any such change must name the edge or face field b, the discrete divergence, and the boundary convention. It must also recompute J_0, M_a, S_a, and the first-moment boundary terms; improvements are not silent gauge moves for boosts or rotations.

Two boundary regimes are admissible compiler inputs:

open box	use explicit $\mathcal{F}_\partial$ terms and test endpoint/face residuals,
periodic box	use no boundary faces, but fix a coordinate branch, sawtooth, or Fourier-interpolation policy before any first-moment test.

The existing Gaussian numerical layer is periodic-symbol first. The next stress-energy numerical suite, S5 in the action plan, should therefore start with open-boundary local continuity tests or with a separately sourced periodic coordinate construction, and should include anisotropic/doubler failure witnesses at the current-density level.

34 First Gaussian Stress-Energy Numerical Suite

34.1 Status, Sources, and Dependencies

Status: checked local-current numerical suite. This shard records the first stress-energy-specific numerical tests. The checked surface is narrow: finite open-chain T_{01} continuity for the CA-29/CA-30 density-current choice, and a one-dimensional nearest-neighbour current-symbol slope residual. It is not a finite boost construction, not a $\mathrm{d}\Gamma(k)$ theorem, not a periodic first-moment test, and not a continuum stress-tensor claim.

Dependencies: CA-29–CA-32 and CONVENTIONS.md (j), (k), (l).

34.2 Finite Open-Chain Continuity

The first suite strengthens the A4 local-current witness instead of replacing it. For each tested open chain, the Julia layer constructs the CA-29 equal-endpoint site energies e_j, the total Hamiltonian

$$H = \sum_{j=1}^{N} e_j,$$

and the edge currents

$$J_{j+1/2} = i[e_j, e_{j+1}], \qquad 1 \le j < N.$$

The residual helper checks

$$i[H, e_j] - (J_{j-1/2} - J_{j+1/2}) = 0$$

with endpoint currents $J_{1/2} = J_{N+1/2} = 0$.

The test cases now cover several chain sizes and onsite/bond choices:

$$(N, a, b) \in \{(2, 0.75, -0.5), (3, 2.0, 1.25), (5, 7.25, -1.5), (7, 0.1, -0.875)\}.$$

For each case the suite asserts:

1. $\sum_j e_j = H$,
2. H_{qq} is the expected tridiagonal matrix with diagonal a and bond b,
3. $H_{pp} = I$,
4. $J_{j+1/2}$ equals the closed form $\frac{b}{2}(q_{j+1}p_j - q_j p_{j+1})$,
5. $J_{j+1/2}$ is independent of the onsite coefficient,
6. all continuity residual matrices vanish,
7. $i[H, e_1] = -J_{3/2}, \quad i[H, e_j] = J_{j-1/2} - J_{j+1/2}, \quad i[H, e_N] = J_{N-1/2}$.

The explicit endpoint tests are the important stress-energy addition: boundary signs are part of the local T_{01} claim, not discarded finite-size errors. The suite also checks that a one-site chain cannot produce a bond current and that an out-of-range open bond index raises an error.

34.3 Current-Symbol Slope Residual

The second suite is the smallest symbol-level current-density failure witness currently justified by CA-31. For the one-dimensional nearest-neighbour symbol

$$\mathcal{J}_\epsilon(k) = -\epsilon b \sin(\epsilon k),$$

the small-k slope is

$$\mathcal{J}_\epsilon'(0) = -\epsilon^2 b.$$

The new helper returns the speed residual

$$\rho_J(b, \epsilon, c) = -\epsilon^2 b - c^2.$$

It validates positive spacing, positive target speed, and real bond data before returning a number.

For the speed-c nearest-neighbour Klein-Gordon bond

$$b = -c^2 \epsilon^{-2},$$

the residual is zero. The tests assert this for several spacings and speeds. They also assert failure witnesses:

$$b = -\epsilon^{-2} \quad \text{fails when } c \neq 1,$$

wrong bond magnitude gives the expected nonzero residual, and reversing the bond sign gives

$$\rho_J(c^2\epsilon^{-2}, \epsilon, c) = -2c^2.$$

These are current-density symbol failures for the CA-31 orientation. They are not Hessian-only failures and they do not use a periodic coordinate branch.

34.4 What This Falsifies

The suite falsifies three classes of local-current mistakes:

1. incorrect open-boundary signs in $i[H, e_j] = J_{j-1/2} - J_{j+1/2}$,
2. wrong orientation in $J_{j+1/2} = i[e_j, e_{j+1}]$,
3. nearest-neighbour current symbols whose low-energy slope does not match the selected c^2.

The onsite-independence checks also catch an accidental insertion of the mass or onsite term into the nearest-neighbour current.

34.5 Open Obligations

The suite deliberately leaves the following undone:

1. finite open-chain wave packets and comparison with $\mathrm{d}\Gamma(k)$,
2. periodic first moments or periodic coordinate generators,
3. higher-dimensional current-density formulas and incidence-sign tests,
4. next-nearest-neighbour or doubler current-density witnesses,
5. momentum-density T_{10} and stress-current T_{11} continuity tests,
6. scaling maps, state choices, domains, and continuum convergence.

The doubler evidence remains the CA-28 coefficient/minimum-data rejection. It should not be described as a current-density failure until a current formula for that non-nearest-neighbour symbol is derived and tested.

35 Qubit Nearest-Neighbour Symmetry Quest

35.1 Status, Sources, and Dependencies

Status: programme shard with checked seed. This shard scopes the next lattice-symmetry quest for qubit systems. CA-35 supplies the first checked finite algebraic obstruction. CA-36 and CA-37 state proposal-level diagnostics for $2+1$-dimensional Poincare tests and $1+1$-dimensional Witt/Virasoro tests.

Dependencies: CA-10–CA-17, CONVENTIONS.md (h), (k), (m), and the review orchestration in `reviews/2026-05-31_qubit_nn_symmetry/`.

35.2 Input Tier

The first input class is a translation-invariant qubit nearest-neighbour density. In the convention of CONVENTIONS.md (m),

$$h = \sum_{\alpha,\beta=0}^{3} h_{\alpha\beta}\, \sigma^{\alpha} \otimes \sigma^{\beta}, \qquad h_{\alpha\beta} \in \mathbb{R},$$

with $\sigma^0 = I$ and $\sigma^1, \sigma^2, \sigma^3 = X, Y, Z$. On a one-dimensional open chain the density h_j acts on sites $(j, j+1)$, so

$$H = \sum_j h_j.$$

In two spatial dimensions the same coefficient format may be used separately on horizontal and vertical edges, but the edge orientation, midpoint, boundary, and bulk-projection conventions are not fixed by this shard.

35.3 Question

The question is not whether a finite qubit lattice has exact Poincare or Virasoro symmetry. CA-15 already forbids that shortcut. The question is whether the finite algebra forced by the candidate generator ansatz gives cheap rejection tests:

$$\begin{gathered}\text{Pauli coefficient data} \longrightarrow \text{finite commutator residuals} \\ \longrightarrow \text{rule out impossible symmetry candidates early.}\end{gathered}$$

The first such test is the adjacent-density current

$$J(h) := i[h_{12}, h_{23}],$$

because CA-12 shows that the first energy moment

$$K = \sum_j x_j h_j$$

obeys $i[H, K]$ equal to the sum of adjacent commutators in the translation-invariant bulk. If this current vanishes, the first-moment boost ansatz produces zero spatial momentum.

35.4 Investigation Plan

One-dimensional Poincare filters. Compute $J(h)$ directly from $h_{\alpha\beta}$. Reject the density as a nontrivial Poincare candidate for the CA-12 first-moment route if the bulk current vanishes while the energy density is non-scalar. Then test stronger finite residuals such as $i[P, K] - H$, but only after boundary, scaling, and normalization policies are named.

Two-dimensional Poincare filters. Use horizontal and vertical edge densities h^x, h^y. Define ramp moments from edge midpoints and compute $P_x = i[H, K_x]$, $P_y = i[H, K_y]$ on open patches. Proposed finite residuals include current nontriviality, $i[H, P_a]$, $i[P_x, P_y]$, isotropy under $x \leftrightarrow y$, and later rotation closure after a local momentum-density split is fixed.

Witt/Virasoro filters. On a periodic one-dimensional chain, form Fourier modes of the density and of the adjacent current. Koo–Saleur supplies the sourced pattern: Hamiltonian-density modes plus commutator corrections. The first rejection test is again current collapse; central terms and finite-size non-closure are not early failure tests.

35.5 Immediate Rule-Outs

The first checked obstruction is deliberately modest. Under the symmetric nearest-neighbour split of a genuine on-site qubit Hamiltonian, $J(h) = 0$. Thus a fully local single-site Hamiltonian cannot provide nonzero bulk momentum through the CA-12 first-moment boost ansatz. The same obstruction also catches commuting classical nearest-neighbour densities, such as pure diagonal ZZ terms without noncommuting fields.

Nonzero $J(h)$ is not sufficient. It is only the first algebraic gate. A density that passes still needs the CA-15 evidence payload: local support policy, state sequence, renormalisation maps, convergence topology, low-energy sector or core, and observable covariance.

36 One-Dimensional Qubit Boost-Current Obstructions

36.1 Status, Sources, and Dependencies

Status: checked local derivation. This shard turns the CA-12 nearest-neighbour identity into explicit polynomial conditions on the two-qubit Pauli coefficient matrix. The obstruction is checked in Julia.

Dependencies: CA-12, CA-34, CONVENTIONS.md (h), (k), (m).

36.2 Adjacent Current in Pauli Coefficients

Let

$$h_j = \sum_{\alpha,\beta=0}^{3} h_{\alpha\beta}\, \sigma_j^{\alpha} \sigma_{j+1}^{\beta}.$$

The only noncommuting overlap between h_j and h_{j+1} is at site $j+1$. With

$$[\sigma^b, \sigma^c] = 2i \sum_{e=1}^{3} \epsilon_{bce} \sigma^e \qquad (b, c \in \{1,2,3\}),$$

one obtains

$$i[h_j, h_{j+1}] = -2 \sum_{\alpha,\delta=0}^{3} \sum_{e=1}^{3} \left(\sum_{b,c=1}^{3} h_{\alpha b} h_{c\delta} \epsilon_{bce} \right) \sigma_j^{\alpha} \sigma_{j+1}^{e} \sigma_{j+2}^{\delta}.$$

Equivalently, define right-leg and left-leg vectors

$$R_\alpha = (h_{\alpha 1}, h_{\alpha 2}, h_{\alpha 3}), \qquad L_\delta = (h_{1\delta}, h_{2\delta}, h_{3\delta}).$$

Then the current coefficients are the cross products

$$i[h_j, h_{j+1}] = -2 \sum_{\alpha,\delta=0}^{3} \sum_{e=1}^{3} (R_\alpha \times L_\delta)_e \sigma_j^{\alpha} \sigma_{j+1}^{e} \sigma_{j+2}^{\delta}.$$

Thus the simple current-collapse condition is

$$R_\alpha \times L_\delta = 0 \qquad \text{for every } \alpha, \delta.$$

When this holds, every adjacent-current contribution to the CA-12 bulk momentum candidate vanishes.

36.3 Fully On-Site Terms

For a genuine translation-invariant on-site density

$$A = \sum_{\alpha=0}^{3} a_\alpha \sigma^\alpha,$$

CONVENTIONS.md (m) embeds it into the two-site bond tier by the symmetric split

$$h_{\text{onsite}} = \frac{1}{2}(A \otimes I + I \otimes A).$$

Then the right-leg and left-leg vectors are all parallel in the only places where they overlap, so

$$i[(h_{\text{onsite}})_{12}, (h_{\text{onsite}})_{23}] = 0.$$

The Julia test checks this directly for a generic real linear combination of I, X, Y, Z.

Consequently the first-moment boost ansatz gives

$$i[H, K] = 0$$

in the translation-invariant bulk. If the target is a nontrivial Poincare representation with $i[H, K] = P$ and $i[P, K] = H$, this first-moment route cannot realise it unless the energy itself is trivial after the relevant compression or subtraction.

36.4 Other Immediate Rejections

The same current-collapse test catches commuting nearest-neighbour density families. A pure diagonal classical interaction such as ZZ has

$$[h_j, h_{j+1}] = 0$$

because the overlapping Pauli factor is the same diagonal matrix. Such a model may still be a well-defined lattice Hamiltonian, but it fails the first nontrivial bulk-symmetry diagnostic: the boost-current candidate supplies no spatial momentum.

By contrast, the tested transverse-Ising-style density with a ZZ bond and a symmetric X-field split has nonzero adjacent current. This does not prove Poincare invariance; it only passes the first obstruction.

36.5 The Next Residual

The next necessary Poincare relation is translation conservation:

$$i[H, P] = 0.$$

For the local candidate $P = \sum_j C_j$, $C_j = i[h_j, h_{j+1}]$, the minimal bulk density after the Jacobi cancellation of outer-overlap terms is

$$A_j = i[h_j + h_{j+1}, C_j].$$

The full infinite-chain equation is not necessarily $A_j = 0$, but

$$A_j = u_j - u_{j+1}$$

for a finite density u. CA-39 writes this and the next boost relation $i[P, K] = v^2 H$ as explicit Pauli-coefficient equations.

This residual is useful for detecting density-split artifacts. An asymmetric encoding of an on-site Hamiltonian,

$$h = U \otimes I + I \otimes V,$$

can create the fake current

$$i[h_{12}, h_{23}] = I \otimes i[V, U] \otimes I.$$

But the checked sentinel with $U = X$, $V = Z$ has nonzero

$$i[h_{12} + h_{23}, i[h_{12}, h_{23}]],$$

so it fails translation conservation even though the adjacent current is nonzero. The CA-39 boost equation then tests the remaining relation after the allowed density-divergence and scalar-energy subtraction have been fixed.

36.6 Boundary

The obstruction is conditional on the density convention. An asymmetric encoding of an on-site Hamiltonian as $A \otimes I + I \otimes B$ can create spurious adjacent commutators when A and B fail to commute. Such currents are density-split artifacts unless a separate convention justifies the split, and the conservation residual above is the first backstop against them. Passing the obstruction is also far weaker than CA-15 acceptance: boundary residuals, scaling maps, low-energy sectors, and observable covariance remain unproved.

37 Two-Dimensional Qubit Plaquette Symmetry Diagnostics

37.1 Status, Sources, and Dependencies

Status: proposal diagnostics. This shard gives a finite algebraic test plan for $2+1$-dimensional qubit lattices. No two-dimensional current or rotation convention is fixed yet, so every condition below is a proposed compiler filter.

Dependencies: CA-13, CA-15, CA-16, CA-32, CONVENTIONS.md (h), (k), (m).

37.2 Edge-Density Input

On an open square patch, use oriented horizontal and vertical edges:

$$h^a_{r,r+\hat{a}} = \sum_{\alpha,\beta=0}^{3} c^a_{\alpha\beta}\, \sigma^\alpha_r \sigma^\beta_{r+\hat{a}}, \qquad a \in \{x, y\}.$$

Let m_e be the midpoint of edge e, and write

$$H = \sum_e h_e, \qquad K_a = \sum_e m^a_e h_e.$$

The ramp-current candidate is

$$P_a := i[H, K_a] = \sum_{\{e,f\}} (m^a_f - m^a_e) i[h_e, h_f],$$

where only overlapping edge pairs contribute. This is the direct higher-dimensional analogue of CA-13's scalar-ramp formula.

37.3 Patch Diagnostics

The smallest useful checks are local:

1. three-site lines for straight x and y currents,
2. 2×2 plaquettes for corner commutators and C_4 covariance,
3. 3×3 open patches for P_x, P_y and $i[H, P_a]$,
4. 4×4 or larger patches for bulk-projected $i[P_x, P_y]$.

Boundary-supported residuals should be projected away or recorded separately. The diagnostic object is the interior Pauli-string residual, not raw equality on the whole open patch.

37.4 Proposed Rejection Filters

Current nontriviality. Both P_x and P_y should be nonzero in the bulk for a nontrivial $2+1$-dimensional Poincare candidate.

Translation conservation. The bulk residual $i[H, P_a]$ should vanish or be controlled in the intended finite algebra. A nonzero interior Pauli-string residual is an early rejection witness for exact finite bulk closure.

Translation commutativity. The bulk part of $i[P_x, P_y]$ should vanish or be a named boundary/irrelevant term before the model is promoted.

Boost-translation bracket. After normalisation, the residual

$$i[P_b, K_a] - \delta_{ab} H$$

is the direct $2+1$-dimensional Poincare diagnostic. This is strong and must be treated as a finite filter, not a theorem.

Isotropy. For scalar qubits with no internal spatial action, a microscopic 90° rotation should relate horizontal and vertical edge data after endpoint reversal. Obvious decoupled stacks, such as $c^y = 0$, fail this filter.

37.5 Rotation Is Not Free

A rotation candidate needs a local momentum-density split

$$P_a = \sum_v p_a(v)$$

before one can write

$$J_{xy} = \sum_v (x_v p_y(v) - y_v p_x(v)).$$

CA-13 already marks this as extra data. A plausible first convention is to assign each edge-pair current to the shared vertex, but that convention must be written and tested before rotation closure is claimed. Only then should one check

$$i[H, J_{xy}], \qquad i[P_x, J_{xy}] - P_y, \qquad i[P_y, J_{xy}] + P_x.$$

37.6 Rule-Out Classes

The proposal immediately targets four classes:

1. fully on-site Hamiltonians: $P_x = P_y = 0$,
2. commuting classical edge densities: all ramp currents vanish,
3. decoupled stacks: one momentum component vanishes,
4. anisotropic edge data: fail the scalar-qubit C_4 filter.

None of these rejections is a continuum theorem. They are finite algebraic filters for the specific candidate-generator route.

38 Qubit Nearest-Neighbour Witt/Virasoro Diagnostics

38.1 Status, Sources, and Dependencies

Status: proposal with local derivation targets. This shard is a finite-scale diagnostic proposal, not a continuum Virasoro claim.

Dependencies: CA-12, CA-14–CA-16, CA-34–CA-35, CONVENTIONS.md (m). CFT normalisations in CONVENTIONS.md (d) remain unfixed, so no central-charge extraction convention is available here.

38.2 Fourier Density and Current Modes

Let a periodic qubit chain of length N have

$$H = \sum_j h_j, \qquad h_j = \sum_{\alpha,\beta=0}^{3} h_{\alpha\beta}\sigma_j^\alpha \sigma_{j+1}^\beta.$$

For $q_n = 2\pi n/N$, define

$$H_n = \sum_j e^{iq_n j} h_j, \qquad J_n = \sum_j e^{iq_n j} j_j, \qquad j_j = i[h_j, h_{j+1}].$$

The nearest-neighbour commutator identity is

$$i[H_m, H_n] = (e^{iq_n} - e^{iq_m}) J_{m+n}.$$

This is the finite Pauli-chain analogue of the Koo–Saleur pattern in which Hamiltonian-density modes and commutator corrections approximate chiral stress-tensor modes.

38.3 First Failure Witness

CA-35 gives the coefficient condition

$$R_\alpha \times L_\delta = 0 \qquad \text{for all } \alpha, \delta,$$

which implies $j_j = 0$ and hence $J_n = 0$ for every Fourier mode. In that case the density modes cannot acquire the Koo–Saleur commutator correction needed to split left-moving and right-moving candidates. Fully on-site Hamiltonians under the symmetric split fail here, as do commuting classical densities.

38.4 Chiral-Rank Diagnostic

A later finite diagnostic may define tentative chiral candidates of the form

$$\ell_n = a_N{:}H_n{:} + \lambda_n J_n + \mu_N \delta_{n0} I, \qquad \bar{\ell}_{-n} = a_N{:}H_n{:} - \lambda_n J_n + \mu_N \delta_{n0} I.$$

Before testing brackets, one should check that H_n and J_n are linearly independent on the chosen low-energy sector P_N. A rank-collapse test such as

$$\det \mathrm{Gram}_{P_N}(H_n, J_n) \neq 0$$

is an early finite witness that the two proposed chiral directions have not collapsed to one.

38.5 Compressed Witt Residuals

The raw full Hilbert-space commutator is the wrong first test. Koo–Saleur already warns that finite-size quantities do not close as Virasoro generators and that commutators interact subtly with the scaling limit. The diagnostic should therefore choose a low-energy projection P_N, an intermediate projection Q_N, and test residuals such as

$$P_N\big(\ell_m Q_N \ell_n - \ell_n Q_N \ell_m - (m-n)\ell_{m+n}\big)P_N$$

for $m+n \neq 0$, together with the barred and cross-chiral versions.

The $m+n=0$ brackets should not be the first rejection test: central terms require a Virasoro normalization, vacuum or ground-state subtraction, and a scaling state sequence. These are not fixed for arbitrary qubit chains.

38.6 Boundary

This shard gives finite algebraic diagnostics only. Passing them does not produce a Virasoro representation. A continuum claim still needs the full CA-15 payload plus Koo–Saleur-style scaling representation data: state sequence, subtraction policy, mode normalization, low-energy sector, operator topology, and convergence theorem.

39 Qubit Local-Algebra Equation Framework

39.1 Status, Sources, and Dependencies

Status: checked local framework. This shard supplies the algebraic language used by CA-39–CA-44. It does not assert that a solution has continuum Poincare or Virasoro symmetry.

Dependencies: CA-11, CA-12, CA-15, CA-34–CA-35, CONVENTIONS.md (m), (n).

39.2 Pauli Word Algebra

Use zero-based Pauli labels $a \in \{0,1,2,3\}$, with $\sigma^0 = I$. For a finite interval, a Pauli word is

$$P_{\alpha_0 \dots \alpha_{\ell-1}} = \sigma^{\alpha_0} \otimes \cdots \otimes \sigma^{\alpha_{\ell-1}}.$$

The local product is encoded by constants M_{ab}^c:

$$\sigma^a \sigma^b = \sum_{c=0}^{3} M_{ab}^c \sigma^c,$$

where

$$M_{0b}^c = \delta_{bc}, \qquad M_{a0}^c = \delta_{ac},$$

and, for $a, b \geq 1$,

$$M_{ab}^0 = \delta_{ab}, \qquad M_{ab}^c = i\epsilon_{abc} \quad (c \geq 1).$$

Thus for two ℓ-site coefficient tensors A, B,

$$(AB)_\gamma = \sum_{\alpha,\beta} A_\alpha B_\beta \prod_{r=0}^{\ell-1} M_{\alpha_r \beta_r}^{\gamma_r}.$$

The commutator coefficient is therefore the finite expression

$$(i[A,B])_\gamma = i \sum_{\alpha,\beta} A_\alpha B_\beta \left(\prod_r M_{\alpha_r \beta_r}^{\gamma_r} - \prod_r M_{\beta_r \alpha_r}^{\gamma_r} \right).$$

When two local densities have different supports, first embed both tensors into their common finite window by padding with Pauli label 0, then apply this formula.

39.3 Bulk Equality Modulo Divergence

Let τ be the unit shift of the infinite chain. A finite density R represents a zero bulk formal sum if it is a coboundary:

$$R = u - \tau(u).$$

If u has n Pauli indices, the coefficient form fixed in CONVENTIONS.md (n) is

$$(Du)_{a_0 \dots a_n} = u_{a_0 \dots a_{n-1}} \delta_{a_n 0} - \delta_{a_0 0} u_{a_1 \dots a_n}. \tag{38.1}$$

Consequently a local residual $R(h)$ passes the bulk equation precisely when there exists a finite tensor u such that

$$R_{a_0 \dots a_n}(h) - (Du)_{a_0 \dots a_n} = 0 \qquad \text{for every Pauli word } a_0 \dots a_n. \tag{38.2}$$

The scalar coefficient of u lies in the kernel of D, so it is a gauge variable. A coefficientwise condition $R(h) = 0$ is allowed as a stronger filter, but it is not the default infinite-chain equation.

39.4 What "Necessary" Means Here

Every equation in the next shards is necessary only for the named generator route: the fixed Pauli density, the fixed first-moment boost ansatz, and the fixed coboundary quotient. Solving these equations is not sufficient for continuum symmetry. Failing one of them is still useful: it excludes the Hamiltonian from this route before any low-energy sector, scaling map, or state-space analysis is attempted.

40 One-Dimensional Qubit Poincare Necessary Equations

40.1 Status, Sources, and Dependencies

Status: local derivation, partly checked. The current, coboundary, and residual tensors are implemented in Julia. The equations are necessary for the fixed CA-12 first-moment route, not sufficient for continuum Poincare symmetry.

Dependencies: CA-11, CA-12, CA-35, CA-38, CONVENTIONS.md (k), (m), (n).

40.2 Input and Current

Let

$$h_j = \sum_{\alpha,\beta=0}^{3} h_{\alpha\beta}\sigma_j^\alpha \sigma_{j+1}^\beta, \qquad h_{\alpha\beta} \in \mathbb{R}.$$

The first-moment route fixes

$$H = \sum_j h_j, \qquad K = \sum_j j h_j, \qquad P = i[H,K] = \sum_j p_j, \qquad p_j = i[h_j, h_{j+1}].$$

The current coefficients are

$$p_{aed} = -2 \sum_{b,c=1}^{3} h_{ab} h_{cd} \epsilon_{bce}, \qquad e \in \{1,2,3\}, \tag{39.1}$$

with $p_{a0d} = 0$. Equivalently the right-leg vector (h_{a1}, h_{a2}, h_{a3}) must cross the left-leg vector (h_{1d}, h_{2d}, h_{3d}). If

$$p_{aed} = 0 \quad \text{for all } a, e, d, \tag{39.2}$$

then the candidate bulk momentum is $P = 0$. For a nontrivial speed $v^2 \neq 0$, the later equation $i[P,K] = v^2 H$ can then hold only if the Hamiltonian density is scalar-trivial in the coboundary quotient. The symmetric fully on-site split in CONVENTIONS.md (m) is checked to have $p = 0$; hence fully local non-scalar Hamiltonians are ruled out for this route.

40.3 Translation Conservation

Let

$$A_j = i[h_j + h_{j+1}, p_j].$$

The outer-overlap terms in $i[H,P]$ cancel in the formal sum by the Jacobi identity, so translation conservation is

$$A = Du \tag{39.3}$$

for some two-site tensor u_{rs}. In Pauli coefficients,

$$\begin{aligned} A_{rst} = i \sum_{\alpha,\beta,a,b} h_{\alpha\beta} p_{abt} \left(M^r_{\alpha a} M^s_{\beta b} - M^r_{a\alpha} M^s_{b\beta} \right) \\ + i \sum_{\alpha,\beta,b,c} h_{\alpha\beta} p_{rbc} \left(M^s_{\alpha b} M^t_{\beta c} - M^s_{b\alpha} M^t_{c\beta} \right). \end{aligned}$$

Thus the explicit degree-three equations are

$$A_{rst} - u_{rs}\delta_{t0} + \delta_{r0} u_{st} = 0 \qquad (r,s,t = 0,\ldots,3). \tag{39.4}$$

40.4 Boost-Translation Relation

Assume (39.3) has been witnessed by a chosen u. The remaining bulk representative of $i[P,K]$ is

$$B_j - u_j, \qquad B_j = i[p_j, h_{j+1}] + 2i[p_j, h_{j+2}].$$

The four-site coefficients of B are

$$\begin{aligned} B_{rstu} = i\delta_{u0} \sum_{b,c,\alpha,\beta} p_{rbc} h_{\alpha\beta} \left(M^s_{b\alpha} M^t_{c\beta} - M^s_{\alpha b} M^t_{\beta c} \right) \\ + 2i \sum_{c,\alpha} p_{rsc} h_{\alpha u} \left(M^t_{c\alpha} - M^t_{\alpha c} \right). \end{aligned}$$

Allow an explicit scalar energy-origin subtraction

$$\bar{h}_{rs} = h_{rs} - e\,\delta_{r0}\delta_{s0}.$$

The necessary speed-v boost equation $i[P,K] = v^2 H$ is that there exists a three-site tensor w such that

$$\begin{aligned} 0 = B_{rstu} - u_{rs}\delta_{t0}\delta_{u0} - v^2 \bar{h}_{rs}\delta_{t0}\delta_{u0} \\ - w_{rst}\delta_{u0} + \delta_{r0} w_{stu} \qquad (r,s,t,u = 0,\ldots,3). \end{aligned} \tag{39.5}$$

Equations (39.1), (39.4), and (39.5), together with real variables u, w, e, v^2, are the advertised explicit finite polynomial conditions on $h_{\alpha\beta}$ for the one-dimensional first-moment Poincare route.

40.5 Boundary

The scalar h_{00} cannot be detected by commutators; it must be handled by the explicit e variable or by a state-dependent normal ordering. The coboundary witnesses are also not unique. These are convention choices, not loopholes: changing the density by a divergence changes $K = \sum_j j h_j$, so it changes the equations.

41 Computer Algebra for One-Dimensional Qubit Residuals

41.1 Status, Sources, and Dependencies

Status: checked. This shard documents the finite algebra engine behind CA-35 and CA-39.
Dependencies: CA-35, CA-38, CA-39, CONVENTIONS.md (m), (n).

41.2 Operator and Coefficient Roundtrip

The file `src/QubitPauliLattice.jl` implements

$$\sum_{\alpha_0,\ldots,\alpha_{n-1}=0}^{3} c_{\alpha_0\ldots\alpha_{n-1}} \sigma^{\alpha_0} \otimes \cdots \otimes \sigma^{\alpha_{n-1}}$$

for $n = 1, 2, 3$ and for general n. The inverse map uses

$$c_\alpha = 2^{-n} \operatorname{Tr}(P_\alpha A).$$

The tests assert coefficient roundtrips for two-site densities and for the five-site residual representatives. The code rejects arrays whose axes are not all length 4, and rejects non-self-adjoint matrix inputs.

41.3 Checked Residual Helpers

The current helper returns

$$p = i[h_{12}, h_{23}]$$

and the direct coefficient helper returns exactly (39.1). The conservation helper

$$A = i[h_{12} + h_{23}, p]$$

is the minimal three-site density for $i[H, P]$ after the Jacobi cancellation of outer-overlap terms.

The file `src/QubitPauliResiduals.jl` adds

$$(Du)_{a_0\ldots a_n} = u_{a_0\ldots a_{n-1}} \delta_{a_n 0} - \delta_{a_0 0} u_{a_1\ldots a_n}$$

and

$$B = i[p_{123}, h_{23}] + 2i[p_{123}, h_{34}].$$

The test suite checks that the coboundary coefficients reconstruct the operator identity

$$Du = u_{12} - u_{23}$$

on a three-site window, including the fact that the scalar $u_{00} I$ is in the kernel.

41.4 Sentinels

The checked examples are deliberately small:

- A generic real one-site density embedded by the symmetric split has $p = 0$, $A = 0$, and $B = 0$, while the embedded Hamiltonian density itself is nonzero. This is the concrete fully-local exclusion witness for CA-39.

- A pure classical ZZ density also has $p = 0$.
- A transverse-Ising-style density has nonzero p and nonzero boost residual density B. This only means it survives the first obstruction.
- An asymmetric fake on-site split creates a nonzero current, but the conservation sentinel is larger than the current and catches the density-split artifact.

41.5 Current Gap

The code does not yet solve the polynomial system (39.1), (39.4), (39.5), and does not build moment matrices. Its role is narrower: it makes every displayed coefficient tensor executable and gives mutation-friendly invariants before an SDP exporter is added.

42 Two-Dimensional Qubit Poincare Equation Schema

42.1 Status, Sources, and Dependencies

Status: proposal-level local schema. This is a finite algebraic target for a future checker. It is not yet a canonical stress tensor or a continuum $2+1$-dimensional Poincare theorem.

Dependencies: CA-13, CA-36, CA-38, CONVENTIONS.md (o).

42.2 Edge Densities and Vertex Currents

On $\mathbb{Z}^2$, write positively oriented edge densities

$$h_r^a = \sum_{\alpha,\beta=0}^{3} c_{\alpha\beta}^a \, \sigma_r^\alpha \sigma_{r+\hat{a}}^\beta, \qquad a \in \{x, y\}.$$

At a vertex v, order incident edges as W, S, E, N, with midpoint offsets

$$m_W = (-1/2, 0), \quad m_S = (0, -1/2), \quad m_E = (1/2, 0), \quad m_N = (0, 1/2).$$

Use shared-leg coefficient tensors

$$q_{\lambda\mu}^W = c_{\lambda\mu}^x, \quad q_{\lambda\mu}^E = c_{\mu\lambda}^x, \quad q_{\lambda\mu}^S = c_{\lambda\mu}^y, \quad q_{\lambda\mu}^N = c_{\mu\lambda}^y.$$

For ordered incident pairs $A < B$ in $W < S < E < N$, the local commutator coefficient on outer leg A, center v, outer leg B is

$$J_{\lambda\kappa\rho}^{AB} = -2 \sum_{\mu,\nu=1}^{3} q_{\lambda\mu}^A q_{\rho\nu}^B \epsilon_{\mu\nu\kappa}. \tag{41.1}$$

The ramp-generated momentum densities are

$$p_a(v) = \sum_{A<B} (m_B^a - m_A^a) J^{AB}(v). \tag{41.2}$$

For $AB = WS, WE, WN, SE, SN, EN$, the x-weights are

$$1/2,\ 1,\ 1/2,\ 1/2,\ 0,\ -1/2,$$

and the y-weights are

$$-1/2,\ 0,\ 1/2,\ 1/2,\ 1,\ 1/2.$$

42.3 Divergence-Relaxed Residuals

Every residual below is a finite Pauli tensor R on a named patch. The strong filter is $R = 0$. The default bulk filter is the existence of finite-support tensors U_x, U_y with

$$R = (1 - \tau_{-\hat{x}})U_x + (1 - \tau_{-\hat{y}})U_y. \tag{41.3}$$

Coefficientwise, this is an explicit linear equation in the auxiliary divergence tensors and a polynomial equation in c^x, c^y, using the Pauli product constants of CA-38.

42.4 Necessary Bulk Equations

Momentum conservation is

$$C_a := \sum_{e:\operatorname{supp} h_e \cap \operatorname{supp} p_a(0) \neq \varnothing} i[h_e, p_a(0)] = \nabla \cdot U^{(C_a)}, \qquad a = x, y. \tag{41.4}$$

The commuting-translation equation is

$$T_{xy} := \sum_{r:\operatorname{supp} p_x(0) \cap \operatorname{supp} p_y(r) \neq \varnothing} i[p_x(0), p_y(r)] = \nabla \cdot U^{(T)}. \tag{41.5}$$

Let $e_0 = h_0^x + h_0^y$ denote the chosen local energy cell. With m_e^a the a-coordinate of the edge midpoint,

$$B_{ab} := \sum_{e:\operatorname{supp} h_e \cap \operatorname{supp} p_b(0) \neq \varnothing} m_e^a \, i[p_b(0), h_e] - \delta_{ab} v_a^2 e_0 = \nabla \cdot U^{(B_{ab})}. \tag{41.6}$$

This is the $i[P_b, K_a] = \delta_{ab} v_a^2 H$ equation after the coordinate weighted conservation term is removed by (41.4).

If one defines the rotation candidate

$$J = \sum_v (v_x p_y(v) - v_y p_x(v)),$$

then the translation-rotation residual is

$$R_c = \sum_r \left(r_x \, i[p_c(0), p_y(r)] - r_y \, i[p_c(0), p_x(r)] \right) - (\delta_{cx} p_y(0) - \delta_{cy} p_x(0)) = \nabla \cdot U^{(R_c)}. \tag{41.7}$$

This last equation is quartic in c^x, c^y before the quadratic subtraction. It depends on the chosen momentum-density split, so it is less canonical than the boost equations.

42.5 Implementation Boundary

CA-41 supplies the finite equations a checker should implement. The next implementation step is to enumerate the patch supports in (41.4)–(41.7), embed every edge tensor into the common patch, and emit Pauli coefficients plus the divergence auxiliaries. Until that exists, the $2+1$-dimensional claims remain proposal-level.

43 Qubit Witt/Virasoro Necessary Equations

43.1 Status, Sources, and Dependencies

Status: exact finite algebra plus proposal residuals. The mode identities are finite local algebra. Their use as Virasoro diagnostics is a necessary filter only.

Dependencies: CA-14, CA-35, CA-37–CA-39, CONVENTIONS.md (d), (m), (n).

43.2 Exact Density-Mode Algebra

On a periodic chain of length N, set $z_n = \exp(2\pi i n/N)$ and

$$H_n = \sum_j z_n^j h_j, \qquad C_j = i[h_j, h_{j+1}], \qquad J_n = \sum_j z_n^j C_j.$$

Then $H_n^* = H_{-n}$, $J_n^* = J_{-n}$, and the nearest-neighbour identity is

$$[H_m, H_n] = -i(z_n - z_m) J_{m+n}. \tag{42.1}$$

This is the exact lattice antecedent of the Koo-Saleur $H(f), P(f)$ commutator. In Pauli coefficients C_j is the tensor p from (39.1).

Define higher-current densities

$$D_{r,j} = i[h_{j+r}, C_j] \quad (r = 0, 1, 2), \qquad F_j^{(r)} = i[C_j, C_{j+r}] \quad (r = 1, 2).$$

Using the Jacobi cancellation of the $r = -1$ term,

$$i[H_m, J_n] = \sum_j z_{m+n}^j \left(D_{0,j} + z_m D_{1,j} + (z_m^2 - z_n) D_{2,j} \right), \tag{42.2}$$

and

$$i[J_m, J_n] = \sum_j z_{m+n}^j \left((z_n - z_m) F_j^{(1)} + (z_n^2 - z_m^2) F_j^{(2)} \right). \tag{42.3}$$

The D-coefficients are degree three polynomials in $h_{\alpha\beta}$; the F-coefficients are degree four.

43.3 Chiral Residuals

A Koo-Saleur-style chiral ansatz has the form

$$\ell_n = A_n {:}H_n{:}_\omega + B_n {:}J_n{:}_\omega + \mu \delta_{n0} I,$$
$$\bar{\ell}_n = A_n {:}H_n{:}_\omega - B_n {:}J_n{:}_\omega + \mu \delta_{n0} I,$$

with $A_{-n} = \overline{A_n}$ and $B_{-n} = \overline{B_n}$. The necessary Virasoro residuals for a chosen finite mode set are

$$R_{mn}^{++} = [\ell_m, \ell_n] - (m - n)\ell_{m+n} - \frac{c}{12} m(m^2 - 1)\delta_{m+n,0} I, \tag{42.4}$$

$$R_{mn}^{--} = [\bar{\ell}_m, \bar{\ell}_n] - (m - n)\bar{\ell}_{m+n} - \frac{c}{12} m(m^2 - 1)\delta_{m+n,0} I, \tag{42.5}$$

and

$$R_{mn}^{+-} = [\ell_m, \bar{\ell}_n]. \tag{42.6}$$

For example,

$$[\ell_m, \ell_n] = -iA_m A_n (z_n - z_m) J_{m+n} - iA_m B_n \mathcal{D}_{mn} + iB_m A_n \mathcal{D}_{nm} - iB_m B_n \mathcal{F}_{mn},$$

where $\mathcal{D}_{mn} = i[H_m, J_n]$ and $\mathcal{F}_{mn} = i[J_m, J_n]$ are given by (42.2)–(42.3).

43.4 Necessary Filter

At finite size, the Koo-Saleur source explicitly warns that the lattice quantities need not close. Therefore the safe condition is not "finite Virasoro symmetry". The safe condition is: for the chosen ansatz and mode set, every Pauli coefficient of R^{++}, R^{--}, R^{+-} outside the named span

$$\mathrm{span}\{I, H_{m+n}, J_{m+n}\}$$

must either vanish, be a registered finite-size correction, or be annihilated in the chosen low-energy/state restriction. CA-43–CA-44 turn the last option into moment constraints.

44 Vacuum Moment Constraints for Qubit Residuals

44.1 Status, Sources, and Dependencies

Status: proposal-level SDP interface backed by sourced state/GNS architecture. The moment equations are finite algebra. The existence of a continuum vacuum or positive-energy representation is not claimed.

Dependencies: CA-15, CA-38–CA-42, CONVENTIONS.md (n).

44.2 Finite Moment Data

Let P_s be a finite Pauli word on $\mathbb{Z}$, padded by identities outside its support. A candidate translation-invariant global state ω has moments

$$y_s = \omega(P_s), \qquad y_\emptyset = 1, \qquad y_{\tau^n s} = y_s. \tag{43.1}$$

For a finite word set $\mathcal{W}$, positivity is the moment-matrix constraint

$$M_{\mathcal{W}}(y)_{u,v} = \omega(P_u^* P_v) = \sum_s \mu_{u,v}^s y_s \succeq 0, \tag{43.2}$$

where the constants $\mu_{u,v}^s$ are computed from the Pauli product table of CA-38. Hermiticity gives

$$M_{\mathcal{W}}(y)_{v,u} = \overline{M_{\mathcal{W}}(y)_{u,v}}.$$

The SDP may be represented over real variables by splitting real and imaginary parts.

44.3 Zero Expectations

The infinite sums H, P, K are not finite observables. The finite data are their local densities and residuals. After choosing the scalar subtraction $\bar{h} = h - eI$, a vacuum-normalized test may impose

$$\omega(\bar{h}_j) = 0, \qquad \omega(p_j) = 0, \tag{43.3}$$

and, for the residuals of CA-39,

$$\omega(A_j - Du_j) = 0, \qquad \omega(B_j - u_j - v^2 \bar{h}_j - Dw_j) = 0. \tag{43.4}$$

These are only the level-zero moment consequences. They are usually too weak to exclude much.

44.4 GNS Relation Constraints

The stronger finite restriction says that a residual vanishes on the vacuum sector as a relation in the GNS representation. For each chosen residual density $R(h)$ and probe words X, Y,

$$\omega(X^* R(h) Y) = 0. \tag{43.5}$$

For fixed h, fixed v^2, e, u, w, and fixed residual coefficients, (43.5) is linear in the moments y_s. The same rule applies to the finite Witt/Virasoro residual densities $r_{mn}^{++}, r_{mn}^{--}, r_{mn}^{+-}$ from CA-42:

$$\omega(X^* r_{mn}^{\pm\pm}(h) Y) = 0, \qquad \omega(X^* r_{mn}^{+-}(h) Y) = 0. \tag{43.6}$$

These equations are the promised finite data extracted from the demand that a global state ω make the generators and relations vanish in expectation on a finite probe space.

44.5 Highest-Weight Caveat

If a chiral vacuum sector is named, one may also impose finite analogues of

$$\ell_n\Omega = 0, \quad \bar{\ell}_n\Omega = 0 \qquad (n > 0),$$

for example

$$\omega(\ell_{-n}\ell_n) = 0. \tag{43.7}$$

But central charge, normal ordering, and finite-volume mode normalisation are still convention-sensitive. Until CONVENTIONS.md (d) is fixed for this block, such constraints are optional diagnostics rather than canonical Virasoro vacuum equations.

45 SDP Exclusion Hierarchy for Qubit Hamiltonians

45.1 Status, Sources, and Dependencies

Status: rigorous proposal. The hierarchy is a finite-state consequence of the sourced state/GNS framework and the local residual derivations. No solver is implemented yet.

Dependencies: CA-38–CA-43, CONVENTIONS.md (n).

45.2 Level Data

Fix a Hamiltonian coefficient matrix $h_{\alpha\beta}$. Also fix, or scan externally, the scalar subtraction e, speed v^2, and any coboundary witnesses u, w used in CA-39. This is important: if those witnesses are unknown variables multiplied by unknown moments, the problem is no longer an SDP without an additional lifting relaxation.

At level N, choose finite sets

$$\mathcal{W}_N = \{\text{Pauli words supported in } [-N, N]\},$$

$$\mathcal{A}_N = \{\text{probe words whose residual products fit the window}\},$$

and residual densities

$$\mathcal{R}_N(h) = \{A - Du,\ B - u - v^2\bar{h} - Dw\}$$

plus any selected finite Witt/Virasoro residuals from CA-42.

45.3 SDP Constraints

The variables are finitely many moments y_s. The constraints are

$$y_\emptyset = 1, \qquad y_{\tau s} = y_s, \tag{44.1}$$

$$M_{\mathcal{W}_N}(y) \succeq 0, \tag{44.2}$$

and optional stationarity constraints for local probes:

$$\omega(\delta_H(X)) = 0, \qquad \omega(\delta_P(X)) = 0, \qquad X \in \mathcal{A}_N. \tag{44.3}$$

The generator-relation constraints are

$$\omega(X^* R Y) = 0, \qquad R \in \mathcal{R}_N(h), \quad X, Y \in \mathcal{W}_N, \tag{44.4}$$

whenever the Pauli product X^*RY fits inside the moment support. Each entry in (44.1)–(44.4) is affine in y for fixed residual coefficients, so this is a semidefinite feasibility problem.

45.4 Exclusion Statement

If a translation-invariant global state ω exists whose GNS representation carries the chosen local residual relations, then every finite restriction of ω satisfies every level N. Therefore:

$$\text{SDP level } N \text{ infeasible} \quad \Longrightarrow \quad h_{\alpha\beta} \text{ is excluded for this generator route.} \tag{44.5}$$

The hierarchy is monotone: increasing N adds moment variables and relation tests, and cannot turn a previously infeasible level into evidence of symmetry.

45.5 Non-Implications

Feasibility at all checked levels does not prove that a global state exists, does not prove uniqueness or clustering, does not prove positive energy, and does not prove a continuum Poincare or Virasoro representation. It only means that the tested finite restrictions have not yet found a contradiction.

When $h_{\alpha\beta}$ is also a variable, the residual coefficients are polynomial in h. Then the feasibility problem becomes a polynomial moment problem over both Hamiltonian and state variables. That is a separate relaxation; the first implementation should instead take h as input and return either an SDP infeasibility certificate or "not excluded at this level".

46 Roadmap for Qubit Symmetry Exclusion

46.1 Status, Sources, and Dependencies

Status: implementation plan. This closes the current qubit necessary-equation expansion and names the next build steps.

Dependencies: CA-34–CA-44, CONVENTIONS.md (m)–(o).

46.2 What We Have Now

The one-dimensional first-moment route now has explicit necessary equations on the Pauli coefficient matrix $h_{\alpha\beta}$:

$$p_{aed} = -2 \sum_{b,c=1}^{3} h_{ab} h_{cd} \epsilon_{bce},$$

$$A(h) = Du, \qquad B(h) - u - v^2(h - eI) = Dw.$$

These equations already rule out fully local symmetric on-site Hamiltonians and commuting classical densities for any nontrivial Poincare route of this form, because their generated momentum current is zero.

The two-dimensional shard gives a square-lattice edge schema, not yet code. The Witt/Virasoro shard gives exact finite mode commutators and residuals, but keeps Koo-Saleur's finite-nonclosure warning active. The vacuum/SDP shards turn "there exists a global state ω" into finite positive moment data and linear relation constraints.

46.3 Implementation Steps

1. Add symbolic coefficient builders for $A(h)$, $B(h)$, and coboundary equations, returning sparse polynomial tensors rather than dense matrices.

2. Add a solver-facing algebraic filter: for a fixed h, check current collapse; optionally solve the linear coboundary witnesses u, w after p, A, B are computed.

3. Build a Pauli-word moment enumerator for intervals $[-N, N]$, including product constants, translation-invariance canonicalisation, and Hermitian realification.

4. Export the fixed-h feasibility problem (44.1)–(44.4) through a local SDP interface. Record every run under `runs/` with the command line, h, level N, parameters e, v^2, u, w, and solver status.

5. Add mutation tests: perturb a known zero-current density to make the current nonzero; perturb the coboundary sign to make the coboundary reconstruction test fail; perturb a residual coefficient and require a small SDP level to change.

46.4 Acceptance for an Exclusion Claim

A future statement "h is excluded at level N" must point to:

- the exact Pauli coefficient matrix $h_{\alpha\beta}$ and density convention;
- the algebraic residual tensors and any witnesses e, v^2, u, w;

- the finite word sets $\mathcal{W}_N, \mathcal{A}_N$;
- the SDP solver transcript or certificate in a run bundle;
- a worklog entry explaining why the infeasible constraints are necessary for the named route.

46.5 Open Gaps

The central charge convention remains unfixed. The two-dimensional edge schema has no implementation. The SDP hierarchy is fixed-h; scanning continuous families of $h_{\alpha\beta}$ requires a separate polynomial relaxation. None of the filters replaces CA-15's scaling-limit obligations. They are designed to cheaply reject impossible candidates before the expensive continuum work begins.

47 Qubit SDP Implementation Contract

47.1 Status, Sources, and Dependencies

Status: checked implementation contract. This shard records the Julia layer that implements the fixed-h hierarchy of CA-43–CA-45.

Dependencies: CA-38–CA-45, CONVENTIONS.md (n).

47.2 Package Split

The implementation is deliberately layered.

- `QubitPauliWords.jl`: exact positioned Pauli words, exact multiplication phases, translation canonicalisation, and coefficient-term extraction.
- `QubitPoincareWitnesses.jl`: linear algebraic witnesses for $A = Du$ and $B - u - \lambda h + \mu I = Dw$, used before invoking an SDP.
- `QubitMomentSDP.jl`: finite moment bases, moment matrices, relation constraints $\omega(X^* RY) = 0$, JuMP model construction, and Mosek-backed solve helpers.
- `QubitHamiltonianScreening.jl`: deterministic sentinel Hamiltonians and current-collapse filters.

The fixed-h boundary from CA-44 is enforced in the API: an SDP instance takes residual terms as fixed coefficients. If h, u, w, and λ are all variables, the problem is no longer this SDP; it becomes a separate polynomial/lifted hierarchy.

47.3 Verdicts

The solver helper returns one of three finite-level statuses:

`:excluded`, `:not_excluded_at_level`, `:solver_unknown`.

Only the first is a mathematical rejection, and only for the named level and named residual constraints. The second is not evidence for symmetry; it means that this finite relaxation did not find a contradiction.

An exclusion claim must cite the coefficient matrix $h_{\alpha\beta}$, the residuals, the moment/probe word sets, the solver status, and a run bundle or test that reproduces the result.

48 Pauli Word Moment Basis for Infinite Qubit Chains

48.1 Status, Sources, and Dependencies

Status: checked. The word algebra is implemented without dense matrices and checked against exact Pauli identities.

Dependencies: CA-38, CA-43, CA-46.

48.2 Positioned Words

A word is stored as sorted non-identity labels on integer sites. Identities are not stored, but gaps are preserved:

$$X_0 Z_2 \quad \leftrightarrow \quad \texttt{sites=(0,2), labels=(1,3).}$$

The empty word is the identity. Multiplication is sitewise with

$$XY = iZ, \quad YZ = iX, \quad ZX = iY,$$

and the reversed products carrying $-i$. The implementation returns

$$P_u P_v = \phi(u, v) P_{u \cdot v}, \qquad \phi(u, v) \in \{1, -1, i, -i\}.$$

The tests assert $XY = iZ$, $YX = -iZ$, $X^2 = I$, and commutation of disjoint words.

48.3 Translation Quotient

The SDP row/column words are not quotiented by translation. They are actual local probes in a finite window. Translation invariance enters only when a moment variable is named:

$$y_s = \omega(P_s) = \omega(P_{\tau^n s}).$$

The canonical representative shifts the first non-identity site to 0. Thus $X_{-3}Z_{-1}$ and $X_0 Z_2$ share the same moment variable.

Because each phase-free Pauli word is self-adjoint, each canonical moment variable is real. Complex numbers enter only as multiplication phases.

49 Moment Matrices and Positivity Constraints

49.1 Status, Sources, and Dependencies

Status: checked. Moment matrices are constructed by `build_qubit_sdp_model`.

Dependencies: CA-43–CA-47.

49.2 Affine Moment Matrix

For a finite actual probe set $\mathcal{W}$, define

$$M_{u,v} = \omega(P_u^* P_v) = \phi(u,v) y_{\mathrm{can}(u \cdot v)}.$$

The identity moment is the constant 1. All other canonical words appearing in products or relation constraints receive real JuMP variables.

Write $M = A + iB$, with A, B real affine matrices. Positivity of the complex Hermitian moment matrix is imposed by the real symmetric cone

$$\begin{pmatrix} A & -B \\ B & A \end{pmatrix} \succeq 0. \tag{48.1}$$

For the one-site probe set $\{I, X, Y, Z\}$, the implementation creates three real moment variables and an 8×8 realified PSD block. This exact dimension is checked in the Julia suite.

49.3 Tiny Feasibility Tests

The zero-residual one-site moment problem solves with Mosek as

`:not_excluded_at_level`.

The artificial relation $\omega(I) = 0$ solves as `:excluded`. These are smoke tests for the solver path, not physical Hamiltonian claims.

50 Compiling Poincare Residuals into Moment Equations

50.1 Status, Sources, and Dependencies

Status: checked local implementation. Dense matrix residual oracles from CA-40 are converted into coefficient terms and then into moment equations.

Dependencies: CA-39–CA-40, CA-43–CA-48, CONVENTIONS.md (n).

50.2 Algebraic Witnesses

For fixed h, the first algebraic gate solves

$$A(h) = Du$$

as a real least-squares linear system after fixing the scalar gauge $u_{00} = 0$. Passing this gate supplies a concrete u.

The next gate solves the linearized boost equation

$$B(h) - u - \lambda h + \mu I = Dw, \tag{49.1}$$

again with the scalar gauge $w_{000} = 0$. Here $\lambda = v^2$ and $\mu = v^2 e$. A positive λ is the nontrivial-speed branch; a solution with $\lambda = 0$ is only the trivial current-collapsed branch.

50.3 Moment Relations

Once residual terms $R = \sum_r c_r P_r$ are fixed, each pair of probe words gives

$$\omega(X^* R Y) = \sum_r c_r \, \phi(X, r, Y) \, y_{\mathrm{can}(XrY)} = 0. \tag{49.2}$$

The implementation splits (49.2) into real and imaginary affine constraints. This is the finite SDP condition promised by CA-43.

50.4 Important Boundary

Witness solving is an algebraic necessary filter. The SDP layer does not optimize over h, u, w, or λ. It checks whether a fixed residual system is compatible with a finite positive moment restriction.

51 SDP Levels and the Mosek Backend

51.1 Status, Sources, and Dependencies

Status: checked implementation. The solver path is exercised by tests and by a recorded smoke run.

Dependencies: CA-44–CA-49.

51.2 Level Data

An implemented level consists of:

$$\mathcal{W}_{\mathrm{psd}}, \qquad \mathcal{W}_{\mathrm{rel}}, \qquad \mathcal{R}.$$

The first set is the row/column basis for the PSD moment matrix. The second set supplies X, Y in $\omega(X^* R Y) = 0$. The third is a list of fixed residual coefficient terms.

The convenience constructor uses full Pauli-word windows:

$$\mathcal{W} = \{P_s : \operatorname{supp} s \subseteq [a, a + L - 1]\}.$$

Optional weight truncation is an implementation diagnostic, not the canonical hierarchy.

51.3 Solver Status

The feasibility problem has no objective. The status map is:

MOI termination status	hierarchy verdict
`OPTIMAL`	`:not_excluded_at_level`
`INFEASIBLE`, `INFEASIBLE_OR_UNBOUNDED`	`:excluded`
anything else	`:solver_unknown`.

This map is intentionally conservative.

51.4 Smoke Run

The run bundle

`runs/2026-05-31-qubit-sdp-smoke/`

records three tiny Mosek solves: zero residual, identity-zero relation, and a forced $ZZ = 0$ relation. The first is not excluded at the level; the latter two are excluded. The ZZ case is artificial: it forces $\omega(ZZ \cdot ZZ) = \omega(I) = 0$.

52 Sentinel Hamiltonians for the SDP Hierarchy

52.1 Status, Sources, and Dependencies

Status: checked sentinels. These examples test the implementation and do not assert physical criticality.

Dependencies: CA-35, CA-39–CA-50.

52.2 Matrices

The package records five 4×4 Pauli coefficient matrices:

- a symmetric fully onsite split of $0.25I + 0.7X - 0.3Y + 0.5Z$;
- a commuting classical ZZ density;
- a transverse-Ising-style density $-ZZ - 0.35(XI + IX)$;
- a generic currentful density;
- the fake asymmetric onsite split $XI + IZ$.

The deterministic current filter gives

$$\text{onsite}, ZZ : \texttt{:current_collapsed},$$

and

$$\text{transverse-Ising}, \text{generic}, \text{fake} : \texttt{:currentful}.$$

52.3 Gate Outcomes

The onsite and ZZ examples are excluded before SDP for the nontrivial first-moment route because $p = 0$. Their boost-witness solve has only the trivial $\lambda = 0$ branch.

The generic and fake examples have nonzero current but fail $A = Du$. This is the second algebraic gate catching more than current collapse.

The transverse-Ising-style example has nonzero current and passes $A = Du$, but the implemented linear boost witness does not solve (49.1) with the tested data. It is therefore a useful near-candidate sentinel: it reaches a deeper gate before failing.

53 Candidate Hamiltonian Selection Roadmap

53.1 Status, Sources, and Dependencies

Status: roadmap. This shard connects the implementation to the north star: identifying Hamiltonians worth expensive continuum analysis.

Dependencies: CA-15, CA-42–CA-51.

53.2 Screening Order

For a fixed nearest-neighbour qubit density $h_{\alpha\beta}$, use:

1. current collapse $p = 0$;
2. conservation witness $A = Du$;
3. boost witness $B - u - \lambda h + \mu I = Dw$ with $\lambda > 0$;
4. finite moment SDP feasibility for the chosen residuals;
5. only then, low-energy/scaling and Koo-Saleur/Virasoro tests.

The point is to reject impossible candidates cheaply. A Hamiltonian surviving all finite checks is not accepted as symmetric; it is merely promoted to the next, more expensive analysis.

53.3 Next Implementation Steps

The immediate next step is a run format for candidate scans: input $h_{\alpha\beta}$, current status, witness data, level data, solver status, and a headline verdict. Continuous families require external parameter scans or a later polynomial hierarchy. Named physical model families need registered sources before the report can claim criticality, universality class, or continuum CFT content.

54 Qubit Candidate Scan Contract

54.1 Status, Sources, and Dependencies

Status: checked implementation. The scan layer packages the CA-46–CA-52 fixed-h gates into reproducible candidate runs.

Dependencies: CA-39, CA-46–CA-52, CONVENTIONS.md (m)–(q).

54.2 Contract

Each scan point is a real 4×4 Pauli coefficient matrix $h_{\alpha\beta}$. The gate order is:

$$p = i[h_j, h_{j+1}], \qquad A = Du, \qquad B - u - \lambda h + \mu I = Dw,$$

followed, only when requested, by finite moment-SDP checks.

The serialized row records the family, parameters, coefficient matrix, current norm, witness residual norms, speed data, terminal gate, and scoped verdict. The scope field records the fixed first-moment route.

54.3 Verdict Boundary

`excluded_current_collapsed` and `excluded_no_conservation_witness` are early algebraic failures for the fixed current route. `excluded_no_boost_witness` means the exact local boost-witness equation failed; it does not reject other lattice stress-tensor constructions such as Koo–Saleur/free-fermion routes. This distinction is essential for sourced critical transverse Ising.

55 Pauli Coefficients for Scan Families

55.1 Status, Sources, and Dependencies

Status: checked implementation convention. The family constructors are implemented in the Julia backend.

Dependencies: CA-34–CA-53, CONVENTIONS.md (m), (q).

55.2 Implemented Maps

The scan uses (I, X, Y, Z) row/column order. A one-site field $f_a\sigma_j^a$ is represented on bonds by

$$h_{a0} = h_{0a} = f_a/2.$$

Thus the self-dual transverse-Ising sample is

$$h = -ZZ - \frac{1}{2}(XI + IX),$$

which is the bond-density version of the sourced transverse-Ising form.

The XXZ constructor uses

$$h = J(XX + YY + \Delta ZZ),$$

and Heisenberg is the $\Delta = 1$ specialization. The transverse-XY constructor uses $J_xXX + J_yYY + h_zZ$ with the same symmetric field split.

55.3 Synthetic Families

XYZ, compass, DM-style, field-deformed Heisenberg, and deterministic dense generic matrices are stress tests. They exercise the gates but carry no criticality or universality-class claim in this lab book.

56 Locally Sourced Spin-Chain Inputs

56.1 Status, Sources, and Dependencies

Status: source audit. This shard records which family names the first scan may treat as physical rather than synthetic.

Dependencies: CA-14, CA-52–CA-54.

56.2 Allowed Physical Labels

The defect-gauging source gives an effective qubit Hamiltonian proportional to $\sum_i (I + Z_i Z_{i+1} + X_i)$, identifies it as transverse Ising, and states that it gives a critical theory. The Ising anyon-chain source independently identifies the localized chain with critical TFIM and the Ising CFT.

The lattice-fermion source supports transverse XY as a Jordan–Wigner image and also records TFIM/Koo–Saleur convergence on the even observable subalgebra. The XXZ source gives a Pauli-chain Hamiltonian and the gapless/free-boson continuum anchors for $-1 < \Delta \leq 1$, with Heisenberg as a special point.

56.3 Excluded From Physical Labeling

The first scan does not make physical claims for XYZ, DM, compass, clock, Potts, or Ashkin–Teller. Potts is qutrit in the local source. The sourced Ising zigzag/Ashkin–Teller relation uses a three-site term, outside the present nearest-neighbour two-qubit input class.

57 Synthetic and Stress-Test Hamiltonian Grids

57.1 Status, Sources, and Dependencies

Status: checked implementation input. These grids are generated by the candidate-scan sample constructor.

Dependencies: CA-53–CA-55, CONVENTIONS.md (q).

57.2 Grids

The synthetic block includes XYZ triples on $\{-1, 0, 1\}^3 \setminus \{0\}$, XY anisotropy, compass XX/ZZ, DM-style antisymmetric $XY - YX$, Heisenberg with one field component, and 24 deterministic dense generic matrices. The generic matrices are generated by fixed sine/cosine formulas in the code rather than by an ambient random number generator.

57.3 Purpose

These samples answer engineering questions: Does the conservation gate reject generic dense inputs? Do structured bilinear families reach the boost gate? Do current-collapsed one-axis cases remain caught? They do not answer whether XYZ or DM chains are critical.

58 Current, Conservation, and Boost Witness Gates

58.1 Status, Sources, and Dependencies

Status: checked scan gates. The implementation calls the CA-40 and CA-49 helpers pointwise. Dependencies: CA-39–CA-40, CA-49, CA-53–CA-56.

58.2 Gates

The current gate checks whether $p = i[h_j, h_{j+1}]$ collapses. The conservation gate solves $A = Du$. The boost gate solves

$$B - u - \lambda h + \mu I = Dw$$

with $w_{000} = 0$ and requires nonzero positive speed for the nontrivial operator route.

58.3 Interpretation

The first full scan has no survivor of the exact boost gate. The self-dual TFIM point, which is locally sourced as critical, passes current and conservation but fails the exact boost witness. Therefore this gate is best read as a test for an exact UV first-moment construction, not as a general criterion for CFT emergence.

59 Optional SDP Levels for Scan Survivors

59.1 Status, Sources, and Dependencies

Status: implemented option, unused by the first broad run. The scan API can call `solve_qubit_sdp` after a nontrivial boost witness.

Dependencies: CA-44, CA-50, CA-53–CA-57.

59.2 Level Data

An SDP row must record the PSD window, relation window, moment count, relation constraint count, optimizer, and solver status. The status vocabulary remains

`:excluded`, `:not_excluded_at_level`, `:solver_unknown`.

59.3 First-Run Boundary

The 99-point run has no nontrivial exact boost witness, so it does not invoke Mosek per point. This keeps the run honest: adding SDP rows before fixing a residual witness would mix the fixed-h SDP with a polynomial variable-witness problem.

60 Reproducible Scan Bundle Format

60.1 Status, Sources, and Dependencies

Status: checked run bundle. The bundle is

`runs/2026-05-31-qubit-candidate-scan/`.

Dependencies: CA-53–CA-58 and AGENTS.md Law 3.

60.2 Files

`inputs.toml` records the Pauli basis, indexing rule, family inventory, source-kind inventory, and sample count. `summary.toml` records verdict counts and notable sourced points. `results.toml` records one row per sample, including coefficients, parameters, terminal gate, residual norms, and scope.

60.3 Acceptance

The run command prints the same headline counts that the TOML stores: 99 samples, 9 current-collapsed, 27 conservation failures, and 63 exact boost-witness failures.

61 First Candidate Scan Results

61.1 Status, Sources, and Dependencies

Status: checked run result. This shard reports only the run bundle named in CA-59.

Dependencies: CA-53–CA-59.

61.2 Headline Table

The 99 samples split as:

verdict	count
`excluded_current_collapsed`	9
`excluded_no_conservation_witness`	27
`excluded_no_boost_witness`	63

No point reaches `not_excluded_algebraic`.

61.3 Notable Points

The self-dual TFIM sample

$$-ZZ - \frac{1}{2}(XI + IX)$$

is currentful and has a conservation witness, but its exact boost residual norm is nonzero in the run. The Heisenberg point and the XX point behave the same way: they reach the boost gate and fail it.

Because TFIM and XXZ have local source anchors as critical or gapless models, the conclusion is about the route: the exact CA-39 boost witness is too strict as a universal candidate filter.

62 From Scan Failures to Scaling Work

62.1 Status, Sources, and Dependencies

Status: roadmap from checked failures. This is the handoff after CA-60.
Dependencies: CA-14, CA-15, CA-42, CA-52–CA-60.

62.2 Queue

Do not retry the same exact boost-witness gate expecting TFIM to pass. The next candidate path should use the sourced Koo–Saleur/free-fermion/OAR evidence chain for TFIM and transverse XY, and the Koo–Saleur/XXZ literature for XXZ. Those routes use low-energy or scaling-limit stress tensors, not an exact finite-support UV boost density satisfying CA-39.

62.3 SDP Upgrade

The state-level hierarchy should next learn how to quotient or relax the unknown Dw term without turning the problem into an uncontrolled polynomial relaxation. Until then, the implemented SDP layer is still useful for fixed residual systems and artificial contradictions, but the broad scan has shown where the exact-local route stops.

63 The Categorical Borchers–Uhlmann Pipeline: a Programme Refinement

63.1 Status, Sources, and Dependencies

Status: proposal/question. This shard records a programme refinement, not a theorem. Its central proposal is that the CA-02/CA-03 indefinite-particle grammar should be reinterpreted first as an observable $*$-algebra of local words, and only later as a Hilbert space after a state and GNS quotient have been chosen.

Dependencies: CA-01, CA-02, CA-03, CA-09, CA-43, CA-44, CA-57, CA-60, CA-61; CONVENTIONS.md (q); companion shards CA-63, CA-64, CA-65, CA-66 (this block).

63.2 Refined North Star

Proposal. The 2026-07-05 refinement of the north star is a recipe, as functorial as possible, which starts from a unitary fusion category or modular tensor category $\mathcal{C}$, together with any conformal data required by the target CFT, and produces a family of microscopic lattice models whose scaling limit realises a chiral or full CFT whose defects and excitations recover the input category.

Proposal. The key reinterpretation is algebra-first. CA-03 builds

$$\mathfrak{F}(H) = \widehat{\bigoplus}_{n\geq 0} H^{\widehat{\otimes} n}$$

as a Hilbert completion and explicitly identifies the underlying tensor algebra $T(H) = \bigoplus_n H^{\otimes n}$. In this shard, the tensor algebra is not primarily the pre-Hilbert finite-particle subspace. It is the proposed free local-word $*$-algebra from which observables are generated. The local anchor for this style is the free unital $*$-algebra of generators $\Phi(f)$ over test functions, quotiented by a two-sided $*$-ideal of relations; the registered source builds exactly this object (it does not use the name Borchers–Uhlmann, which we keep as informal terminology).

Proposal. Candidate lattice models emerge only after three extra choices: a reference state ω, approximate discrete Virasoro or Koo–Saleur-type residual relations, and quotienting by the state-null relations. In symbols, the intended order is

$$\text{local-word } *\text{-algebra} \quad \xrightarrow{\omega,\mathcal{R}} \quad \mathfrak{N}_{\omega,\mathcal{R}} \quad \xrightarrow{\text{GNS}} \quad (\mathcal{H}_\omega, \pi_\omega, \Omega_\omega),$$

not "choose a Hilbert space first, then search for relations".

63.3 Five Stages

Stage 0: input data. Status: prior-shard question. The input is a unitary fusion category or modular tensor category, plus conformal/OPE data when the target CFT needs more than $\mathcal{C}$. CA-01 states this as the global programme map. CA-09 gives the first checked kinematic example: Fibonacci fusion admissibility produces a sector projector, but full anyons still need coherent F/R-data.

Stage 1: free algebra of anyonic words. Status: proposal. Replace the Hilbert-first reading of CA-03 by an observable-algebra reading: generate formal local anyonic words from the categorical variable-N sectors

$$\bigoplus_{N,c,\text{configs}} \operatorname{Mor}(X_c, X_{a_1} \otimes \cdots \otimes X_{a_N}).$$

The precise algebra generators, involution, locality grading, and completion policy are fixed in CA-65 and CA-66 (companion shards, this block).

Stage 2: categorical relations. Status: mixed checked/proposal. Fusion admissibility is already a sector-projector relation in CA-09. Additional categorical relations should include defect/MPO relations. Hollands supplies the finite-size anchor: the bipartite anyonic-chain MPO algebra is isomorphic to the CFT defect algebra at finite size, while the persistence of that algebra in the scaling limit is explicitly left as an analytic question.

Stage 3: reference state and relaxed residuals. Status: proposal, currently exclusion-mode. The state must be chosen before a Hilbert space is claimed. CA-43 and CA-44 provide the present finite architecture: translation-invariant moments, positivity, and residual relation constraints of the form

$$\omega(X^* R Y) = 0.$$

In the GNS reading, these are the finite probe shadows of the null-relation condition. CA-63 (companion shard, this block) is the construction-mode state existence route; CA-64 (companion shard, this block) is the relaxation route.

Stage 4: GNS quotient and lattice model. Status: sourced scaffold plus proposal. Once $(\mathcal{A}, \omega)$ and a chosen residual set $\mathcal{R}$ are fixed, the candidate Hilbert space and operator representation are produced by the GNS quotient. The local OAR sources support this ordering: build scaling maps and an inductive-limit algebra, construct compatible/projective limit states, then perform GNS to get Hilbert-space representations and compatible representation systems.

Stage 5: continuum limit and fidelity. Status: precedent plus question. The continuum-limit technology is anchored by OAR and the free-fermion Koo–Saleur theorem: lattice Virasoro approximants and correlation functions converge in that sourced free-fermion setting. The anyonic precedent is Stottmeister's braiding-RG note, where braiding defines a real-space RG for anyonic chains and the Ising-chain example reaches the Ising CFT vacuum. The general shape of the open problem is now also locally sourced: Zini–Wang define scaling limits of quantum theories as double colimits and conjecture that every unitary minimal model VOA arises as a strong scaling limit with approximating lattice Virasoro observables. The categorical fidelity target remains a question:

$$\text{DHR/defect content of the limit} \simeq \mathcal{C}?$$

Hollands makes this a precise open problem by proving a finite algebraic defect correspondence and asking whether it persists under scaling limits.

63.4 Two Load-Bearing Constraints

Constraint 1: the Jones no-go changes the order. The locally registered no-go theorem states that the block-spin/Temperley–Lieb semicontinuous Hilbert space does not support a chiral CFT whose Hamiltonian generates circle translation as a continuum limit of lattice rotations; in the main theorem, triadic-rotation-rigid vectors in the triadic limit of an annular TL representation are necessarily zero. The OAR discussion corroborates the programme reading: nontrivial Thompson implementers are not implementable in the presence of nonzero central charge, and the Koo–Saleur strategy succeeds in those situations. The consequence is load-bearing: do not expect a fixed inductive-limit Hilbert space with dyadic rotations to carry the continuum geometry first. The

state comes first; geometry and symmetry are recovered through state-dependent scaling limits and generator convergence.

Constraint 2: exact UV relations are too strict. CA-57 defines the current, conservation, and exact boost-witness gates. CA-60 reports the checked 99-point scan: no point reaches `not_excluded_algebraic`, and the locally sourced self-dual TFIM point fails the exact boost witness. CA-61 and CONVENTIONS.md (q) record the route-scoped reading: this rejects the exact first-moment route, not the Ising scaling limit. Therefore Stage 3 must use quantified state-level or GNS-norm residuals, not exact coefficient identities, as the default relation language.

63.5 Functoriality and Fidelity Assessment

Proposal. Stages 1 and 2 should be functorial in principle: free algebras and imposed categorical relations are natural constructions once the generator and convention data are fixed. This is not yet a theorem; CA-65 records the intended functor on pairs of a category and a site object.

Proposal with sourced scaffold. GNS should be treated functorially on compatible $(\mathcal{A}, \omega)$-pairs. The local source support is not an abstract category theorem; it is the OAR architecture in which projective nets of states induce compatible nets of GNS representations and Hilbert-space connecting maps.

Question. The state assignment is the non-functorial crux. The realistic target is canonicity by a universal property: define a compact or closed convex set of states satisfying the relaxed residual constraints, then prove non-emptiness, uniqueness or controlled faces, and fidelity of the resulting limit. The strongest intended theorem would say that this state set exists and its continuum DHR/defect category recovers the input $\mathcal{C}$.

63.6 Boundary

This shard does not fix the categorical local-word algebra; the F/R-symbol gauge, object indexing, bracketing, and locality/completion conventions are fixed in CA-65/CA-66, not here.

It does not prove existence of the reference state. CA-43/CA-44 are exclusion-mode finite constraints; CA-63 is the companion construction-mode shard.

It does not fix the relaxation convention. The exact-relation scan failure only says exact UV boost witnesses are too strict for the current route; CA-64 must define the norm, scale, tolerance, and limiting quantifiers.

It does not prove a new continuum theorem. The only continuum precedents cited here are the sourced free-scalar/OAR, free-fermion Koo–Saleur, and Ising braiding-RG examples; the Zini–Wang minimal-model statement is cited as a conjecture, not a result.

It does not claim that SDP feasibility or finite residual smallness implies symmetry. CA-44 already records that feasibility does not prove a global state, positive energy, or a continuum Poincare/Virasoro representation.

64 All-Level Moment Feasibility Gives a Qubit Reference State

64.1 Status, Sources, and Dependencies

Status: derived compactness theorem, conditional on standard C*-algebra facts. This upgrades CA-44 only for the exact monotone full-window hierarchy. It does not apply to optional weight truncations, arbitrary implemented diagnostic levels, solver-unknown statuses, or nonmonotone residual lists.

Dependencies: CA-43, CA-44, CA-62, CONVENTIONS.md (n), CONVENTIONS.md (p).

64.2 Canonical Full-Window Levels

Let

$$I_N = [-N, N], \qquad \mathcal{A}_N = \bigotimes_{j \in I_N} M_2(\mathbb{C}),$$

and let $\mathcal{W}_N$ be the set of all Pauli words supported in I_N. For this shard, a level means the full-window problem:

$$y_\emptyset = 1, \qquad y_s = y_{\tau^k s} \tag{63.1}$$

whenever both words fit in the level window,

$$M_N(y)_{u,v} = \omega(P_u^* P_v) \succeq 0, \qquad u, v \in \mathcal{W}_N, \tag{63.2}$$

and every fixed residual relation

$$\omega(X^* R Y) = 0 \tag{63.3}$$

whose product support fits in I_N. The residual family is fixed in advance, or is enlarged monotonically. This monotonicity is part of the theorem; without it the compactness argument below has no nesting premise.

64.3 Finite Moments Are Finite States

Given a feasible y at level N, define a linear functional on $\mathcal{A}_N$ by

$$\omega_N(P_s) = y_s$$

and extend linearly in the Pauli-word basis. If $X = \sum_{u \in \mathcal{W}_N} c_u P_u$, then

$$\omega_N(X^* X) = \sum_{u,v} \overline{c_u} c_v \, \omega_N(P_u^* P_v) = c^* M_N(y) c \geq 0. \tag{63.4}$$

Since every positive element in a finite matrix algebra has a square root, (63.4) makes ω_N positive. The normalization $y_\emptyset = 1$ makes it unital, hence a state on $\mathcal{A}_N$.

Equivalently, with $d_N = 2^{|I_N|}$,

$$\rho_N = d_N^{-1} \sum_{s \in \mathcal{W}_N} y_s P_s \tag{63.5}$$

satisfies $\omega_N(A) = \mathrm{Tr}(\rho_N A)$, the density-matrix presentation of the finite state. Thus $M_N(y) \succeq 0$ if and only if $\rho_N \succeq 0$.

64.4 Compactness

Let $\mathcal{A}$ be the quasi-local spin C*-algebra, the norm closure of the local algebras. The normalized product trace outside I_N gives a unital completely positive conditional expectation

$$E_N : \mathcal{A} \to \mathcal{A}_N.$$

Therefore $\omega_N \circ E_N$ is a global state and agrees with ω_N on $\mathcal{A}_N$. This proves nonemptiness of the level set below whenever the level moment problem is feasible.

Let $K_N \subset S(\mathcal{A})$ be the set of global states whose restriction to $\mathcal{A}_N$ satisfies the affine level-N translation and residual constraints. Positivity is automatic for elements of the state space. Each constraint is a finite linear equation in evaluations $\varphi \mapsto \varphi(A)$, hence K_N is weak-* closed. Since $S(\mathcal{A})$ is weak-* compact, each K_N is compact.

The full-window levels are nested. Every level-N translation equality and every level-N residual test also appears at level $N+1$. Hence

$$K_{N+1} \subseteq K_N. \tag{63.6}$$

If every full-window level is feasible, the K_N are nested nonempty compact sets, so

$$\bigcap_{N \geq 0} K_N \neq \emptyset. \tag{63.7}$$

Choose ω in this intersection.

For any Pauli word s and any integer shift k, choose M so large that the intermediate translates from s to $\tau^k s$ fit in I_M. Since $\omega \in K_M$, (63.1) gives

$$\omega(P_s) = \omega(P_{\tau^k s}).$$

By linearity, density of the local Pauli algebra, and continuity of the shift automorphism, $\omega \circ \tau^k = \omega$ on all of $\mathcal{A}$.

64.5 Residual Relations in the GNS Representation

Let $(\mathcal{H}_\omega, \pi_\omega, \Omega_\omega)$ be the GNS representation. Fix one residual density R. For local Pauli words X, Y, choose N so that X^*RY fits the level window. Because $\omega \in K_N$,

$$\langle \pi_\omega(X)\Omega_\omega, \pi_\omega(R)\pi_\omega(Y)\Omega_\omega \rangle = \omega(X^*RY) = 0. \tag{63.8}$$

The vectors $\pi_\omega(A)\Omega_\omega$, with A local, are dense in the GNS Hilbert space. Since $\pi_\omega(R)$ is bounded, (63.8) on this dense subspace implies

$$\pi_\omega(R) = 0. \tag{63.9}$$

Translation-invariant states implement the lattice shift by a GNS unitary, so the translated residual densities vanish as well.

64.6 Exact Alternative

Conversely, if a translation-invariant state ω satisfies all residual relations, then $y_s = \omega(P_s)$ is feasible at every full-window level. Indeed, $y_\emptyset = 1$, translation invariance gives (63.1), positivity of ω gives (63.2), and the residual hypothesis gives (63.3).

Thus, for fixed exact residual data, there is no third mathematical possibility:

$$\text{some full-window level is infeasible} \quad \text{or} \quad \text{a global reference state exists.} \tag{63.10}$$

This is an exact compactness statement, not a finite-time algorithm.

64.7 Non-Implications

The state in (63.7) need not be unique. It need not be a ground state, a positive-energy state, a clustering state, or a scaling-limit state. The argument does not construct a continuum limit and does not produce a projective-unitary Poincare or Virasoro representation. It only proves that the imposed finite residual relations have a genuine C*-algebraic state model.

64.8 Boundary

Finite feasibility at checked levels still proves no existence theorem. The compactness result requires exact feasibility at every monotone full-window level. Numerically, exclusion remains semi-decidable by finding a certified infeasible finite level; existence is not certified by any finite prefix of feasible solver runs. The finite witnesses for the proof ingredients (density-matrix reconstruction, extension moment preservation, level nesting, and a positivity mutation) are checked in the Julia testset "qubit full-window moment compactness witnesses".

65 Quantified Relaxations of the Qubit Symmetry Gates

65.1 Status, Sources, and Dependencies

Status: proposal with checked finite pieces. The definitions below are finite Pauli-coefficient linear algebra plus fixed-residual moment SDPs. This shard answers the CA-61 obligation to relax or quotient the boost relation instead of requiring an exact local witness, and it supplies the Stage-3 relation language of CA-62.

Dependencies: CA-39, CA-42, CA-43, CA-44, CA-57, CA-60, CA-61, CA-62, CONVENTIONS.md (m), (n), (p), (q).

65.2 Coefficient Norm and Gauge

Let V_L be the Pauli coefficient space of L-site densities, with

$$\langle r, s\rangle_\tau = \sum_a \overline{r_a} s_a.$$

This is the normalized tracial Hilbert–Schmidt inner product. Padding by trailing identities is denoted $\iota_{L,N} : V_L \to V_N$.

The coboundary map of CONVENTIONS.md (n) is

$$(Du)_{a_0\cdots a_L} = u_{a_0\cdots a_{L-1}}\delta_{a_L 0} - \delta_{a_0 0} u_{a_1\cdots a_L}.$$

Residuals are distances to affine subspaces in this norm. Scalar witness components are gauge directions; they are fixed to zero for reported witnesses, but the distance itself is gauge-invariant.

65.3 Algebraic Profiles

The current gate remains hard for the nonzero-speed first-moment route:

$$\|p\|_\tau = 0 \quad\Longrightarrow\quad \text{fail this route.}$$

For $A = i[h_j + h_{j+1}, p_j]$, define the conservation profile

$$\epsilon_c(L; h) = \operatorname{dist}\big(\iota_{3,n} A, \iota_{L+1,n} \operatorname{im} D_L\big), \qquad n = \max(3, L+1).$$

The reported dimensionless value is

$$\widehat{\epsilon}_c(L; h) = \epsilon_c(L; h) / \max(\|A\|_\tau, \varepsilon_{\text{mach}}).$$

For candidates with exact conservation at support L, define the exact-conservation boost profile

$$\epsilon_b^0(L; h, \lambda, e) = \min_{u,w:\, D_L u = A} \|\iota B - \iota u - \lambda\, \iota(h - eI) - D_L w\|_\tau,$$

with $B = i[p_j, h_{j+1}] + 2i[p_j, h_{j+2}]$ and $\lambda = v^2$. Letting u float freely in the boost residual is wrong: it can absorb boost error while no longer witnessing the conservation coboundary. For candidates without exact conservation, use the joint profile

$$\epsilon_{cb}^2 = \min_{u,w} \|\widehat{A - Du}\|_\tau^2 + \|B - u - \widehat{\lambda(h - eI)} - Dw\|_\tau^2.$$

The raw profiles are nonincreasing in L, because a support-L witness is also a support-$L+1$ witness after padding by an identity. The speed λ is optimized only by external scan or bounded quadratic programming, with $\mu = \lambda e$ as the linear scalar variable; the sign of the optimizing λ is a recorded diagnostic, not an assumption.

65.4 State-Level Residuals

For a fixed residual density R, the GNS quantity is

$$\|\pi_\omega(R)\Omega_\omega\|^2 = \omega(R^*R).$$

At moment level N, define

$$v_N(R) = \min_y \omega_y(R^*R)$$

subject to the CA-43–CA-44 moment PSD, translation-invariance, normalization, and optional stationarity constraints. For fixed R, this is an SDP because $\omega_y(R^*R)$ is affine in y. Normalize by

$$\nu_N(R) = v_N(R)/\tau(R^*R),$$

when $\tau(R^*R) > 0$; the tracial value $\tau(R^*R) = \sum_s |r_s|^2$ is computable exactly from Pauli coefficients. A dual (certified) lower bound $\nu_N(R) \geq \delta > 0$ is an exclusion certificate for the relaxed route; $\nu_N(R)$ small is survival, never symmetry evidence.

Do not optimize u, w, λ, e jointly with the moments in this SDP tier. For $R(a) = R_0 + \sum_i a_i R_i$, the expression $\omega(R(a)^*R(a))$ is quadratic in a with coefficients affine in y, hence bilinear jointly. That joint minimization is not a plain SDP. Alternation gives upper-bound candidates; certified lower bounds require fixed witnesses or a bounded polynomial/sum-of-squares lift recorded as a new convention before use.

65.5 Scaling Reading

A continuum claim is not a fixed-N claim. A scaling statement must name: state sequence, renormalisation maps, vacuum subtraction, and operator topology, as required by CA-14 and CA-15. The target form is that normalized GNS residuals vanish on a growing long-wavelength window:

$$\sup_{|m|,|n|\leq K_N} \nu_N(r^{\sigma\sigma'}_{mn}) \to 0, \qquad K_N/N \to 0,$$

applied to the CA-42 Witt/Virasoro residual densities $r^{++}_{mn}, r^{--}_{mn}, r^{+-}_{mn}$. Modes near the Brillouin scale are not expected to close first; long-wavelength modes are the diagnostic window.

65.6 Sentinel Expectations

These are predictions to be checked by the relaxed re-scan, not results. The self-dual TFIM point is currentful, has an exact conservation witness, and fails the exact boost gate with recorded residual norm 1.8257418583505538; the relaxed verdict should therefore be survival into a GNS/scaling queue rather than terminal exclusion. Fully on-site and classical commuting densities still fail the hard current gate. Generic dense samples must show actual profile decay to survive; a single small value at one support is not a pass.

65.7 Boundary

No continuum theorem is claimed. The profiles and SDP objectives are finite diagnostics; converting decay of ν_N into a continuum symmetry statement requires the scaling payload of CA-15 and remains open. The longer-support TFIM profile values computed during design are not quoted here because their run bundle is not yet registered; only the recorded scan artifact is cited.

66 Anyonic Word Algebra from a Site Object

66.1 Status, Sources, and Dependencies

Status: proposal shard from the adopted W4 design. The finite local algebras are source-local C*-algebra objects; the free presentation and functoriality clauses are the adopted interface and still need implementation tests.

Dependencies: CA-02, CA-03, CA-09, CA-62, CA-63, CA-64, and CONVENTIONS.md (a), (b) (fixed with this block), (c), (f), (g), and (r).

66.2 Site Object First

The input to the word algebra is a pair $(\mathcal{C}, O)$, not the category $\mathcal{C}$ alone. Here $\mathcal{C}$ is a unitary fusion category and $O \in \mathcal{C}$ is the one-site object. The default mobile hard-core choice is

$$O = \mathbf{1} \oplus X,$$

where X is the allowed non-vacuum species object. For Fibonacci, $X = \tau$; for an all-species hard-core site, $X = \bigoplus_{a \neq \mathbf{1}} X_a$.

For a finite open interval I with $|I| = L$, fix the left-associated tensor power from CONVENTIONS.md (c),(r):

$$O_I := (((O \otimes O) \otimes O) \cdots) \otimes O.$$

The local anyonic observable algebra is

$$\mathcal{A}_I(\mathcal{C}, O) := \mathrm{End}_{\mathcal{C}}(O_I), \qquad f^* := f^\dagger.$$

Because endomorphism algebras in a unitary category are finite-dimensional unitary *-algebras, each $\mathcal{A}_I$ is a finite-dimensional C*-algebra.

Choose representatives $\{X_c\}_{c \in \mathrm{Irr}(\mathcal{C})}$, with $\mathbf{1}$ first by CONVENTIONS.md (a) as fixed in this block, and define

$$m_c(L) := \dim_{\mathbb{C}} \mathrm{Hom}_{\mathcal{C}}(X_c, O_I).$$

After choosing orthonormal bases of these Hom spaces,

$$\mathcal{A}_I(\mathcal{C}, O) \cong \bigoplus_{c \in \mathrm{Irr}(\mathcal{C})} M_{m_c(L)}(\mathbb{C}).$$

This matrix form is a presentation. The abstract algebra $\mathrm{End}_{\mathcal{C}}(O_I)$ is basis-free; fusion-tree bases and F-symbols enter only when we write matrices.

66.3 Placed Matrix Blocks

Let $w, w' \in \mathrm{Hom}_{\mathcal{C}}(X_c, O_I)$ be basis isometries in the same charge block and let $f \in \mathrm{End}_{\mathcal{C}}(X_c)$. The corresponding placed matrix block is

$$E_I(w', f, w) := w' \circ f \circ w^\dagger \in \mathcal{A}_I(\mathcal{C}, O).$$

For basis vectors u, v, w, z in compatible charge blocks,

$$E_I(u, g, v)\, E_I(w, f, z) = \delta_{v,w}\, E_I(u, g \circ f, z),$$

and

$$E_I(w', f, w)^* = E_I(w, f^\dagger, w').$$

For simple X_c, f is a scalar multiple of id_{X_c}, so these are ordinary matrix units inside the c-block. This is the categorical replacement for positioned Pauli words in CA-47: rows and columns are not bare strings of letters, but charge-resolved fusion-tree basis vectors.

66.4 Inclusions, Shift, and AF Limit

For $I \subset J$, the local inclusion is obtained by tensoring with identities on the added site factors, using the fixed bracketing convention:

$$f \longmapsto \mathrm{id}_O^{\otimes \ell} \otimes f \otimes \mathrm{id}_O^{\otimes r} .$$

On a one-sided chain the shift is

$$\alpha(f) = \mathrm{id}_O \otimes f.$$

The quasi-local observable algebra is the inductive-limit C*-algebra

$$\mathcal{A}(\mathcal{C}, O) := \overline{\bigcup_I \mathcal{A}_I(\mathcal{C}, O)}.$$

It is AF by construction: it is the norm closure of an increasing union of finite-dimensional C*-algebras.

The Bratteli diagram is the fusion graph of O. If $O_I \cong \bigoplus_c m_c(L) X_c$, then

$$m_d(L+1) = \sum_c m_c(L)\, N_{c,O}^d, \qquad N_{c,O}^d := \dim \mathrm{Hom}_{\mathcal{C}}(X_d, X_c \otimes O).$$

Sanity check: for $\mathcal{C} = \mathrm{Hilb}_{\mathrm{fd}}$ and $O = \mathbb{C} \oplus \mathbb{C}^d$, this gives

$$\mathcal{A}_L \cong M_{(d+1)^L}(\mathbb{C}),$$

and the inductive limit is the ordinary UHF spin-chain algebra with a $(d{+}1)$-dimensional maybe-spin site.

66.5 The BU-Style Free Pre-Layer

The adopted free layer is the tensor *-algebra

$$\mathcal{BU}_0(\mathcal{C}, O) := T(V), \qquad V := \bigoplus_{j \in \mathbb{Z}} \bigoplus_{a \neq \mathbf{1}} \mathrm{Hom}_{\mathcal{C}}(X_a, O)_j.$$

The subscript j records the lattice site. The involution is the formal dual/adjoint operation fixed by CONVENTIONS.md (b),(r); in a fusion-tree matrix presentation the unitary gauge is mandatory so that $*$ is entrywise conjugate transpose.

The intended presentation map is

$$\pi : \mathcal{BU}_0(\mathcal{C}, O) \twoheadrightarrow \mathcal{A}(\mathcal{C}, O),$$

with

$$J_{\mathrm{fus}} := \ker \pi.$$

Thus

$$\mathcal{BU}_0(\mathcal{C}, O)/J_{\mathrm{fus}} \cong \mathcal{A}(\mathcal{C}, O).$$

The ideal J_{fus} is kinematic. It contains the hard-core site relations, fusion admissibility, F-coherence, and the matrix-block relations above. This mirrors the AQFT construction in which one starts from a free unital *-algebra and quotients by a two-sided *-ideal of relations.

The load-bearing split is therefore

$$\mathcal{BU}_0 \xrightarrow{\pi \text{ (kinematic fusion)}} \mathcal{A}(\mathcal{C}, O) \xrightarrow{\text{state/GNS}} H_\omega.$$

Fusion is not deferred to the state. The GNS null space may impose dynamical or scaling relations, but category fidelity lives already in J_{fus}.

66.6 Defect Algebra Is a Different Layer

Hollands' double triangle algebra is a finite-dimensional vector space built from Hom spaces such as $\mathrm{Hom}_M(cd, \bar{a}b)$, with its own vertical product. That algebra is not the local observable algebra $\mathcal{A}_I(\mathcal{C}, O)$. In this programme it is the deferred defect/MPO/symmetry layer represented on, or acting by endomorphisms of, the observable tower. A later shard must specify that representation and its compatibility with the shift.

66.7 Functoriality Target

For a unitary monoidal dagger functor $F : \mathcal{C} \to \mathcal{C}'$ and a chosen unitary $F(O) \cong O'$, the expected finite-level map is

$$\mathrm{End}_{\mathcal{C}}(O^{\otimes L}) \longrightarrow \mathrm{End}_{\mathcal{C}'}((O')^{\otimes L}),$$

given by applying F and conjugating by the chosen site-unitary tensor power. The coherence data make this pseudofunctorial rather than strictly functorial on the nose. The free construction $\mathcal{BU}_0$ is natural at the generator level, and π should be natural once J_{fus} is implemented.

66.8 Lamport Dependency Skeleton

$D65.1$: Input pair $(\mathcal{C}, O)$, $\quad O = \mathbf{1} \oplus X$ by default.
$D65.2$: $O_I = O^{\otimes |I|}$ with fixed left bracketing.
$D65.3$: $\mathcal{A}_I(\mathcal{C}, O) = \mathrm{End}_{\mathcal{C}}(O_I), \quad f^* = f^\dagger$.
$D65.4$: $m_c(L) = \dim \mathrm{Hom}_{\mathcal{C}}(X_c, O^{\otimes L}), \quad \mathcal{A}_L \cong \bigoplus_c M_{m_c(L)}(\mathbb{C})$.
$D65.5$: $E_I(w', f, w) = w' f w^\dagger$.
$D65.6$: $\mathcal{BU}_0 = T\Big(\bigoplus_{j, a \neq \mathbf{1}} \mathrm{Hom}(X_a, O)_j\Big)$.
$D65.7$: $J_{\mathrm{fus}} = \ker(\mathcal{BU}_0 \twoheadrightarrow \mathcal{A})$.
$T65.1$: $E(u, g, v)E(w, f, z) = \delta_{v,w} E(u, g \circ f, z)$.
$T65.2$: $E(w', f, w)^* = E(w, f^\dagger, w')$.
$T65.3$: $m_d(L+1) = \sum_c m_c(L) N_{c,O}^d$.
$T65.4$: $\mathcal{A}(\mathcal{C}, O) = \overline{\bigcup_I \mathcal{A}_I}$ is AF.

66.9 Boundary

This shard does not construct the dynamical vacuum, prove a scaling limit, or derive Virasoro generators. Approximate Virasoro relations becoming exact in a continuum limit remain a programme conjecture. The categorical Markov trace and its naturality under spherical/unitary functors are not used here because the local source chain for canonical spherical structure is still open. The exact generator-level implementation of $\mathcal{BU}_0$, the endpoint-closing rule for charged words, and the double-triangle representation on $\mathcal{A}$ are implementation tasks, not completed results. "'

67 Anyonic States and Variable-N Fock Space

67.1 Status, Sources, and Dependencies

Status: proposal shard from the adopted W4 design. The finite carrier and Fibonacci arithmetic are local derivations from sourced fusion rules; no Julia invariant test has been added yet.

Dependencies: CA-02, CA-03, CA-06, CA-09, CA-62, CA-63, CA-64, CA-65, and CONVENTIONS.md (a), (b) (fixed with this block), (c), (f), (g), and (r).

67.2 The Finite Hard-Core Carrier

Fix the site object $O = \mathbf{1} \oplus X$ from CA-65. For L open sites, define

$$H_L(\mathcal{C}, O) := \bigoplus_{c \in \mathrm{Irr}(\mathcal{C})} \mathrm{Hom}_{\mathcal{C}}(X_c, O^{\otimes L}).$$

The inner product on each Hom space is the categorical one: if s, t have irreducible source X_c, then $s^\dagger t$ is a scalar multiple of id_{X_c}. The direct sum over charges is orthogonal.

Expanding the maybe-object gives the variable-particle grading. For one species X,

$$O^{\otimes L} = (\mathbf{1} \oplus X)^{\otimes L} \cong \bigoplus_{S \subseteq \{1,\dots,L\}} X^{\otimes S},$$

where S records the occupied sites and the tensor factors are kept in increasing site order. Hence

$$H_L(\mathcal{C}, O) \cong \bigoplus_{N=0}^{L} \bigoplus_{|S|=N} \bigoplus_{c \in \mathrm{Irr}(\mathcal{C})} \mathrm{Hom}_{\mathcal{C}}(X_c, X^{\otimes S}).$$

This is the sourced variable-N target from `literature/SYNTHESIS.md`. The hard-core condition is built into the site object: each site contributes either $\mathbf{1}$ or one non-vacuum summand, never two particles at the same site.

For multiple species, take $X = \bigoplus_{a \neq \mathbf{1}} X_a$. The same formula refines by a species labelling of the occupied set S.

67.3 Action of the Word Algebra

The finite algebra

$$\mathcal{A}_L(\mathcal{C}, O) = \mathrm{End}_{\mathcal{C}}(O^{\otimes L})$$

acts on every charge sector by postcomposition:

$$T \cdot w := T \circ w, \qquad T \in \mathcal{A}_L, \quad w \in \mathrm{Hom}_{\mathcal{C}}(X_c, O^{\otimes L}).$$

Thus H_L is the direct sum of the defining modules for the matrix blocks

$$\mathcal{A}_L \cong \bigoplus_c M_{m_c(L)}(\mathbb{C}), \qquad m_c(L) = \dim \mathrm{Hom}(X_c, O^{\otimes L}).$$

This is the finite carrier on which the kinematic quotient from CA-65 acts.

A charge-diagonal finite state can be written after choosing one minimal projection e_c in each nonzero charge block. Let

$$p := \bigoplus_c e_c.$$

Using the ordinary finite matrix trace in the chosen block presentation, set

$$\omega_{L,p}(T) := \frac{\mathrm{Tr}(pTp)}{\mathrm{Tr}(p)}.$$

The corresponding GNS quotient is intended to recover one cyclic column from each charge block, hence a space of dimension $\sum_c m_c(L) = \dim H_L$. The choice of e_c is not canonical when $m_c(L) > 1$; coherent choices in L are an open point in this construction.

67.4 Birth and Death Operators

Let a be a simple object with conjugate $\bar{a}$, and choose the raw cup

$$v_a : \mathbf{1} \to a \otimes \bar{a}.$$

The raw normalization is

$$v_a^\dagger v_a = d_a \, \mathrm{id}_{\mathbf{1}} .$$

At adjacent sites $j, j+1$, define the local birth map

$$b^a_{j,j+1} := (\iota_a \otimes \iota_{\bar{a}}) \circ v_a \circ (\pi_{\mathbf{1}} \otimes \pi_{\mathbf{1}}),$$

where $\iota_a : a \to O$ is the site summand inclusion and $\pi_{\mathbf{1}} : O \to \mathbf{1}$ is the vacuum summand projection. Its adjoint

$$d^a_{j,j+1} := (b^a_{j,j+1})^\dagger$$

is the death map. These maps change the occupation number by $N \leftrightarrow N \pm 2$, and annihilate configurations not matching the two-vacuum or $a, \bar{a}$ local pattern.

The default convention is raw cups with explicit d_a bookkeeping:

$$(b^a_{j,j+1})^\dagger b^a_{j,j+1} = d_a \, P_{\mathbf{11}}.$$

The normalized cup $d_a^{-1/2} v_a$ would make the birth map a partial isometry on the two-vacuum corner, but that is not the default convention.

67.5 Hollands' Dense Chain as a Corner

Hollands' chain Hilbert space V^L is built from sequences of intertwiners, and the corresponding operators are relative commutants. In the present tower it is recovered by two compressions. First take the fixed-$N = L$ corner that drops the vacuum summand at every site:

$$O = \mathbf{1} \oplus \tau \quad \rightsquigarrow \quad P_\tau O^{\otimes L} \cong \tau^{\otimes L}.$$

Then take the total-charge-$\mathbf{1}$ corner:

$$e_{\mathbf{1}} P_\tau H_L \cong \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, \tau^{\otimes L}).$$

Hollands' Jones projections are local elements in this compressed corner. The extra Q-system/subfactor data used in the dense-chain source are not supplied by a bare UFC input and remain deferred.

67.6 Fibonacci Test Targets

For Fibonacci, take

$$\mathrm{Irr}(\mathrm{Fib}) = \{\mathbf{1}, \tau\}, \qquad \tau \otimes \tau = \mathbf{1} \oplus \tau, \qquad O = \mathbf{1} \oplus \tau.$$

With $F_0 = 0$, $F_1 = 1$, the multiplicities are

$$(m_{\mathbf{1}}(L), m_\tau(L)) = (F_{2L-1}, F_{2L}) \qquad (L \geq 1).$$

Therefore

$$\dim H_L = m_{\mathbf{1}}(L) + m_\tau(L) = F_{2L+1},$$

and

$$\dim \mathcal{A}_L = m_{\mathbf{1}}(L)^2 + m_\tau(L)^2 = F_{4L-1}.$$

The first arithmetic targets are

$$\dim H_2 = 5, \quad \dim H_3 = 13, \quad \dim H_4 = 34,$$

and

$$\dim \mathcal{A}_2 = 13, \quad \dim \mathcal{A}_3 = 89, \quad \dim \mathcal{A}_4 = 610.$$

For the dense-chain corner $O = \tau$, the same recurrence reduces to CA-09's (a_n, b_n) path-count sequence.

For the raw Fibonacci cup

$$v : \mathbf{1} \to \tau \otimes \tau,$$

Penneys gives

$$v^\dagger v = \phi \, \mathrm{id}_{\mathbf{1}}, \qquad \phi = \frac{1+\sqrt{5}}{2}.$$

Thus the local cup-cap invariant for the raw birth/death pair is ϕ, not 1. The unitary Fibonacci recoupling matrix is

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} \phi^{-1} & \phi^{-1/2} \\ \phi^{-1/2} & -\phi^{-1} \end{pmatrix} = (F_\tau^{\tau\tau\tau})^\dagger .$$

For this nontrivial block the matrix is also involutory, but the adopted convention is still the unitary gauge so that matrix presentations use the ordinary adjoint.

67.7 Lamport Dependency Skeleton

$D66.1$: $H_L(\mathcal{C}, O) = \bigoplus_c \mathrm{Hom}(X_c, O^{\otimes L})$.
$D66.2$: $O = \mathbf{1} \oplus X \Rightarrow H_L \cong \bigoplus_{N,S,c} \mathrm{Hom}(X_c, X^{\otimes S})$.
$D66.3$: $\mathcal{A}_L = \mathrm{End}(O^{\otimes L})$ acts by $T \cdot w = T \circ w$.
$D66.4$: $p = \bigoplus_c e_c$, $\omega_{L,p}(T) = \mathrm{Tr}(pTp)/\mathrm{Tr}(p)$.
$D66.5$: $b^a_{j,j+1} = (\iota_a \otimes \iota_{\bar{a}}) v_a (\pi_{\mathbf{1}} \otimes \pi_{\mathbf{1}})$, $d = b^\dagger$.
$T66.1$: $(b^a)^\dagger b^a = d_a P_{\mathbf{11}}$ for raw cups.
$T66.2$: $O = \mathbf{1} \oplus \tau$: $(m_{\mathbf{1}}(L), m_\tau(L)) = (F_{2L-1}, F_{2L})$.
$T66.3$: $\dim H_L = F_{2L+1}$, $\dim \mathcal{A}_L = F_{4L-1}$.
$T66.4$: Hollands fixed-N, charge-$\mathbf{1}$ chain is a corner of H_L.

67.8 Boundary

This shard does not prove existence or uniqueness of a dynamical vacuum. The finite state $\omega_{L,p}$ is a useful charge-diagonal target, not a canonical vacuum. The categorical Markov trace is the expected natural "infinite-temperature" state only after the spherical/pivotal convention is locally sourced. The residual choice of e_c for $m_c > 1$ is open.

The general statement "unitary UFC implies canonical spherical structure and positive quantum dimensions" is still a source gap for this block. The Fibonacci self-duality sign $\kappa_\tau = +1$ is also not locally sourced. Approximate Virasoro relations becoming exact in the scaling-limit GNS null space remains a programme conjecture, not a result of this shard.

68 Refinement-Map Requirements and Obstructions

68.1 Status, Sources, and Dependencies

Status: source-local synthesis plus derived checklist. Every requirement and failure mode below is anchored in a registered local source; the checklist assembly and the dense-chain diagnosis are local readings.

Dependencies: CA-14, CA-15, CA-62, CA-65, CA-66; CONVENTIONS.md (r).

68.2 Strict OAR Requirements

The sourced strict architecture asks of a refinement family $\alpha^N_{N'} : \mathcal{A}_N \to \mathcal{A}_{N'}$:

1. unital and completely positive (states pull back to states);
2. a *-morphism, injective, hence isometric;
3. the semigroup property $\alpha^{N'}_{N''} \circ \alpha^N_{N'} = \alpha^N_{N''}$;
4. bounded, scale-independent support growth (locality);
5. smearing: a lattice field maps to a smeared continuum field, not a point-like one.

On the state side, a scaling limit is a projectively consistent family $\omega^{(N)} = \omega^{(N')} \circ \alpha^N_{N'}$; on the dynamics side, the refined evolutions must form Cauchy sequences in the scaling-limit GNS representation, and the source stresses that success depends sensitively on the choice of scaling maps.

68.3 Fallback and Soft Relaxations

Two sourced fallbacks weaken the strict demands without abandoning the continuum target. First, when the *-morphism property or regularity fails (block-spin, point-localization), the algebra-level limit is lost but the vacuum can still be recovered through Wightman two-point functions; continuum local operators are then only indirectly accessible by reconstruction.

Second, the soft-inductive-system formalism replaces exact structure by asymptotic structure: connecting maps need only be contractions with asymptotic transitivity; operations converge iff they *asymptotically intertwine* the connecting maps,

$$\lim_{n \gg m} \left\| (\tilde{\jmath}_{nm} T_m - T_n j_{nm}) x_m \right\| = 0,$$

and C*-structure survives under an asymptotic multiplicativity condition. For symmetry recovery the same source is explicit: Thompson-group implementers fail for nonzero central charge, and the successful strategy is Koo–Saleur approximation of conformal symmetries. **Consequence [derived]:** for a CFT target with $c \neq 0$, "compatible with the Virasoro action" can only mean *asymptotic* intertwining of normal-ordered Koo–Saleur generators on a suitable core — there is no exact-implementer formulation to aim for.

68.4 The No-Go Mechanism as a Checklist

The Jones no-go concerns the semicontinuous limit built from a single scale-invariant tree-local refinement element. The registered source identifies the root cause: the underlying dangling binary tree does not feel the circle topology, and the pulled-back smallest rotation is a long product of local elements controlled by a norm-contracting renormalisation (squaring) map, whose iterates vanish — forcing rotation matrix elements weakly to zero. Kliesch–Koenig extend this: discontinuity has

full measure among tree isometries, and they isolate an easily verified necessary condition for a continuous limit.

Checklist [derived reading]. A refinement family risks the no-go mechanism when:

NG1	it is generated by a single scale-invariant tree-local element, decoupled from the circle;
NG2	its induced squaring/renormalisation map is norm-contracting on the relevant subspace;
NG3	it embeds point-like, without low-pass smearing;
NG4	it operates in the large-loop-parameter regime where the contraction estimate bites;
NG5	it demands exact unitary implementers of the discrete rotation/conformal action on a fixed limit space.

Any candidate family must either avoid each mode or verify the Kliesch–Koenig necessary condition explicitly. The wavelet precedent shows what success looks like: translation-covariant smearing recovers continuous translations, while Lorentz/conformal covariance is separate further work.

68.5 Where Fixed Anyon Number Enters the Braiding RG

The braiding-RG refinement for anyonic chains is sourced as: attach new through-strings to a string-box diagram and braid their ends into position, *increasing the local density of σ-anyons*; the basis carries a σ-anyon at every site. **Diagnosis [derived].** Three fixed-N commitments follow: (i) the braid needs a strand at every site it acts on, so the map is undefined on empty sites; (ii) the identification of local operators with Temperley–Lieb generators presupposes exactly one anyon per site; (iii) the refinement doubles the strand count deterministically. For the CA-65/CA-66 site object $O = \mathbf{1} \oplus X$, occupation is a quantum variable, so a refinement must be defined on the whole word space including empty sites — the braiding-RG step does not port. Conversely, dense chains have no refinable vacuum: every refined site must carry an anyon, so any fine-graining changes the physical content. The variable-N tower is exactly what removes this obstruction, at the price of the new proof obligations recorded in CA-68.

68.6 Boundary

This shard proves no convergence theorem and constructs no refinement; it fixes the requirement set and the failure modes. The Kliesch–Koenig necessary condition is registered but its verification for any variable-N family is future work. The Koo–Saleur compatibility target is stated precisely in CA-68; nothing here claims it holds.

69 Refinement Maps for the Variable-N Anyonic Word Algebra

69.1 Status, Sources, and Dependencies

Status: proposal plus local derivations. The vacuum-insertion map is an exact finite-level construction in the CA-65/CA-66 tower. Its Virasoro compatibility is an open scaling problem, not a theorem.

Dependencies: CA-62, CA-63, CA-64, CA-65, CA-66, CA-67, and CONVENTIONS.md (a), (b), (c), and (r).

69.2 Vacuum Insertion

Definition [choice]. Fix $O = \mathbf{1} \oplus X$. Let $\iota_{\mathbf{1}} : \mathbf{1} \to O$ be the empty-site summand inclusion. The canonical site-doubling refinement is

$$V_L : O^{\otimes L} \longrightarrow O^{\otimes 2L}, \qquad \text{old site } j \mapsto 2j-1, \quad \text{fine site } 2j \mapsto \mathbf{1}.$$

Equivalently, in each two-site fine cell the local factor is $\mathrm{id}_O \otimes \iota_{\mathbf{1}} : O \to O \otimes O$, with the fixed left-associated bracketing of CONVENTIONS.md (r).

Lemma [derived]. V_L is an isometry: $V_L^\dagger V_L = \mathrm{id}_{O^{\otimes L}}$. Each inserted factor contributes $\iota_{\mathbf{1}}^\dagger \iota_{\mathbf{1}} = \mathrm{id}_{\mathbf{1}}$, and tensor products of isometries are isometries.

Lemma [derived]. The induced map on charge sectors

$$J_{L,c} : \mathrm{Hom}(X_c, O^{\otimes L}) \to \mathrm{Hom}(X_c, O^{\otimes 2L}), \qquad w \mapsto V_L \circ w,$$

is isometric and preserves the total charge c. It is strictly local: the cellwise factors act on disjoint fine two-site cells.

69.3 Algebra Map

Observation [derived]. Let $P_L = V_L V_L^\dagger$. The map

$$\theta_L(T) = V_L T V_L^\dagger, \qquad T \in \mathcal{A}_L := \mathrm{End}_{\mathcal{C}}(O^{\otimes L}),$$

is a *-isomorphism from $\mathcal{A}_L$ onto the corner $P_L \mathcal{A}_{2L} P_L$. It is not a unital embedding into $\mathcal{A}_{2L}$, because $\theta_L(1) = P_L \neq 1$.

Boundary [open]. Strict OAR, as locally sourced, wants unital injective *-morphisms of algebras. Thus θ_L is not yet a strict OAR scaling map on the full $\mathcal{A}_{2L}$. A unital extension needs complement data that this shard does not canonically supply; the honest alternatives are a unital completely positive completion inside a *soft* C*-system with asymptotic multiplicativity (CA-67), or the state/GNS corner formalism.

69.4 State Consistency

Exact corner consistency [derived]. If a state ω_{2L} on $\mathcal{A}_{2L}$ satisfies $\omega_{2L}(P_L) = 1$, then $\omega_L := \omega_{2L} \circ \theta_L$ is a state on $\mathcal{A}_L$. For the CA-66 charge-diagonal states, exact consistency is obtained by choosing fine minimal projections $e_c^{(2L)} = V_L e_c^{(L)} V_L^\dagger$.

Dynamical-vacuum condition [open]. A physical scaling vacuum should satisfy exact or asymptotic GNS consistency $J_L \Omega_L \approx \Omega_{2L}$. Bare vacuum insertion supplies a candidate reference subspace, not a dynamical-vacuum theorem.

69.5 Density Reading

Observation [derived]. A word with N occupied sites refines to a word with the same N occupied sites on $2L$ fine sites: the site-filling fraction halves while the *physical* particle number at fixed physical volume is unchanged as the lattice spacing halves. Dense chains cannot do this: every refined site must carry the chain anyon, which is the CA-67 diagnosis of why the braiding-RG route does not port.

Boundary [open]. This reading fits fixed-particle and finite physical-density sectors. Whether a *critical* vacuum sector is reachable depends on the dressed state and coupling flow below.

69.6 Virasoro Compatibility Target

Target [open]. Let $:L^{(L)}_{k,\pm}(g_L):$ denote normal-ordered Koo–Saleur approximants at renormalized coupling g_L (vacuum subtraction per CA-14). A refinement family W_L should satisfy, on a low-energy GNS core and a growing long-wavelength window,

$$\sup_{|k|\le K_L} \big\| :L^{(2L)}_{k,\pm}(g_{2L}): W_L\psi - W_L :L^{(L)}_{k,\pm}(g_L):\psi \big\| \longrightarrow 0, \qquad K_L/L \to 0.$$

This is the CA-67 asymptotic-intertwining condition specialised to Koo–Saleur modes, the anyonic analogue of the sourced free-fermion convergence theorem. Per CA-67 there is no exact-implementer formulation to aim for when $c \neq 0$; asymptotic intertwining in a soft system is the target statement.

Proposal [open]. Bare V_L is not expected to satisfy the target for a fixed local Hamiltonian. Use a dressed family

$$W_L = U_L V_L,$$

with $U_L \in \mathcal{A}_{2L}$ a bounded-depth, charge-preserving local circuit, scale- and coupling-dependent. W_L remains local, isometric, and charge-preserving; the role of U_L is to create the short-range entanglement that bare empty-site insertion lacks, playing the part of the wavelet low-pass filter (and of the braiding disentangler in the dense setting) so that NG1–NG3 of CA-67 are avoided.

69.7 Dead Prior Art: the Deficit/Pair-Creation Precedent

The archived predecessor project (dead, not canonical; re-derive before use) validated two fine-graining isometries numerically for Fibonacci/Ising at small size under D4 wavelets: a *normalised product map* (spread each anyon through a one-particle wavelet isometry, column-normalise the diagonal Gram matrix) and an experimental *number-changing isometry* $V_0 + V_2$ in which the isometry deficit $I - V_0^\dagger V_0$ of the raw fixed-N wavelet map is filled by a pair-creation channel into the $N{+}2$ sector. **Reading [derived].** The deficit phenomenon is structural evidence for the present programme: fixed-N wavelet refinement fails isometry exactly where indefinite anyon number can repair it, using the CA-66 birth maps. The archived construction was never reconciled with any algebraic framework, never connected to Virasoro data, and its existence conditions were not proved; those are re-derivation targets for this block, not inherited results.

69.8 Fibonacci $H_2 \to H_4$ Test

Data [derived]. For Fibonacci $O = \mathbf{1} \oplus \tau$, $(m_{\mathbf{1}}(2), m_\tau(2)) = (2, 3)$ and $(m_{\mathbf{1}}(4), m_\tau(4)) = (13, 21)$.

Matrix target [derived]. In the occupied-subset path basis, V_2 sends $S \subseteq \{1,2\}$ to $2S - 1 \subseteq \{1,2,3,4\}$: the $\mathbf{1}$-block maps $\emptyset \mapsto \emptyset$, $\{1,2\} \mapsto \{1,3\}$; the τ-block maps $\{1\} \mapsto \{1\}$, $\{2\} \mapsto \{3\}$, $\{1,2\} \mapsto \{1,3\}$. The charge-block matrices are 13×2 and 21×3 zero-one isometries. Julia invariants:

$V_c^\dagger V_c = I$ on both charge blocks, $P = VV^\dagger$ idempotent self-adjoint, occupation covariance $Vn_j = n_{2j-1}V$ and $n_{2j}V = 0$.

Birth covariance [derived]. For the CA-66 adjacent birth $b^a_{j,j+1}$,

$$V_L b^a_{j,j+1} = \widetilde{b}^a_{2j-1,2j+1} V_L,$$

where $\widetilde{b}$ creates the pair on fine sites $2j-1, 2j+1$ and keeps fine site $2j$ in the vacuum. The naive adjacent-fine-site covariance is false for this insertion convention; a mutation test must pin this.

69.9 Comparison and Boundary

The braiding RG refines dense chains by increasing string density through braiding; the present family refines a vacuum-capable word algebra by adding empty sites. This shard does not claim to evade all rotation obstructions: it avoids the specific fixed TL semicontinuous Hilbert space of the no-go, and any future discrete-rotation action on this tower must be checked against the CA-67 checklist, including the registered Kliesch–Koenig necessary condition. The expected algebraic home of the Fibonacci word tower — a dilute Temperley–Lieb-type diagram algebra — and any Koo–Saleur construction for *dilute* loop models are open research targets; the dilute-algebra sources are registered but not yet used. No continuum limit, Virasoro representation, or category-fidelity theorem is proved here. The output is a local, isometric, charge-preserving refinement candidate, its dressing proposal, and its proof obligations.

70 The Dilute Temperley–Lieb Algebra and the Anyonic Word Algebra

70.1 Status, Sources, and Dependencies

Status: source-local registration plus derived finite arithmetic. The algebra data are registered from the local sources; the evaluation map is a proposal; the non-surjectivity obstruction and the small-L dimension matches are checked local arithmetic.

Dependencies: CA-65, CA-66, CA-68; CONVENTIONS.md (b), (r).

70.2 The Dilute Algebra

The dilute Temperley–Lieb algebra $\mathrm{dTL}_n(\beta)$ is spanned by dilute n-diagrams: n points on each vertical side, an even number of which may be *vacancies*, all remaining points joined pairwise by non-intersecting strings. The product is concatenation, with a factor β per closed loop and zero whenever a floating string fails to close. A generating set is $\{a_i, a_i^t, b_i, b_i^t, e_j, x_j\}$, with $e_i + x_i = \mathrm{id}$ and the dense Temperley–Lieb generator recovered as $u_i = b_i^t b_i$. Two facts are load-bearing for this block:

Parity. The vacancy numbers on the two sides of a diagram always share parity, and the even/odd spans are two-sided ideals: $\mathrm{dTL}_n = \mathrm{edTL}_n \oplus \mathrm{odTL}_n$. Equivalently, a dilute diagram never changes the occupied-site number N by an odd amount: strings pair endpoints, so occupied counts on the two sides are congruent mod 2.

Dense corners. For any subset A of positions, the idempotent $\pi_A = \prod_{i\in A,\, j\notin A} x_j e_i$ satisfies $\pi_A\, \mathrm{dTL}_n\, \pi_A \cong \mathrm{TL}_{|A|}$: the fully occupied corner is the ordinary dense algebra.

The dimension is a Motzkin number:

$$\dim \mathrm{dTL}_n = \sum_{k=0}^{n} \binom{2n}{2k} \dim \mathrm{TL}_k = M_{2n},$$

so $\dim \mathrm{dTL}_n = 1, 2, 9, 51, 323$ for $n = 0, \ldots, 4$. For q not a root of unity ($\beta = q + q^{-1}$) the algebra is semisimple.

70.3 The Evaluation Map

Proposal. Let $\mathcal{C}$ be a unitary fusion category with self-dual simple X of quantum dimension d, and $O = \mathbf{1} \oplus X$ the CA-65 site object. Define

$$\rho_L : \mathrm{dTL}_L(d) \longrightarrow \mathrm{End}_{\mathcal{C}}(O^{\otimes L})$$

by reading a vacancy as the $\mathbf{1}$-summand, an occupied point as the X-summand, a through-string as the identity on X transported between its endpoint sites, and cups/caps as the raw coevaluation/evaluation of CONVENTIONS.md (r) (Fibonacci: $v^\dagger v = \varphi\, \mathrm{id}_{\mathbf{1}}$, so closed loops evaluate to $d = \varphi$, matching the diagrammatic β). The Frobenius–Schur flag of (r) is a recorded obligation before ρ_L is promoted beyond proposal status for general self-dual X.

70.4 The Obstruction: No Surjection for Fibonacci

Checked arithmetic. For Fibonacci, CA-66 gives $\dim \mathrm{End}(O^{\otimes L}) = F_{4L-1}$ (checked in the Julia testset "fibonacci anyonic word algebra and vacuum insertion"), while $\dim \mathrm{dTL}_L = M_{2L}$:

L	1	2	3	4
$\dim \mathrm{dTL}_L = M_{2L}$	2	9	51	323
$\dim \mathrm{End}(O^{\otimes L}) = F_{4L-1}$	2	13	89	610

Since $M_{2L} < F_{4L-1}$ for $L \geq 2$, no map from dTL_L can be surjective, and passing to any quotient (e.g. by a Jones–Wenzl-type ideal) only shrinks the image. The dense intuition $\mathrm{TL}_L(\varphi) \twoheadrightarrow \mathrm{End}(\tau^{\otimes L})$ does *not* extend to the dilute pair.

Derived mechanism. The failure is exactly the parity ideal structure: dilute diagrams preserve occupied-number parity, while Fibonacci has $N_{\tau\tau}^{\tau} = 1$, so the word algebra contains fusion morphisms changing N by one (the trivalent vertex $\tau \to \tau \otimes \tau$). At $L = 2$ the missing $13 - 9 = 4$ dimensions are precisely the matrix units linking the $N = 1$ and $N = 2$ states of the charge-τ block.

Derived dimension coincidence. Splitting each CA-66 charge block by occupied-number parity, the parity-preserving subalgebra of $\mathrm{End}(O^{\otimes L})$ has dimension

$$L = 2: \ (2^2) + (2^2 + 1^2) = 9, \qquad L = 3: \ (4^2 + 1^2) + (5^2 + 3^2) = 51,$$

matching M_{2L} exactly in both cases. **Conjecture (checked at the dimension level for $L \leq 3$; open beyond):** for Fibonacci, ρ_L is an isomorphism onto the parity-even–preserving subalgebra of $\mathrm{End}(O^{\otimes L})$ for small L, with a nontrivial kernel expected only once dense corners reach the first negligible Jones–Wenzl projector (TL_4 at $\beta = \varphi$), i.e. from $L = 4$. At $L = 4$ the counts 323 versus the parity-preserving dimension must be computed, not assumed; this is a CA-71-tier numerical obligation.

70.5 Consequences for the Programme

Two tiers separate cleanly. *Tier 1 (dilute TL proper):* Hamiltonians built from the dTL generators — occupancy projectors, dense exchange u_i, hopping $a_i + a_i^t$, pair creation/annihilation $b_i + b_i^t$ — act on the word algebra through ρ_L and conserve N-parity. The sourced anyonic t-J models are of this shape, and the dilute Koo–Saleur programme (CA-70) lives here. *Tier 2 (beyond dTL):* the full word algebra needs the fusion vertex; densities using N-parity-violating terms leave the dilute algebra and require category-specific data. This is the precise algebraic sense in which the variable-N word algebra is strictly richer than the dilute diagram algebra.

70.6 Boundary

No Hamiltonian, criticality, or Virasoro statement is made here. The evaluation map is a proposal pending the Frobenius–Schur convention; its injectivity on $\mathrm{dTL}_L(\varphi)$ for $L \leq 3$ is a dimension-level argument only and must be confirmed by an explicit rank computation (CA-71). The affine/periodic dilute algebra needed by a periodic Koo–Saleur setup is not defined in the registered sources and remains an acquisition/derivation target.

71 The Dilute Koo–Saleur Ansatz

71.1 Status, Sources, and Dependencies

Status: proposal. To our knowledge (2026-07-05 survey, recorded in the review bundle) no Koo–Saleur construction exists for any dilute loop model or the dilute Temperley–Lieb algebra; this shard fixes the candidate generators and their proof obligations. Nothing here is a convergence claim.

Dependencies: CA-14, CA-42, CA-64, CA-69; CONVENTIONS.md (r).

71.2 What the Dense Template Requires

The sourced dense-TL recipe takes: (i) a critical Hamiltonian $H_N = \sum_j h_j$ with density built from algebra generators, normalized so the sound velocity is one; (ii) the ground-state energy density e_∞ subtracted termwise; (iii) a momentum density from the nearest-neighbour commutator of Hamiltonian densities; (iv) periodic boundary conditions; and (v) the $c/24$ zero-mode shift. The generators are Fourier modes of the subtracted density plus the current correction. The known frontier is recorded in the same source: even in the dense case, the limit of commutators agrees with the commutator of limits only up to a modified central term, and parts of that statement remain conjectural.

71.3 The Dilute Ansatz

Proposal. Let h_j be a self-adjoint, translation-invariant two-site density in the dilute algebra: a real linear combination of occupancy projectors, the dense exchange $u_j = b_j^t b_j$, hopping $a_j + a_j^t$, and pair creation/annihilation $b_j + b_j^t$ (all dilute diagrams, hence occupied-parity-preserving by CA-69). On the periodic chain — the algebraic setting being the registered periodic dilute Temperley–Lieb algebra — define

$$E_n^{(N)} = \frac{N}{2\pi v} \sum_{j=1}^{N} e^{inj2\pi/N} \, (h_j - e_\infty), \qquad p_j = i[h_j, h_{j+1}],$$

$$P_n^{(N)} = \frac{N}{2\pi} \sum_{j=1}^{N} e^{in(j+\frac{1}{2})2\pi/N} \, p_j,$$

$$L_n^{(N)} = \tfrac{1}{2}\big(E_n^{(N)} + \tfrac{1}{v} P_n^{(N)}\big) + \tfrac{c}{24}\,\delta_{n,0}, \qquad \bar{L}_{-n}^{(N)} = \tfrac{1}{2}\big(E_n^{(N)} - \tfrac{1}{v} P_n^{(N)}\big) + \tfrac{c}{24}\,\delta_{n,0},$$

where v and e_∞ are the model's sound velocity and ground-state energy density, and c its central charge. This is the direct dilute transcription of the sourced dense formula; every ingredient except the algebra membership of h_j is inherited. The candidate modes live in the even (parity-preserving) dilute ideal, consistently with CA-69.

71.4 Two Sourced Critical Anchors

Integrable anchor (dilute O(n)/Izergin–Korepin). The registered sources give exact branch central charges, $c = 1 - \frac{3\phi^2}{\pi(\pi-2\lambda)}$ (branches 1, 2) and $c = \frac{3}{2} - \frac{3\phi^2}{2\pi\lambda}$ (branches 3, 4), and the periodic dilute algebra carries the loop transfer matrices. The explicit quantum-chain Hamiltonians (the Hamiltonian limits of these transfer matrices, with their v and e_∞) are in a journal-only source flagged above; until it is registered, the integrable anchor supports regime selection but not a shard-ready density. A recorded warning: at least one regime of the $A_2^{(2)}$ model has a *non-compact*

continuum limit (registered source VernierJacobsenSaleur2014), so regime selection is part of the input data, not a detail.

Anyonic anchor (itinerant t-J). The locally sourced anyonic t-J models are dilute in exactly the CA-69 sense: anyon hopping plus a dense projector exchange term, with hard-core occupancy — an h_j in the image of the dilute generators under the CA-69 evaluation map, acting on the CA-66 Fibonacci sectors. Their central charges are numerically identified ($c = 7/10$ or $4/5$ for Fibonacci, $1/2$ for Ising; entanglement gives $1 + c$ with the charge sector). This anchor is shard-ready on the input side but its criticality is numerical, not Bethe-exact.

71.5 Proof Obligations

1. $e_\infty,\ v,\ c$ must be sourced (integrable anchor) or estimated with recorded finite-size protocols (anyonic anchor);
2. the central-term/limit gap is inherited from the dense case and remains conjectural there; no dilute claim may be stronger;
3. periodic dilute algebra: the CA-69 evaluation map must be extended to the periodic setting before $L_n^{(N)}$ acts on word sectors;
4. gaplessness alone is insufficient: a dilute-looking density can be gapless with dynamical exponent $z \neq 1$ (no local source yet; registered as an external caution for CA-71 acceptance tests);
5. diagnostics run through CA-64: normalized GNS residuals ν_N of the dilute Witt relations on long-wavelength windows.

71.6 Validation Plan (CA-71 Tier)

Exact diagonalization in the Fibonacci $O = \mathbf{1} \oplus \tau$ charge sectors (bases already implemented and tested for CA-65–CA-68): build the dense projector from the sourced unitary F-matrix, hopping across vacancies, and raw-cup birth/death; estimate e_∞, v, and c from finite-size spectra; construct $L_n^{(N)}$ and test low-energy commutator closure and the CA-64 residuals. Acceptance: dilute relations hold exactly; sourced or estimated (e_∞, v, c) reproduce the anchor values; current modes do not collapse; residuals decrease on the named window. Failure witnesses: current collapse, wrong central charge, residuals bounded away from zero, or non-conformal ($z \neq 1$) finite-size scaling.

71.7 Boundary

No convergence theorem, no continuum Virasoro representation, and no claim that the ansatz closes even at the level of the dense case's conjectures. The integrable anchor's Hamiltonian is a registered acquisition gap. The anyonic anchor's criticality is numerical input, not a theorem. The extension of the CA-69 evaluation map to the periodic dilute algebra is future work.

72 The Refinement Placement Category and Exact Composition

72.1 Status, Sources, and Dependencies

Status: local derivations plus sourced structure theory. The placement definitions, isometry, charge preservation, and the exact composition theorem are finite-level derivations checked in Julia. The forest-functor identification and the Ore dichotomy are sourced. No dressing, no state selection, no dynamics, and no continuum statement appears here.

Dependencies: CA-65, CA-66, CA-67, CA-68; CONVENTIONS (a), (b), (c), (r), (s), (t). This shard executes item W1.1 of `reviews/2026-07-06_pipeline_consistency_scoping/PLAN.md`.

72.2 Placement Injections and Bare Refinements

Definition [choice]. For $k \le l$, a *placement* is an order-preserving injection

$$\varphi : \{1, \dots, k\} \hookrightarrow \{1, \dots, l\},$$

recorded as its strictly increasing slot list $(\varphi(1), \dots, \varphi(k))$. Placements compose as functions; the identity placement is $k = l$, $\varphi = \mathrm{id}$. We write Pl for the category with objects $\mathbb{Z}_{>0}$ and morphisms the placements.

Definition [choice]. Fix the site object $O = \mathbf{1} \oplus X$ of CONVENTIONS (r) and let $\iota_{\mathbf{1}} : \mathbf{1} \to O$ be the empty-site summand inclusion. The *bare refinement* attached to a placement $\varphi : [k] \to [l]$ is

$$V_\varphi : O^{\otimes k} \longrightarrow O^{\otimes l}, \qquad \text{coarse site } i \mapsto \text{fine slot } \varphi(i), \quad \text{slots } s \notin \operatorname{im}\varphi \mapsto \iota_{\mathbf{1}}(\mathbf{1}),$$

with the left-associated bracketing of CONVENTIONS (c)/(r). CA-68's dyadic vacuum insertion is the placement $\varphi_{\mathrm{dyad}}(j) = 2j - 1$.

Lemma [derived]. V_φ is an isometry, $V_\varphi^\dagger V_\varphi = \mathrm{id}_{O^{\otimes k}}$: every inserted factor contributes $\iota_{\mathbf{1}}^\dagger \iota_{\mathbf{1}} = \mathrm{id}_{\mathbf{1}}$, and tensor products of isometries are isometries (the CA-68 argument, verbatim, for an arbitrary slot pattern).

Lemma [derived]. On charge sectors, $w \mapsto V_\varphi \circ w$ maps $\mathrm{Hom}(X_c, O^{\otimes k})$ isometrically into $\mathrm{Hom}(X_c, O^{\otimes l})$ for every charge c; the total charge is preserved because composing with V_φ does not touch the source object X_c.

Observation [derived]. In the occupied-set/fusion-path basis of CONVENTIONS (r) and `src/AnyonicWordAlgebra.jl`, V_φ acts by relabelling: the occupied set S maps to $\varphi(S)$, and the cumulative-charge path repeats the running charge at every vacuum slot. This is exact, with no F-move corrections, because every associator moved past an inserted factor carries the unit label, and unit-label F-symbols are identities in the unitary gauge (CONVENTIONS (b)). Consequently the matrix of V_φ on each charge block is a zero-one matrix whose columns are distinct standard basis vectors.

72.3 Exact Composition

Theorem [derived]. For composable placements $\varphi : [k] \to [l]$ and $\psi : [l] \to [m]$,

$$V_\psi \circ V_\varphi = V_{\psi \circ \varphi}.$$

Proof sketch. Both sides place coarse site i at fine slot $\psi(\varphi(i))$ and the vacuum summand everywhere else: a slot of $[m]$ is outside $\operatorname{im}(\psi \circ \varphi)$ iff it is outside $\operatorname{im}\psi$ or is the ψ-image of a slot outside $\operatorname{im}\varphi$, and in either case it carries $\iota_{\mathbf{1}}(\mathbf{1})$. All bracketing moves involve unit labels only, so the two zero-one matrices agree column by column. □

Corollary [derived]. The map $\varphi \mapsto V_\varphi$ is a functor Pl $\to$ Hilb (isometries), and $\theta_\varphi(T) = V_\varphi T V_\varphi^\dagger$ satisfies $\theta_\psi \circ \theta_\varphi = \theta_{\psi\circ\varphi}$ exactly: each θ_φ is a $*$-isomorphism of $\mathcal{A}_k$ onto the corner $P_\varphi \mathcal{A}_l P_\varphi$, $P_\varphi = V_\varphi V_\varphi^\dagger$ (the CA-68 corner statement, now closed under composition).

Corollary [derived]. The dyadic tower composes on the nose: $V^{(2L\to 4L)}_{\varphi_{\mathrm{dyad}}} \circ V^{(L\to 2L)}_{\varphi_{\mathrm{dyad}}} = V^{(L\to 4L)}_{\varphi}$ with $\varphi(j) = 4j - 3$. This settles the semigroup question left open in the 2026-07-06 consistency map (seam S2 at finite level): the reindexing is the composite placement itself.

72.4 Cellwise Refinement and Cell-Shift Compatibility

Definition [choice]. For a chain of M cells of k sites, the uniform cellwise refinement applies one fixed cell placement $\varphi : [k] \to [l]$ in every cell; the result is the whole-chain placement $\Phi_M(\varphi) : [Mk] \to [Ml]$, cell m occupying fine slots $(m-1)l + 1, \ldots, ml$.

Lemma [derived]. On its open-chain domain, $V_{\Phi_M(\varphi)}$ intertwines whole-cell shifts: shifting the coarse chain by one cell (k sites) corresponds to shifting the fine chain by one cell (l sites). No exact fine counterpart of a *single-site* coarse shift exists when $l/k \notin \mathbb{Z}$, and none is used: cross-scale position-weighted comparisons go through the cell-shift subgroup only (CONVENTIONS (t)).

72.5 The Forest-Functor Reading

Sourced identification. The registered Brothier–Stottmeister construction assigns to any covariant functor Φ from the category of binary forests to a concrete category the inductive limit $X_\Phi = \varinjlim_t X_t$ over the directed set of trees, together with the *Jones action* of Thompson's group F on X_Φ; a *tensor* functor is characterized by the single seed $R = \Phi(Y) \in \mathrm{Hom}(\Phi(1), \Phi(1)\otimes\Phi(1))$, any pair (A, R) conversely defines one, and for tensor functors the F-action always extends to Thompson's V; the targets of record are Hilbert spaces with isometries and C*-algebras with injective $*$-morphisms.

Observation [derived]. The vacuum caret

$$R = \mathrm{id}_O \otimes \iota_{\mathbf{1}} : O \longrightarrow O \otimes O$$

is such a seed, and the forest generator $f_{j,n}$ maps to $\mathrm{id}^{\otimes j-1} \otimes R \otimes \mathrm{id}^{\otimes n-j}$ = "insert one vacuum slot after site j". Since every placement is a composite of single vacuum insertions, the placement functor of this shard is exactly the tensor functor generated by the vacuum caret; distinct forests mapping to the same placement record insertion histories, so Pl is the image of the forest category under this functor. CA-68's refinement tower therefore carries, for free and at the level of sourced theorems, an inductive-limit space and a Thompson-F (indeed V) action. **Boundary [open]:** whether that action is continuous or physically relevant is precisely the no-go territory of CA-67 (NG1–NG5, Kliesch–Koenig); nothing here claims continuity, and the Kliesch–Koenig applicability question for this charge-graded carrier is deliberately deferred to the W2.4 audit.

72.6 The Ore Dichotomy and Fallback Homes

Sourced dichotomy. Fraction groups exist for a category of refinement diagrams iff it is left-cancellative and satisfies Ore's property (morphisms with a common target admit a common extension). The registered forest-skein source proves: the free forest-skein category over a colour set S is always left-cancellative; one colour recovers Thompson's F; with more than two colours Ore's property *always fails*, and skein relations are the mechanism that can restore it.

Consequence [derived reading]. Homogeneous refinement schedules (one caret type, e.g. pure vacuum insertion) retain a Thompson-like fraction-group symmetry. Mixed-radix or multi-caret

schedules — the kind that Wave 2's content-creating dressings and filling-targeted $k \to l$ schedules will need — generically lose the group and retain only the refinement *groupoid*. The two sourced fallback homes are: divisibility-directed inductive systems over scales ordered by divisibility (the AF/hyperfinite pattern over matrix building blocks), and soft inductive systems over an arbitrary directed set, where exact transitivity is relaxed to asymptotic transitivity — the recorded destination for dressed families $W = UV$ (CA-67, CA-68).

72.7 Julia Invariants

`src/RefinementPlacements.jl` implements placements (validated strictly increasing slot lists), composition, the charge-block matrices of V_φ in the CA-66 basis, and uniform cellwise placements. The testset "refinement placement category (CA-71)" pins: the bridge $V_{\varphi_{\mathrm{dyad}}} =$ `vacuum_insertion_matrix` at $L = 1, 2, 3$; isometry on both charge blocks for all placements with $k \le 3$, $l \le 5$; exact functoriality $V_\psi V_\varphi = V_{\psi \circ \varphi}$ over an exhaustive composable sweep and the dyadic $2 \to 4 \to 8$ chain against the direct $j \mapsto 4j - 3$ placement; occupation covariance $V_\varphi n_i = n_{\varphi(i)} V_\varphi$ and $n_s V_\varphi = 0$ for $s \notin \operatorname{im} \varphi$; and the Fibonacci multiplicity table $(m_{\mathbf{1}}, m_\tau)(L) = (F_{2L-1}, F_{2L})$ at the sizes used. Each invariant is mutation-proved per Rule 6 (composition off-by-one, wrong path repetition, unmapped occupied set), with the mutation-to-failing-test record kept at the head of the testset.

72.8 Boundary

This shard defines no dressing U, selects no state, and makes no dynamical or continuum claim; the asymptotic cocycle condition for dressed families and the filling-flow constructions are Wave-2 targets (W2.1). The cell-shift lemma is an open-chain statement; periodic chains await the CONVENTIONS (r) periodic entry. The Thompson action inherited from the forest functor is an algebraic fact about the finite tower and its inductive limit; its continuity is not claimed and is exactly what the Jones and Kliesch–Koenig no-gos constrain (CA-67). Nothing here defines Koo–Saleur modes; the categorical residual set and the scale ledger of CONVENTIONS (s) are consumed by the following shards (W1.2, W1.3).

73 GNS Descent and the Corner Calculus of Placement Refinements

73.1 Status, Sources, and Dependencies

Status: local derivations with checked finite shadows, plus two recorded interface contracts. Every lemma below is an elementary finite-dimensional derivation, checked in Julia. The two items that are *not* theorems here — the residual-compatibility contract and the BU-level square — are stated as proposals with their consuming shards named. This shard settles the residual-free part of seam S1 of the 2026-07-06 consistency map; the residual part waits for the categorical residual set (W1.3).

Dependencies: CA-63, CA-65, CA-66, CA-67, CA-68, CA-71; CONVENTIONS (r), (s), (t). Executes item W1.2 of `reviews/2026-07-06_pipeline_consistency_scoping/PLAN.md`.

73.2 Setting

Fix a placement $\varphi : [k] \to [l]$ with bare refinement V_φ, corner projection $P_\varphi = V_\varphi V_\varphi^\dagger$, and corner morphism $\theta_\varphi(T) = V_\varphi T V_\varphi^\dagger$ (CA-71). States on $\mathcal{A}_l = \bigoplus_c M_{m_c(l)}(\mathbb{C})$ are block density pairs $\omega_l = (w_c, \rho_c)$, $\omega_l(T) = \sum_c w_c \operatorname{tr}(\rho_c T_c)$ (CA-66). Throughout, Ω_ω denotes the GNS cyclic vector and $[T]_\omega$ the GNS class of T.

73.3 The Corner Calculus

Lemma 72.1 [derived]. θ_φ is an injective $*$-morphism onto the corner $P_\varphi \mathcal{A}_l P_\varphi$, and

$$\theta_\varphi(S)^\dagger \theta_\varphi(T) = \theta_\varphi(S^\dagger T), \qquad \theta_\varphi(S)\,(1 - P_\varphi) = 0 = (1 - P_\varphi)\,\theta_\varphi(S).$$

Proof. $V^\dagger V = 1$ gives $VS^\dagger V^\dagger V T V^\dagger = VS^\dagger T V^\dagger$; and $VSV^\dagger(1 - VV^\dagger) = VS(V^\dagger - V^\dagger) = 0$. □

Lemma 72.2 [derived] (pullback functionals). $\omega_l \circ \theta_\varphi$ is a positive linear functional of total mass $\omega_l(P_\varphi)$; it is a state iff $\omega_l(P_\varphi) = 1$ (the CA-68 exact corner case). For $\omega_l(P_\varphi) > 0$ the *normalized pullback*

$$\omega_k := \frac{\omega_l \circ \theta_\varphi}{\omega_l(P_\varphi)}$$

is a state on $\mathcal{A}_k$. **Boundary:** translation invariance of ω_l does *not* imply translation invariance of ω_k in any sense finer than the cell-shift subgroup of CONVENTIONS (t); no invariance is claimed here.

73.4 The GNS Descent Lemma

Theorem 72.3 [derived]. Let $\omega_l(P_\varphi) > 0$ and let ω_k be the normalized pullback. Then

$$J\,[T]_{\omega_k} := \omega_l(P_\varphi)^{-1/2}\,[\theta_\varphi(T)]_{\omega_l}$$

is a well-defined isometry $\mathcal{H}_{\omega_k} \to \mathcal{H}_{\omega_l}$ satisfying, for all $S, T \in \mathcal{A}_k$,

$$J\,\pi_{\omega_k}(S) = \pi_{\omega_l}(\theta_\varphi(S))\,J, \qquad \big\| J\Omega_{\omega_k} - \Omega_{\omega_l} \big\|^2 = 2\Big(1 - \sqrt{\omega_l(P_\varphi)}\Big).$$

In particular $J\Omega_{\omega_k} = \Omega_{\omega_l}$ iff $\omega_l(P_\varphi) = 1$.

Proof. Isometry and well-definedness: $\langle J[S], J[T]\rangle = \omega_l(P)^{-1}\omega_l(\theta(S)^\dagger\theta(T)) = \omega_l(P)^{-1}\omega_l(\theta(S^\dagger T)) = \omega_k(S^\dagger T)$ by Lemma 72.1, so ω_k-null classes map to ω_l-null classes. Intertwining: $J[ST] = \omega_l(P)^{-1/2}[\theta(S)\theta(T)] =$

$\pi_{\omega_l}(\theta(S))J[T]$. Vacuum defect: $\langle J\Omega_k, \Omega_l\rangle = \omega_l(P)^{-1/2}\,\omega_l(P^\dagger 1) = \sqrt{\omega_l(P)}$, hence $\|J\Omega_k - \Omega_l\|^2 = 1 + 1 - 2\sqrt{\omega_l(P)}$. □

Reading. Every placement refinement descends to the GNS level *unconditionally*: multiplicativity survives the corner because $V^\dagger V = 1$. What is conditional is only whether the refinement is *pointed* (vacuum-preserving), and that is measured by the single scalar $1 - \omega_l(P_\varphi)$ — the same scalar that controls the completion defect below. This closes the residual-free part of seam S1.

73.5 Residual Descent: the Contract for the Categorical Residual Set

Lemma 72.4 [derived]. In the setting of Theorem 72.3, for any $R' \in \mathcal{A}_k$ and $T \in \mathcal{A}_k$,

$$\big\|\pi_{\omega_k}(R')\,[T]_{\omega_k}\big\| = \big\|\pi_{\omega_l}(\theta_\varphi(R'))\,J[T]_{\omega_k}\big\|.$$

Hence $\pi_{\omega_k}(R') = 0$ iff $\pi_{\omega_l}(\theta_\varphi(R'))$ vanishes on $\operatorname{ran} J$; in particular $\pi_{\omega_l}(\theta_\varphi(R')) = 0 \Rightarrow \pi_{\omega_k}(R') = 0$: coarse relations are inherited from fine ones through θ_φ.

Contract [proposal, consumed by W1.3]. A scale family of residual sets $\{\mathcal{R}_k\}$ is *descent-compatible* for (ω_l, φ) when $\pi_{\omega_l}(\theta_\varphi(R'))\,J = 0$ for every $R' \in \mathcal{R}_k$ — e.g. when $\theta_\varphi(\mathcal{R}_k)$ lies in the π_{ω_l}-null structure generated by $\mathcal{R}_l$. Once W1.3 defines the categorical residual densities, this is the equation their scale coherence must satisfy; nothing about it is claimed here.

Lemma 72.5 [derived] (independent families). If instead ω_k is *given* (a projective family, not the pullback), then $J[T]_{\omega_k} := [\theta_\varphi(T)]_{\omega_l}$ is well-defined and bounded iff the Gram domination $\omega_l(\theta(T)^\dagger\theta(T)) \le C\,\omega_k(T^\dagger T)$ holds for some C; the pullback case is $C = \omega_l(P_\varphi)$ with equality. This is the finite Gram condition to check when comparing two independently constructed state families.

73.6 The Unital CP Completion and Its Exact Defect

Strict OAR wants unital injective $*$-morphisms; θ_φ is not unital. For a state χ on $\mathcal{A}_k$ define the *χ-completion*

$$\Phi_\chi(T) := \theta_\varphi(T) + \chi(T)\,(1 - P_\varphi).$$

Lemma 72.6 [derived]. Φ_χ is unital and completely positive (sum of a $*$-morphism and the CP map $T \mapsto \chi(T)(1 - P_\varphi)$), with state pullback the convex mixture

$$\omega_l \circ \Phi_\chi = \omega_l(P_\varphi)\,\omega_k + \big(1 - \omega_l(P_\varphi)\big)\,\chi.$$

Theorem 72.7 [derived] (exact multiplicativity defect). For all $S, T \in \mathcal{A}_k$,

$$\Phi_\chi(ST) - \Phi_\chi(S)\Phi_\chi(T) = \big(\chi(ST) - \chi(S)\chi(T)\big)\,(1 - P_\varphi),$$

and consequently

$$\big\|\Phi_\chi(ST) - \Phi_\chi(S)\Phi_\chi(T)\big\|_{\omega_l} = \big|\chi(ST) - \chi(S)\chi(T)\big|\,\sqrt{\omega_l(1 - P_\varphi)}.$$

Proof. Expand $\Phi_\chi(S)\Phi_\chi(T)$; the cross terms vanish by Lemma 72.1, leaving $\theta(ST) + \chi(S)\chi(T)(1 - P)$; subtract from $\Phi_\chi(ST) = \theta(ST) + \chi(ST)(1 - P)$. The seminorm follows from $(1 - P)^\dagger(1 - P) = 1 - P$. □

Corollary 72.8 [derived reading]. On the matrix-block algebras $\mathcal{A}_k$ no state is multiplicative on all of $\mathcal{A}_k$ (unless a block is one-dimensional and χ is concentrated there), so the completion family $\{\Phi_\chi\}$ satisfies the soft-C*-system asymptotic homomorphism property (the sourced eq. (29) condition) *along a scale sequence precisely when* $\omega_l(1 - P_\varphi) \to 0$ in the relevant GNS seminorms. The

corner scalar of Theorem 72.3 is thus also the exact modulus of softness of the unital completion: strict corner ($\omega(P) = 1$) gives an exactly pointed descent, and $\omega(P) \to 1$ gives an asymptotically multiplicative unital CP system. Whether a physically relevant (critical) fine-state family can achieve $\omega_l(P) \to 1$ for dressed placements is exactly the W2.1 dressed-exact-corner question; nothing is claimed here.

73.7 The BU-Level Square

Proposal [contract, blocked on W1.5]. On the free word layer $\mathcal{BU}_0$ (CA-65), a placement acts on generators by slot relabelling, $\widehat{R}_\varphi(v_{a,j}) := v_{a,\varphi(j)}$, extended freely. The required compatibility square is

$$\pi_l \circ \widehat{R}_\varphi = \theta_\varphi \circ \pi_k \quad \text{on local words,} \qquad \widehat{R}_\varphi(J_{\mathrm{fus},k}) \subseteq J_{\mathrm{fus},l},$$

so that refinement is defined already at the presentation level and descends through the kinematic quotient. Its verification needs the endpoint-closing rule (the CA-65 generator-to-block bridge, scheduled as W1.5); until then the square is an interface contract, not a result.

73.8 Julia Invariants

`src/GnsCornerCalculus.jl` implements block states, corner weights $\omega_l(P_\varphi)$, normalized pullbacks, explicit finite GNS data (Gram matrices, left actions, cyclic vectors), the descent intertwiner J, and the completion Φ_χ. The testset "gns descent and corner calculus (CA-72)" pins: the corner-calculus identities of Lemma 72.1 exactly on integer matrices; the CA-68 corner regression ($\omega_l(P_\varphi) = 1$ and pullback recovery for fine states built on the corner — a regression by construction, not a probe); isometry $J^\dagger G_l J = G_k$, the intertwining relation, and the vacuum-defect scalar $2(1 - \sqrt{\omega_l(P_\varphi)})$ for faithful fine states; the residual-descent norm identity of Lemma 72.4; the exact defect identity of Theorem 72.7 with its GNS-norm equality and the convex-mixture pullback of Lemma 72.6; and one exact rational pin, $\omega_l(P_\varphi) = 5/34$ for the carrier-trace state $\mathrm{Tr}_{H_4} / \dim H_4$ under the dyadic placement (corner ranks $2 + 3$ against $\dim H_4 = 13 + 21 = 34$; the implementing agent caught and corrected the orchestrator's wrong draft pin here — recorded in the worklog). Each invariant is mutation-proved per Rule 6 (dropped J-normalization, P in place of $1 - P$ in the completion, $V^\dagger V$ in place of $VV^\dagger$), with the record at the head of the testset.

73.9 Boundary

This shard proves nothing about dynamics, translation invariance, criticality, dressings, or continuum limits; it fixes the exact finite-L calculus through which any of those must pass. The completion state χ is a non-canonical choice, and no canonical choice is proposed. The residual-descent contract and the BU-level square are named interfaces for W1.3 and W1.5. Whether $\omega_l(P) \to 1$ is achievable on physical state families (and with which dressed placements) is the W2.1 question; the Kliesch–Koenig applicability audit remains W2.4. Seam S1 is closed here only in its residual-free part, as the plan demands.

74 The Categorical Residual Set on the Anyonic Word Tower

74.1 Status, Sources, and Dependencies

Status: derived port plus one new lemma and one definition. The residual calculus below is the CA-38/CA-39/CA-42 qubit framework transported to the anyonic word tower; the transport lemma and the coherent residual family are new local derivations. The densities are the CA-70 Tier-1 proposal. Everything is stated on the *open* chain per the Wave-1 commitment of the 2026-07-06 plan; periodic mode calculus is gated behind decision D1. Nothing here claims that residual annihilation implies symmetry (CA-44/CA-64 semantics are preserved verbatim).

Dependencies: CA-38, CA-39, CA-42, CA-43, CA-63, CA-64, CA-69, CA-70, CA-71, CA-72; CONVENTIONS (n), (r), (s), (t). Executes item W1.3 of `reviews/2026-07-06_pipeline_consistency_scoping/`

74.2 Setting

Fix $(\mathcal{C}, O)$ with $O = \mathbf{1} \oplus X$ per CONVENTIONS (r), the open-chain tower $\mathcal{A}_L = \mathrm{End}_{\mathcal{C}}(O^{\otimes L})$, and a self-adjoint two-site density $h \in \mathcal{A}_2$ placed at bonds by the shift, $h_j \in \mathcal{A}_L$ for $1 \le j \le L-1$, so that $[h_j, h_k] = 0$ for $|j-k| \ge 2$ (disjoint supports). The Tier-1 dilute density of CA-70 is the default instance: a real combination of occupancy projectors, the dense exchange $u_j = b_j^t b_j$, hopping $a_j + a_j^t$, and pair terms $b_j + b_j^t$, all parity-even by CA-69. The qubit block is the $\mathcal{C} = \mathrm{Hilb}$, $O = \mathbb{C}^2$ degenerate case, which is what makes the port below a specialization-in-reverse rather than an analogy.

74.3 The Ported Local Residual Calculus

Definition [port]. For a local density u supported on m sites at position j, the coboundary is $(Du)_j = u_j - u_{j+1}$. The local residual densities are, verbatim from CONVENTIONS (n) but now as elements of $\mathcal{A}_L$:

$$p_j = i[h_j, h_{j+1}], \qquad A_j = i[h_j + h_{j+1},\, p_j], \qquad B_j = i[p_j, h_{j+1}] + 2i[p_j, h_{j+2}],$$

with the conservation and boost residual equations

$$R_j^{\mathrm{cons}}(h;u) = A_j - (Du)_j, \qquad R_j^{\mathrm{boost}}(h;u,w,\lambda,e) = B_j - u_j - \lambda\,\bar{h}_j - (Dw)_j, \qquad \bar{h} = h - e\,\mathbf{1},$$

where λ is the unconstrained multiplier of CONVENTIONS (s) (not a squared velocity) and the scalar parts of u, w are gauge, exactly as in CA-38. **Derived [port]:** nothing in the CA-38/CA-39 derivations uses the Pauli basis — only bilinearity, locality, and the shift — so every identity holds in $\mathcal{A}_L$; the Pauli *coefficient* equations of CA-38 are recovered by expanding in the matrix-unit or diagram basis of the charge blocks $\bigoplus_c M_{m_c(L)}(\mathbb{C})$.

74.4 Open-Window Mode Calculus

Definition [choice]. Fix the phase base $z_n = e^{2\pi i n/L}$ on the length-L open chain (CONVENTIONS (t): index phases at own scale). For a window $W = [j_0, j_1] \subset [1, L-1]$ define $H_n^{(W)} = \sum_{j \in W} z_n^j h_j$ and $J_n^{(W)} = \sum_{j \in W} z_n^j p_j$.

Lemma [derived]. The exact periodic identity (42.1) ports to the open window with an edge remainder:

$$[H_m^{(W)}, H_n^{(W)}] = -i(z_n - z_m)\, J_{m+n}^{(W)} + \partial_{mn}^{(W)},$$

where $\partial^{(W)}_{mn}$ is supported in the two edge cells of W (the telescoping of CA-42's derivation fails only where the window ends). The same holds for the ported (42.2)–(42.3) with their D- and F-density expansions. The Julia suite asserts the support statement directly; it is the strong form of "bulk identity modulo boundary". Periodic modes — the setting of CA-70's $L^{(N)}_n$ and of the registered periodic dilute algebra — are deliberately *not* defined here (decision D1; CA-70 obligation 3).

74.5 Witt/Virasoro Residuals with Two Central Constants

Definition [port + CONVENTIONS (s)]. For a state ω let $:X:_\omega = X - \omega(X)\mathbf{1}$ termwise on densities. The chiral ansatz on a window is

$$\ell_n = A_n{:}H^{(W)}_n{:}_\omega + B_n{:}J^{(W)}_n{:}_\omega + \mu\,\delta_{n0}\mathbf{1}, \qquad \bar\ell_n = A_n{:}H^{(W)}_n{:}_\omega - B_n{:}J^{(W)}_n{:}_\omega + \mu\,\delta_{n0}\mathbf{1},$$

and the closure residuals carry the *two-constant* central term of CONVENTIONS (s):

$$r^{++}_{mn} = [\ell_m,\ell_n] - (m-n)\ell_{m+n} - \tfrac{\delta_{m+n,0}}{12}\big(m^3 c^*_L - m\,c_L\big)\mathbf{1},$$

with r^{--}_{mn}, $r^{+-}_{mn} = [\ell_m,\bar\ell_n]$ analogous. Setting $c^*_L = c_L$ is a measured outcome or theorem target, never an input. The estimator pair $(c_L(m), c^*_L(m))$ is defined by solving $\omega([\ell_m,\ell_{-m}] - 2m\ell_0) = \frac{1}{12}(m^3c^*_L - mc_L)$ at two mode values, relative to a declared (ω, W).

74.6 The Categorical Residual Set

Definition. The scale-L residual family of the pair (h, window data) is

$$\mathcal{R}_L(h) = \big\{R^{\mathrm{cons}}_j,\ R^{\mathrm{boost}}_j\big\}_j \cup \big\{r^{\sigma\sigma'}_{mn}\big\}_{|m|,|n|\le K},$$

as elements of $\mathcal{A}_L$. This is the object whose absence was seam S3 of the 2026-07-06 consistency map: the moment/GNS constraints $\omega(X^\dagger RY) = 0$ of CA-43/CA-63, and the relaxed profiles and ν-objectives of CA-64, are now well-posed on the categorical tower by substitution. **Boundary:** CA-63's existence theorem is proved for the qubit UHF algebra; its transport to the AF tower is W1.6's categorical corollary, not a claim of this shard.

74.7 Exact Transport and the Stretched Dichotomy

Lemma 73.1 [derived] (exact transport). Let θ_φ be the corner morphism of a placement (CA-71, CA-72). Since θ_φ is a $*$-morphism (Lemma 72.1), for every noncommutative $*$-polynomial F,

$$\theta_\varphi\big(F(h_{j_1},\dots,h_{j_r})\big) = F\big(\theta_\varphi(h_{j_1}),\dots,\theta_\varphi(h_{j_r})\big).$$

Hence the *entire* residual calculus transports exactly: the θ_φ-image of every element of $\mathcal{R}_k(h)$ is the same residual built from the *stretched densities* $\theta_\varphi(h_j)$, which for the dyadic placement are supported on fine sites $\{2j-1, 2j, 2j+1\}$ with the inserted site pinned to vacuum (the CA-68 stretched-covariance phenomenon at the density level).

Observation [derived] (the dichotomy). The stretched density is *not* the fine density: $\theta_\varphi(h_j) \neq h^{(l)}_{j'}$ for any placement of the same h at fine scale. Consequently the CA-72 descent-compatibility contract *fails* for the naive pairing $(\mathcal{R}_k(h), \mathcal{R}_l(h))$: annihilating the fine residuals does not annihilate the θ-transported coarse ones. This is the precise residual-level content of "bare vacuum insertion is naive".

Definition + Proposition 73.2 [derived] (coherent family). The *stretched-augmented* residual family assigns to the fine scale

$$\mathcal{R}_l^{\mathrm{coh}} := \mathcal{R}_l(h) \ \cup \ \bigcup_{\varphi} \theta_\varphi\big(\mathcal{R}_k(h)\big)$$

over the placements φ of the chosen refinement schedule. If a state ω_l annihilates $\mathcal{R}_l^{\mathrm{coh}}$ in the CA-63 sense ($\pi_{\omega_l}(R) = 0$ for all $R \in \mathcal{R}_l^{\mathrm{coh}}$), then the normalized pullback ω_k annihilates $\mathcal{R}_k(h)$, by Lemma 72.4 applied to $\pi_{\omega_l}(\theta_\varphi(R')) = 0$. Descent-compatibility thus holds *by construction* for the coherent family. **Price [recorded]:** the fine moment problem gains the stretched constraints — a strictly stronger feasibility demand on ω_l. **Alternative [recorded, W2.1]:** a dressing U may instead be chosen so that $\mathrm{Ad}_U \circ \theta_\varphi$ carries stretched residuals asymptotically into the null structure of $\mathcal{R}_l(h)$; quantifying that is precisely the dressed-refinement task, now stated as an equation rather than a slogan.

74.8 Julia Invariants

`src/FibonacciLocalOperators.jl` implements the two-site local-operator embedding on the CA-66 path basis (the single nontrivial Fibonacci F-matrix anchored to the registered Trebst source, unit-label moves trivial per CONVENTIONS (b)) and the Tier-1 generators (occupancies, hop, raw-cup pair terms with $u^2 = \varphi u$, dense e_j). `src/CategoricalResiduals.jl` implements the ported residual densities, coboundary distances (pinv/nullspace per the CA-64 lesson), the open-window mode identity with its edge-support assertion, and the transport tests of Lemma 73.1 against the CA-71/CA-72 machinery. The testset "categorical residual set (CA-73)" pins: disjoint-support commutation; the occupancy cross-check against the independent `occupation_number_matrix`; $e_j^2 = \varphi e_j$ on the full space and the braid relation $e_j e_{j\pm1} e_j = e_j$ *on the dense corner only* (on the full dilute space $e_1 e_2 e_1 = 0$, because $u_j = b_j^t b_j$ annihilates states with an empty neighbour — the checked computational shadow of CA-69's corner statement $\pi_A \,\mathrm{dTL}\, \pi_A \cong \mathrm{TL}$); source-anchored dilute relations; parity commutation of every Tier-1 generator (the shadow of CA-69's parity finding); the edge-support lemma in its sharp form (the bulk telescopes and exactly one top-edge current survives); a checked physics value — the pure-hopping density's conservation residual is a coboundary to 10^{-14} (hopping conserves momentum) — and the dilute-image dimension data at $L = 2, 3$ (pinned to CA-69's $9, 51$) together with the computed $L = 4$ values, independently re-run by the orchestrator — recorded here as *data for CA-74*, whose interpretation belongs to that shard, not this one.

74.9 Boundary

Tier-2 (parity-odd, fusion-vertex) densities are not defined here; they are the S4/W3.1 item. Periodic modes and the periodic evaluation map remain gated behind D1 (CA-70 obligation 3; interchiral caveat recorded in the plan). No estimation protocol for $(e_\infty, v_L, c_L, c_L^*)$ is fixed here (W2.2). Residual annihilation, exact or relaxed, is never symmetry evidence (CA-44, CA-64). The transport lemma is exact but says nothing about convergence; the coherent family makes descent hold by fiat at the cost of stronger constraints, and whether any physical state family satisfies them is precisely the open state-existence frontier (W1.6, W2.1).

75 The Jones–Wenzl Kernel Decision for the Dilute Evaluation Map

75.1 Status, Sources, and Dependencies

Status: derived, with one numerical input. This shard interprets the CA-73 decision data and settles the CA-69 conjecture at $L = 4$. The combinatorics and the kernel identification are local derivations from registered sources; the single essential numerical input is the computed rank $\operatorname{rank} \rho_4 = 322$ (CA-73, re-run here). Executes item W1.4 of `reviews/2026-07-06_pipeline_consistency_scoping` orchestration record in `reviews/2026-07-06_ca74_jw_kernel_decision/`.

Dependencies: CA-66, CA-69, CA-71, CA-72, CA-73; CONVENTIONS (b), (r).

75.2 Setting and the Decision Data

Fix Fibonacci $(\mathcal{C}, O)$, $O = \mathbf{1} \oplus \tau$, per CONVENTIONS (r), the tower $\mathcal{A}_L = \mathrm{End}_{\mathcal{C}}(O^{\otimes L})$, and the CA-69 evaluation map $\rho_L : \mathrm{dTL}_L(\varphi) \to \mathcal{A}_L$. CA-73 computed (and this session re-ran) the decision data: the image of the Tier-1 dilute generators has dimension 9, 51, 322 at $L = 2, 3, 4$, equal at every L to the dimension of the parity-preserving subalgebra, and at $L = 4$ exactly one short of $\dim \mathrm{dTL}_4 = M_8 = 323$. This shard answers the two questions those numbers pose: *why* 322, and *which* element spans the kernel.

75.3 Parity-Refined Multiplicities

Derived. Write $m_c^{\pm}(L)$ for the number of CA-66 basis paths of $O^{\otimes L}$ with total charge $c \in \{\mathbf{1}, \tau\}$ and even (+) or odd (−) occupied number N. Appending a site keeps N (vacancy summand) or increments it and fuses the charge (τ summand), so

$$m_{\mathbf{1}}^{\pm}(L+1) = m_{\mathbf{1}}^{\pm}(L) + m_{\tau}^{\mp}(L), \qquad m_{\tau}^{\pm}(L+1) = m_{\tau}^{\pm}(L) + m_{\mathbf{1}}^{\mp}(L) + m_{\tau}^{\mp}(L),$$

with seed $(m_{\mathbf{1}}^{+}, m_{\mathbf{1}}^{-}, m_{\tau}^{+}, m_{\tau}^{-})(1) = (1, 0, 0, 1)$. The alternating sums $M_c^{-}(L) = m_c^{+}(L) - m_c^{-}(L)$ obey the Fibonacci recursion generated by $I - T$, $T = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, giving $M_{\mathbf{1}}^{-}(L) = F_{L+1}$, $M_{\tau}^{-}(L) = -F_L$ and the closed forms

$$m_{\mathbf{1}}^{\pm}(L) = \tfrac{1}{2}\big(F_{2L-1} \pm F_{L+1}\big), \qquad m_{\tau}^{\pm}(L) = \tfrac{1}{2}\big(F_{2L} \mp F_L\big),$$

consistent with $m_{\mathbf{1}} = F_{2L-1}$, $m_{\tau} = F_{2L}$ (CA-66, T66.2). Charge $\mathbf{1}$ is even-heavy (the all-vacuum state and the $\tau \otimes \tau \to \mathbf{1}$ channel), charge τ odd-heavy (single-τ states); the asymmetry $\sim \varphi^L$ is subdominant to $m_c \sim \varphi^{2L}$.

L	$m_{\mathbf{1}}^{+}$	$m_{\mathbf{1}}^{-}$	m_{τ}^{+}	m_{τ}^{-}
1	1	0	0	1
2	2	0	1	2
3	4	1	3	5
4	9	4	9	12
5	21	13	25	30
6	51	38	68	76

The $L = 4$ row is the observed split $13 = 9 + 4$, $21 = 9 + 12$. The parity-*preserving* subalgebra $\bigoplus_c \big(M_{m_c^{+}} \oplus M_{m_c^{-}}\big)$ has dimension

$$P(L) = \sum_c \big((m_c^{+})^2 + (m_c^{-})^2\big) = 9,\ 51,\ 322,\ 2135,\ 14445 \quad (L = 2, \ldots, 6),$$

with complement check $\dim \mathcal{A}_L - P(L) = \sum_c 2m_c^+ m_c^-$ ($610 - 322 = 288$ at $L = 4$). This derives the 322: the CA-73 datum says ρ_4 is *onto* the parity-preserving subalgebra, the maximum permitted by the parity obstruction of CA-69.

75.4 The Kernel Theorem

Convention bridge (Rule 3, recorded). SRC-TL-JONES works with $\mathrm{TL}_n(q)$, loop parameter $-(q + q^{-1})$, and $\ell = |q^2|$. Our diagram algebra has loop value $\beta = +\varphi = 2\cos(\pi/5)$ (CONVENTIONS (r); $e_j^2 = \varphi e_j$ checked in CA-73), i.e. $q = e^{i\pi/5}$, $\ell = 5$; quantum integers $[1], \ldots, [5] = 1, \varphi, \varphi, 1, 0$. The loop sign is a gauge choice ($\kappa_\tau = +1$ is sourced) and touches none of the dimension, trace, or radical statements used below; the identification is pinned by the source's own $\ell = 5$ example, whose module dimensions are exactly our path counts. In its notation the first Jones–Wenzl idempotent at the root is $E_{\ell-1} = E_4 \in \mathrm{TL}_4$; we write p_4.

Theorem 74.1 [derived; one numerical input]. Let $\iota : \mathrm{TL}_4(\varphi) \cong \pi_A \, \mathrm{dTL}_4 \, \pi_A$ be the fully-occupied corner isomorphism ($A = \{1, 2, 3, 4\}$). Then

$$\ker \rho_4 = \mathrm{span}\{\iota(p_4)\},$$

and ρ_4 maps $\mathrm{dTL}_4(\varphi)$ onto the parity-preserving subalgebra of $\mathcal{A}_4$. *Proof.* (i) p_4 *exists and is negligible:* at $\ell = 5$ the trace form on TL_4 is degenerate with one-dimensional radical generated by the unique idempotent p_4 satisfying $e_i p_4 = p_4 e_i = 0$ ($i = 1, 2, 3$), and $\mathrm{tr}_4(p_4) = 0$ [sourced :63–69, :135–141, :818]. (ii) $\iota(p_4) \neq 0$*:* p_4 has coefficient 1 on the identity diagram [sourced :792], diagrams are a basis, and ι is injective [sourced corner iso]. (iii) $\rho_4(\iota(p_4)) = 0$*:* the corner restriction $\rho_4 \circ \iota$ is the dense evaluation $\mathrm{TL}_4(\varphi) \to \mathrm{End}_{\mathcal{C}}(\tau^{\otimes 4})$ (CA-69), whose kernel is exactly the radical $\mathrm{span}\{p_4\}$: the quotient $Q_4(5) \cong \mathrm{End}(\Delta_q(1)^{\otimes 4})$ has dimension $2^2 + 3^2 = 13 = C_4 - 1$ [sourced :74–80; ledger derived]. Independently: $\rho_4 \circ \iota$ is a unital $*$-representation in the unitary gauge, so $\rho_4(\iota(p_4))$ is an orthogonal projection whose quantum trace is $\mathrm{tr}_4(p_4) = 0$; positivity of quantum dimensions (CONVENTIONS (r)) forces it to vanish. (iv) *Rank–nullity:* $\dim \mathrm{dTL}_4 = M_8 = 323$ [sourced CA-69] and $\mathrm{rank}\, \rho_4 = 322$ [numerical input, CA-73 datum, re-run this session], so the kernel is one-dimensional and (ii)+(iii) identify it. Since $322 = P(4)$ and the image lies in the parity-preserving subalgebra (CA-69), the map is onto it. □

Every ingredient except $\mathrm{rank}\, \rho_4 = 322$ is analytical and locally sourced; the numerical rank is the surjectivity half, and the Julia suite additionally confirms $\rho_4(\iota(p_4)) = 0$ to machine precision.

75.5 The Upgraded Conjecture

Conjecture 74.2 [proposal; refines CA-69]. For all $L \geq 1$, ρ_L maps $\mathrm{dTL}_L(\varphi)$ onto the parity-preserving subalgebra of $\mathcal{A}_L$, with kernel the two-sided ideal generated by the corner Jones–Wenzl embeddings, $\ker \rho_L = \langle \iota_A(p_4) : A \subseteq [L], |A| = 4 \rangle$. Exact for $L \leq 3$ (zero kernel) and $L = 4$ (Theorem 74.1; a two-sided ideal inside a one-dimensional kernel equals it). Under the conjecture,

$$\dim \ker \rho_L = M_{2L} - P(L) = 0,\ 0,\ 1,\ \mathbf{53},\ \mathbf{1066} \quad (L = 2, \ldots, 6);$$

because image $\subseteq$ parity-preserving is *proved*, these are unconditionally rigorous *lower bounds* on $\dim \ker \rho_L$ — the open half at $L \geq 5$ is surjectivity alone. For comparison, the dense kernels $\ker(\mathrm{TL}_n(\varphi) \to \mathrm{End}(\tau^{\otimes n}))$ have dimensions $C_n - F_{2n-1} = 1, 8, 43$ ($n = 4, 5, 6$); the dilute kernel is strictly larger because dTL_L glues binomially many dense corners [sourced :2314–2318, the dilute-radical decomposition]. A within-budget $L = 5$ check must be combinatorial — either exhibiting dilute words spanning all $P(5) = 2135$ parity-blocks, or computing $\dim\langle \iota_A(p_4) \rangle = 53$ through the dilute cellular structure; the 2188-dimensional rank computation is out of budget (AGENTS.md platform note).

75.6 Julia Invariants

`src/JonesWenzlKernel.jl` builds the corner unit Π_A, the corner-compressed dense generators E_j, and the Wenzl recursion $p_{r+1} = p_r - ([r]/[r+1])\, p_r E_r p_r$ on the CA-66 path basis, carrying the formal word expansion alongside the matrices. Because the local extraction of the recursion truncates its diagram factor (source note above), the testset "jones-wenzl kernel decision (CA-74)" pins the *defining* properties instead of the formula: idempotency, self-adjointness, two-sided annihilation $E_j p_k = p_k E_j = 0$ for $j < k$, and unit-word coefficient exactly 1. It further pins: $\operatorname{tr} \Pi_A = (2, 3)$ on the two charge blocks; $e_j^2 = \varphi e_j$ and the braid relation in the corner; $[1..5] = (1, \varphi, \varphi, 1, 0)$ with $[5] = 0$ at 10^{-14}; survival $\|\rho(\iota(p_2))\| = \|\rho(\iota(p_3))\| = 1$; **the decision** $\|\rho(\iota(p_4))\|_{\mathrm{op}} = 2.3 \times 10^{-15} \leq 10^{-12}$; the CA-73 cross-checks `dilute_image_dimension`$(4) = 322 =$ `parity_even_dimension`(4) (reused, not forked); and the full parity table above for $L = 2..5$ by pure path counting. 113 assertions, standalone green, re-run independently by the orchestrator; three mutations (Wenzl coefficient, recursion β, corner unit) each RED (14/25/8 failures), reverted.

75.7 Gram-Rank Deferral and Boundary

The Gram-rank kernel dictionary (descendant Gram matrices, JW $\leftrightarrow$ null-vector reading) is W2.3 and stays behind the endpoint-closing ledger W1.5/T1: Gram data need a canonical fusion-tree ONB in each fixed-S sector, which no shard yet fixes; CA-74's statement is rank-only and basis-independent, which is why it could proceed (PLAN.md W1.4). Not claimed here: any continuum or Koo–Saleur statement; a closed form for the dilute ideal $\langle \iota_A(p_4) \rangle$ at general L; surjectivity beyond $L = 4$. Recorded gaps (acquisition queue): a source with the explicit algebraic Wenzl recursion (e.g. Kauffman–Lins or Morrison arXiv:1503.00384) and the categorical negligible-morphism formalism (e.g. EGNO Ch. 8 / Goodman–Wenzl); one line of the ILZ extraction (`:666`, "$\dim Q_8 = 233$") contradicts the paper's own $\ell = 5$ formulas (610) and is treated as an extraction artifact — only $\dim Q_4(5) = 13$ is used, triple-checked.

76 The CFT Machine and Its Quantitative OAR Contract

76.1 Sources, Dependencies, and Status

This shard records Tobias's 2026-09-22 formulation of the programme and the new workspace `cft_machine/`. It does not assert a universal construction. Dependencies: CA-62, CA-67, CA-72, CA-73; CONVENTIONS (u)–(y).

76.2 The Actual Construction Target

Programme. Discover an algorithm, or functor, from categorical data to concrete lattice systems and microscopic Virasoro generators, whose limit through operator-algebraic renormalisation (OAR) and Wilson's triangle is a rigorous CFT with the input modular data. Indeterminate-number anyon spaces are one mechanism; they do not restrict the search.

A successful output names the microscopic algebras and Hamiltonians, refinement maps, renormalized states, symmetry generators, limiting local CFT, and modular-data comparison. Any additional conformal input must be explicit. Matching a finite spectrum, producing limiting commutators, or checking a finite moment problem does not supply the complete output.

The new workspace investigates a source-controlled fermion seed, a graph-norm convergence criterion, and equivariant fixed-point constructions alongside the anyonic tower. The source-grounded comparison and scoped claims are in `cft_machine/research/`.

76.3 A Quantitative Wilson-State Theorem

Theorem WT-1 [local derivation]. Assume unital C*-algebras A_k and coherent unital injective $*$-maps $\alpha_n^k : A_k \to A_n$ for $n \geq k$. Let ω_n be microscopic states. Suppose for each fixed k there are nonnegative $d_{k,n}$ with

$$\|\omega_{n+1} \circ \alpha_{n+1}^k - \omega_n \circ \alpha_n^k\| \leq d_{k,n}, \qquad \sum_{n \geq k} d_{k,n} < \infty.$$

The norm is the dual operator norm. Set $t_k(n) = \sum_{j \geq n} d_{k,j}$. **Prove** there is a unique state ω on the inductive algebra whose restrictions are the horizontal row limits, and

$$\|\omega \circ i_k - \omega_n \circ \alpha_n^k\| \leq t_k(n).$$

⟨1⟩1. Every row is norm Cauchy. BY telescoping adjacent differences and summability; for $m > n$ the difference is bounded by $\sum_{j=n}^{m-1} d_{k,j}$.

⟨1⟩2. Its limit ω_k is a state with the stated tail bound. BY completeness of the dual, closure of positivity and normalization under norm limits, and ⟨1⟩1.

⟨1⟩3. The limiting family is compatible: $\omega_l \circ \alpha_l^k = \omega_k$. BY coherence, the analogous equality holds for every microscopic scale $n \geq l$; take row limits.

⟨1⟩4. Define $\omega(i_k(a)) = \omega_k(a)$ on the dense algebraic union. It is well-defined, bounded, positive and unital. BY ⟨1⟩2–⟨1⟩3; continuous extension gives existence and density gives uniqueness.

⟨1⟩5. The GNS maps $[a]_k \mapsto [\alpha_l^k(a)]_l$ are compatible isometries preserving cyclic vectors. BY multiplicativity, unitality, and $\omega_l(\alpha_l^k(a^*a)) = \omega_k(a^*a)$.

⟨1⟩6. QED BY ⟨1⟩2–⟨1⟩5.

Effective corollary WT-2. If expectations and refinements are computable and the tail has a computable modulus, an observable $a \in A_k$ with a supplied computable bound $B \geq \|a\|$ can be evaluated in the limit to error η: choose n with $t_k(n)B < \eta/2$ and compute its row expectation to error $\eta/2$. The modulus is substantive analytic input; finitely sampled decay does not provide it.

76.4 The Generator Contract and Its Boundary

Conditional local theorem VR-1. The companion proof in `cft_machine/research/virasoro_routes.md` constructs limiting modes and all their finite products under four explicit hypotheses: compatible isometric refinements with a positive auxiliary control operator, uniform bounds between successive graph norms, summable refinement defects in those norms, and vanishing weighted closure residuals. Its proof uses complete graph spaces and a direct telescoping identity for products.

The finite identity embeds as a projection P_N, with $P_N \to 1$ on the common smooth domain. A corner embedding is not identified with a unital observable map. Both central coefficients of CONVENTIONS (s) remain explicit. Adjoint pairing on the domain does not prove essential self-adjointness.

These two contracts must use the *same* state and representation system. WT-1 alone constructs no dynamics; VR-1 alone constructs no local OAR algebras. Positive energy, integration of smeared modes, local-net axioms, rationality, and modular fidelity need separate proofs or carefully verified applications of existing theorems.

Numerical scope. The fermion benchmark separately checks a generator-refinement square and the pullbacks of finite Dirac ground-state covariances. Its compatible limiting chiral sea is not substituted for these nontrivial Wilson rows. The corresponding source and run paths are listed in `INDEX.md` and the benchmark README.

77 A Supported Ising Modular-Data Compiler

77.1 Sources, Dependencies, and Scope

Status: explicit supported construction using existing continuum theorems. The code and structured derivations live in `cft_machine/compiler/`; the source-controlled numerical backend is `cft_machine/benchmarks/fermion/`. This is not a constructor for arbitrary categories. Dependencies: CA-81; CONVENTIONS (u), (w), (y), (z).

77.2 Exact Input and Concrete Output

In order $(1, \sigma, \psi)$ the supported seed has

$$S = \frac{1}{2}\begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix}, \qquad \Theta = \mathrm{diag}(1, e^{\pi i/8}, -1), \qquad c = \frac{1}{2}.$$

The compiler compares S, twists, fusion multiplicities and the selected real c exactly, allowing unit-preserving relabelling. Rational polynomial arithmetic modulo $z^8 + 1$ represents $z = e^{\pi i/8}$ and $\sqrt{2} = z^2 - z^6$. Character T retains its separate vacuum phase $e^{-2\pi i c/24}$; it is not coerced into this smaller cyclotomic field. Explicit factor lists declare products; no general factorization algorithm or reconstruction of arbitrary F/R data is claimed.

At even size M, take antiperiodic momenta $r = -M/2 + 1/2, \ldots, M/2 - 1/2$ and $\epsilon_M = 2\pi/M$. The finite microscopic mode matrix is the source-modified weighted shift

$$A_{M,n} e_r = \cos^2(\epsilon_M n/4) \frac{\sin(\epsilon_M (r - n/2))}{\epsilon_M} e_{r-n}$$

when both labels are in the window, and zero otherwise. The normal-ordered self-dual quadratic $Q_M(A_{M,n})$ includes the factor $1/2$ and the vacuum scalar; the recipe emits its actual CAR terms. The $n = 0$ sine Hamiltonian is nearest-neighbour in the real spatial basis, with an antiperiodic seam. Its positive-momentum occupation energies are positive, so the declared sea is its finite ground state. This chiral Hamiltonian is distinct from the auxiliary two-component Dirac benchmark.

The algebra map refines CAR generators by the isometric zero-padding map, $\Psi(f) \mapsto \Psi(Jf)$. It is unital and preserves conjugation and parity. A one-particle corner matrix JAJ^* is not substituted for this CAR algebra map.

77.3 The Local Observable Theory

The selected finite observable algebra is the even self-dual CAR algebra. For k factors take independent parity fixed points $(\mathbb{Z}_2)^k$. Total parity alone would retain odd-times-odd fields and give a different observable extension. Refinements are equivariant and the sea states are compatible. Thus these initial chiral states give constant Wilson rows; CA-81's nonconstant Dirac rows are a separate example of the same formalism.

The local reconstruction uses norm limits of even products of smoothly sampled, interval-supported real antiperiodic test sections. Its represented weak closure is the output interval algebra. This prescription defines the sampled-generator local net; it does not assert equality with a maximal construction using every arbitrary bounded scaling sequence.

Local convergence argument [structured derivation]. Assume a smooth real test section f supported inside a fixed interval chart, sampled as $v_M(x) = \sqrt{\epsilon_M} f(x)$. **Prove** its sharp Fourier embedding converges to f in $L^2(dx)$.

⟨1⟩1. Every fixed Fourier coefficient converges. By the Riemann sum for the coefficient in the orthonormal basis $e^{irx}/\sqrt{2\pi}$.

⟨1⟩2. The norms converge to $\|f\|_2$. By unitarity of the discrete Fourier transform and the Riemann sum $\epsilon_M \sum_x |f(x)|^2 \to \int |f|^2$.

⟨1⟩3. Weak convergence and convergence of norms give strong convergence. By density of finite Fourier sums, ⟨1⟩1, boundedness from ⟨1⟩2, and expansion of the squared Hilbert-space difference.

⟨1⟩4. QED by ⟨1⟩3.

CAR norm continuity then gives convergence of every fixed even field product. Density of supported smooth tests identifies the generating local even CAR algebra. The NS vacuum representation and the sourced even-net identity identify its vacuum interval net with $\mathrm{Vir}_{1/2}$. The universal global circle observable algebra is not identified with a plain even CAR algebra; the source explicitly distinguishes them.

77.4 Symmetry and Modular Fidelity

The source's modified-KS theorem supplies the strong finite-energy limit and convergence of its unitary implementers in the Majorana representation. The finite Schwinger computation separately checks the self-dual central coefficient $(n^3 - n)/24$; it is not a central term of a finite matrix commutator without normal ordering.

The complete-net sector theorem at minimal-model parameter $m = 3$ gives three sectors, with weights $0, 1/16, 1/2$, Ising fusion and the displayed twists and S. This supplies fidelity of the identified local net rather than guessing its sectors from finite spectra.

For products, the compiler retains each factor's stress family and their placed sum. The sum alone is not the full tensor-product net. The sourced tensor-sector completeness result and factorwise braiding give product modular data and additive central charge. The output explicitly records the relation between input labels and canonical product labels.

This construction establishes a supported branch. General category inputs, other Ising chiralities, arbitrary central-charge lifts, and general tensor functors remain outside its implemented domain. The broader machine must expand that domain with additional constructions and proofs.

78 Coset Lattice Stress Generators with Exact Core Stabilization

78.1 Sources, Dependencies, and Scope

Status: source-based recipe with a local stabilization proof and exact finite witnesses. This constructs stress generators inside an ambient fermion theory. It does not yet construct the local coset lattice theory itself. Dependencies: CA-81, CA-82; CONVENTIONS (z).

78.2 An Explicit Microscopic Polynomial

Take $k+1$ two-flavor complex fermion copies, $k \geq 1$, with half-integer momenta in the finite window and the filled negative sea. Let $t^a = \sigma_a/2$ and use the no-wrap one-particle shifts

$$j^a_{M,p}(e_r \otimes v) = e_{r-p} \otimes t^a v$$

when both labels are in the window, and zero otherwise. Normal-ordered complex second quantization gives finite quadratic currents $J^{u,a}_{M,p}$. Let I sum the first k copies, J denote the last, and $D = I + J$. For a level-l family X define

$$\mathsf{S}^{[l]}_{M,n,R}(X) = \frac{1}{l+2} \sum_{a=1}^{3} \sum_{\substack{|p| \leq R \\ |n-p| \leq R}} :X^a_{M,n-p} X^a_{M,p}:,$$

$$C_{M,n,R} = \mathsf{S}^{[k]}(I) + \mathsf{S}^{[1]}(J) - \mathsf{S}^{[k+1]}(D).$$

The normal ordering orders current modes, keeping their contraction terms when the displayed products are expanded. Thus C is an explicit CAR polynomial of degree at most four, not an unspecified continuum operator.

The infinite current cocycle has coefficient $p\delta_{ab}/2$ at level one: exactly p Fermi crossings contribute $\mathrm{Tr}(t^a t^b) = \delta_{ab}/2$. Independent copies add levels. The sourced GKO result gives

$$c = \frac{3k}{k+2} + 1 - \frac{3(k+1)}{k+3} = 1 - \frac{6}{(k+2)(k+3)}.$$

This fixes the normalization before any finite computation.

78.3 Exact Stabilization on Free-Energy Cores

Theorem CR-2 [local derivation]. Assume the no-wrap finite currents above, compatible sea GNS embeddings, and auxiliary excitation energy

$$E_M = \sum_{u,f,r>0} r\, n_{u,f,r} + \sum_{u,f,r<0} |r|(1 - n_{u,f,r}).$$

Let P_e project onto auxiliary energy at most e. **Prove** the embedded $C_{M,n,R}$ agrees with the continuum coset mode C_n on P_e whenever

$$R > e + |n|, \qquad M/2 - 1/2 > e + 2R.$$

⟨1⟩1. $[E_M, J_{M,p}] = -pJ_{M,p}$. By each nonzero bilinear moving occupation from r to $r-p$; the scalar subtraction has $p = 0$. Consequently $J_{M,p}P_e = 0$ for $p > e$ by nonnegativity of E_M. The same holds in the infinite sea.

⟨1⟩2. Every omitted normally ordered current pair annihilates P_e. BY its indices summing to n: if either absolute value exceeds R, one index is positive and greater than e. Normal ordering places that mode on the right; use ⟨1⟩1.

⟨1⟩3. Retained products agree with continuum products in the safe window. BY deviations from the sea lying within $|r| \le e$, one current action involving only $|r| \le e + |p|$, and two retained actions involving only $|r| \le e + 2R$.

⟨1⟩4. The finite retained sum equals the continuum sum on P_e. QED BY ⟨1⟩2–⟨1⟩3.

The choice $R_M = \lfloor M/8 \rfloor$ eventually meets both conditions for every fixed e, n. A fixed word of modes uses the enlarged core $e + \sum_j |n_j|$ and therefore also stabilizes exactly. The symmetric two-index cutoff preserves finite adjoints; same-sign pairs must be grouped before taking adjoints, and zero-index pairs use the exact same-component commutation of J_0^a with J_p^a.

Vacuum corollary CR-3. For $n \ge 1$, the smaller conditions $R \ge n$ and $M/2 - 1/2 > n$ suffice for $C_{M,-n,R}\Omega = C_{-n}\Omega$. Indeed, positive and zero currents annihilate the sea, so only two negative indices of total magnitude n contribute; all involved orbitals have $|r| \le n$.

78.4 Exact Witnesses and the Remaining Interface

The sparse-CAR producer evaluates the actual polynomials with exact complex rational amplitudes and canonical fermion signs. No many-body matrix is allocated. At $M = 8, 16$ and $R = 3, 4$, it gives

k	c	$\|C_{-2}\Omega\|^2$	$\|C_{-3}\Omega\|^2$
1	1/2	1/4	1
2	7/10	7/20	7/5

The independent oracle is $c(n^3 - n)/12$, not a fitted central charge. The run distinguishes the general CR-2 window from the smaller CR-3 vacuum window. CAR signs, level addition, adjoint pairings and refined descendants are checked separately; coefficient and embedding-sign mutations are rejected.

What remains is a coherent *local coset observable* construction. The ambient fermion tower retains additional sectors. A stress difference does not remove them, and finite commutants or vacuum-cyclic compressions have not been shown to commute with refinement. Global finite-scale positivity and microscopic ground states of the quartic Hamiltonian are also unproved. The result solves a generator and cutoff-control problem; it does not yet enlarge the completed modular-data compiler beyond CA-82.

79 Coset Positivity and a Scalar-Augmented OAR Tower

79.1 Sources, Dependencies, and Scope

Status: local derivations and exact finite witnesses. Dependencies: CA-83, CH-1–CH-3 in `cft_machine/research/coset_hamiltonian.md`, CO-1–CO-3 in `cft_machine/research/coset_observable_system.md`; CONVENTIONS (aa).

79.2 The Raw Finite Zero Mode Is Not Positive

For $k = 1$, fill flavor 1 at every momentum in both copies and leave flavor 2 empty. Let v be this normalized polarized occupation vector. For even M and $0 \le R < M$, the raw coset polynomial of CA-83 obeys

$$\langle v, C_{M,0,R} v \rangle = -\frac{(M-R)(M-R-1)}{12}.$$

Proof CH-1 [structured calculation].

⟨1⟩1. In each copy, $\sum_a \|J_0^a v\|^2 = M^2/4 + M/2$: the third current has eigenvalue $M/2$, and each of the other two flips one of M flavors with amplitude magnitude $1/2$. The two-copy diagonal current has squared norm $M^2 + M$.

⟨1⟩2. The zero-mode Sugawara difference is $2(M^2/4 + M/2)/3 - (M^2 + M)/4 = -M(M-1)/12$.

⟨1⟩3. For $1 \le p \le R$, each copy has $M - p$ retained flavor-flip transitions. Counting the two normally ordered terms gives a coset contribution $(M-p)/6$.

⟨1⟩4. Sum ⟨1⟩3 and add ⟨1⟩2. QED by finite summation.

Thus the sea, whose raw energy is zero, is not a ground state when $R < M - 1$. No diagonalization is needed. Exact checks give

M	$R = 3$	$R = 4$
8	$-5/3$	-1
16	-13	-11

The witness has auxiliary energy $M^2/2$, so it does not contradict stabilization on a fixed low-energy subspace as M grows.

79.3 Computable Vacuum-Descendant Spaces

Construction CH-2. Let V_b be the span of the vacuum and words $C_{-n_1} \cdots C_{-n_t}\Omega$ with $n_j \ge 1$ and $\sum_j n_j \le b$. Construct them using the finite CAR polynomials at

$$R_b = 2b + 1, \qquad M_b = 16(b+1).$$

There are finitely many words. At every intermediate grade at most b, CR-2's conditions hold: $R_b > 2b$ and $M_b/2 - 1/2 = 8b + 15/2 > b + 2R_b$. Thus the finite vectors agree exactly with their continuum representatives.

Exact Gram elimination removes linear dependencies. If the retained column vectors form U_b and $G_b = U_b^* U_b$, then $P_b = U_b G_b^{-1} U_b^*$ is their orthogonal projector. Refinement identifies the same words across scales, so the isometries $J_{Bb} : V_b \to V_B$ compose exactly. On these spaces both E and C_0 act by total grade. The grade-zero space is the vacuum line.

The implemented preview through grade three uses $(\Omega, C_{-2}\Omega, C_{-3}\Omega)$ when present, with exact Gram diagonal $(1, c/2, 2c)$ and grades $(0, 2, 3)$. It uses the smaller CR-3 vacuum windows at $M = 8, 16$, not a large computation of the general schedule.

Positive Hamiltonians CH-3. The ambient operator

$$H_b = P_b E P_b + (b+1)(1 - P_b)$$

is positive and has unique vacuum ground state: its first block has nonnegative grades and one zero vector, while its other block is strictly positive. This is a different Hamiltonian from the raw quartic zero mode. Its global projection currently has no proved local fusion-space realization.

79.4 An Independent Scalar Gives Genuine Unital OAR

Define $A_b = B(V_b) \oplus \mathbb{C}$ and, for $B \geq b$,

$$\alpha_{Bb}(a, \lambda) = \big(J_{Bb} a J_{Bb}^* + \lambda(1 - J_{Bb} J_{Bb}^*), \lambda\big).$$

The scalar is independent of a; replacing it with a state expectation would generally preserve only complete positivity, not multiplication.

Proof CO-1.

⟨1⟩**1.** The two first-coordinate summands have orthogonal supports. Multiplication and adjoints are preserved, and $(1, 1)$ maps to $(1, 1)$.

⟨1⟩**2.** Compressing by J_{Bb}^* recovers a; the second coordinate recovers λ. The map is injective.

⟨1⟩**3.** The complementary projection splits into the newly added shell and the transported previous shell. Substitution proves $\alpha_{DB}\alpha_{Bb} = \alpha_{Db}$.

⟨1⟩**4.** The states $\omega_b(a, \lambda) = \langle \Omega, a\Omega \rangle$ are compatible. Their Wilson rows are constant. The norm inductive limit is $K(\mathcal{H}_{\text{vac}}) + \mathbb{C}1$, and the vacuum GNS is $\mathcal{H}_{\text{vac}} = \overline{\bigcup_b V_b}$, since finite-rank operators act cyclically on Ω.

⟨1⟩**5.** QED by ⟨1⟩1–⟨1⟩4.

The finite Hamiltonian $(E|_{V_b}, b+1)$ is positive. Its dynamics intertwine exactly: the old subspace and the new shell are energy-reducing, and the embedded coarse algebra is scalar on the new shell. The Hamiltonians themselves need not intertwine.

79.5 What the Construction Does Not Identify

This is a new constrained observable prescription. It is generally not the restriction of the original unital CAR refinement, which retains all new-mode multiplicities. The new map instead inserts the selected sea when it transports the matrix block.

Likewise the ambient penalized resolvents tend to $P(E + i)^{-1}P$, where P projects onto the limiting vacuum module. This has a kernel on the discarded ambient sectors and is not a self-adjoint resolvent on that entire ambient Hilbert space. The new vacuum GNS representation is essential.

Global constraints can be intrinsic fusion constraints. Their presence is not an obstruction to locality. What remains unproved is the identification of these descendant spaces with categorical fusion spaces, together with a local realization of the Hamiltonian and refinement operations on them.

80 Smooth Coset Symmetry Limits and Local Reconstruction

80.1 Sources and Scope

Dependencies: CA-83, CA-84; SC-1/SC-2 in `cft_machine/research/smeared_coset_limits.md`; CO-2/CO-3 in `cft_machine/research/coset_observable_system.md`; CONVENTIONS (ab).

80.2 Three Cutoffs and an Ambient Operator Theorem

For a real smooth function f on the circle, set $\widehat{f}_n = (2\pi)^{-1}\int f(x)e^{-inx}\,dx$ and

$$\|f\|_{3/2} = \sum_n |\widehat{f}_n|(1+|n|^{3/2}).$$

The microscopic quartic operator is

$$T_M(f) = \sum_{|n|\le K_M} \widehat{f}_n C_{M,n,R_M}, \qquad K_M = \lfloor M/16\rfloor, \quad R_M = \lfloor M/8\rfloor.$$

Its lift $A_M(f)$ uses the unital CAR refinement, not a one-particle or many-body corner extension. Each finite lift is bounded self-adjoint; its norm is not assumed uniformly bounded in M.

Common-core lemma SC-1. The algebraic finite-free-energy space decomposes orthogonally into unitary highest-weight coset modules. At the lowest unused free grade, positive modes annihilate; diagonalize the Hermitian C_0 there and generate descendants. Adjoint pairing preserves orthogonal complements. Finite free-grade multiplicity makes this procedure exhaust the space. The positive norm identity $\|C_{-1}v\|^2 = 2h\|v\|^2$ gives nonnegative lowest weights.

The sourced Carpi–Weiner bound is uniform in lowest weight at fixed c:

$$\|T(f)\xi\| \le r_c\|f\|_{3/2}\|(1+C_0)\xi\|.$$

It therefore applies to the module sum. The self-adjoint closures and their resolvents form direct sums, and the finite-free-energy vectors are a common core. This modulewise argument does not declare the whole ambient sector sum to be a single projective vacuum representation: its rotation phases can differ.

Theorem SC-2 [structured proof]. Prove $A_M(f)$ converges to $T(f)$ on that core, in strong resolvent sense, and through its one-parameter unitaries.

⟨1⟩**1.** On P_e, every retained microscopic mode agrees exactly with its continuum mode once $R_M > e+K_M$ and $M/2-1/2 > e+2R_M$. The displayed schedules meet both conditions eventually; $M \ge 32(e+1)$ suffices.

⟨1⟩**2.** Hence the remaining error is only a continuum Fourier tail, bounded by

$$r_c\|(1+C_0)\xi\| \sum_{|n|>K_M} |\widehat{f}_n|(1+|n|^{3/2}) \longrightarrow 0.$$

⟨1⟩**3.** For core vectors,

$$(A_M - i)^{-1}(T-i)\xi - \xi = (A_M-i)^{-1}(T-A_M)\xi \longrightarrow 0.$$

The resolvent norms are at most one, and $(T-i)$ maps a graph core densely into the Hilbert space. Extend the limit by density; similarly at $-i$.

⟨1⟩4. The self-adjoint strong-resolvent theorem gives strong convergence of e^{itA_M}, uniformly on compact t intervals. Finite unitary products converge by telescoping and unit norm.

⟨1⟩5. QED by ⟨1⟩1–⟨1⟩4.

This is an ambient microscopic symmetry limit. It does not by itself remove the extra current and charge sectors or prove positivity of the raw finite Hamiltonian, which CA-84 disproves.

80.3 Local Fields in the Descendant OAR Representation

In the vacuum module of CA-84, grade selection gives the exactly finite compression

$$T_b(f) = P_b T(f) P_b = \sum_{|n|\le b} \widehat{f}_n P_b C_n P_b.$$

The CAR construction computes every retained matrix element using the safe schedule of CH-2. A computable smearing supplies only finitely many requested Fourier coefficients at each stage; no large many-body exponentiation was performed in the numerical work.

The algebra element $(e^{itT_b(f)}, 1)$ is represented as identity off V_b. Its strong limit is $e^{itT(f)}$: the zero-extended finite compressions converge on the essential core, and the preceding resolvent argument applies. The complementary scalar matters; a bare corner unitary would omit identity on the remaining Hilbert space.

Let $\mathcal{B}(I)$ be the sup-norm C*-algebra of sequences generated by these bounded unitary approximants with real smooth f supported in I. The strong-star limit map is a $*$-homomorphism on this generated algebra. Its kernel is a closed ideal, and the quotient identifies with the C*-algebra generated by the limiting stress unitaries. Vacuum GNS weak closures give the sourced minimal-model Virasoro net and its complete sector data.

This represented field construction is additional to the genuine strict OAR norm limit $K(\mathcal{H}_{\text{vac}}) + \mathbb{C}1$. Strong-star-null sequences are not generally norm-null, so the quotient is not relabelled as the norm-soft inductive limit of the source.

80.4 The Local Fusion-Space Question

The finite descendant matrices currently do not retain spatial intervals: finitely many Fourier moments of functions supported in any nonempty interval can be independently specified. Nor can refinement preserve every same-smearing compression exactly, since a smearing can have all low moments zero and higher moments nonzero.

These facts concern this specific encoding, not global fusion constraints in general. The next task is to identify a categorical fusion-space basis and local interactions/refinements that realize the required matrices and limits. No such identification or bounded-range implementation is claimed in the present work.

81 String-Net Histories as Trotter Expansions: the Spacetime Dictionary

81.1 Status, Sources, and Dependencies

Status: derived. Lemma 86.1 and Proposition 86.4 are checked local derivations; the face weights, branch data and central charges they use are source-local; the final subsection is a proposal. Reproducibility: the trigonometric identities of Proposition 86.4 were evaluated numerically for $L = 2, \dots, 5$ on both branches (worklog chunk 016); a Julia port is an open item.

Dependencies: CA-65, CA-66, CA-69, CA-74; CONVENTIONS.md (a), (b), (r), (ak).

81.2 The Levin–Wen Rules and What They Fix

Source-local. Levin and Wen specify a fixed-point wave function Φ on string-net configurations by four local linear relations: (1) topological invariance, (2) a closed type-i loop contributes d_i, (3) a bubble on a string is δ_{ij}, (4) the F-move. Rules (2)–(3) are motivated by scale invariance of an RG fixed point, rule (1) by the expectation that topological phases have topologically invariant fixed points, and rule (4) is the simplest completing constraint. The plaquette coefficients $a_s = d_s / \sum_i d_i^2$ are not chosen: they are the unique choice with a smooth continuum limit. The lesson carried forward is that a *principle* (scale invariance of the fixed point) fixes the model.

81.3 The Trotter Dictionary

Setting (CONVENTIONS (r), (ak)). $\mathcal{C}$ unitary fusion, $O = \mathbf{1} \oplus X$, $A_L = \mathrm{End}_{\mathcal{C}}(O^{\otimes L})$. A nearest-neighbour Hamiltonian is $H = -\sum_j h_j$ with $h_j \in \mathrm{End}_{\mathcal{C}}(O \otimes O)$ acting on sites $j, j+1$. Fix a basis of *elementary tiles* D_α of $\mathrm{End}_{\mathcal{C}}(O \otimes O)$ consisting of morphism diagrams (vacancy projectors, through-strings, cup, cap, hops, and for non-loop categories the fusion vertices), and expand $h_j = \sum_\alpha c_\alpha D_\alpha^{(j)}$ with $c_\alpha \in \mathbb{C}$. Brick-wall Trotter with step $\delta\tau$:

$$U(\delta\tau) = \prod_{j \text{ odd}} e^{\delta\tau h_j} \prod_{j \text{ even}} e^{\delta\tau h_j}, \qquad e^{-\beta H} = U(\delta\tau)^M + O(\beta\,\delta\tau) \quad (\beta = M\delta\tau)$$

on a finite chain, by the Lie–Trotter formula. A *history* Γ assigns to each plaquette of the M-row brick lattice either the identity tile or one elementary tile $D_{\alpha(p)}$. Composing the tiles row by row gives a morphism $\mathrm{Ev}_{\mathcal{C}}(\Gamma) \in A_L$, the $\mathcal{C}$-evaluation of the spacetime string-net drawn by Γ.

Lemma 86.1 [derived; tile expansion]. To first order per plaquette,

$$U(\delta\tau)^M = \sum_\Gamma \Big(\prod_{p:\, \alpha(p) \neq \mathrm{id}} \delta\tau\, c_{\alpha(p)} \Big)\, \mathrm{Ev}_{\mathcal{C}}(\Gamma) \;+\; R,$$

where R collects every term carrying a second-order factor $O(\delta\tau^2 \|h_j\|^2)$ on at least one plaquette. **Proof.** $e^{\delta\tau h_j} = 1 + \delta\tau h_j + O(\delta\tau^2\|h_j\|^2)$ per plaquette. Expanding $\prod_p (1 + \delta\tau h_{j(p)})$ and each h_j in tiles gives exactly the sum over histories with one tile per plaquette; composition of the tiles in the brick order is the composition in A_L, which is $\mathrm{Ev}_{\mathcal{C}}(\Gamma)$. The remainder collects all terms with a second-order factor on at least one plaquette. □

Corollary 86.2 [the Levin–Wen rules are automatic]. $\mathrm{Ev}_{\mathcal{C}}(\Gamma)$ obeys rules (2), (3), (4): a closed X-loop evaluates to d_X by the raw cup normalisation $v^\dagger v = d_X\, \mathrm{id}_{\mathbf{1}}$ of (r); a bubble between simple labels is δ by Schur; recoupling is the F-move. Rule (1), isotopy, holds for $\mathrm{Ev}_{\mathcal{C}}$ alone (planar

isotopy of morphism diagrams in a pivotal category) and fails for the full amplitude, which carries the geometric factor $\prod_p \delta\tau\, c_{\alpha(p)}$ depending on how many plaquettes each string traverses. Hence

$$\text{amplitude}(\Gamma) = \underbrace{\mathrm{Ev}_{\mathcal{C}}(\Gamma)}_{\text{topological}} \times \underbrace{\textstyle\prod_p w(p)}_{\text{geometric}}, \qquad w(\mathrm{id}) = 1,\ w(\alpha) = \delta\tau\, c_\alpha.$$

This is the spacetime reading made precise: rules (2)–(4) intact, rule (1) replaced by anisotropic tile weights (identity tiles weigh 1, all others $O(\delta\tau)$); isotropy can only emerge in the continuum.

Proposition 86.3 [derived; exactness on an integrable line]. Suppose the tiles carry weights $\rho_\alpha(u)$ forming a face operator $X_j(u) = \sum_\alpha \rho_\alpha(u) D_\alpha^{(j)}$ with $X_j(0) = N(0)\cdot 1$, whose diagonal-to-diagonal (brick-wall) transfer matrices $T(u) = \prod_{j\ \mathrm{odd}} X_j(u) \prod_{j\ \mathrm{even}} X_j(u)$ commute for all u, as the sourced $\hat\tau_D$ does. Then $h_j = X_j'(0)/N(0) - (N'(0)/N(0))\,1$ satisfies $1 + \delta\tau h_j = X_j(\delta\tau)/N(\delta\tau) + O(\delta\tau^2)$, the Trotter product is $T(\delta\tau)^M$ up to normalisation, and H commutes with $T(u)$ for every u. In particular the ground state of H is a common eigenvector of the transfer matrices at every u, including the isotropic point.

Proof. The first two claims are Taylor expansion. For the third, the Taylor coefficients of a commuting one-parameter family commute with every member of the family, and H is the first coefficient of $-\log T(u)$. □ On an integrable line no perturbative matching of rates is needed: u interpolates exactly between the Hamiltonian limit and the isotropic point.

81.4 Dense Calibration

Take X with $d = d_X$ and $H = -\sum_j e_j$, $e_j = b_j d_j$ the cup-cap generator of CA-66 ($e_j^2 = d\, e_j$). Because $e_j^2 \propto e_j$,

$$e^{\beta e_j} = 1 + w(\beta)\, e_j, \qquad w(\beta) = \frac{e^{\beta d} - 1}{d} = \beta + O(\beta^2),$$

so the brick-wall expansion is *exact* with two tiles per plaquette and

$$\langle \mathrm{out}|U^M|\mathrm{in}\rangle = \sum_\Gamma w^{\#\{e\text{-tiles}\}}\, d^{\#\{\text{closed loops}\}}\, \langle \mathrm{out}|\Gamma_{\mathrm{open}}|\mathrm{in}\rangle.$$

This is the sourced diagonal Temperley–Lieb transfer matrix at small spectral parameter: with $d = \sqrt{Q} = 2\cos\gamma$, $\log(Q^{-L/2}\epsilon_y^L \hat\tau_D(u)) = (u/\sin\gamma)\sum_j e_j + O(u^2)$, a commuting family, so Proposition 86.3 applies. For $d = \varphi$, $\gamma = \pi/5$, the antiferromagnetic sign is the golden chain, critical with $c = 7/10$ by the exact RSOS map. The dense chain has no continuous parameter at all: the category fixes γ and the remaining choice is a sign.

81.5 Dilute Calibration

Tiles (CONVENTIONS (ak)). The dilute face operator is a linear combination of nine tiles. With in-nodes on the left and bottom edges (sites j, $j+1$) and out-nodes on the top and right edges, the tiles are, in the source's numbering: 1: both vacant, P_{00}; 2, 3: one through-string, P_{X0}, P_{0X}; 8: two through-strings, P_{XX}; 9: cap-then-cup, e_j; 4, 5: cap d_j and cup b_j; 6, 7: hops $h_j^{\pm}$. The sourced weights, normalised so that $\rho_{1,2,3,8}(0) = 1$ and $\rho_{4,\ldots,7,9}(0) = 0$, are

$$\rho_1 = 1 + \tfrac{\sin u\,\sin(3\lambda - u)}{\sin 2\lambda \sin 3\lambda}, \quad \rho_{2,3} = \tfrac{\sin(3\lambda-u)}{\sin 3\lambda}, \quad \rho_{4,5} = \tfrac{\sin u}{\sin 3\lambda},$$
$$\rho_{6,7} = \tfrac{\sin u\,\sin(3\lambda-u)}{\sin 2\lambda \sin 3\lambda}, \quad \rho_8 = \tfrac{\sin(2\lambda-u)\sin(3\lambda-u)}{\sin 2\lambda \sin 3\lambda}, \quad \rho_9 = -\tfrac{\sin u\,\sin(\lambda-u)}{\sin 2\lambda \sin 3\lambda},$$

with loop fugacity $n = -2\cos 4\lambda$. The $u = 0$ values confirm the tile assignment: $X_j(0) = P_{00}+P_{X0}+P_{0X}+P_{XX} = 1$, matching the sourced factorisation into identity-arc triangles. The Hamiltonian limit $H_{\rm dil} = -\sum_j X_j'(0)$ is

$$H_{\rm dil} = \sum_j \Big[-\tfrac{P_{00}}{\sin 2\lambda} + \cot 3\lambda\,(P_{X0}+P_{0X}) + \tfrac{\sin 5\lambda}{\sin 2\lambda \sin 3\lambda} P_{XX} - \tfrac{b_j+d_j}{\sin 3\lambda} - \tfrac{h_j^+ + h_j^-}{\sin 2\lambda} + \tfrac{\sin\lambda}{\sin 2\lambda \sin 3\lambda}\, e_j \Big].$$

Every coupling is a function of λ alone. For a Temperley–Lieb type category with $d = 2\cos(\pi/(L+1))$ the equation $n = d$ has the two solutions $\lambda = \frac{\pi}{4}\frac{L}{L+1}$ (branch 1) and $\lambda = \frac{\pi}{4}\frac{L+2}{L+1}$ (branch 2), both critical, with $c = 1 - 6/(h(h-1))$, $h = L+2$ resp. $h = L+1$. Fibonacci is $L = 4$: branch 1 gives $c = 4/5$, branch 2 gives $c = 7/10$.

Proposition 86.4 [derived; honeycomb reduction at $u = \lambda$]. At $u = \lambda$ the nine-tile model is the honeycomb $O(n)$ loop model with edge fugacity $x = 1/(2\cos\lambda)$. On branch 1 this is Nienhuis's critical point $x_c(n) = 1/\sqrt{2+\sqrt{2-n}}$; on branch 2 it is the second-branch value $1/\sqrt{2-\sqrt{2-n}}$.

Proof. Split each square tile along the diagonal from the bottom-left to the top-right corner into two trivalent vertices, A with edges $\{\text{top}, \text{left}, \text{diag}\}$ and B with $\{\text{bottom}, \text{right}, \text{diag}\}$. Give each occupied half-edge of a tile weight $x^{1/2}$ and the internal diagonal weight x. Then the tiles weigh: $1 \mapsto 1$; $2, 3 \mapsto x$ (an arc inside A or B); $4, 5, 6, 7 \mapsto x^2$ (an arc through the diagonal); $8 \mapsto x^2$; $9 \mapsto 0$ (it would use the diagonal twice). At $u = \lambda$: $\rho_9 = 0$; $\rho_4 = \rho_5 = \rho_6 = \rho_7 = \rho_8 = \sin\lambda/\sin 3\lambda$; $\rho_{2,3} = \sin 2\lambda/\sin 3\lambda$; $\rho_1 = 1 + \sin\lambda/\sin 3\lambda = 2\sin 2\lambda\cos\lambda/\sin 3\lambda$. Hence $\rho_2/\rho_1 = 1/(2\cos\lambda) =: x$ and $\rho_4/\rho_1 = \sin\lambda/(4\sin\lambda\cos^2\lambda) = x^2$, consistent. Finally $x^{-2} = 4\cos^2\lambda = 2 + 2\cos 2\lambda$ and $\sqrt{2-n} = \sqrt{2+2\cos 4\lambda} = 2|\cos 2\lambda|$, so $x = x_c$ when $\cos 2\lambda \geq 0$ (branch 1, $\lambda < \pi/4$) and $x = 1/\sqrt{2-\sqrt{2-n}}$ when $\cos 2\lambda < 0$ (branch 2). □

Checked arithmetic (Fibonacci, $L = 4$).

branch	λ	$x = 1/(2\cos\lambda)$	Nienhuis value	c	couplings $(-\rho_1', -\rho_2', -\rho_4', -\rho_6', -\rho_8', -\rho_9')$
1	$\pi/5$	0.618034	$x_c = 0.618034$	4/5	$(-1.0515,\ -0.3249,\ -1.0515,\ -1.0515,\ 0,\ +0.6498$
2	$3\pi/10$	0.850651	$\tilde{x}_c = 0.850651$	7/10	$(-1.0515,\ -3.0777,\ -3.2361,\ -1.0515,\ -3.4026,\ +2.7$

The same identities hold for $L = 2, 3, 5$ (worklog 016). Note the sign of the e_j term: the integrable dilute chain has a *repulsive* cup-cap term on both branches, opposite to the dense golden chain, and on branch 1 the P_{XX} coupling vanishes because $\sin 5\lambda = \sin\pi = 0$.

81.6 What the Calibration Says About the Principle

For Temperley–Lieb type input the machine closes with no continuous choice. The category fixes $n = d$, hence λ up to a branch; the couplings of $H_{\rm dil}$ are functions of λ; the Hamiltonian, the isotropic point $u = 3\lambda/2$ and the honeycomb point $u = \lambda$ are one commuting family; the limit is critical with a sourced c. Isotropy is restored along the family, not imposed. The geometric parameter is the anisotropy u, not a coupling. The residual freedom is the branch: two critical points for $u > 0$, two more with an Ising factor for $u < 0$. Whether a principle selects among branches is open.

81.7 Beyond Loops: the Fibonacci Net Gas (Proposal)

For Fibonacci on $O = \mathbf{1} \oplus \tau$, $O \otimes O = 2\cdot\mathbf{1} \oplus 3\cdot\tau$, so $\mathrm{End}(O\otimes O) = M_2 \oplus M_3$ has dimension 13, while $\mathrm{dTL}_2(\varphi)$ has dimension 9 (CA-69). The four missing dimensions are the vertex tiles $\tau\otimes\tau \leftrightarrow \tau$ with one vacancy, which CA-74 proves lie outside the dilute loop algebra. The general reflection-symmetric nearest-neighbour Hamiltonian therefore has, beyond the loop couplings above, a vertex coupling v:

$$h = \big[\mathrm{diag}(\epsilon_{00}, \epsilon_1) + w(b+d)\big]_{\mathbf{1}\text{-block}} \oplus \big[\mathrm{diag}(\mu, \mu, \epsilon_\tau) + t(h^+ + h^-) + v(\nu + \nu^\dagger)\big]_{\tau\text{-block}},$$

with $\nu : |\tau\tau \to \tau\rangle \mapsto |\tau\mathbf{1}\rangle + |\mathbf{1}\tau\rangle$ built from the vertex morphism of $\mathcal{C}$ (normalisation to be fixed in CONVENTIONS before use). Lemma 86.1 turns any such h into a *trivalent net gas* in spacetime with per-tile weights $\delta\tau \cdot$ (coupling) and the $\mathcal{C}$-evaluation of the net. Two corners are calibrated on the quantum side: $\mu \to -\infty$ freezes every site to τ and recovers the dense chain ($c = 7/10$ or $4/5$ by sign); the $v = 0$ hyperplane is the parity-preserving dilute loop sector of CA-74, which contains H_{dil}. No integrable line is known once $v \neq 0$: the sourced statement is that beyond Temperley–Lieb, integrability requires further fine-tuning. Proposition 86.3 is therefore unavailable, and the comparison of rates is genuinely first-order: the open problem is to find the locus in (ϵ, μ, w, t, v) at which the net gas is critical. The first analytical tool to point at it is a lattice holomorphicity condition on a $Z(\mathcal{C})$-spin defect endpoint, which is the next shard.

82 The Net Case: the Dilute Fibonacci Chain Is the Golden Chain Blocked

82.1 Status, Sources, and Dependencies

Status: derived and checked. Lemma 87.1 is a categorical identity; Theorem 87.2 is derived by hand and checked by the reproducible run `runs/2026-09-23-net-tl-representation/` (script `scripts/python/net_tl_representation_check.py`, INDEX.md); Proposition 87.3 assembles three sourced identities; the last subsection is a proposal.

Dependencies: CA-65, CA-66, CA-69, CA-74, CA-86; CONVENTIONS.md (b), (r), (aj), (ak).

82.2 Blocking: When Dilute Is Dense in Disguise

Lemma 87.1 [derived]. Let $\mathcal{C}$ be a unitary fusion category and $O = \mathbf{1} \oplus X$ the site object of (r). If there is a simple Y with $Y \otimes \bar{Y} \cong \mathbf{1} \oplus X$, then for every L

$$O^{\otimes L} \;\cong\; (Y \otimes \bar{Y})^{\otimes L}, \qquad \mathrm{End}_{\mathcal{C}}(O^{\otimes L}) \;\cong\; \mathrm{End}_{\mathcal{C}}\left((Y\bar{Y})^{\otimes L}\right),$$

and a nearest-neighbour operator on O-sites $j, j+1$ is a range-four operator on the alternating $Y\bar{Y}$ chain, aligned to pairs. For self-dual Y the right-hand side is the dense Y-chain on $2L$ sites.

Proof. Tensor the isomorphism $O \cong Y \otimes \bar{Y}$ L times; End is functorial; a morphism on $O \otimes O \cong (Y\bar{Y})^{\otimes 2}$ has support on four consecutive Y-sites. □

Instances. Fibonacci: $\tau \otimes \tau = \mathbf{1} \oplus \tau$, so the dilute $\mathbf{1} \oplus \tau$ chain of L sites is the golden chain of $2L$ sites. The dimensions agree: CA-66 gives $\dim \mathcal{H}_L = F_{2L+1}$ for the dilute chain and $\dim \mathrm{End}(O^{\otimes L}) = F_{4L-1}$, which are the fusion-path count and $\sum_c m_c^2$ of $\tau^{\otimes 2L}$. Ising: $\sigma \otimes \sigma = \mathbf{1} \oplus \psi$. $SU(2)_k$: $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$, so the dilute spin-1 chain is the dense spin-$\frac{1}{2}$ chain. Haagerup H_3 with $X = \rho$: the sourced rule $\rho^2 = \mathbf{1} + (\mathbf{1} + a + a^2)\rho$ gives $d_\rho^2 = 1 + 3d_\rho$, so $d_\rho = (3 + \sqrt{13})/2$ and the invertibles have dimension 1; no simple Y has $d_Y^2 = 1 + d_\rho = (5 + \sqrt{13})/2$, so no such Y exists.

Consequence for CA-69/CA-74. The correct two-site algebra of the dilute Fibonacci chain is $\mathrm{End}(\tau^{\otimes 4})$, the Fibonacci quotient of $\mathrm{TL}_4(\varphi)$, of dimension 13; the dilute algebra $\mathrm{dTL}_2(\varphi)$ of dimension 9 is its parity-preserving subalgebra. The "vertex tile outside dTL" of CA-74 is a Temperley–Lieb generator acting across a pair boundary. For Fibonacci there is no net gas beyond loops: every trivalent history is a loop history of the blocked chain.

82.3 The Temperley–Lieb Generators in Tile Form

Write $\mathrm{TL}_{2L}(\varphi)$, $e_i^2 = \varphi e_i$, $e_i e_{i\pm1} e_i = e_i$, on the dilute chain as

$$e_{2j-1} = \varphi\, P_0^{(j)} \;(j = 1, \ldots, L), \qquad e_{2j} = F_{j,j+1} \;(j = 1, \ldots, L-1),$$

with $P_0^{(j)}$ the vacancy projector of site j (the pair $\tau_{2j-1}\tau_{2j}$ fused to $\mathbf{1}$) and $F \in \mathrm{End}(O \otimes O)$ unknown. In the pair basis $\{|\mathbf{11}\rangle, |\tau\tau; \mathbf{1}\rangle\} \oplus \{|\mathbf{1}\tau\rangle, |\tau\mathbf{1}\rangle, |\tau\tau; \tau\rangle\}$ of (aj) the relations $e_{2j\pm1}e_{2j}e_{2j\pm1} = e_{2j\pm1}$ force the vacancy-diagonal entries $\langle \mathbf{11}|F|\mathbf{11}\rangle = \langle \tau\mathbf{1}|F|\tau\mathbf{1}\rangle = \langle \mathbf{1}\tau|F|\mathbf{1}\tau\rangle = \varphi^{-1}$; $F^2 = \varphi F$ with reflection symmetry then forces rank one in each block.

Theorem 87.2 [derived; checked]. Up to the sign of the vertex entry, the unique reflection-symmetric solution is

$$F = \varphi\, |u\rangle\langle u| \;\oplus\; \varphi\, |w\rangle\langle w|, \qquad u = (\varphi^{-1},\ \varphi^{-1/2}), \qquad w = (\varphi^{-1},\ \varphi^{-1},\ \pm\varphi^{-3/2}),$$

with the cup sign + tied to the F-gauge of (b). All Temperley–Lieb relations, including $[F_{j,j+1}, F_{j+1,j+2}] = 0$, which involves the F-move, hold to machine precision for $L = 3, \ldots, 6$; the opposite cup sign fails

at 7.6×10^{-1}; a 1% mutation of the vertex entry fails at 3×10^{-3}. Moreover $H = -\sum_i e_i$ on the dilute chain of L sites and the golden chain $-\varphi \sum_j P_{\mathbf{1}}^{(j,j+1)}$ on $2L$ sites have identical dimension F_{2L+1} and identical spectra:

$N = 2L-1$	dim	E_0	first gaps
5	13	−6.5141330104	0.746587, 1.891335, 2.463767
9	89	−11.4453171319	0.483796, 1.270064, 1.624217
15	1597	−18.8535471512	0.316696, 0.841769, 1.066473

in both representations, digit for digit (run bundle). This is Lemma 87.1 made explicit: $F_{j,j+1} = \varphi P_{\mathbf{1}}^{(2j,2j+1)}$ rewritten in the pair basis. In tile language every one of the six couplings of CA-86 is on:

$$F = \tfrac{1}{\varphi} P_{00} + \tfrac{1}{\varphi}(b+d) + P_{\mathbf{1}}^{\tau\tau} \;\oplus\; \tfrac{1}{\varphi}(P_{0\tau} + P_{\tau 0}) + \tfrac{1}{\varphi}(h^+ + h^-) + \varphi^{-7/4}(\nu^R + \nu^L + \text{h.c.}) + \varphi^{-2} P_{\tau}^{\tau\tau},$$

with the raw cup b of (r) and the (aj)-normalised vertex tiles.

Colour reading. $u_1^2 = \varphi^{-2} = 1/Q$, $u_2^2 = \varphi^{-1} = (Q-1)/Q$, $w_1^2 = 1/Q$, $w_3^2 = \varphi^{-3} = (Q-2)/Q$ with $Q = \varphi + 1$: these are the probabilities that a Potts spin flipped uniformly over Q colours stays, creates a wall pair, hops a wall, or splits a wall into two. F is the flip generator of the $Q = \varphi + 1$ Potts chain in domain-wall variables, and H is that chain at its self-dual point. The gap ratios at $N = 15$, 2.658 and 3.368, sit near 8/3 and 10/3, the ratios $(3/5+1)/(3/5)$ and $2/(3/5)$ expected from the weights 3/5 and the level-two vacuum descendant of the sourced golden-chain spectrum.

82.4 The Isotropic Net Gas Is the $Q = \varphi + 1$ Potts Model

Proposition 87.3 [derived from sourced identities]. With the vertex normalisation (aj), the isotopy-invariant evaluation of a planar Fibonacci net N with V_3 trivalent vertices is

$$\langle N \rangle = \varphi^{-2}\, \varphi^{\frac{3}{4} V_3(N)}\, \chi_{\widehat{N}}(\varphi + 1),$$

$\chi_{\widehat{N}}$ the chromatic polynomial of the dual graph. Hence the isotropic net gas with edge fugacity x and vertex fugacity $y = \varphi^{-3/4}$ is

$$\sum_N x^{E(N)} y^{V_3(N)} \langle N \rangle = \varphi^{-2}\, Z_{\text{Potts}}(Q = \varphi + 1,\; K = x),$$

the low-temperature expansion of the $Q = \varphi + 1$ Potts model with domain walls on the net lattice and spins on its dual.

Proof. The sourced formula reads $\langle G \rangle = \varphi^{-5} \varphi^{\frac{3}{2} F(G)} \chi_{\widehat{G}}(\varphi + 1)$ for a connected net, $F(G)$ its number of faces. Euler with $3V_3 = 2E$ gives $F = 2 + \frac{1}{2} V_3$, so $\langle G \rangle = \varphi^{-2} \varphi^{\frac{3}{4} V_3} \chi_{\widehat{G}}$. Checks: a loop ($V_3 = 0$, $\chi = Q(Q-1) = \varphi^3$) gives φ; a theta ($V_3 = 2$, $\chi = Q(Q-1)(Q-2) = \varphi^2$) gives $\varphi^{3/2}$, the theta value of (aj). For k components $\langle N \rangle$ is multiplicative while $\widehat{N}$ is a k-fold one-point union, so $\chi_{\widehat{N}} = \prod_i \chi_{\widehat{N}_i} / Q^{k-1}$; the factors φ^{-2k} and $Q^{k-1} = \varphi^{2(k-1)}$ combine to φ^{-2}. The sourced low-temperature expansion $Z_{\text{Potts}} = \sum_N K^{E(N)} \chi_{\widehat{N}}(Q)$ then matches term by term at $y = \varphi^{-3/4}$. □

Since $Q = \varphi + 1 < 4$, the model has a critical point, in the universality class of completely packed loops with weight $\sqrt{Q} = \varphi$, i.e. the dense branch of CA-86 with $c = 7/10$. The sourced condition $e^{\beta_c J} - 1 = \sqrt{Q}$ is the square-lattice self-dual value; the triangular-lattice value for honeycomb domain walls needs a local source (open). Two remarks. Chromatic polynomials at non-integer Q

take both signs, so the net-gas weights are not positive: positivity lives in the fusion-path basis of the blocked chain, not in the net basis, exactly as the strange-correlator source warns. And the critical net gas is, by Lemma 87.1, the golden chain again, with the Potts critical point as its self-dual point.

82.5 What This Does to the Principle

Every calibrated critical point of the Fibonacci $\mathbf{1} \oplus \tau$ chain is now a point of the pair-aligned range-four golden chain family:

point	identity	c
dense, antiferromagnetic	golden chain	7/10
dense, ferromagnetic	Z_3 parafermion (sourced)	4/5
dilute $A_2^{(2)}$ branch 1 (CA-86)	range-four integrable point	4/5
dilute $A_2^{(2)}$ branch 2 (CA-86)	range-four integrable point	7/10
Potts / isotropic net gas	golden chain, self-dual	7/10
hard-hexagon strange correlator (sourced)	tau-projected string-net	4/5

Two universality classes, both minimal models with Fibonacci defect lines. For Fibonacci the trivalent vertex adds nothing that Temperley–Lieb does not already contain, and Intuition 3's "forced" creation and annihilation is the statement that pairs of τ fuse to the vacuum. The net-gas question is therefore not void in general but void for Fibonacci.

Proposal. The genuine net case is a category whose site object is not a product: Haagerup with $O = \mathbf{1} \oplus \rho$, for which $\mathrm{End}(O \otimes O)$ is not a Temperley–Lieb quotient and the vertex tiles $\rho \otimes \rho \to \rho$, $\alpha\rho$, $\alpha^2\rho$ carry three independent couplings. The blocked-chain trick is unavailable, no integrable line is known, and the isotropic net evaluation is not a chromatic polynomial. This is the concrete form of the acid test: the first computation is the Haagerup analogue of Theorem 87.2, namely whether any tile combination in $\mathrm{End}(O \otimes O)$ generates a known algebra with a critical Hamiltonian, and the second is the lattice holomorphicity condition of the sweep. Open items: Julia port of the run bundle; a local source for the triangular-lattice Potts critical coupling; and whether the range-four golden-chain family has any critical point outside $\{7/10, 4/5\}$.